

# EXHALE: Update Log (stage 2, from 2026-09-05)

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This log starts fresh on 2026-09-05. The two earlier logs are kept unchanged under their new names:

- docs/Update\_EXHALE\_stage0.{tex,pdf}: the early phase (the ATES fork up to 2026-08-15).
- docs/Update\_EXHALE\_stage1.{md,tex,pdf}: sections 1-171, 2026-06 to 2026-09-05. Every "Update\_EXHALE section N" reference in the source comments and in the older memos points into that file; the references were rewritten to the new name on 2026-09-05.

Each entry below is one dated section: what changed, why, which files, the gate that was run and its result, and the movement it produced against the scratch baseline or the goldens. Section numbers restart at 1. The plan the entries execute is docs/development\_plan\_20260905\_rev3.md (sections 1-10) with the order of docs/development\_plan\_20260905\_execution.md; item labels (A1, A8, 10.1-1, D0, C1 ...) are the plan's.

## 1. Plan, audit and decisions that open stage 2 (2026-09-05)

State of the tree at the start: HEAD 35d9dd5 (clean apart from docs/), binary rebuilt from it with gfortran 16.2 (md5 16cde610347c10c0a48db5c2a7fa6cbb). All twelve regression goldens differ from the case outputs (they are the pre-section-170 snapshots); by decision they are not refreshed until the end of Phase 1 (below).

Documents written today, all under docs/:

- physics\_numerics\_audit\_20260905.md (external audit of 35d9dd5), its diagnostics in audit\_20260905/.
- development\_plan\_20260905.md, \_rev1.md, \_rev2.md, \_rev3.md and the three reviews \_review.md, \_review2.md, \_review3.md: the plan and its revisions after each review. Rev 3 is the plan of record; its section 10 holds the independent five-group source audit (three new P0, twelve new P1, about fifty P2), with the reports and the verification probe in audit\_20260905/source\_audit/.
- development\_plan\_20260905\_execution.md: Steps 0-7, who acts, gates.

Decisions taken by the user today (recorded in rev 3 sections 8 and 10.5 and in the execution document, Step 0):

1. Golden are refreshed once, at the end of Phase 1; until then a scratch baseline (backup/regression/baseline\_post170\_<date>/, read through a REGRESSION\_GOLDEN\_DIR variable to be added to run\_check.sh) is the control for byte-identity. golden/ is not touched before that.
2. ROE interface: speed\_estimate\_ROE returns a status and, in the vacuum case, the two gas-edge speeds; Num\_flux gains an HLLE vacuum branch. Signature shown before coding.
3. One spectrum type builds every band. Power-law input: power law on the whole grid, XUV and below 13.6 eV alike (the present construction is the stated meaning of the key). A new Spectrum type: Planck gives the Planck photosphere from Stellar Teff and Stellar radius on the whole grid. Load: the table on the whole grid. The Balmer continuum follows the run's type. The setup report states the sub-13.6 eV source and the integrated grid flux.
4.  $m_{\text{He}}/m_{\text{H}} = 3.9715$  in the species table;  $R_J = 7.1492e9$  cm (IAU nominal equatorial) in the code and the Wind-AE bridge alike ( $R_0$  of every run +2.26%); MJ and MSUN to the IAU nominal values.
5. The du stop uses the signed spread of  $\rho v r^2$  over the mean, as flux\_spread does.
6. Every planet folder, benchmark and planet example runs on the SED its reference literature used or assumed, one file per planet under inputdata/sed/ with provenance; the input switch is a configuration change, the reruns stay instruction-gated.
7. A loaded SED that ends above the band an active absorber needs (He 2<sup>3</sup>S at 2583 Å, a low-IP metal at its threshold) stops the run; the message offers the two remedies (turn the absorber off in the input, or supply a covering SED). No continue key.

Open: decision 4 (A3 retry design: carrier substep retry or outer step controller); 2, 5, 6 inside D0; 7 before B2; 10 in Phase 5.

Documentation restructuring done today (this section): Update\_EXHALE.\* renamed to Update\_EXHALE\_stage1.\*, Update\_EXHALE\_early\_phase.\* to Update\_EXHALE\_stage0.\*; every reference in src/, docs/, python/, README\*.md, the notebooks and the workspace CLAUDE.md rewritten to the new names. Check: `git diff --numstat` shows additions equal to deletions in every one of the 112 touched files (no line moved; the three exceptions are the renamed logs themselves, which git sees as deleted paths). The rebuilt binary's md5 moved from 16cde610347c10c0a48db5c2a7fa6cbb to e779298d0b482bfb40ae2f9aa80d60a because four of the rewritten references are inside printed message strings (EXHALE\_main.f90 2602, ionization\_equilibrium.f90 2948, input\_read.f90 1004, diffusion\_tests.f90 1292), which is text the binary carries; no executable statement changed. The pointers in README.md, README\_HOWTO.md and the workspace CLAUDE.md name this file as the current log and the stage 1 file as the history.

Scratch baseline created: backup/regression/baseline\_post170\_20260905/, the Hydro\_ioniz.txt, Ion\_species.txt, \*\_adv.txt and run.log of the twelve cases as they sat in the case directories (48 files, 26 MB; provenance in the headers: git=6d07d48afd41 tree=dirty, the working tree that became 35d9dd5, runs of 2026-09-05). It is the byte-identity control for Phase 1 and is deleted after the golden refresh at the end of Phase 1.

## 2. Phase 0: the tests exist and are red on the known defects (2026-09-05)

Plan reference: `development_plan_20260905_execution.md` Step 1. Four workers, disjoint files; every result below was re-run by the advisor after the worker's report (times are the advisor's runs).

### 2.1 `src/tests/physics_probe/` (W1): 41 assertions, 15 PASS, 26 FAIL, 3.9 s

Derived from the frozen diagnostics in `docs/audit_20260905/` (which stay); links the production modules through a computed use closure (`source_closure.py`), so no module can be left out of the link by hand. Drivers named for the physics: `riemann_wave_speeds.f90` (LLF signal speed 1 against the required 3, mass flux -3.5 against -4.5, mirror symmetry; ROE normalization invariance broken by 0.661 and 3.162 in the two branches; NaN star speeds on the equal-pressure expansion; the production ROE flux leaves thermal energy -2.93 after one Euler update on the vacuum-producing expansion), `h2_channel_detector_ratio.f90` ((S+D)/M 0.2545 against the measured 0.2827, D/(S+D) 1/9 against 0.20; channel sum and stoichiometry GREEN), `h3p_cooling_limits.f90` (cooling independent of  $n(\text{H}_2)$  at 0, 1, 100  $\text{cm}^{-3}$ ; joins +0.430, -0.00113, -0.0235 at 300, 800, 1800 K; factor jump  $1.5\text{e-}3$  at  $1\text{e}14$ ), `h2_rovibrational_identity.f90` (chemistry  $T^2 \text{dln}Z/\text{dT}$  against the EOS  $u_{rv}$ : -0.63% to -8.29% over 300-8000 K), `photoevent_energy_ledger.f90` (H event at 20 eV closes exactly through the production photoelectron\_share; the H2 double event at 80 eV accounts for 60.28 of 80 eV, 19.72 eV unassigned), `radiation_exchange_ledger.py` (two-cell ownership rule, GREEN by construction).

### 2.2 `src/tests/grid_and_gates/` (W2): 6 programs, all RED, 13.4 s

`grid_width_identity.f90` (production `define_grid`): `dr_j(j)` against `r_edg(j) - r_edg(j-1)` off by 1.0145 on the Mixed grid at  $r_{\text{max}} = 10$ , 1.0100 at  $r_{\text{max}} = 2$ , 1.0046 on Stretched, exact on Uniform (where every cell has the same width, which is why the defect was invisible). `photon_grid_quadrature.f90` (production `set_energy_vectors`, `J_inc`, cross sections; `wasp_full` spectrum): code/exact P\_HI 1.0951 (floor 4.8 eV), 1.0890 (5.139 eV), 1.0003 (13.6 eV); P\_HeI 1.0157 and P\_HeII 1.0299 in every configuration; heating 0.99984. New from this test: P\_HeI carries the same threshold-bin defect (24.6 eV is a grid point whatever the floor), and the X-ray block ends at 1184.19 eV, not 1240 (share of P\_HI  $1.9\text{e-}10$ ). `mass_flux_spread_functional.py`: the coded `du` and the signed `flux_spread` on the baseline states (`lower_profile` 823.4 vs 3.61 with two sign changes in the window; `mol_base_handoff` 1.29 vs 0.696 with none; `wasp_full`  $9.85\text{e-}4$  vs  $9.85\text{e-}4$ ), and `du = 0` on a synthetic +F/-F window. `output_state_consistency.sh` (200-step `mol_base_handoff`): header `sec_ion=T`, `heat_total/heat` median 0.2752, largest difference 0.822. `base_level_single_statement.sh` (`lower_profile`, 2 steps): setup  $2.0035\text{e-}5$  bar against the profile's  $1.00\text{e-}6$  bar, factor 20.035. `restart_grid_guard.sh` (roundtrip on Mixed, restarted on Uniform with the same N): exit 0, the restart's  $r$  equals the file's Mixed radii exactly and sits 0.60 relative from the run's own grid.

### 2.3 `inputdata/sed/` (W4): the SED inventory of decision 16

Four files built from data already in the workspace, each with a provenance header, a build script under `inputdata/sed/build/`, and the coverage test `sed_coverage_and_order_check.py` (31 assertions, all PASS): WASP-52 b (eps Eri, MUSCLES v22, reproduces the existing planet-folder file to round-off), hot Uranus (the bench-

mark's Ribas band table continued to 2995 Å from the Gueymard composite it came from), HD 209458 b (WHI 2008 solar minimum at 0.04707 au, the p-winds file reproduced exactly), LHS 1140 b (GJ 1132 proxy). Not built, data missing: WASP-121 b (Huang 2023's LLmodels F5V spectrum above 1700 Å) and HD 189733 b (Salz 2016's composite; its solar table is not here). `inputdata/scaled_solar_hd189.sed` judged an orphan (no provenance, hardness inconsistent with its own normalization, ends at 1899.5 Å). Verbatim paper quotes and the open items are in `inputdata/sed/README.md`; rev 3 decision 16 records the status. Also found: WASP-52b/`input.inp` points at a directory that no longer exists (EXHALE/), the top-level Koskinen benchmark runs at 0.048 au against the paper's 0.05, HD209458b/`input.inp` at 0.048 au against the literature 0.047.

## 2.4 Regression harness and new cases (W3)

`run_check.sh`: REGRESSION\_GOLDEN\_DIR (default `golden/`; golden mode still writes only to `golden/`); the rebuild is fatal with its log kept in `backup/regression/.build.log`; the summary counts only the cases of the run; the binary copy is status-tested; FILES gains the two `_adv.txt` profiles (the post-processing pass was outside the regression); a file without a reference is MISS, not FAIL, and a run that compared nothing prints NO VERDICT; a restart case keeps its IC pair in `<case>/IC/` and the script restores it after clearing output/; `phase0-cases` prints the new case list. Makefile: `make check CASES=...`; `make test` runs `element_census_tests`, `diffusion_tests`, `residual_determinism` and every executable `src/tests/*/run.sh` found at run time, every suite even after a failure, nonzero exit if any failed. New cases, not in DEFAULT\_CASES (PHASE0\_CASES): `hydrostatic_column` (hot-Uranus gravity, irradiation 18 orders below normal, no key exists to switch cooling off so the gas is held where it is negligible; 300 steps, isothermal to 2%, but draining through the outflow boundary at 0.8 Mach, so it is a configuration for the Phase 4 D2 measurement, not a hydrostatic state), `oxygen_chemistry` (`examples/18_oxygen_chemistry` verbatim, 1000 steps, 3 min 52 s single-threaded under load), `hp_front`, `hp_zero_seed`, `hp_trace_seed` (restarts of the `mol_base_handoff` state with `Ionization transport: True`; the H II column set to 0 and to  $1e-12$  of the H nuclei by moving the protons into H I, since `load_IC` refuses a changed He/H; the three 100-step states are mutually distinct, so the seed reaches the solver). No dump hook asserts the seed at the entry of `solve_carriers`; the README names the two lines where one belongs. Measured (advisor re-run of two cases: 6.2 s, byte-identical on all four files): `check wasp_full mol_base_handoff` against the scratch baseline is byte-identical on all four files per case (W3, 13 min); the five new cases reproduce their own baseline copies; `make check CASES=mol_base_handoff` against the stale `golden/` reports the known  $8.3e-4$  and  $3.6e-4$  movements and MISS on the two `_adv` files; `make test` (31 s) runs five suites: `diffusion_tests` and `residual_determinism` PASS, `grid_and_gates` and `physics_probe` RED on the known defects as intended, and **element\_census\_tests does not compile at HEAD** (pre-existing: `carrier_state` gained `nrho` and `wfac` in section 158 and the test's two calls still pass ten arguments; `carrier_write_back` likewise takes `nrho` where the test passes `ntot`). Fixed in section 3 below.

Also found by W3: with the flux made negligible, the He recombination coupling warning (`ionization_equilibrium.f90` 947, 952) fires in every cell with a ratio of  $1e80$  because both sides sit at their guards; diagnostic only.

### 3. element\_census\_tests compiles again (2026-09-05)

src/tests/element\_census\_tests.f90: the two carrier\_state calls pass the nrho and wfac outputs the routine has had since stage 1 section 158, and the two carrier\_write\_back calls pass nrho (the density the carrier fraction is a fraction of) in place of ntot. Test only; no production source. Result: builds, 20 assertions PASS (E1 stoichiometry and charge, E2 elemental closure through the write-back including the CO collapse, E3 the H2 photoevent nuclei ledger, E4 the base H2 fraction). Until now the suite had been unbuildable, so its assertions had not been checked since section 158.

### 4. Phase 1 batch 2a: bounded corrections with no golden movement (2026-09-05)

Seven items, seven workers on disjoint files in one tree (private OBJDIR/EXE builds, no shared regression runs); one rebuild and one gate by the advisor afterwards. Plan references: rev 3 sections 4.1 (A2), 10.2 items 7, 8, 10, 12, 10.3, 10.5 decisions 3, 13, 17.

- **A2, ROE estimator and caller** (speed\_estimate\_ROE.f90 rewritten; Num\_Fluxes.f90 ROE branch; input\_read.f90; EXHALE\_main.f90): docs/a2\_roe\_interface.md implemented as approved: roe\_star\_state, status OK/VACUUM/INADMISSIBLE, the vacuum test first, PVRS kept only when admissible, two-rarefaction exponent  $1/z$ , two-shock densities times the upstream densities, HLLE flux on faces without an admissible star state with a printed counter, refusal of Numerical flux: ROE with molecular chemistry and the ladder EOS. Four deviations accepted: a positivity guard on the input states at the top of the estimator; the refusal placed in the consistency block after the optional keys (where thereis\_mol is known); the old p\_min floor removed (the guard supersedes it); the counter printed for ROE runs only. Test: 30 ROE assertions GREEN (scale invariance, equal-pressure expansion  $c_* \cdot 0.9576611$ , stationary contact, asymmetric shock and rarefaction against exact Riemann solutions at 5%, vacuum edges  $-6.127017/+6.127017$ , positive thermal energy after one Euler update).
- **2a-SED, decision 17** (sed\_read.f90): a table that stops above the band an active absorber needs stops the run with the file, the missing band in eV and A, the absorbers (He 2<sup>3</sup>S; each active neutral metal below the file minimum, K I at 4.341 eV included) and the two remedies; no continue key. Test sed\_coverage\_stop.sh (six assertions) GREEN; no configuration of record trips it (the Koskinen runs have the metastable off).
- **2a-OPAC** (util\_ion\_eq.f90 PH\_heat\_H/PH\_heat\_HHe, opacity\_models.f90): model P's factor multiplies the shared flux weight  $\text{int}_f$  of every absorption integrand (rates, heating, secondary ionization, acc\_q), so the beam's loss equals the local absorption; test opacity\_model\_p\_ledger.f90: a cell with  $f = 10$  closes energy and H I ionizations to  $3e-8$  (was a factor 10 off).
- **2a-LYA** (lya\_rt.f90, util\_ion\_eq.f90 fuv\_lw\_photon\_field, water\_photolysis.f90 header, plus the two call sites in ionization\_equilibrium.f90 and diffusive\_photochemistry.f90 that the new nHI argument required): band B2 is attenuated by  $\text{lya\_stellar\_beam\_transmission}(T, \tau) = \text{erfc}(x1/(\text{sqrt}(2) X_s))$ , the expression jlya\_escape\_prob carried inline and now calls (bitwise equal at 4941 test points). Test lya\_band\_transmission.f90 (ten assertions) GREEN. Movement, oxygen\_chemistry 1000-step snapshot: du 903 -> 596, log10 Mdot 9.62 -> 9.61, B2 transmission  $1e-291$  at the base,  $1e-3$  at  $1.07 R_p$ , 0.98 at  $1.5 R_p$ ;

H<sub>2</sub>O moves up to 6.2x and OH 7.3x in the layer. Scratch baseline of that case re-snapshotted (previous copy kept).

- **2a-WAE** (wae\_exhale\_input.f90): mass fractions from the same m\_He/m\_H the solver is handed; test: reconstructed He/H equals the input to 1e-12 (was 1.0071696x). RJ/MJ/MSUN untouched (decision 14 is 2c).
- **2a-HPFIX** (constrained\_chemical\_equilibrium.f90): hp\_is\_fixed pins species\_fixed(is\_HII) like H<sub>2</sub> and the oxygen carriers; new suite src/tests/constrained\_network\_layout/ (eight assertions) GREEN. The three hp\_\* cases never enter the continuation (every cell is a class-1 root), so no run exercises the fix yet.
- **2a-PLANCK** (J\_inc.f90, set\_energy\_vectors.f90, input\_read.f90, write\_setup\_report.f90, excited\_hydrogen.f90, docs/input\_schema.md): Spectrum type: Planck (requires Stellar Teff and Stellar radius) builds the whole grid from  $\pi B_{\nu}(T_{\text{eff}}) (R_{\text{star}}/a)^2$ ; the Balmer routines call the same shared function (unchanged result); the setup report states the spectrum type, the source of the band below 13.6 eV and the integrated grid flux against the nominal J\_XUV. New suite src/tests/spectrum\_type/ (12 assertions) GREEN. Measured on the wasp\_full star: Planck/power-law 1642 at 4.8 eV, 358 at 6.2 eV, 2.9 at 10 eV, 0.014 at 13.6 eV; He 2<sup>3</sup>S rate 45.2 s<sup>-1</sup> (Planck) vs 0.100 (power law); integrated grid flux / nominal 1.42 (power law, floor 4.8 eV), 0.995 (floor 13.6), 112.8 (Planck).
- **2a-GUARDS** (define\_grid.f90, set\_IC.f90, EXHALE\_main.f90, viscous\_conduction.f90, steady\_newton.f90, input\_read.f90, diffusive\_photochemistry.f90, water\_photolysis.f90, oxygen\_rates.f90, lyman\_werner.f90, parameters.f90, molecular\_infrared\_data.f90): cell\_nearest\_radius (the bare minloc read r(503) = 0 out of bounds on the default grid), j\_min >= 1, dead report\_unmet\_steady\_gates removed, Mdot guarded for small N and non-outflow, conduction floor counted and reported, okres read in the ray test, correct sweep count, get\_word initialized, dead close(unit=1) removed, adv\_corr zeroed on allocation, dead Tcode argument removed from four carrier routines and their callers, carrier\_mass\_amu through the named species indices, wp\_ready used as the guard it names, dead constants and stale headers fixed, molecular\_infrared\_data's i private. Test grid\_window\_indices (seven assertions) GREEN.

Gate, advisor's single rebuild (md5 b52c79ebb15ee9014858425bfea4cb1c): make test runs seven suites; every assertion added in this batch is GREEN and only the known defects of later items remain RED (two test drivers had linked the program object element\_census\_tests.o from build/; their link lists now exclude program objects). Regression, sixteen cases against the scratch baseline, single-threaded: fifteen byte-identical on all four files (the eleven of DEFAULT\_CASES, hydrostatic\_column, the three hp\_\*); oxygen\_chemistry moved as stated under 2a-LYA.

Follow-ups recorded for batch 2b and the docs pass: carrier\_mass\_amu returns 0 for a species missing from the table (needs a stop); write\_output.f90 806-816's FUV ledger closed form carries no line transmission for B2 (its rel\_diff reads 1.0 now; the LW row was already 0.84); inputdata/sed/README.md describes the old warning in six places; docs/input\_schema.md row 11a should state the coverage requirement; docs/a2\_oxygen\_option\_design.md 402 describes the pre-2a B2 field; the physics\_probe/README.md summary counts are stale; J\_XUV (used by the Wind-AE bridge), the FUV band fluxes and the monochromatic flux do not yet follow the spectrum type (decision 13 is enforced for the photon grid only); backup/regression/parse\_golden/ was already stale at 35d9dd5; the three audit probe scripts under docs/audit\_20260905/ no longer compile against the tree and are marked as frozen records.

## 5. Phase 1 batch 2b: bounded corrections that move some outputs (2026-09-05)

Six workers (five code items and one documentation pass), disjoint files, private builds; one rebuild (md5 d2173f0210abdc84f41cb6fb5076e18) and one gate by the advisor. Plan references: rev 3 sections 4.1 (A1, A3 increment (a), A8) and 10.3 (k\_lw cell mean, He I singlet, He II recombination cooling).

- **A1, LLF signal speed** (Num\_Fluxes.f90, RK\_rhs.f90, EXHALE\_main.f90):  $a1 = \max(\text{abs}(vL)+aL, \text{abs}(vR)+aR)$ , the spectral radius of the Euler flux Jacobian, replacing  $\max(\text{abs}(vL+aL), \text{abs}(vR+aR))$  (1 instead of 3 on the inflow test pair); the header's positivity claim now describes the coefficient the code carries. Reporting: calls of the positivity repair, repaired interfaces, repaired interfaces in accepted steps, and LLF faces, all printed even when zero. Test: the three LLF assertions GREEN. Movement: none (no case selects LLF and the repair path is never entered in the matrix: flux correction: 0 positivity repair call(s) in every run).
- **A3 increment (a), carrier reference scales and status** (diffusive\_photochemistry.f90, write\_setup\_report.f90, EXHALE\_main.f90): one carrier\_element\_reference\_density(ic, nH, n0, nC) (H2 and H+ hydrogen, OH and H2O oxygen, CO min(O, C)) read by both the Jacobian step (carrier\_source\_derivative\_step) and the residual floor; the reported residual is that of the state after limit\_to\_element\_budget; solve\_carriers returns converged / iteration cap / line search failed and the caller stops on a failed status in a carrier the limiter did not move (rejection and retry remain increment (b), decision 4 open); a cumulative CO-ceiling record (cells, applications, molecules removed) in the run summary and EXHALE\_resolved.out; carrier\_mass\_amu stops on a species missing from the table. New suite src/tests/carrier\_reference\_scales/ (14 assertions; before the change the proton row's diagonal derivative was exactly zero at  $H^+ = 0$  and  $1e-12$  n\_H because the step had collapsed to  $1e-300$ ). Movement of this item alone: hp\_zero\_seed up to  $7.8e-4$  at the ionization front, hp\_trace\_seed  $2.8e-5$ , hp\_front round-off, mol\_carrier and mol\_base\_handoff byte-identical. Cost of the extra residual assembly +1.5% wall time on hp\_zero\_seed.
- **A8, Chung detector inversion** (h2\_photo\_channels.f90): Table II is the detector event ratio  $(N_S+N_D)/N_M$  (Chung et al. 1993 p. 886, one pulse per event, quoted at the code site),  $q_D = N_D/(N_S+N_D) = 0.20$  at 80 eV;  $\sigma_M = (1-f_{di}) \sigma_i$ ,  $\sigma_S = (1-q_D) f_{di} \sigma_i$ ,  $\sigma_D = q_D f_{di} \sigma_i$ ; yan\_rho converted with  $q_D = \rho_D/(1+\rho_D)/f_{di}$ ; an inadmissible  $q_D$  stops the run instead of being clipped; frac\_H2\_double\_of\_protons renamed frac\_H2\_double\_of\_proton\_events. At 80 eV: D 0.02254 -> 0.04408, M 0.7972 -> 0.7796, S 0.1803 -> 0.1763. Test: 18 assertions GREEN including the stored-channel and yield integration checks; E1 test 4 prints the same  $6.951e-3$ . Movement of this item alone: mol\_base\_handoff n\_e up to  $1.8e-2$  at  $r = 1.018$ , x(H+)  $1.9e-2$  at the base, T  $4.6e-4$ ; mol\_sec\_ion at the  $1e-3$  to  $5e-3$  level, with a 0.078 cm/s velocity shift moving the  $v = 0$  crossing by one cell at  $r = 1.024$ - $1.026$  and flipping the advection-correction branch of the \_adv files there. Mdot unchanged.
- **k\_lw cell mean** (lyman\_werner.f90, util\_ion\_eq.f90 fuv\_lw\_photon\_field): new lyman\_werner\_dissociation\_rate\_cell\_mean(F, N\_out, N\_in, T, n\_H, tau\_out, tau\_in), a composite three-point Gauss rule on segments of at most 0.05 dex of column and 0.5 of continuum depth (the

level-resolved cross section is a piecewise power law with slope up to 3.3, so the water bands' exponential mean does not apply; measured accuracy  $2.5\text{e-}4$  worst over 1500 random cells); the star-ward face carries  $\text{NH2col}(j+1)$ . Test: seven assertions GREEN (the old inner-face rate was 3.18x too small on a cell spanning 1.12 dex). Movement: `mol_lyman_werner k_LW` +1.4% to +28.1% (median +2.7%), `n(H2)` -20% and `n(H3+)` -20% at the front  $r = 1.071$ ,  $T$  up to  $8\text{e-}3$ ,  $n_e$   $1.7\text{e-}2$ ; `mol_base_handoff` and `wasp_full` identical (no LW flux). Cost under 0.5%.

- **He I singlet and He II recombination cooling** (`util_ion_eq.f90 eval_cool/eval_cool_cells`, `Cool_coeff.f90`, `ionization_equilibrium.f90` comment): the 24.6 eV collisional ionization and the Cen (1992) excitation of He I are charged to the ground singlet `he_ground_singlet_density(nhei, nheiTR)` (the metastable had paid both levels); `lambda_rec_HeII = kT alpha_rec_HeII_total(T)` with `alpha_rec_HeII_total = alpha_11S + alpha_23S` when the metastable is tracked (the coefficient the balance removes  $\text{He}^+$  at) and `alpha_B` otherwise; the coefficient is built inside `eval_cool` from  $T$  and the run's flags so that `energy_semi_implicit`'s finite-difference  $dC/dT$  sees one function (a coefficient-passing form would have left an offset the derivative multiplies by  $1\text{e}5$ ). At  $1\text{e}4$  K the He II coefficient rises  $2.807\text{e-}13 \rightarrow 3.640\text{e-}13$  (+30%); the He recombination coupling's own weight (`y_net alpha_1`) is still not charged (about 10% of this channel, stated at the code site). Test: 15 assertions GREEN. Movement of this item alone: `wasp_full` median  $2\text{e-}5$  / max  $1.7\text{e-}4$  in  $\rho$ ,  $T$  max  $5.9\text{e-}4$ , Ca I and Na I up to  $4.7\text{e-}3$ ; `mol_base_handoff` cool column median 1% / max 2.6% (the recombination channel is 64% of the cooling there), state  $1\text{e-}4$ .
- **Documentation pass:** `inputdata/sed/README.md` (seven places, the stop), `docs/input_schema.md` (row 11a coverage rule; rows 11, 11b, 11d, K3, K4 line references corrected), `docs/a2_oxygen_option_design.md` (B2 row), `src/tests/physics_probe/README.md` (two summaries: 41/15/26 at 35d9dd5; 83 assertions, 69 PASS, 14 FAIL after 2a with the LLF fix already in the tree) and `grid_and_gates/README.md`, rev 3 section 10.2 and 10.5 "landed" notes, the execution document Steps 1 and 2a marked done, `README.md` and `README_HOWTO.md` (Spectrum type: Planck, the coverage rule, make test, the regression section's case list corrected from eight to eleven).

Gate (advisor): `make test` runs eight suites; every 2b assertion GREEN, RED only on the 2c and Phase 3 targets. Regression, sixteen cases against the post-2a scratch baseline, single-threaded: `wasp_he23off` and `hydrostatic_column` byte-identical (metastable off, no carriers); the other fourteen moved, `Mdot` unchanged to two decimals in all sixteen. Largest relative differences over all cells and columns (single cells; the large values are  $v$  near a zero crossing or trace species in one cell, see the items):

| case                          | Hydro_ioniz      | Ion_species      | Hydro_ioniz_adv  | Ion_species_adv  |
|-------------------------------|------------------|------------------|------------------|------------------|
| <code>wasp_full</code>        | $1.06\text{e-}3$ | $4.63\text{e-}3$ | $1.37\text{e-}3$ | $4.63\text{e-}3$ |
| <code>wasp_full_newton</code> | $1.48\text{e-}3$ | $4.69\text{e-}3$ | $1.91\text{e-}3$ | $4.69\text{e-}3$ |
| <code>mol_base_handoff</code> | $2.54\text{e-}2$ | $1.99\text{e-}2$ | $5.58\text{e-}2$ | $4.21\text{e-}2$ |
| <code>mol_metals</code>       | $2.16\text{e-}2$ | $2.20\text{e-}2$ | $2.70\text{e-}2$ | $2.20\text{e-}2$ |
| <code>mol_lyman_werner</code> | $1.25\text{e-}1$ | $2.06\text{e-}1$ | $5.63\text{e-}2$ | $2.06\text{e-}1$ |
| <code>mol_diffusion</code>    | $2.54\text{e-}2$ | $1.99\text{e-}2$ | $5.86\text{e-}1$ | $1.95\text{e-}2$ |

| case             | Hydro_ioniz                 | Ion_species | Hydro_ioniz_adv | Ion_species_adv           |
|------------------|-----------------------------|-------------|-----------------|---------------------------|
| mol_ir_bands     | 2.16e-2                     | 2.20e-2     | 9.65e-2         | 2.20e-2                   |
| mol_sec_ion      | 1.70 (v at a zero crossing) | 6.31e-3     | 1.70            | 1.0 (trace ion, one cell) |
| mol_carrier      | 2.52e-2                     | 1.98e-2     | 3.87e-2         | 1.95e-2                   |
| lower_profile    | 5.38e-1                     | 5.19e-2     | 5.38e-1         | 4.96e-2                   |
| oxygen_chemistry | 1.94                        | 1.0         | 1.94            | 1.0                       |
| hp_zero_seed     | 8.13e-2                     | 9.54e-2     | 4.20e-2         | 9.54e-2                   |
| hp_trace_seed    | 4.25e-1                     | 1.0         | 4.24e-2         | 1.0                       |
| hp_front         | 2.59e-2                     | 6.30e-3     | 4.42e-2         | 6.29e-3                   |

The scratch baseline was re-snapshotted from these outputs; the post-2a state of every case is kept in backup/regression/baseline\_post2a\_20260905/.

Follow-ups recorded: T\_equation.f90 (the temperature root of the advection post-process) carries no He 2<sup>3</sup>S cooling channel and charges the He I ground-state coefficients to the summed neutral column, so the \_adv temperature and the hydro temperature are not solved with one cooling function (Phase 3 A5/A6 or the \_adv consistency item of Phase 6); post\_process\_adv.f90 392 calls eval\_cool without nheitr; cross\_sec.f90 519-528, parameters.f90 475-501 and input\_read.f90 856-858 still describe the H+ column as a proton count and quote a "factor 1.6" between the two double-ionization models that is now reversed and much smaller; write\_output.f90 200-203 says the f\_shield column is the factor the rate carries (the rate is now the cell mean); pct\_limited counts only overshoots above 1e-10 while the clamp acts at any size (same class as the CO counter, now recorded cumulatively); the give-up path of retry\_step leaves the loop through the same exit as an accepted step (decision 4); set\_oxygen\_coeffs is called with strided sections that create array temporaries in the carrier Jacobian; docs/EXHALE\_user\_manual.tex is stale on the spectrum types and the SED stop.

## 6. Phase 1 batch 2c: corrections that move every output (2026-09-05)

Fourteen worker items on disjoint files plus advisor edits, every one measured alone against a frozen snapshot of the tree as it stood when the item started (private builds; the items therefore chain, and the "before" of one item is the "after" of those that landed earlier). One rebuild by the advisor with the conda gfortran 16.2 and its OpenBLAS (md5 5845df587aa7948153f1bed3becb28ac), make test, and one regression run against the post-2b scratch baseline. Plan references: rev 3 sections 10.1 (items 1-3), 10.2 (items 1-6, 9, 11), 10.5 (decisions 13-15, 17, 2c-CONST). Movement is quoted as the largest relative change over the 500 physical cells of wasp\_full (WASP-121b, He 2<sup>3</sup>S + metals, du stop, no Newton finish, so its Mdot carries the path-dependent spread of that stop) and mol\_base\_handoff (hot Uranus, 12000-step relaxation snapshot, not steady, so its outer-tail and base-cell numbers compare two points of one transient).

- **2c-DRJ, the cell width is the cell's own width** (define\_grid.f90, set\_IC.f90):  $dr\_j(j) = r\_edg(j) - r\_edg(j-1)$ , an exact identity of the finite-volume grid; the stored width had been the width of cell  $j+1$ , 0.7

- to 1.25 percent too large over the stretched region. Every consumer confirmed to use `dr_j(j)` for cell `j`; none changed. `grid_width_identity` GREEN. `wasp_full`: `T` 2.0e-3, `v` 5.4e-3 (base cell), `n_e` 6.3e-3, `x(H+)` 1.1e-2, mass flux +0.65 percent, printed `log10 Mdot` 13.24 both. `mol_base_handoff`: bulk profiles 1e-2 class with the transient tail at 1e-1, mass flux -0.96 percent, printed `Mdot` 10.52 -> 10.51.
- **2c-QUAD, thresholds on bin edges** (`set_energy_vectors.f90`): the photon grid is built from bin edges, `e_v` the geometric center of each bin and `de_v` its exact width; H I, He I, He II, the metastable edge and `e_mid`, `e_top` are edges, and the X-ray block reaches `e_top`. `photon_grid_quadrature` GREEN except P\_HeI at 1.2e-3 (closed by 2c-THR). `wasp_full`: `T` 7.5e-2 at 1.12 `Rp`, `n_e` 7.8e-2, `x(H+)` 0.23, base `v` 0.47, mass flux +0.82 percent, `log10 Mdot` 13.24 -> 13.25. `mol_base_handoff`: `T` 4.7e-2 (tail), `n_e` 2.5e-2, mass flux +0.08 percent, `Mdot` 10.51 both. This is the single largest movement of the batch after the Jupiter radius, and it is the removal of the 9.5 percent overcount of P\_HI found by the audit.
  - **2c-BALMER, the Balmer continuum follows the run's spectrum type** (`excited_hydrogen.f90`, `J_inc.f90`; decision 13): the  $n = 2$  photoionization rate and heating integrate the field the photon grid uses, the power law for a power-law run, the Planck field for Planck, the loaded table for Load (`stellar_flux_ev`, `spectrum_covers_ev`); a table that does not reach 3647 Å stops the run with the decision-17 message naming H( $n=2$ ). `balmer_continuum_field` GREEN (11 assertions). `wasp_full` ( $n = 2$  coupling armed): `gamma_2` falls by a factor 2939, `n_e` at 1.02 `Rp` -12.5 percent, `x(H+)` at the base -27 percent, `T` 9.6e-3, wind above 1.2 `Rp` and `Mdot` unchanged (13.25). `mol_base_handoff`: byte-identical (no stellar lines).
  - **2c-KICK, the momentum kick removed** (`EXHALE_main.f90`): the `u(2,:) + 1e-16` diagnostic aid is gone; the relative step change skips cells whose old momentum is exactly zero. `momentum_row_from_fluxes_only` GREEN. Movement round-off on `wasp_full` (1e-9), 4e-6 on `mol_base_handoff`, 4.4e-3 in `v` on `lower_profile` (cold-start snapshot with a sign-changing base flux amplifies it); `Mdot` unchanged everywhere.
  - **2c-STATE, one state per output file** (`EXHALE_main.f90`): when the secondary-ionization coupling never switched on during the march, the final write no longer forces it on; the header says `sec_ion=F`, and the `_adv` files and the heating breakdown are built from the relaxed state. `Hydro_ioniz.txt` / `Ion_species.txt` byte-identical in every case; the `_adv` files and `Heating_breakdown.txt` of the fixed-step snapshots whose flip never fired (`mol_base_handoff`, `lower_profile`, and the other molecular gates) change by up to 0.97 in `n(H II)_adv` at the base. `output_state_consistency` stays RED at 1.2e-5: the heat column is assembled from the pre-sweep rates and the breakdown re-evaluates on the post-sweep composition (Phase 3, together with the `excited_H_update` call between the loop and the final write).
  - **2c-DU, the signed mass-flux spread** (`EXHALE_main.f90`; decision 15): one functional `mass_flux_spread`,  $(\max - \min) / \max(|\text{mean}|, 1e-30)$  of the signed  $\rho v r^2$  over `j_min..N`, read by the stop, the Newton arming and the stage flip, and written to `run.log` at the final step (`mass_flux_spread_functional.py` GREEN). `wasp_full` byte-identical (every threshold crossing on the same step); `mol_base_handoff` byte-identical (fixed steps; the reported `du` falls from 1.29 to 0.70); `wasp_full_newton` flips and hands off one step earlier, solution moves by 7e-8.
  - **2c-RESTART, a restart keeps the run's grid** (`load_IC.f90`): the file's cell centers are compared with the run's over the physical cells and a relative difference above 1e-10 refuses the restart, naming both grids.

- restart\_grid\_guard.sh GREEN; roundtrip and the three hp\_\* cases byte-identical on the same grid. Consequence: after the grid-moving items of this batch (RJ +2.26 percent of the domain in R\_p units, dr\_j) the shipped hp\_\* IC pairs are refused, and are regenerated with backup/regression/make\_hp\_initial\_conditions.py from a fresh mol\_base\_handoff output (below).
- **2c-BASE, one base level with a lower-atmosphere profile** (input\_read.f90, write\_setup\_report.f90): with a profile, n0 follows from p\_match\_bar exactly as from p\_base of base.inp; a density key beside a profile is refused unless it agrees within 1 percent (base\_level\_single\_statement.sh GREEN). lower\_profile starts at the profile's level: base density  $1.0e14 \rightarrow 5.0e12 \text{ cm}^{-3}$ , its 12000-step snapshot changes entirely (printed log10 Mdot 8.44  $\rightarrow$  7.92, ionization front inward, base n\_e x8). The case README and examples/17\_lower\_profile/input.inp drop the density key. Nothing else moves.
  - **2c-CI, the C I Voronov row** (Cool\_coeff.f90):  $P = 0$ ,  $X = 0.193$ ,  $K = 0.25$ ,  $A = 6.85e-8$ ,  $dE = 11.26$  (Voronov 1997 Table 1; the transcription had X and K in the wrong slots, 2.56x at 2000 K, 1.69x at 1e4 K). voronov\_carbon\_ionization GREEN. wasp\_full 1e-5 class, x(C I) 6e-4, Mdot -1e-5 dex; mol\_metals bitwise identical (collisional ionization of C I is negligible at its temperatures).
  - **2c-MASS and 2c-MASS2, helium mass and Jupiter radius** (decision 14; species\_table.f90, parameters.f90, wae\_exhale\_input.f90, wae\_ic\_writer.f90, binary\_element\_diffusion.f90, diffusive\_photochemistry.f90, lower\_column.f90, utilities.f90, examples/exhale\_io.py and the Python tools): He I / He II / He III / He 2<sup>3</sup>S carry 3.9715 m\_H (HeH+ 4.9715), RJ = 7.1492e9 cm, MJ = 1.8982e30 g, Msun = 1.98842e33 g (IAU 2015 nominal), one definition each, the Python tools importing RJ\_CM, MJ\_G, MSUN\_G from exhale\_io. Tests atomic\_mass\_and\_radius\_constants, jupiter\_radius\_python\_tools.py, suite species\_masses GREEN. Measured as two sub-steps: the helium mass alone moves wasp\_full by 2.3e-3 in T, 1.1e-2 in x(H+), Mdot +0.095 percent, and mol\_base\_handoff by 1.2e-2 (tail), Mdot -0.004 percent; the Jupiter radius then rescales every physical length (R0, the domain top r\_max in R\_p units, the gravity) and moves wasp\_full by 7.3e-2 in T, 0.16 in n\_e, 0.34 in x(H+), raw Mdot +11.0 percent (printed 13.24  $\rightarrow$  13.29), and mol\_base\_handoff raw Mdot +15.8 percent (printed 10.52  $\rightarrow$  10.58). MASS2 alone (the remaining copies): mol\_diffusion 2.8e-5, mol\_carrier 6.5e-8, wasp\_full byte-identical.
  - **2c-THR, one definition of each threshold** (parameters.f90, cross\_sec.f90 and the files that restated a threshold): e\_th\_HI = 13.598434599, e\_th\_H2 = 15.425927, e\_th\_HeI = 24.587389, e\_th\_HeII = 54.417765, e\_th\_HeTR = 4.767775 eV (NIST ASD), every cross section turning on exactly there; sigma(E, Z) takes the ion's measured potential as an optional argument so He II is not restated as 13.6 Z<sup>2</sup>. ionization\_threshold\_turn\_on and threshold\_literal\_uniqueness.py GREEN; the P\_HeI residual of QUAD closes (0.99878  $\rightarrow$  0.99995). wasp\_full 7e-4 at most, mass flux +0.044 percent; mol\_base\_handoff 1.3e-3 (tail); printed Mdot unchanged.
  - **2c-N2FLOOR, H(n=2) is an absorber of the grid-floor rule** (sed\_read.f90 photon\_grid\_floor\_eV, J\_inc.f90, set\_energy\_vectors.f90, excited\_hydrogen.f90): with the excited-hydrogen coupling armed the grid floor is e\_th\_HI/4 = 3647 A, a loaded SED is read down to it and the decision-17 stop is about the file, not the type (WASP-52b with the eps Eri table to 3700 A now runs). Tests sed\_coverage\_stop.sh and

photon\_grid\_n2\_floor GREEN: the 20 bins added below the lowest bulk-gas absorber change no rate the grid computes, exactly. wasp\_full 2e-4 at most (the edge written as e\_th\_HI/4 instead of 3.40 eV), Mdot 0.0000 dex; wall clock +2.3 percent for the 20 bins.

- 2c-CONST, the particle masses and the astronomical constants in one place** (parameters.f90, species\_table.f90, lower\_column.f90, Cool\_coeff.f90, binary\_element\_diffusion.f90, diffusive\_photochemistry.f90, lyman\_werner.f90, oxygen\_rates.f90): m\_He\_atom = 6.6464790722e-24 g (4.002603254 u x CODATA 2018 u; the brief's 6.6464731e-24 was 9.2e-7 low and is not the helium-4 atom), amu, m\_e, m\_p in parameters.f90, m\_He\_over\_m\_H and amu\_over\_m\_H derived, bsp\_mass built from them, melem\_A = melem\_A\_u \* amu/mu. This closed a UNIT ERROR: the standard atomic weights in u had been added to mass sums carried in hydrogen atoms, every metal nucleus 0.78 percent too heavy in rho, the mean molecular weight, the base composition, the census closure and the metal element diffusion. The advisor closed the remaining local copies (Cool\_coeff line-center opacity now takes the atomic weight in u and uses the global amu; amu\_C/N/O = melem\_A\_u(iel\_\*); the local m\_amu\_g and kb\_erg/m\_h2 copies of the diffusion, Lyman-Werner and oxygen modules replaced by the globals). wasp\_full 1.0e-3 at most (base v), mass flux +3.2e-5; mol\_base\_handoff 2.8e-5; printed Mdot unchanged.
- 2c-METALEDGE, every active threshold a bin edge** (set\_energy\_vectors.f90): each of the six bands keeps its bin budget and is cut at every ionization threshold strictly inside it (H2 when the molecular chemistry is on, every photo-ionizable stage of an element with nonzero abundance), the budget shared over the sub-intervals in proportion to their logarithmic width with at least one bin each; NL is derived from the partition and equals the previous count in every configuration (240 wasp\_full, 220 mol\_metals, 220 mol\_base\_handoff). A band with no interior threshold keeps exactly the partition it had. photon\_grid\_threshold\_edges GREEN: seventeen thresholds at relative distance 0 from an edge (eighteen with H2), P\_m of C I 1.045 -> 1.000, Ca I 0.933 -> 0.997, Na I 1.030 -> 0.998, Fe I 0.965 -> 0.994 of a 400001-point reference, total H2 photoabsorption 0.985 -> 1.000. wasp\_full: T 9.6e-3, n\_e 5.2e-3, x(C I) 0.14 at the base, mass flux -0.17 percent, log10 Mdot 13.2860 -> 13.2852. mol\_metals: bulk 2e-3, n\_e 4.5e-2 in the neutral layer at 1.1 Rp (ionization degree 1e-6), Mdot unchanged. mol\_base\_handoff: 1e-4 class, by the H2 edge alone. Open (decision needed, below): Mg I, Ca I, Na I, Fe I still miss 1e-3 by 0.25 to 0.58 percent, the rectangle-rule error of the 20-bin sub-Lyman band.
- Advisor edits after the worker:** the H2 channel thresholds e\_th\_H2\_di (18.076 eV) and e\_th\_H2\_dd (51.4 eV) join the edge set when the molecular chemistry is on, because the channel shares of one cross section switch on there (the 32-41.5 eV neutral-dissociation window is continuous and is not cut; the comment at the code site says so); the setup report prints the grid floor as the lower edge of the first bin (photon\_grid\_floor\_eV, or the table floor for a loaded SED) instead of the first bin center, which the new partition had moved to 4.95 eV against a floor of 4.768 eV. The documents that restated the changed quantities were brought in line: docs/input\_schema.md (profile level, R\_J), docs/a2\_oxygen\_option\_design.md (molecular masses in m\_H units), rev 3 and the execution document (landed marks).

**Decision handed to the user: the sub-Lyman band budget**

The band [4.768, 13.598] eV has 20 bins (edge ratio 1.054), 4.4 times coarser in  $d \ln E$  than the pure-H I band (50 bins, 1.012), and it is where every low-IP metal is photoionized; Fe I's cross section falls by a factor 2.5 across the single bin above its threshold. Measured ladder (code/exact):

| num_TR         | 20      | 30      | 40      | 50      | 80      |
|----------------|---------|---------|---------|---------|---------|
| Mg I           | 0.99753 | 0.99871 | 0.99915 | 0.99955 | 0.99982 |
| Ca I           | 0.99737 | 0.99877 | 0.99937 | 0.99961 | 0.99984 |
| Na I           | 0.99752 | 0.99912 | 0.99958 | 0.99967 | 0.99987 |
| Fe I           | 0.99417 | 0.99685 | 0.99805 | 0.99883 | 0.99953 |
| Nl (wasp_full) | 240     | 250     | 260     | 270     | 300     |

**Decided by the user 2026-09-05:** num\_TR = 89, the budget that gives the band the bin ratio of the pure-H I band (0.0118 in  $\ln E$ ). Measured with the same driver: Mg I 0.999853, Ca I 0.999870, Na I 0.999902, Fe I 0.999615, so the four rates are asserted at the  $1e-3$  of every other ion. Nl becomes 309 for wasp\_full (was 240) and 289 for the hot-Uranus cases (was 220); the H(n=2) band keeps its 20 bins. Every He 2<sup>3</sup>S and metals-on case moves again by this, and the gate below was re-run on the 89-bin binary (md5 b0a4ada9f84ee8410c84fda6780ba097) before the baseline re-snapshot. The user also decided the LHS1140b/examples/photochem\_profile\_base/input.inp density key: deleted (the profile fixes the level).

**Gate (advisor)**

make test on the single rebuild: every assertion of this batch GREEN (threshold\_edges, n2\_floor, balmer\_continuum\_field, atomic\_mass\_and\_radius\_constants, species\_masses, ionization\_threshold\_turn\_on, threshold\_literal\_uniqueness, voronov\_carbon\_ionization, grid\_width\_identity, photon\_grid\_quadrature, restart\_grid\_guard, base\_level\_single\_statement, mass\_flux\_spread\_functional, momentum\_row\_from\_fluxes\_only, sed\_coverage\_stop); diffusion\_tests had 13 of 35 assertions fail because its own references carried the helium mass as  $4 m_H$  ( $7.2e-4 = 1 - 3.9715/4$ ), and now derives every helium mass from  $m_{He\_over\_m_H}$  (35 of 35, no tolerance changed, the T9 ambipolar targets written as functions of the mass ratio). RED, as intended, only on the Phase 3 targets: output\_state\_consistency ( $1.2e-5$ ), h3p\_cooling\_limits, h2\_rovibrational\_identity, h2\_double\_photoevent\_energy\_sum, helium\_collisional\_ionization\_by\_level.

Regression, sixteen cases against the post-2b scratch baseline, single-threaded: the thirteen cases that ran all moved beyond the  $1e-3$  tolerance, as every item of this batch predicts; the three hp\_\* restart cases were refused by the new grid guard (their IC pairs sat on the old grid: domain top 4.7253 -> 4.6215  $R_p$  from  $R_J$ , base spacing from  $dr_j$ ) and were regenerated with make\_hp\_initial\_conditions.py from the fresh mol\_base\_handoff/output (nuclei conserved row by row; H II as written / 0 /  $1e-12$  of the nuclei) and re-run. Movement of the thirteen (the cell-wise maxima compare the same cell index, whose radius moved by up to 2.2 percent with the domain, so they overstate the change at fixed radius; the Mdot column is the run's printed value):

| case               | log10 Mdot<br>before -> after | mass flux at<br>cell N-20 | max rel T | max rel v                                              | max rel rho | domain top<br>[R_p] |
|--------------------|-------------------------------|---------------------------|-----------|--------------------------------------------------------|-------------|---------------------|
| wasp_full          | 13.24 -> 13.29                | +6.2 %                    | 6.7e-02   | 7.6e-01                                                | 4.1e-01     | 1.6014 -><br>1.5666 |
| wasp_he23off       | 13.24 -> 13.29                | +6.2 %                    | 6.7e-02   | 7.7e-01                                                | 4.1e-01     | 1.6014 -><br>1.5666 |
| wasp_full_newton   | 13.21 -> 13.26                | +6.7 %                    | 1.3e-01   | 1.9e-01                                                | 3.4e-01     | 1.6014 -><br>1.5666 |
| mol_base_handoff   | 10.52 -> 10.57                | +8.3 %                    | 2.5e-01   | 1.1e-01                                                | 4.5e-01     | 4.7253 -><br>4.6215 |
| mol_metals         | 10.52 -> 10.58                | +8.3 %                    | 2.4e-01   | 1.1e-01                                                | 4.4e-01     | 4.7253 -><br>4.6215 |
| mol_lyman_werner   | 10.50 -> 10.55                | +8.2 %                    | 1.0e-01   | 1.2e-01                                                | 2.5e-01     | 4.7253 -><br>4.6215 |
| mol_diffusion      | 10.52 -> 10.57                | +8.3 %                    | 2.5e-01   | 1.1e-01                                                | 4.5e-01     | 4.7253 -><br>4.6215 |
| mol_ir_bands       | 10.52 -> 10.58                | +8.3 %                    | 2.4e-01   | 1.1e-01                                                | 4.4e-01     | 4.7253 -><br>4.6215 |
| mol_sec_ion        | 10.47 -> 10.52                | +8.2 %                    | 2.1e-01   | 3.7e+02 (sign<br>change of a<br>stagnant base<br>cell) | 3.8e-01     | 4.7253 -><br>4.6215 |
| mol_carrier        | 10.52 -> 10.58                | +8.4 %                    | 2.5e-01   | 1.1e-01                                                | 4.6e-01     | 4.7253 -><br>4.6215 |
| lower_profile      | 8.44 -> 7.95                  | -68.9 %                   | 2.0e+00   | 1.5e+03 (sign<br>change)                               | 9.7e-01     | 4.1487 -><br>4.0577 |
| hydrostatic_column | 8.25 -> 8.27                  | -0.1 %                    | 5.7e-02   | 4.3e-01                                                | 4.9e-01     | 2.9600 -><br>2.9595 |
| oxygen_chemistry   | 9.61 -> 9.60                  | -6.9 %                    | 7.3e-01   | 3.1e+01 (sign<br>change)                               | 8.9e-01     | 4.1487 -><br>4.0577 |

The +0.05 dex of every WASP-121b and hot-Uranus case is the Jupiter radius (item MASS, +11 and +16 percent in the raw flux at fixed cell, partly compensated by the shrunken domain at cell N-20); lower\_profile is item BASE (the run now starts at the profile's 1 microbar level); the oxygen\_chemistry and hydrostatic\_column moves are the grid and the threshold edges. The baseline was re-snapshotted after the hp\_\* re-run (the post-2b copy kept as baseline\_post2b\_20260905/).

#### Golden refresh: prepared, awaiting the user's command

golden/ was archived to backup/regression/golden\_pre170\_20260905/ with a README (provenance from the headers, unknowns marked), roundtrip was re-run on the post-2c binary, and DEFAULT\_CASES of run\_check.sh now

lists the five Phase 0 cases as well. The refresh itself, `run_check.sh` golden over the seventeen cases followed by `make check` and one `run_fcheck.sh`, is the user's command to give; until it is given the reference of record is the pre-2c state and `make check` reports every case as moved.

#### Gate on the 89-bin band (2026-09-06)

Fourteen cases re-run against the post-2c (20-bin) baseline with md5 `b0a4ada9f84ee8410c84fda6780ba097`: `hydrostatic_column` byte-identical (no sub-Lyman absorber), every other case within  $1.3\text{e-}3$  in any cell (`mol_metals`  $\rho$   $1.3\text{e-}3$ , `wasp_full`  $v$   $1.3\text{e-}3$  at the base) and printed `Mdot` unchanged, the size the 20  $\rightarrow$  89 ladder predicts; the `hp_*` IC pairs were regenerated on the 89-bin `mol_base_handoff` output and the three cases run (`Mdot` 10.56).

**oxygen\_chemistry stopped on the 89-bin grid, diagnosed.** At step 13, cell 7, the carrier Newton reached its iteration cap (30) with a relative OH row residual of 0.63 and the status stop of item 2b-A3a ended the run. Diagnosis (worker, measured on frozen snapshots): the Newton is correct (a root exists; the same solver with a larger budget reaches it in 141 iterations, the last three  $5.4\text{e-}2$ ,  $3.1\text{e-}9$ ,  $3.9\text{e-}12$ ) and its residual scale and the element limiter are not involved. The stiffness comes from upstream: at step 12 the ionization sweep accepted a non-root at cell 7 under the relaxation amnesty (`info` 4, normalized residual  $5.2\text{e}4$ ), which put  $n_e = 3.5\text{e}10$  and  $n(\text{H}^+) = 2.6\text{e}8$  into the frozen background where the previous step had  $1.1\text{e}3$  and 0, so the  $\text{H}^+ + \text{H}_2$  channel asked for two thirds of the cell's  $\text{H}_2$  in one 2.85 s step. On the 20-bin grid cell 7 never took that path. The " $7.8\text{e-}2$  carrier row" of the 20-bin summary is the STEADY residual of a relaxation snapshot (`dt`  $\rightarrow$  infinity), not the Newton's; the backward-Euler Newton converged to below  $6\text{e-}12$  on all 999 steps there. Two things landed: the iteration budget is 200 (the comment at the code site states the measurement), and the attribution test of the status stop, which asked "did an element constraint act" by carrier over the whole grid, now asks it by cell (the CO ceiling, acting in 160 cells of this case, had made any CO failure anywhere excusable, and had left OH rows it moved by 49 percent through the oxygen closure reported as the solver's); new suite `src/tests/carrier_constraint_attribution/`, 7 assertions, RED before. Open for the user: decision 4 (reject and retry the step on a shorter `dt`, the structure that also covers a step without a root) and the ionization sweep's relaxation amnesty, which handed the carrier system a non-root (acceptance policy, `ionization_equilibrium.f90` class 4).

#### Golden refresh executed (2026-09-06, on the user's instruction)

`run_check.sh` golden over sixteen cases: the eleven of the previous `DEFAULT_CASES`, `hydrostatic_column`, the three `hp_*` restart cases and `roundtrip`, all from the 89-bin binary `b0a4ada9f84ee8410c84fda6780ba097` (conda `gfortran` 16.2, `OpenBLAS`); the previous golden is `golden_pre170_20260905/.golden/` now carries the `*_adv.txt` files as well. `oxygen_chemistry` has no golden entry yet (it stops, above); it stays in `DEFAULT_CASES`, so `make check` reports it as the one failing case until the carrier solver question is settled. `make check` and one `run_fcheck.sh` follow in this section.

#### After the refresh, same day (2026-09-06): the loaded-SED grid, the carrier budget, hygiene, the SED files

- **2c-SEDGRID, the photon grid of a loaded SED** (`sed_read.f90`, `J_inc.f90`, `parameters.f90`, `set_energy_vectors.f90`; item C5 of the D0 draft): the Load path kept the table rows as bin points with

central-difference widths, so no threshold was a bin edge there, on the path every planet calculation of record uses (decision 16). Now the bins are built from edges (geometric means of neighboring rows, the table span as the two end edges), every active ionization threshold inside the span is inserted as an edge with the split bin's two halves carrying the row's flux (exact for the integrated flux), `e_v` the geometric center, `de_v` the exact width; `read_sed` receives the threshold list from `ionization_thresholds_active`, one definition. The table rows themselves are kept in `e_sed_node/F_sed_node` (undiluted), which `stellar_flux_eV`, `loaded_table_floor_eV` and `spectrum_covers_eV` now read, so the loaded field no longer depends on the binning. Test `sed_edges` (`src/tests/spectrum_type/`, 15 assertions) on the `eps Eri` table of WASP-52 b: thresholds at distance 0, flux identity to  $2e-16$ , rate deviations from the exact table integral  $P_{\text{HI}} 1.8e-3 \rightarrow 4.6e-7$ ,  $P_{\text{HeI}} 1.1e-3 \rightarrow 1.1e-6$ ,  $P_{\text{HeII}} 8.4e-3 \rightarrow 8.5e-6$ , worst metal  $3.2e-3 \rightarrow 5.5e-4$  (Mg II, the table's own 1 Å resolution). Movement: `benchmarks/wasp52` 2000-step marching stop, mass flux at cell N-20  $2.3e-6$ ,  $\log_{10} \dot{M}$  9.68 both, profiles up to  $2e-2$  at a relaxation point; `wasp_full` (power law) byte-identical on seven output files. No regression case loads a table, so no golden moves. Noticed, not fixed: `e_mid` is not an edge of the loaded grid (the LX/LEUV split it feeds moved 0.02 percent); `stop_if_balmer_band_unstated` prints bin centers as the band.

- **Carrier Newton budget** (`diffusive_photochemistry.f90`): `newton_maxit` 30  $\rightarrow$  200 after the diagnosis above; `oxygen_chemistry` runs its 1000 steps on the 89-bin grid ( $\log_{10} \dot{M}$  9.60) and is added to `golden/` below. The constraint attribution of the status stop, by cell, and its test suite `carrier_constraint_attribution` (7 assertions) landed with it.
- **Hygiene, D0 items C16 and C4**: the dummy `alpha`, threaded through `reconstruction_continuation_rhs`, `RK_rhs` and `Num_flux` and never read, is removed from the three signatures and ten call sites (`hydrostatic_column` byte-identical against a same-compiler control); `output_state_consistency.sh` and its README describe the rate lag (measured RED  $1.19e-5$  with the header now reading `sec_ion=F`) instead of the removed post-loop block. `docs/audit_20260905/roe_flux_vacuum_probe.f90` is an archived probe that no longer compiles against `speed_estimate_R0E`; left as the frozen record it is.
- **Document sync** (`docs/EXHALE_user_manual.tex` + PDF rebuilt, `docs/p44_base_sawtooth.md`, `write_output.f90` and `util_ion_eq.f90` comments, `write_setup_report.f90`): the manual carries the Planck type, the decision-17 stop, the six-band edge grid, the IAU constants and NIST thresholds, the signed `du`, the profile base level and the restart grid guard; NLTR is now counted on the Load path (the "no grid point below 13.6 eV" sentence had been printed for every loaded SED); `src/utis/glob.py`, which shadowed the standard library module, is `src/utis/EXHALE_interface_state.py` with its two importers updated. Reported, not changed: the GUI offers no Planck type; `global_parameters` exports an integer named `count` that shadows the intrinsic.
- **Stellar SED files** (`inputdata/sed/`, decision 16, source rule of 2026-09-06: MUSCLES data normalized as the planet's reference paper did): `wasp121b_solar_huang2023.txt` (recommended; WHI 2008 solar shape below 1700 Å,  $F(1-912 \text{ Å}) = 1.6e6$  and  $L_{\text{ya}} = 1.0e5$  as Huang et al. 2023 section 2.2 prescribe, 6459 K photosphere above; the  $1.6e6$  confirmed as the  $\lambda < 912 \text{ Å}$  band by Salz et al. 2019 section 7), `wasp121b_wasp17_huang2023.txt` (MUSCLES WASP-17 alternative; its 1700-2583 Å photosphere

is 1.8x fainter than the blackbody, the size of the He  $2^3S$  photoionization bias of the solar candidate), hd189733b\_epseri\_salz2016.txt (eps Eri, MUSCLES v22, normalized to Salz et al. 2016 Table 3; the proxy pairing is the one Salz et al. 2018 section 4.1.1 make), and hd189733b\_bourrier2020.txt (the released MOVES III tables, mean of Visits B-E, measured to be the flux at 1 au, scaled to 0.031 AU, eps Eri photosphere above 1600 Å identical to the Salz file's; 30 percent fainter in the Lyman-Werner band, 1.4x brighter in the EUV). Each file was read by EXHALE.x for two steps on the final binary without error. User decision: HD 189733 b runs at  $a = 0.031$  AU (Salz 2016 Table 2, the modeling paper of the comparison); the planet folder, benchmarks/hd189, examples 01-11 and 13, and params\_table.txt carry it (the regression inputs jfnk\_hd189\*, tpm\_hd189 and the archived copies keep 0.0330). No input.inp has been switched to a loaded SED yet (2d).

### Phase 1 closing gate (2026-09-06)

make test on the final binary b2c15e53c888d09fa90ffdad80443c3a: ten suites, eight PASS; RED only on the Phase 3 targets (output\_state\_consistency 1.19e-5, h3p\_cooling\_limits, h2\_rovibrational\_identity, h2\_double\_photoevent\_energy\_sum, helium\_collisional\_ionization\_by\_level). make check: sixteen DEFAULT\_CASES against the refreshed golden/, all four files byte-identical in every case (REGRESSION PASS); the sixteen goldens taken from the earlier 89-bin binary reproduce on the final one, which is the measured statement that the loaded-SED grid, the carrier budget, the alpha removal and the report edits leave the analytic-spectrum matrix untouched. run\_fcheck.sh result recorded below. run\_fcheck.sh: the -fcheck=bounds, do, mem build ran 1500 bounded HD 209458 b steps with no runtime trap (fcheck regression: CLEAN, log10 Mdot 9.59), and the production build was restored to b2c15e53c888d09fa90ffdad80443c3a. **Phase 1 is closed.** Open for the user before Phase 3: decision 4 (carrier step rejection/retry), the relaxation amnesty of the ionization sweep that handed the carrier system a non-root, the D0 draft (docs/d0\_governing\_system\_20260906.md, 33 consistency items, 15 questions) for acceptance, the switch of the planet inputs to their loaded SEDs (2d), the WASP-121 b SED choice (solar shape, WASP-17 shape, or the two joined at 1700 Å), and the sub-Lyman band budget of the Load grid (the table's own resolution is kept; Mg II stays at  $5.5e-4$ ).

## 7. PLAN\_20260906\_rev2, Step A: acceptance contracts and run modes (2026-09-06)

Plan of record for this stage: docs/PLAN\_20260906\_rev2.md (rev 0 and rev 1 and their two reviews are kept beside it; the execution document of 2026-09-05 points at rev 2). The decisions of To\_be\_determined\_by\_user\_20260906.md were judged against To\_be\_determined\_by\_user\_recommend\_20260906.md and adopted as rev 2 section 2 states.

Documents written (advisor): docs/a0\_run\_mode\_contract\_20260906.md (the three run states; Time stepping: Local and PTC are initialization only; physical time advances only by accepted global steps; headers gain mode= and t\_phys=; the proposed Run mode: init | phys key awaits the user's confirmation before A0-impl); docs/a2\_certification\_contract\_20260906.md (one evaluator, four contexts: probe, stationary trial, physical step, certification; statuses not\_applicable / evaluated / unavailable; one condition per active balance; what replaces what). Documents drafted by workers and verified for structure and wording:

docs/b1a\_active\_equation\_inventory\_20260906.md (flags, equations per flag, constraints, five validity states with producers, the prerequisite table for eight configurations; finding: only the hydro rows have a state-consistent evaluator; the composition projection, the rho rewrite and thirteen closures with stated domains are recorded nowhere) and docs/b1\_target\_system\_20260906\_draft.md (energy ledger with the shared H2 zero at  $v=0$ ,  $J=0$ ; chemistry mass-preserving; Lyman-alpha loss and return from one 2p balance, an escape-probability multiplier on the existing cooling explicitly not the target; the zero-net-current closure derived from Koskinen 2022 B9/B10 and shown to give exactly the ambipolar field binary\_element\_diffusion already computes; the trace-metal counter-flux; the rollback table of fourteen state-changing operations showing `u_old` is not a checkpoint; CO alternative A blocked on rate literature, alternative B specified and recommended; ten open questions).

- **A1 (carrier half)** (`diffusive_photochemistry.f90`; suite `carrier_returned_state_acceptance`, 36 assertions, plus one in `carrier_constraint_attribution`): the carrier update is accepted on the RETURNED state, every unconstrained row of every cell,  $|res|/\max(\text{row\_terms}, 1e-300) \leq 1e-8$  with the absolute floor inside the scale, finiteness of every row, independent of the termination branch; the relative-drop exit is `carrier_solve_stagnated`, a diagnostic; the decision is `carrier_returned_state_verdict`, a side-effect-free function returning verdict and reason for A3 to consume. Defect found and fixed: `limit_to_element_budget` marked a cell constrained only when the clamp exceeded `limit_report = 1e-10`, so smaller clamps left cells presented as Newton-measured. Impact: `mol_carrier` byte-identical (no changed branch reachable: all 499 transport steps of the first 500 ended at the floor); **oxygen\_chemistry refuses at step 4**: the CO thermal ceiling, a clamp without a rate, leaves a neighboring cell's CO row at  $1.0e-7$  of its terms (accepted steps before it:  $1.1e-9$ ,  $4.2e-9$ ), and the code no longer certifies it. That is the plan's position (B1 alternative B: a ceiling-active atmosphere is outside the validated set); the case can run only in A0's initialization mode once A0-impl exists, so it fails make check until then.
- **A1 (class half)** (`steady_residual.f90`, `steady_newton.f90`, `ionization_equilibrium.f90` comment block; suite `acceptance_classes`, 12 assertions): `n_cells_without_chemical_root(acc_n) = acc_n(4)`, one definition for the trial handling (a local-closure validity check, no stationary condition) and for the three points that declare a steady state solved; a class-4 cell in the state handed back is written but reported "NOT certified". Impact: `wasp_full_newton` and `mol_base_handoff` identical (no class-4 or class-5 acceptance in any ledger). Open: the marching stop cannot pass the count because the loop never receives the sweep ledger; A2 closes it.
- **C1** (in progress) and **A2** (steps 1-3 of the contract, in progress).
- **C1** (`inputdata/sed/README.md`, `build/sed_coverage_and_order_check.py`, suite `spectrum_type` driver `loaded_sed_table_semantics`, 78 assertions): a table value is a bin average on every shipped file (MUSCLES const-res, WHI, Gueymard: 1 A or 1 nm bins; Bourrier 0.016 A samples with no stated convention) except the blackbody part of the solar WASP-121 b candidate (point sample); the reader treats a row as a histogram in energy, exact for the band energy of a bin-averaged table, second order in the bin width otherwise (MEASURED order 2.0 to 2.9 through the production consumers). Coverage is now checked per configuration from the active

threshold inventory (2600.5 A metastable, 3647 A when both stellar keys arm H(n=2), the lowest active metal), 12 configurations. Findings closed the same day (advisor): sed\_read.f90 single-precision 1e-8 at two sites (6e-9 relative shift of every loaded energy); the HD 209458 b WHI file rebuilt to 5000 A (its 3000 A cut stopped the run in read\_sed naming H(n=2)); WASP-52b/input.inp SED path from before the 2026-07-14 move; the two solar build scripts' headers (2583 A / 4.80 eV). Remaining: the WASP-17 file begins at 10.0 A because the MUSCLES v24 table does (the grid top becomes 1239.84 eV; recorded).

- **BALMERQ** (excited\_hydrogen.f90; suite spectrum\_type driver balmer\_band\_quadrature, 21 assertions; balmer\_continuum\_field Planck references refined): the Balmer-continuum rate and heating integrals follow the field's own nodes in closed form (power-law segments of a loaded table and of the power-law type; 5-point Gauss on a geometric subdivision for Planck), replacing a fixed 400-point uniform rule that stepped over emission lines. On the eps Eri table of benchmarks/wasp52 the old rule was 2.9 percent low in the rate and 2.7 percent high in the heating (the Lyman-alpha line at 10.20 eV lies in the band); on the power law it carried 9.5e-5 / 6.0e-5 of truncation. Movement: benchmarks/wasp52 gamma\_2 +2.99 percent, n\_e 5.3e-4 at 1.09 Rp; wasp\_full 1e-8 class (sub-0.1 percent, no golden refresh). One worker reported two scratch runs killed by a pkill matching a shared basename: the common brief now says kill by PID after checking /proc/<pid>/cwd, never by name.
- **A2, steps 1-3** (src/modules/time\_step/certification.f90 new, steady\_residual.f90, steady\_newton.f90, EXHALE\_main.f90 call sites, utilities.f90 header, diffusive\_photochemistry.f90 save/restore pair; suite certification, 18 assertions): one evaluator builds a 19-entry inventory from the configuration flags and returns per entry a status (not\_applicable / evaluated / unavailable), the row measure (max over physical cells of |res|/max(scale, floor)), the volume average, the worst cell, finiteness and the scale text; hydro rows and carrier balances are evaluated (carriers inside an isolated workspace that saves and restores every module array carrier\_steady\_residual writes, asserted after every call), elemental transport, level equations and the seven System\_HeH\_\* closures are unavailable until step 4. The certification context runs at the marching stop (which now receives the sweep ledger, closing the A1 open item), the JFNK final and the PTC declaration; probe and trial decisions read one record, the merit rule unchanged. Exit status (user decision): a run that DECLARED a stationary state and failed certification exits 2 (stop 2); a run ending on a cap makes no claim, writes certified=F cert\_reason=no\_stationary\_claim and exits 0; run\_check.sh never tests the status. Five shell gates of grid\_and\_gates that asserted exit 0 from a full run now accept 2 with the reason stated at the code site (advisor). Impact: numeric content of wasp\_full, wasp\_full\_newton, mol\_carrier identical; headers gain certified=. **No configuration certifies today**, by construction (the closures are unavailable), and two measured findings stand on their own: the JFNK info = 0 is set at the loop top and never revised when the returned state is re-measured at its own composition (wasp\_full\_newton energy row 4.1e-4 against the 1e-5 target, 40x), and the du stop accepts hydro rows of order one on their own terms (wasp\_full at du = 1.0e-3: mass 1.3e-2, momentum 1.8, energy 0.72; at du = 2e-2 the same rows read 1.7, 2.0, 0.96, so the spread and the rows measure different things). Not implemented in this item (it would change the solve): the probe rule that a class-4 chemistry makes a Newton direction unusable. Tolerances inherited; the convergence study is owed.

- **B1 accepted (2026-09-06).** The user decided the fourteen open questions of docs/b1\_target\_system\_20260906\_draft.md as recommended (metal reservoir zero; electron mass convention; ioniz\_eq density intent(in); the residual H I excitation cooling under the Lyman-alpha ledger; trace metals in the diffusion matrix; H+ in the B9 set; CO exclusion adopted with the destruction model held for rate literature; H3+ low-collider policy; one H2 partition function; rollback of diagnostic extrema; identity row placement; the CO ceiling's validity category; O(1D) reservoir; the refused \_adv product rule). The accepted specification is docs/b1\_target\_system\_20260906.md (the draft kept and marked superseded). This is the Phase 2 gate of PLAN rev 2; B2 (the residual-controlled temperature solve) started the same day.
- **A3**(diffusive\_photochemistry.f90; suite carrier\_retry, 32 assertions): the carrier subsystem's checkpoint (carrier\_checkpoint\_take / restore / matches: carrier fractions, the module arrays of the A2 save/restore enumeration, every constraint and solve diagnostic, the verdict, the cumulative CO record), and the controller: a refused returned state restores the substep's entry state, halves the carrier substep with the background explicitly frozen (listed at the routine), covers the interval with accepted substeps, and on exhaustion (nine refusals, floor  $dt/2^8$ ) restores the ENTRY state and stops; f\_sp is written only for a covered interval. Two CO records (accepted, restored on refusal; attempted, never restored) and controller attempt statistics with the A0 ledger-family hook. Retried result equals the explicitly subdivided integration bitwise; step doubling gives first order (ratio 2.001). Impact: mol\_carrier (0 retried intervals in 11999), mol\_base\_handoff, oxygen\_chemistry (3 steps) byte-identical. **Finding:** the retry recovered the oxygen\_chemistry refusal ( $1.04e-7 \rightarrow 7.9e-9$  after ten halvings) because the row scale carries the  $1/dt$  time term, so any fixed clamp imbalance passes a fixed relative floor at a short enough substep; that is a defect of the acceptance measure shared by A1 and A2, fixed next (A1scale: the returned-state and stationary measures are judged against the physical terms of the row, transport and reaction, with the absolute floor; the time term stays only in the Newton's own iteration test).
- **B2**(energy\_semi\_implicit.f90; suite energy\_update, 20 assertions): the implicit temperature step of a cell,  $R(T) = u(T) - u(T_{old}) - a [\text{heat} - \text{cool}(T)] = 0$  with  $u(T)$  from the run's equation of state (caloric H2 included), is solved by a safeguarded Newton-bisection hybrid on a physical bracket ( $0.01 T_0$  to  $5e4$  K with H2,  $1e7$  K otherwise) to  $|R|/\text{scale} \leq 1e-9$ , the cooling written back is  $\text{cool}(T_{returned})$ , and every non-converged outcome is a named failure (ITER\_CAP / NO\_BRACKET / FLOOR / NONFINITE) with the cell, the residual and the missing volumetric source; no clamp remains, a floor is a failure, and the interim action is a stop that B3a replaces by rejecting the attempted step. The caller is untouched (optional status argument, energy\_update\_last\_status). RED kept in the suite: the two-iteration update on a falling cooling branch left  $|R|/\text{scale} = 2.9e-2$  and a 6.9 percent temperature error. Impact: wasp\_full transient  $5.4e-5$ , converged (same stop step 16257,  $\log_{10} Mdot$  13.29)  $1.6e-6$ ; mol\_base\_handoff  $4.9e-5$ ; hydrostatic\_column identical; no floor or failure fired anywhere; cost +9 percent end to end on wasp\_full. Follow-ups assigned: the identical clamp in the conduction stage (viscous\_conduction.f90 572-589; B2b), the certification's counting of energy-floor hits as an accepted correction (A2 step 4), the stale end-of-run floor report (A0-impl).
- **A1scale**(diffusive\_photochemistry.f90; carrier\_retry 42 assertions): the returned-state acceptance and the stationary carrier measure are judged against the row's PHYSICAL terms ( $|transport| + |reaction| +$

floor, the floor in rate units without the  $1/dt$  form); the time term  $|n - n_{old}|/dt$  stays only in the Newton's own iteration scale (row\_terms, bitwise unchanged), beside which row\_terms\_phys is accumulated. RED: a residual of  $1e-7$  of a row's physical terms read  $3.3e-8$  at  $dt$  and  $5.6e-9$  at  $dt/16$  on the old scale (crossing the  $1e-8$  floor with the defect unchanged); now  $1.0e-7$  at every  $dt$ . Impact: mol\_carrier, mol\_base\_handoff identical; oxygen\_chemistry is refused at step 2 (CO row  $5.5e-8$  beside a clamped cell; 37 retry attempts to the floor, all refused, entry state restored), and the old measure only postponed that refusal to step 88 on today's tree. Finding carried forward (A1scale2, needs its own impact measurement): with a  $dt$ -independent acceptance there is a shortest certifiable substep, since the Newton's stopping test is still relative to the full terms and the round-off it leaves grows as  $1/dt$  against the physical terms; the same measure in both places closes it. Also: cterms in steady\_newton.f90 (534, 645) is written and never read. **Consequence for the matrix:** oxygen\_chemistry cannot run under the physical acceptance while the rateless CO ceiling is active (B1 decision 7: exclusion); once A0-impl lands, the exhaustion action reads the run mode (initialization: record the refusal in the attempted ledger, write the state flagged, continue; physical: stop), which is the item A3b, and the case runs in init mode as the diagnostic it is.

- **B3a and B4 designs** (docs/b3a\_attempted\_step\_controller\_design\_20260906.md, docs/b4\_spatial\_operator\_design\_20260906.md, read-only workers; advisor decisions in their last sections). B3a: eight of the fourteen step operations have no returned-state verdict today (RK/BC, element diffusion, excited H, ioniz\_eq, the projection, Apply\_BC, conduction, Shapiro); checkpoint about 2.1 kB per cell. B4: species mass fluxes are not consistent with the mass flux today (equilibrium species are not transported and rewrite rho; carrier advection is a cell-centered product; element diffusion is in advective form; no species enthalpy transport); increments B4-1 to B4-6 with their expected movement. **Advisor correction to the B4 draft:** the formation-energy flux  $F_{form} = \sum_s \epsilon_s F_s$  is the RESERVOIR's transport and enters no thermal equation (photoheating already carries  $h_{nu} - I$ ; recombination emits  $I$ ); the draft had written it as a thermal source, which would heat a cell by the ionization energy leaving it. B4-2 becomes the diagnostic reservoir balance that the ledger identity T1.5 tests, plus the face-composition enthalpy (T-B4.5). The B4-2 pre-measurement (src/utis/formation\_energy\_flux\_diagnostic.py, section 7 of the design) stands as the size of that reservoir transport: on wasp\_full\_newton 28 percent of the absorbed heating leaves the column as advected ionization energy (median 14 percent of the local heating, 52 percent at the ionization front at 1.23 Rp); the present energy face flux is  $v(E + p)$  with the caloric H2 internal energy already inside  $E$ , addressed by the owning cell (T-B4.5). Findings: lower\_profile's golden mass flux is not flat ( $1.1e-2$ ), so continuity-based diagnostics on it discriminate nothing until it is Newton-finished; benchmarks/hd209/output integrates to cooling 24 percent above heating on its saved state.
- **B2b** (viscous\_conduction.f90; energy\_update suite second driver conduction\_floor\_tests, 24 assertions): the Crank-Nicolson conduction stage no longer clamps; a solved temperature at or below the shared EOS floor (imported from energy\_semi\_implicit, one declaration) is a FAILURE with status, cell, the solved temperature and the energy the floor would have injected; the tridiagonal solve refuses a zero or non-finite pivot; no partial state is written. The operator is in flux form and conserves thermal energy exactly on a closed column

- (MEASURED 0.0); as boundary-conditioned in a run it exchanges heat through the base Dirichlet anchor by design ( $3.4\text{e-}15$  up to that flux). Impact: byte-identical on `wasp_full`, `mol_base_handoff`, `hydrostatic_column` and on two conduction-on variants in which the stage runs; zero floor activations anywhere. B1 section 7.4 rows 9 and 11 updated (no clamp state left). Noticed: `benchmarks/koskinen2022_model_a/*` cannot restart (stored IC on the pre-2c grid,  $4.5\text{e-}4$  off; the same class as the `hp_*` regeneration).
- **B4-5 measurement** (`src/tests/grid_and_gates/hydrostatic_residual.f90`, design section 8): on the analytic isothermal column, through the production `Reconstruct`, `Num_flux`, `RK_rhs` and `Source`, the pressure-gravity pair is at design order in the interior ( $\text{dr}^{\wedge}1.96$  PLM,  $\text{dr}^{\wedge}1.98$  WENO3 at  $r = 2$ ; base cell  $1\text{e-}4$  of its weight, first order in the base spacing), and the  $O(1)$  hydrostatic imbalance of the mechanical column sits at the OUTER free-outflow cell: 0.75 of its own weight at every N under PLM (the zero-gradient ghost zeroes the MC forward difference, so cell N carries no pressure gradient),  $6.4\text{e-}2$  falling at order 1.4 under WENO3. A well-balanced pair (B4-5b) would not remove the drain; the outer boundary fill is the item (B4-4a, started). Two latent defects found: `define_grid`'s `tol = 1.0` initializer implied SAVE, so a second call in one process skipped the stretch-factor Newton (production calls it once; fixed by the advisor: `tol` reset before the loop); the case README's single `Grid cells` ladder does not refine the base cell (README updated).
  - **A2 step 4** (`certification.f90`, `binary_element_diffusion.f90`, `excited_hydrogen.f90`, `ionization_equilibrium.f90`; certification suite +7 driver and +15 shell assertions): the five entries that were unavailable are measured on the state without solving or writing anything (He/H partition and trace-element transport balances, He  $2^3\text{S}$  and H( $n=2$ ) level balances, the eliminated-species closure as the sweep's own normalized reaction residual at the given composition), each restoring every module array it installs, asserted on every call. Output content identical on seven cases. **What refuses certification now:** the three hydrodynamic rows on every marching case; the energy row alone on `wasp_full_newton` ( $4.1\text{e-}4$  at cell 376, the JFNK `info = 0` finding); the H2 carrier row on `mol_carrier`; and, new, the elemental transport on the diffusive runs (`mol_diffusion` partition  $6.4\text{e-}5$  at cell 203; `lower_profile` partition  $4.8\text{e-}3$ , trace metals  $1.2\text{e-}2$ , the first measurement of what the Picard alternation reaches). The closures and level balances are within tolerance everywhere ( $1\text{e-}16$  to  $1\text{e-}12$  of their turnover), so the equations that lacked an evaluator are the solved ones. Both floors are reported as attempt diagnostics (0 everywhere). `cert_tol_element` is inherited, not measured, and is now decisive for the diffusive runs: the convergence study of contract section 4 is owed.
  - **A1scale2** (`diffusive_photochemistry.f90`; `carrier_retry` 51 assertions): the carrier Newton stops when its residual is within its own tolerance of the full terms AND within the acceptance tolerance of the physical terms, or when the arithmetic forbids the second (residual at  $64 \text{ epsilon}$  of the full terms), a named outcome; the interim reason `unreachable_at_substep` for a refusal whose rows are all at round-off was superseded the same day by A1scale3 (below); the stop message reports the halvings made. Option (a) (stopping on the physical scale alone) was built, measured (one extra iteration every step, no rescue of the short substep) and declined. Correction of the A1scale report: the Newton reached one epsilon in two iterations on that column; the  $7.7\text{e-}5$  was the ninth attempt's state. Impact: `mol_carrier`, `mol_base_handoff` identical; `oxygen_chemistry` still refused after step 2 (now the unreachable reason, OH at cell 90, 15 attempts). Decision taken from its finding

(A1scale3, started): a row at the arithmetic round-off of its own terms is not evidence of a defect and is accepted as "round-off limited" with a flag and a count, so a trace carrier at the element floor does not refuse an interval; the physical-terms acceptance stays for every other row.

- **A0-impl** (input\_read.f90, parameters.f90, EXHALE\_main.f90, utilities.f90, load\_IC.f90, write\_setup\_report.f90, ionization\_equilibrium.f90 ledgers; suite run\_mode, 22 assertions): the run mode is in. Run mode: init | phys, default init in every configuration (advisor decision replacing the draft rule: the ordinary use is a stationary solution reached by marching and certified, which is initialization in the contract's terms; a physical integration is asked for); phys with Time stepping: Local refused at startup; in phys the handoff state is tested at step 0, the clock advances only after a complete accepted step (measured equal to the sum of accepted dt to the last bit, with and without forced rejections), two ledger families (ledger\_family in global\_parameters), headers gain mode= and, in phys, t\_phys=; a restart of an init snapshot in phys mode is the handoff at  $t = 0$  (contract section 3 clarified), and only a mode=phys header without a finite t\_phys is refused; init never exits nonzero for certification. Impact: no input changed; wasp\_full, hp\_front, hydrostatic\_column, mol\_base\_handoff bitwise identical. Both floor reports of EXHALE\_main.f90 now state the B2/B2b behavior.
- **COLit** (docs/co\_destruction\_rates\_literature\_20260906.md; publisher PDFs of Westley 1980 NSRDS-NBS 67, Tsai 2017, Wakelam 2012 added to references/): B1's CO choice is settled on physics: every neutral CO destruction channel carries a barrier of 6.7 eV or more, so the one-sided destruction model's  $\tau_{\text{dest}} \ll \tau_{\text{res}}$  fails at the ceiling's onset at every density (factor 9 to  $1.5e8$ ) and the exclusion (decision 7) is the physics. Findings: molecular\_reaction\_heat.f90 (H2, H+, He+, H2+, H3+, HeH+) and the oxygen Shomate table of oxygen\_rates.f90 carry different reference zeros (T1.9 one-table item, assigned as B3b-ENTH); CO2 is not in the species table (removes the one evaluated CO channel below 2500 K and an OH sink); the workspace VULCAN network's CO reactions all produce species EXHALE does not carry, VULCAN's stated validity is 500 to 2500 K, KIDA's 10 to 300 K. Requests for the user: the A&A papers Visser et al. 2009 (503, 323; the CO shielding functions), McElroy et al. 2013 (550, A36  
  - RATE12 file), Venot et al. 2012 (546, A43) and 2020 (634, A78), plus Moses et al. 2011 (ApJ 737, 15) and Tsai et al. 2021 (ApJ 923, 264), which this host cannot download.
- **A1scale3** (diffusive\_photochemistry.f90; carrier\_retry 61 assertions): a row at the arithmetic round-off of its own full terms (64 epsilon) is counted "round-off limited", flagged per row, and left out of the acceptance decision; every other row is judged on the physical terms at  $1e-8$  as before; the controller halves only for genuine refusals; the stop prints both scales for every refused row. Impact: mol\_carrier, mol\_base\_handoff bitwise identical (0 round-off limited rows); oxygen\_chemistry still refused at the same step and row (CO at cell 281,  $|res|/phys = 3.2e-8$ ,  $|res|/full = 2.2e-12$ , four orders above the round-off bound); the element limiter's imbalance one cell beyond a clamped cell, correctly refused. The count's accessor exists; its line in the run summary and the certification carrier entry are B3a's (owner of both files).
- **B3b-H2Q** (caloric\_eos.f90, mol\_rates.f90; h2\_rovibrational\_identity 19 assertions): one H2 partition

function over the observed Roueff et al. 2019 ladder (302 levels, zero at  $v=0, J=0$ ) gives  $u_{rv}(T)$  as its first moment and the chemistry's  $K_{eq}$  as its free energy; the Huber-Herzberg model constants are deleted. The RED (0.6 to 8 percent in  $u_{rv}$ ) was the model ladder's level placement (no centrifugal distortion:  $J^4$  signature, 100 percent of the gap to 2000 K), not a defect of the sum; independent check:  $K_{eq}$  against NIST-JANAF (Shomate) now 0.9997 to 1.0008 over 600 to 5000 K (was 1.010 to 1.101). Impact: `wasp_full` identical; `mol_base_handoff` T 1.3e-5, `n(H2)` 3.1e-4; `mol_diffusion` T 3.2e-5; `mol_metals` at 400 steps `n(H2)` 5.6e-3 at 1.17 Rp; `Mdot` unchanged. Findings: a third copy of the ladder sum in `molecular_infrared_cooling.f90` 212-215 (to fold into `h2_partition_function`); `element_census_tests` does not compile against the live tree (no `_free` argument, another worker's signature change; to fix at the next single build); the load average reached 38 with the parallel long runs, so future briefs pace the 12000-step impact runs.

- **A3b** (`diffusive_photochemistry.f90`; `carrier_retry` 89 assertions): the carrier controller's exhaustion reads the run mode and returns a status (`carrier_interval_covered` / `exhausted`) the caller reads. `phys`: unchanged (restore, print, stop until B3a takes the status). `init`: the entry state is restored, the interval is recorded in the attempts ledger (count, first/last step, worst row and both scales), the carrier history is marked not certifiable (`carrier_history_certifiable()`), and the march continues from the entry carriers. The accepted CO record is indexed by A0-impl's `ledger_family`. Impact: `mol_carrier`, `mol_base_handoff` bitwise identical; `oxygen_chemistry` (`init`, its default) now runs its 1000 steps, exit 0, with 998 exhausted intervals (first at step 2, worst CO at cell 313), `log10 Mdot` 9.61 against the 20-bin 9.60, and a final certification refusing on four carrier rows at 1.0: its carriers are frozen at the step-2 values for the whole run, which is exactly what the mark and the ledger say. The refused row is the real imbalance beside a clamped cell (`|res|/full` 4e-12 against a 1.4e-14 round-off bound), open physics of the excluded CO model, not a controller defect. Its golden is re-taken at the end of this series with the mark noted in the case README.
- **B5pre** (`steady_newton.f90`; suite `steady_completion_flag`): the JFNK and PTC completion flag is decided AFTER the certification of the returned state, from the same record the certification prints; a solve that met the gate at the loop top and not at the returned state ends `info = 2` naming the row. Measured on `wasp_full_newton`: loop-top `||R|| = 5.5e-6`, returned state 4.2e-4; the existing own-composition consistency pass opens the gap (0, 1, 2, 3, 8 sweeps: 5.5e-6, 1.9e-4, 3.2e-4, 3.9e-4, 4.2e-4), so the hydro-only Newton converges to a state that is not a root of the coupled hydro + composition system: the loop-top gate measures a lagged residual. Consequence today: EXHALE\_main treats `info = 2` as a solver failure and falls back to marching, which ends on a du plateau with a state that certifies worse than the JFNK state; assigned to B3a (`info = 2` = a failed stationarity claim, written uncertified, exit 2, no fallback) and to B5a (`steady_newton.f90`: the residual evaluated on the self-consistent composition of the current iterate, so gate, merit and returned state measure one thing). Also found: the JFNK stop message prints an unset `resid_th` (B3a); `certification_row_measure` and `relnorm_over_cells` are one expression written twice.
- **B3b-PE** (`h2_photo_channels.f90`; `photoevent_energy_ledger` 105 assertions): the 19.725 eV missing from the H2 double photoionization event is the kinetic energy release of the two protons: 51.4 eV is a vertical threshold (Yan et al. 1998 section 4), the asymptotic product energy is  $D_0(\text{H}_2) + 2 I(\text{H}) = 31.675$  eV, and energy

conservation at the threshold fixes the fragment kinetic energy with no free parameter (Franck-Condon check  $e^2/R_e = 19.42$  eV, 1.5 percent). Second defect found: the 33 to 41 eV neutral window dissociates into  $H(2p) + H(2s)$  (Chung et al. 1993 p. 887), so 20.398 eV per event is internal excitation leaving as prompt Lyman-alpha and two-photon continuum, which the integrand charged to heat. `h2_channel_energy_recipients` (reservoir, electron, fragment kinetic, radiated; sums to  $h\nu$ ) with the per-channel table and sources. Channels M and S already closed; every atomic absorber and all 17 metal ions correct (`wasp_full` byte-identical). Impact: `mol_sec_ion`  $T 4.2e-4$ , `heat_H2`  $4.5e-2$ ; `mol_base_handoff`  $T 2.1e-5$ ; `Mdot` unchanged. The four-line wiring of the two new terms into the heating integrand of `util_ion_eq.f90` is handed to B4-6 (owner of that file). Open: suprathreshold  $H_2$  dissociation credits heat but destroys no  $H_2$ ; the double channel has no absorber row of its own; channel S has no source for a fragment kinetic split.

- **B3b-H3** (`h3p_cooling.f90`; `h3p_cooling_limits` 17 assertions): the Miller et al. 2013 fits as the paper defines them: Table 5 is the LTE emission of one molecule per steradian ( $\Lambda = n(H_3^+) 4\pi E(T) s$ ), Table 6's  $s(T, n_{H_2})$  is scaled LTE with  $H_2$  the only collider (Oka and Epp coefficient; the paper's upper-limit caveat at the code site); below the table's  $1e6\text{ cm}^{-3}$  the non-LTE factor follows the exact collisional limit (linear in  $n_{H_2}$ , zero at zero), replacing the clamp at the lowest column; above  $1e14$  the table edge is held; the four Table 5 segments do not join (measured with the paper's coefficients: +43 percent at 300 K, -0.11 percent at 800 K, -2.4 percent at 1800 K) and the code joins them by a stated linear blend in  $\log_e E$  over the 5 percent below each join (Table 6's 300 K value supports the upper polynomial); B1 T6.3 reworded to say so. Published values reproduced per segment to the paper's precision. Impact: the  $H_3^+$  cooling peak unmoved;  $T$  moves  $1e-5$  to  $2.7e-4$  in the three molecular cases; the clamp removal changes cells beyond 1.22 Rp where  $n_{H_2} < 1e6$  by more than 50 percent at  $1e-10$  of the peak (emission from gas with no collision partners, gone); `wasp_full` identical. Four domain counters exposed (B6 wiring is the certification owner's).
- **COLit, completed with the user's PDFs** (Visser 2009, McElroy 2013, Venot 2012, 2020; `docs/co_destruction_rates_literature_20260906.md`, 1419 lines): CO photodissociation as Visser et al. define it (37 bands inside EXHALE's 912 to 1110 Å beam, unshielded  $2.6e-10\text{ s}^{-1}$  in the Draine field, shielding tables in  $N_{CO} \times N_{H_2}$  for four parameter sets,  $T_{ex}$  raised only to 512 K because no data exist for dissociation out of  $v'' = 1$ ); evaluated on the oxygen\_chemistry run's own base columns ( $\log N_{CO}$  18.96,  $\log N_{H_2}$  21.80) it gives  $\tau_{dest} = 2.8e8\text{ s}$ , twelve decades faster than the barrier channels and still two to six decades short of  $\tau_{dest} \ll \tau_{res}$  at the base; zero in `mol_base_handoff`, whose spectrum carries no flux longward of 911.8 Å. RATE12's paper prints no entries (the data file is still needed for the ion channels); Venot 2012 is valid 300 to 2500 K and 0.01 to 100 bar (EXHALE's base at  $1e-6$  bar is four decades below); none of the six sources covers the 3000 to 4000 K band where the ceiling fires. The CO exclusion (B1 decision 7) stands, now quantitatively. If a CO photodissociation term is ever added it is a fourth absorber on the existing Lyman-Werner beam with a second column ( $N_{CO}$ ) and the cell-mean form of `lyman_werner_dissociation_rate`.
- **TESTINFRA** (`grid_and_gates/run.sh`, `physics_probe/run.sh`, their READMEs): both suites take EXHALE\_OBJDIR, EXHALE\_EXE, EXHALE\_TEST\_OUT with the semantics of `spectrum_type/run.sh`; default be-

havior byte for byte as before; against a private build both run without touching build/ objects (the six grid\_and\_gates shell tests still place their run copies under build/tests/). A dead fallback that wrote a build stamp the closure never compiled is fixed. Also (advisor): the two Python source scanners now skip .ipynb\_checkpoints/ directories, which hold editor auto-save copies of four modules (git-ignored) and had made threshold\_literals\_written\_once red on a stale copy of cross\_sec.f90; the Makefile source list is explicit and never reached them.

- **B3b-ENTH** (molecular\_reaction\_heat.f90, oxygen\_rates.f90; 78 Fortran
  - 37 Python assertions): one table species\_formation\_energy(isp) for every species column (H/He stages, 27 metal stages cumulative from the neutral, H<sub>2</sub>/H<sub>2</sub><sup>+</sup>/H<sub>3</sub><sup>+</sup>/HeH<sup>+</sup>, OH/H<sub>2</sub>O/CO, plus free electron, H(n=2), O(1D)) from the single B1 reference; the 17 reactions of the H<sub>2</sub>/He network are a stoichiometry table and each heat a difference of table entries; the Shomate table stays with its reference stated, every use of it being a balanced-reaction difference (the zero cancels), and the oxygen and carbon entries of the one table are differences of it; one D0(H<sub>2</sub>) (mol\_rates::h2\_dissociation\_energy\_ev, reproduced by the fit to 1.5 meV). Physics defect fixed: the oxygen photolysis thresholds were 298.15 K reaction enthalpies; they are now 0 K dissociation energies (the lowest Shomate interval's F coefficient is the 0 K enthalpy), 0.9 to 1.0 percent lower, deposited heat +0.8 percent at the oxygen base cell. Impact: mol\_base\_handoff, mol\_sec\_ion, mol\_metals byte-identical; oxygen\_chemistry T +0.4 percent at 1.02 Rp, Mdot 9.61. Open for B3c: the undeposited collisional oxygen heats and the He 2<sup>3</sup>S associative branch (D0 C25; the table exists); the B4-2 Python diagnostic lacks four oxygen entries; formation\_energy\_report has no caller; a stale threshold comment in water\_photolysis.f90:281.
- **C3** (six input.inp and READMEs; inputdata/sed/README.md): the planet folders and benchmarks of HD 209458 b, HD 189733 b and WASP-121 b load their literature SED files (Spectrum type: Load from file...; Power-law index removed, LX/LEUV kept because mandatory and overwritten by the grid integral, which the setup report shows); every README records the configuration of record with the existing power-law products listed by mtime and the reruns left to a separate instruction; read\_sed accepts every file at its configuration's floor (the H(n=2) edge in all six). log F<sub>XUV</sub> at the orbit moves 3.193 -> 3.127 (HD 209458 b), 4.438 -> 4.311 (HD 189733 b; the recomputed LX/LEUV reproduce Salz Table 3), 6.205 -> 6.204 (WASP-121 b). Advisor follow-ups: HD 209458 b's orbital distance set to 0.04707 AU (Oklopčić and Hirata 2018, the reference the run is compared with; the input said 0.0480 while the file was scaled to 0.04707); the setup report's flux ratio format (F8.3 overflowed when the sub-Lyman band dominates); the three benchmark directories added to the coverage script's inventory (all pass). Known: Load IC? True in the two planet folders is broken by the grid guard (pre-rebaseline IC), the same class as the hp\_\* regeneration.
- **COLit, revised with RATE22, Moses 2011, Tsai 2021** (document now 1956 lines): RATE22 (Millar et al. 2024, A&A 682, A109) carries 'He+ + CO -> C+
  - O + Heat 1.6e-9 cm<sup>3</sup> s<sup>-1</sup>, measured, accuracy A, all products EXHALE species; it has noH+ + CO, CO + MorCO + H2destruction entry, and itsTu = 41000on 6562 entries is

an undocumented "no upper bound" marker. On the oxygen\_chemistry profile the He+ channel with the Visser shielding gives  $\tau_{\text{dest}}/\tau_{\text{res}}$  from  $6.7 \times 10^2$  at the base to  $9.6 \times 10^{-5}$  at  $1.14 R_p$ : \*\*the one-sided destruction model is constructible above the helium ionization front and not below it\*\*; the boundary is compositional, not the ceiling's thermal switch. Decision 7 (exclusion) stays until T5.1 is implemented; B1 section 5.2 revised, B3b-CO scheduled after B3c. Moses 2011: no stated envelope, one fit interval 500 to 2500 K, no shielding. Tsai 2021: 0.1 nm continuum opacities, no CO self-shielding; EXHALE's `slyman_werner.f90` is the working precedent. The literature requests are closed except Baulch 1976 vol. 3, which blocks nothing.

- **B4-6** (`utilities.f90` `cell_mean_attenuation`, `util_ion_eq.f90`; `xuv_cell_mean_attenuation` 56 assertions): the photoionization and photoheating sums of a cell use the exact cell mean of the attenuation,  $\exp(-\tau_{\text{out}}) (1 - \exp(-d\tau))/d\tau$  (composite Gauss when the 2D correction is on), instead of the inner-face value the top-down columns gave; one definition for H, He, He 2<sup>3</sup>S, H<sub>2</sub>'s four channels, the metals, the single-absorber rates, `q_abs` and the secondary-ionization integrands. Inner-face error: 0.5 percent at  $d\tau = 0.01$ , 42 percent at

1. Photon conservation on the written states: cell mean 1.000000000 of the incident flux, inner face 0.9725 (`wasp_full`), 0.9843, 0.9849 (the loss sat at the ionization front). Impact: `wasp_full` (24000 steps)  $x(H^+)$   $8.3 \times 10^{-2}$  at  $1.32 R_p$ ,  $\dot{M}$  +0.74 percent; `mol_base_handoff`  $\dot{M}$  +0.93; `mol_sec_ion` +1.01; `hydrostatic_column` unchanged. B3b-PE's two heating terms wired into the integrand (its own numbers reproduced cell for cell). Advisor: the single-precision  $1.0 \times 10^{-18}$  cross-section unit literal (33 sites,  $4 \times 10^{-8}$  relative on every rate) made double. Finding to resolve at the gate: `wasp_full` on the mid-series tree plateaus at  $du = 4.2 \times 10^{-3}$  from step 22000 and never reaches  $1 \times 10^{-3}$  (the golden stopped at 16230), identical in the item's before binary, so it belongs to an in-flight change (B3a edits `EXHALE_main.f90`) or to an earlier landed one; measured after B3a.

- **A2tol** (`measurement`; `a2_certification_contract_20260906.md` section 9): the `du` stop and the hydrodynamic row measures are not measures of one thing (over a decade of `du` on `wasp_full` the mass row falls at order 0.2, the momentum row rises, the energy row is not monotone; `mol_carrier`'s carrier row falls at order 0.08 over 12000 to 48000 steps); the JFNK returns an energy row 41 to 47 times its target at every rung of a two-decade `Resid tol` ladder (structural; B5a); the elemental transport rows cannot be measured on the marching route (the operator advances  $1 \times 10^{-6}$  of the way per CFL step) and the diffusive cases never reach the steady route (`du` stalls at 5 to  $6 \times 10^{-2}$  against the  $1 \times 10^{-2}$  hand-off). Round-off floors on the converged JFNK state: mass  $1.0 \times 10^{-13}$ , momentum  $5.1 \times 10^{-12}$ , energy  $1.6 \times 10^{-11}$  (residual bitwise reproducible). Recommended and adopted (applied by B3a in `certification.f90`, provisional until B5a): `cert_tol_mass`  $3 \times 10^{-12}$ , `cert_tol_momentum`  $1 \times 10^{-8}$ , `cert_tol_energy`  $5 \times 10^{-5}$ , `cert_tol_element`  $1 \times 10^{-8}$  (with the homopause regime caveat), carrier, closure and level tolerances kept. At these values `wasp_full_newton` certifies with `Resid tol`:  $1.0 \times 10^{-7}$  (marginally at  $1 \times 10^{-6}$ ) and nothing as shipped. Noted for the user: the shipped default `Resid tol`  $1 \times 10^{-5}$  is one to two decades looser than the solve reaches at no cost.

- **B5a** (steady\_newton.f90; suite steady\_selfconsistent\_residual): eval\_residual eliminates the composition AT the state it evaluates (the equilibrium sweep iterated at fixed conserved variables until the composition stops moving), so the loop-top gate, the line-search merit, the returned state and the certification measure one object; the gate equals the certification to  $1e-12$  by construction. Finding: the eliminated residual's Jacobian carries the summed Picard response  $(I - M)^{-1}$ , and a finite-difference probe with a convergence test stops after one pass (the seed already is the state's composition), so the Newton direction ascended; the probes now run a FIXED sweep count taken from an elimination seeded at a different state (about 15 terms), and the "frozen-composition Jacobian" the brief offered as a fallback is exactly the arrangement that fails. wasp\_full\_newton: info = 0 at  $\|R\| = 4.4e-7$  in 9 iterations and CERTIFIED (was info = 2 at  $4.8e-5$  after 26), state moved  $1.0e-4$  at most,  $\log_{10} \text{Mdot}$  13.27, cost 1.49x. mol\_base\_handoff identical. Open: Resid tol values were calibrated on the lagged measure; the other Solver: Newton cases and the PTC route are measured at the series gate; n\_selfconsistent\_max comment stale (B3a); the duplicated row-measure expression (B3a).
- **HYG-absorbed** (water\_photolysis.f90, utilities.f90 comment; driver absorbed\_fraction\_switch, 5 assertions): the FUV band rates take the XUV beam's absorbed\_fraction\_per\_unit\_depth by use association; the local copy was bitwise the same over 1501 points and is deleted; the series switch and its error bound (about  $1e-8$  on the closed-form side of  $d = 1e-8$ ) stated once; mol\_base\_handoff, oxygen\_chemistry byte-identical. Advisor: the B4-6 driver's reference carried the same single-precision  $1.0e-18$  the production code had; made double with the production change, the driver is green again (51 assertions had turned red on the  $4e-8$  mismatch).
- **B3a** (EXHALE\_main.f90, new attempted\_step.f90, certification.f90, parameters.f90; suite attempted\_step, 87 assertions): a marching step is one trial of the fourteen operations from a checkpoint covering everything rows 1 to 12 write, evaluated at one adoption boundary before update\_map\_end\_step, retaken at half dt from the checkpoint when refused (cap 8, floor  $dt/2^8$ , a reduced dt bounding the next step); the clock and the accepted-step count move only there; exhaustion stops in phys and continues flagged in init. Impact: wasp\_full (300 and its du stop at 14496 steps,  $\log_{10} \text{Mdot}$  13.33), hydrostatic\_column, mol\_base\_handoff, mol\_carrier bitwise identical against a control built from the same snapshot; cost within run-to-run spread. Measured and reported, not gating: the time-discrete hydro row (1.92 at a cold-start base where the momentum scale vanishes) and the energy identity T1.5 (-0.20 to -0.74 on wasp\_full, the size of the row-8 projection, the RED of AT-1a); the step-doubling estimate gives order 0.89 for the whole step (the operator split is first order), cost 12 percent at n\_err\_every = 20. Two defects of the item caught by its own impact runs and fixed. The mid-item requests landed (A2tol tolerances, info = 2 written uncertified with no fallback, the ledgers in the summary, the H3+ domain counters, one row-measure definition, the n\_selfconsistent\_max comment). The B4-6 "du plateau" was the mid-edit tree; the final tree reaches its du stop. Advisor decision on the exit status: a run that DECLARES a stationary state and fails certification exits 2 in every mode (A0's init exemption covers physical-step claims only); the mode gate is removed (A2exit). Small items of its report assigned (HYG-B3a: operator status on every path, conduction accessor, step-level carrier verdict, optional volume measure) and the -00 floating-point exception on hydrostatic\_column under diagnosis (O0FPE).

- **B4-6b** (`lya_rt.f90`, `excited_hydrogen.f90` field evaluation; driver `lya_beam_cell_mean`, 37 assertions): the stellar Lyman-alpha depth was a trapezoid between cell centers (neither the cell's face depth nor its own, every depth short by half the outermost cell); it is now the rectangle rule of `calc_column_dens` on the cell's own opacity, with the cell's own depth and its star-ward face as outputs. The beam transmission is  $\text{erfc}(c \sqrt{\tau})$ , not exponential, so its cell mean is a 3-point Gauss rule in  $u = \sqrt{\tau}$  (mean and mean of the square, since the trapping makes the field quadratic in the transmission); the photosphere factor's mean is closed form with  $\log_{10}$ . `jlya_escape_prob` keeps the cell center by Neufeld 1990 section II (the escape fraction is a point property of where the photon is created). Impact: `hydrostatic_column`, `mol_base_handoff` byte-identical; `wasp_full` reaches its du stop at step 14442 on both binaries, wind unmoved ( $1e-7$ ), the field  $1.6e-3$  and the Balmer proton source  $6.5e-4$ ; the face rule over-counted the column's scatterings by 2.4 to 2.7 percent. Carried forward: the FUV band B2 line transmission in `util_ion_eq.f90` still reads a face value (B4-6c).
- **HYG-B3a** (`diffusive_photochemistry.f90`, `certification.f90`, `steady_newton.f90`, `viscous_conduction.f90`; suites `carrier_retry` 97, `energy_update` 48, `certification` 29): the carrier operator sets its interval status on every return path (already true since A3b; now stated and tested); `carrier_last_verdict_of_run` renamed for what it is, plus a step-level `carrier_step_verdict()` reset at every transport step; `conduction_floor_cell_hits` exported like the energy update's; the volume-weighted companion of `certification_row_measure` optional and no longer computed where discarded. All impact cases bitwise identical (`wasp_full_newton` finish: `info = 2` at  $||R|| = 7.3e-6$  against the new `cert_tol_momentum = 1e-8`, re-anchored by A2tol2). Left for the gate (advisor): the now-unneeded pre-set in `EXHALE_main.f90` and a stale field comment in `attempted_step.f90`.
- **A2exit** (`EXHALE_main.f90` exit block; `run_mode` suite 30 assertions, `certification` shell gate): the certification exit status follows the CLAIM, not the mode: a run that declared a stationary state (du stop, residual gate, JFNK finish) and was refused exits 2 in every mode with its outputs written; a capped or plateaued run exits 0. `wasp_full_newton` now exits 2 with `info = 2`; `run_check.sh` never reads the status. Advisor: `run_newton.sh` and `run_baseline.sh` (set -e loops) now report a status 2 instead of aborting. Open: the `EXHALE_PTC=1` route stops before the certification and so claims a stationary state with no verdict (B5).
- **B4-4a** (`Apply_BC.f90`; `hydrostatic_residual` +1 and new `free_outflow_boundary` 18 assertions): the outer free-outflow ghost is the isothermal hydrostatic continuation of the last cell ( $\rho_g = \rho_N \exp[-(\phi_g - \phi_N)/(p_N/\rho_N)]$ ,  $p_g$  by the same factor,  $v_g = v_N$ ), one rule for PLM and WENO3, replacing the zero-gradient copy (PLM) and the linear extrapolation (WENO3). The outer cell's momentum residual on the analytic column falls from 0.750 of its weight at every N to  $4.4e-3$  to  $1.0e-4$  over N = 250 to 2000 (order 1.8 PLM, 1.6 WENO3); interior and base unchanged; four ghost rules measured, this one 30 to 40x better. Velocity: the copy (a continuity-exact rule stopped fastest but divides by a ghost density the stratification makes small and was the one rule under which `wasp_full_newton` segfaulted at -O3). Two premises of the brief corrected: `wasp_full`'s edge is subsonic (Mach 0.18 to 0.96; `mol_base_handoff` is the supersonic case), and a supersonic edge blocks the ghost only through the Riemann solution, not the stencil, so byte identity is not available. Impact: `mol_base_handoff` one cell above  $1e-4$ ; `hydrostatic_column`  $3.5e-3$  at cell N; `wasp_full` du stop 16208 ->

17135 steps with the base-breathing phase moved (11 percent in rho at 1.27 Rp, mean mass flux above  $r_{\text{flux}}$  0.078 percent); `wasp_full_newton` Mdot 13.27 both. Removing the outer residual does not by itself stop the marched hydrostatic column's drain (still 0.79  $c_s$  at 300 steps). Reported: a reproducible -O3-only segfault reachable from the rejected velocity rule (latent; O3SEGV to diagnose); `EXHALE_main.f90` 2538-2540 ghost-counter comment stale (B3c); B4 design section 8.3 (ii) historical.

- **O0FPE** (`util_ion_eq.f90`; shell test `fpe_traps` in `grid_and_gates`; `run_fcheck.sh` header): the -O0 failure was two defects. (1) An integer division by zero in `EXHALE_main.f90` about 1234: `mod(count, n_err_every)` with `n_err_every = 0` in init mode, reached because `.and.` does not short-circuit (-O1 and above happen to skip it); fix handed to B3c (owner of the file). (2) A real floating-point overflow in `he_rec_coupling` (`util_ion_eq.f90`:2813):  $1100 \exp(1.2/T4) \sqrt{T4}$  at 17 K is 3.9e309, an Inf consumed as `ne/Inf = 0` into the correct low-density limit; the product is now formed only above  $T4 = 1.71e-3$  and the limit value taken below, byte-identical on `hydrostatic_column`, `wasp_full`, `mol_base_handoff`. A third site of the same class (the stall detector's huge sentinel) also handed to B3c. `run_fcheck.sh` states why it builds at -O1.
- **B4-6c** (`util_ion_eq.f90` `fuv_lw_photon_field`; driver `lya_band_transmission` rewritten, 16 assertions): band B2's line factor is the cell mean of the stellar Lyman-alpha transmission over the cell's own line-center depth (product of the two means with the continuum mean formed inside `water_photolysis_rate`; the covariance shortfall 6.0e-6 against a 1.4e-4 bound, the face rule 1.2e-2 high on the budget); the top cell carries its own mean instead of a neutral 1. The face rule was 1.0018x on the `oxygen_chemistry` state, 1.095x on an exponential H I layer and 18.2x in the base cell of a molecular base whose line-center column reaches  $1e7$  in one scale height. Consumers: the H2O and OH photolysis in B2 (both `ionization_equilibrium` call sites and the carrier photolysis), the output columns, the FUV heating column; the H2 channel reads LW only. Impact: `wasp_full`, `mol_base_handoff`, `mol_sec_ion` identical; `oxygen_chemistry` moves by factors at 45 steps, shown by a control (a 4e-5 perturbation of the face value diverges the same way) to be the case's transient amplification, not the size of the change; the case is not usable as a controlled point until it converges. Finding: `write_output.f90`'s FUV band ledger closed form  $N_{\text{ph}} (1 - \exp(-\tau))$  carries no line factor for B2 and its other rows are 0.79 to 0.98 off (LEDGER-FUV assigned).
- **O3SEGV** (diagnosis, no change): the -O3-only segfault B4-4a reported was a null F0 argument of `jv_product` in a `pgmres` clone the compiler had specialized for a null pointer, i.e. a defect of the source text of one intermediate build (13:08) between two edits of `steady_newton.f90`; every later build carries a plain `pgmres`, HEAD plus the rejected ghost rule completes `wasp_full_newton` three times, AddressSanitizer finds nothing over a full run. Correction to B4-4a's premise: `-g` changes no generated code (`.text` bitwise identical at -O3 with and without it). Noted: at the time of the diagnosis the working tree stopped `wasp_full_newton` at step 0 on the energy floor (B3c in progress).
- **LEDGER-FUV** (`write_output.f90` `write_oxygen_chemistry`; suite `fuv_band_ledger`, 15 assertions): the FUV band ledger is three labeled statements: (a) closure of the rated absorptions against the beam's loss between the cell faces (holds for any rate form,  $1e-16$ ); (b) the beam budget with each band's own line factor and the unrated share named per band (none for B1/B3/B4, which is therefore the cell-mean test; H I resonance scattering

for B2, a share; for LW a sign-changing disagreement); (c) energy; (d) state drift. Two defects of the ledger fixed: it multiplied post-sweep densities by a pre-sweep field, and summed the two inner ghost cells. Data rows of every output identical; only the FUV\_bands.txt trailer changes. **Finding (LW-NORM assigned):** the LW absorbers take 45 percent MORE photons than the beam loses ( $1.27e13$  against  $8.73e12$ ) and  $249 \text{ erg cm}^{-2} \text{ s}^{-1}$  against an incident 171.5: two independent normalizations of one absorption, the level-resolved dissociation table over p\_lw\_absorbed against the DB96 eq. 39 band share (lyman\_werner.f90, h2\_self\_shielding\_table.f90). Also: docs/a2\_oxygen\_option\_design.md 882 documents columns that no longer exist; the a2\_m2 kinetics check asserts the beam identity for every band.

- **A2tol2** (measurement and decision; a2\_certification\_contract\_20260906.md section 10): the section-9 anchors re-measured on the self-consistent JFNK residual of B5a. The returned row is now 0.36 to 0.93 of the solve's target (section 9 measured 41 to 47 above it). wasp\_full\_newton at Resid tol:  $1.0e-8$  certifies (info = 0, 14 iterations, 658 s, log10 Mdot 13.27; rows mass  $2.6e-13$ , momentum  $1.3e-9$ , energy  $6.275e-10$ ); the PTC route stalls from a cold start; wasp\_he23off\_newton, newton\_rsw01, newton\_rsw05 at the default  $1e-5$  refuse on the mass and momentum rows at cell 499; the four Load IC Newton cases (jfnk\_hd189, jfnk\_hd189\_tight, ptc\_warm, armD\_D2\_newton) are stopped by the grid guard and need their IC regenerated on the current grid before they can be measured. **Applied:** cert\_tol\_energy  $5e-5 \rightarrow 1e-6$  in certification.f90 ( $5e-5$  had been anchored on the lagged-gate energy row  $4.700e-06$  and could refuse nothing;  $1e-6$  is a decade above  $1.212e-07$ , the largest energy row of any state whose solve met its own target,  $5.7e4x$  above the floor  $1.765e-11$ ); mass  $3e-12$  and momentum  $1e-8$  kept; the report header now names sections 9 and 10. backup/regression/wasp\_full\_newton/input.inp gains Resid tol:  $1.0e-8$  as its configuration of record, the only setting at which that case certifies (Mdot unchanged at this precision, cost +11 percent); its golden moves at the series gate with everything else. Verified: private build, certification suite green (29 assertions). Not changed: the other three Newton cases stay at the default so that the matrix keeps a refused claim (exit 2) under test.
- **LW-NORM** (diagnosis; no executable statement changed): the 45 percent excess of the FUV band ledger is two WAVELENGTH BANDS, not two fits. The self-shielding table's line list runs to 1200 Å while its sigma\_pump is normalized per photon of 912-1110 Å; the beam loss is the DB96 eq. (39) equivalent width of 912-1110 Å. MEASURED: the column integral of sigma\_diss/p\_eff reaches 1.51538 at the top of the column axis against 1.519288 for the photon content of 912-1200 Å over 912-1110 Å of a flat spectrum (0.26 percent), is exactly density-independent, and reproduces the shipped oxygen\_chemistry ratio 1.4544 at its base column where eq. (39) clamps to 1. p\_eff already separates dissociations from absorptions, so the heating takes fragment energy only (not cause 2). Landed: lyman\_werner.f90 header section 3a with the invariant and the measurement, its section 3 "2 percent" statement corrected to the live table's thin-limit cross section  $2.6396e-17 \text{ cm}^2$ ; lyman\_werner\_cell\_mean probe gains two assertions and a photon-conservation diagnostic (9/9); fuv\_band\_ledger/run.sh comment. Advisor follow-up: the table's generator inputs (nine abs\_T####.npz, 27 CLOUDY rate files, the line-by-line scripts, the Meudon level and line data, 29 MB) moved from the 2026-09-03 scratch directory into src/utis/h2\_shielding\_lbl/ with a

README, and h2\_shielding\_table\_line\_by\_line.py's template brought in line with the hand-renamed axis dummy of h2\_shield\_locate: the generator now reproduces the shipped module BYTE FOR BYTE (MEASURED). The repair, table on 912-1110 A or beam widened to 912-1200 A with the B1 flux joined, is item 6 of To\_be\_determined\_by\_user\_20260906.md (recommendation: widen), because the second redefines the Stellar LW flux key. Still to correct after B3c releases write\_output.f90: the ledger's LW label says "two fits" and should say "two bands" (with the header\_names\_lw\_fits assertion of fuv\_band\_ledger).

- **HYG-HeCI** (test hygiene, advisor): physics\_probe was red on three helium\_collisional\_ionization\_by\_level assertions (1e-13 at 1e4 K to 1.0e-4 at 5e4 K). Not a code defect: batch 2c replaced the single-precision literal 3.940e-11 erg (24.6 eV) of eval\_cool with the global e\_th\_HeI\_erg (24.587389 eV), and the test's reference still carried the old literal with a comment calling it "a separate open item". The reference now uses e\_th\_HeI\_erg and e\_th\_HeTR\_erg; physics\_probe PASSED on the current tree (private build). The LW-NORM worker's attribution of this to a concurrent Cool\_coeff.f90 edit was wrong: that file's uncommitted diff is the landed 2c-CI / MASS2 work.
- **LW-NORM-B** (user decision 2026-09-06: recommendation (B) adopted; landed): ONE Lyman-Werner band, 912-1201 A, with one normalization.
  - **The band.** oxygen\_rates.f90 carries four FUV bands, not five: the Lyman-Werner interval is 912-1201 A and band B1 (1110-1201 A) is gone into it. 1110 A was an edge because Draine & Bertoldi end the Solomon process there for interstellar  $v = 0$  gas; at the 700-3200 K of a planetary base the vibrationally excited levels pump longward of it, and the self-shielding table's line list always ran to 1200 A. The two halves carried the same H2O yield triplet, so no quantum yield moves; the band-averaged cross sections are re-formed over the whole interval and MEASURED again from the same Photochem files:  $\sigma(\text{H}_2\text{O})$  8.51337e-18,  $\sigma(\text{OH})$  4.74371e-18 cm<sup>2</sup>, (H2O) 12.0526 eV, (OH) 11.3829 eV,  $= 2hc/(912+1201 \text{ \AA}) = 11.7354 \text{ eV}$ . They agree to seven digits with the merge of the two shipped halves, and they reproduce the values the band carried before it was split in 2026-08.
  - **The key.** Stellar LW flux now means the 912-1201 A band flux at the orbit; Stellar FUV B1 flux is retired and a file that states it stops the run with a message naming the merge and asking for the two numbers added. Spectrum type: Load integrates 912-1201 A. The resolved dump loses fuv\_band\_B1\_flux. The inputs that carried both keys were restated to their sum: 343.0 + 137.9 = 480.9 for the HD 209458 b spectrum (Update\_EXHALE\_stage1.md records the 912-1201 A integral of that spectrum as 481.0 and its halves as those two numbers), and the LW-only cases (mol\_lyman\_werner, armA\_LW, armD\_D2\_LW, armD\_D2\_LW\_newton) from 343.0 to 480.9 so that they keep the same star. The README\_HOWTO HD 189733 b recipe goes from LW 600.1 + B1 648.4 to LW 1248.5.
  - **The table.** src/utils/h2\_shielding\_lbl/lbl\_table.py normalizes per photon of 912-1201 A (wl\_HI, wl\_max, and E\_LW\_PHOTON\_ERG the mean photon energy of that band); F\_BAND\_ERG is restated as  $343.0 \times 289/198 = 500.6$  with the derivation in the file, and the file now records that it cancels out of sigma\_pump exactly. The nine abs\_T####.npz were recomputed (run\_abs.py, 23 to 251 s each) and h2\_self\_shielding\_table.f90 regenerated. MEASURED: the generator reproduced the shipped mod-

ule byte for byte before the change, so the only difference is the band.  $\sigma_{\text{pump}}$  at the bottom of the column axis is  $1.701\text{e-}17$  (700 K) to  $1.887\text{e-}17$   $\text{cm}^2$ , which is DB96's own  $2.557\text{e-}17$  per 912-1110 Å photon restated per 912-1201 Å photon ( $1.676\text{e-}17$ ) to 1-13 percent, the same agreement as before.

- **The beam loses what the table rates.** The band share A is no longer the DB96 eq. (39) equivalent width; it is the column integral of the pump cross section  $\sigma_{\text{diss}}/p_{\text{eff}}$  of the same table, in closed form ( $h2\_lw\_band\_photon\_fraction\_absorbed$ , exact for the module's own interpolant knot interval by knot interval). MEASURED: A at the top of the column axis is 0.6903 to 0.9974 over the whole (T, n<sub>H</sub>) grid, where it used to reach 1.5154; the closed form agrees with an independent 20000-step quadrature of the two public accessors to  $2\text{e-}10$ . The band ledger's photon count is now the cell mean of the same pump cross section ( $lyman\_werner\_band\_absorption\_rate\_cell\_mean$ ), on the quadrature the dissociation rate already used, and not the cell mean of  $\sigma_{\text{diss}}$  divided by  $p_{\text{eff}}$  at one column: on cells spanning 2.5 H<sub>2</sub> scale heights the second stood 8.0 percent above the beam loss and the first closes on it to  $1\text{e-}6$  ( $physics\_probe/lyman\_werner\_cell\_mean$  sec. 6).  $h2\_band\_equivalent\_width$  and eq. (39) are removed; the two published self-shielding fits stay, called by nothing, for the comparison of docs/h2\_self\_shielding\_cloudy.md.
- **The ledger.** The LW row of output/FUV\_bands.txt is a closure like B3 and B4, and its label says so. MEASURED on the oxygen\_chemistry state, 5 steps:  $unrated\_frac$   $-3.62\text{e-}1$  before,  $-1.18\text{e-}2$  after, and the LW absorbed energy goes from 233.5 against an incident 171.5 (36 per cent over the band) to 202.3 against 200.0. The residual left is the part of the column above the top of the table's column axis: this case reaches  $6.4\text{e}21$   $\text{cm}^2$  against an axis top of  $5\text{e}21$ , where the clamped edge cross section drives A past 1 and the beam is capped while the rate is not. The run already warns about that column.
- **Impact.**  $mol\_lyman\_werner$  at its own 12000 steps, before at 343.0 against after at 480.9:  $\log_{10} M_{\text{dot}}$  10.53 → 10.54, T and  $\rho$  under  $5\text{e-}3$ , cooling  $6.5\text{e-}2$ , the molecular densities and the shielding  $1.2\text{e-}1$  at the H<sub>2</sub> front ( $r = 1.075$ ), the base H<sub>2</sub> column  $3.3875\text{e}20$  →  $3.4067\text{e}20$   $\text{cm}^2$  and the base  $x_{\text{H}2}$  0.96288 → 0.96365. The optically thin rate at the top of the column moves by  $480.9/500.6 = 0.9606$  exactly: the band's photon content grew 46 percent and the cross section fell by the same factor, so what is left is that the real spectrum gives 480.9 over 912-1201 Å where the flat continuation of the 912-1110 Å deck would give 500.6. Deeper cells move -4.7 to -6.0 percent, the wider band's extra lines shielding differently; slightly less destruction leaves slightly more H<sub>2</sub>.
- **Not repaired here, and it blocks the fuv\_band\_ledger suite.** With its restated 480.9 the oxygen\_chemistry case stops at step 0 on the energy floor. MEASURED, it is not the band merge: the same case stops with the OLD binary at the same restated flux, and both binaries stop or run at chance along the flux axis (old: runs at 343.0 and 440.0, stops at 360.0, 400.0 and 480.9; new: runs at 343.0 and 400.0, stops at 360.0, 440.0 and 480.9). That is the step-0 acceptance knife edge of items B3a and 2 of To\_be\_determined\_by\_user\_20260906.md. Every ledger assertion is GREEN on the same case at a flux where it runs.
- Also:  $p\_lw\_absorbed$  is now written by the sweep and checkpointed but read by nothing, since the ledger takes the pump rate directly. Removing it needs attempted\_step.f90.

- B3b-IR** (molecular\_infrared\_cooling.f90; h2\_infrared\_line\_populations and h2\_partition\_sum\_uniqueness probes): the third copy of the H2 ladder sum (212-215) is gone; the H2 infrared line populations are normalized by caloric\_eos's h2\_partition\_function, the one Boltzmann sum over the Roueff et al. 2019 ladder whose energy the equation of state carries and whose free energy sets the  $H + H \leftrightarrow H_2$  equilibrium. The probe that greps the tree for a second sum went RED (1) -> GREEN (0). Impact (advisor measured, 12000-step snapshots, OMP\_NUM\_THREADS=1): mol\_ir\_bands (9 output files) and mol\_base\_handoff (7 files) BYTE-IDENTICAL before and after, Mdot 7.99 both; the retired copy had an  $\exp(-\min(E/T, 700))$  clamp whose only difference from the shared sum's  $\exp(-E/T)$  is below double rounding. No golden moves. The worker's draft report stopped at "impact runs in progress"; this entry closes it.
- LW-NORM-B, advisor verification** (2026-09-06): private build of the landed tree, physics\_probe PASSED (554 assertions incl. the rewritten section 6: column integral of sigma\_pump at the top of the axis 0.69 to 0.997 on the whole grid, was 1.515; photons spent / photons lost 0.999999), spectrum\_type PASSED (154), fuv\_band\_ledger NOT RUNNABLE: its case is oxygen\_chemistry, whose Stellar LW flux restated to 480.9 (the same spectrum over 912-1201 Å) stops at step 0 with energy update FAILURE at cells 85-88 (T 2.1e4 K, ionization sweep NON-ROOTs accepted by the relaxation amnesty with residual 1 to 4, then the balance asks for T below the 14.5 K floor). MEASURED by the worker along the flux axis with both binaries: 343 runs, 360 stops, 400 runs (after) / stops (before), 440 stops (after), 480.9 stops: a step-0 acceptance knife edge of the OLD code at cells with no H2, items 1 and 2 of To\_be\_determined\_by\_user\_20260906.md (decision 4: reject-and-retry; amnesty cap), not of the band. The flux is NOT tuned to make the case run; the case and fuv\_band\_ledger stay red until those two decisions land. Code read:  $a\_lines = \min(A, 1)$  with transmission  $1 - A$  (a band share, not an optical depth) from the closed-form power-law integral of the tabulated  $\sigma\_diss/p\_eff$ ; both the rate and the share are read at the local T and  $n\_H$  of the cell for the whole column above it (homogeneous-column table), a comment in util\_ion\_eq.f90 now says so. Accepted. Goldens untouched; every LW case moves at the series gate (mol\_lyman\_werner 500-step pair: T 3.3e-3, H2 0.125 at the front,  $k\_LW$  thin-limit  $\times 0.9606 = 480.9/500.6$ ).
- ACCEPT-1/2** (user decisions 2026-09-06, items 1 and 2 of To\_be\_determined\_by\_user\_20260906.md, both as recommended; brief\_plan\_rev2\_brief\_ACCEPT12.md, to start when B3c releases EXHALE\_main.f90, energy\_semi\_implicit.f90, ionization\_equilibrium.f90): (1b) a failed energy update refuses the attempted step in both modes and the controller retakes it at half dt, exhaustion exits 2; (2a) the relaxation amnesty accepts a non-root only below a residual cap measured on the matrix, above it the cell keeps the composition it entered the sweep with (new acceptance class 6). Motivation: oxygen\_chemistry at its restated LW flux dies at step 0 on exactly this pair (LW-NORM-B verification entry above).
- B3b-IR impact runs, advisor correction** (2026-09-06): the worker's before/after pairs were run with binaries built from the tree as it stood at 12:30, when the B3a controller still refused on the time-discrete hydrodynamic row in init mode: MEASURED from their run logs, 12000 steps accepted of 104389 attempted (mol\_ir\_bands) and 104594 (mol\_base\_handoff), 103697 refusals all "the Runge-Kutta stages and the fluxes: the time-discrete hydrodynamic row was out of tolerance", dt collapsed and T stayed within 1 K of the 1140 K initial condition over the whole grid (golden mol\_ir\_bands reaches 2553 K at 2.18 Rp). The byte-identity of the pair therefore

says little about the layer; the statement that the IR refactor changed no arithmetic rests on the emission assertions at five temperatures with identical digits and on  $\text{retired}/Q - 1 = 0$  at every probe temperature, which stand. On the current tree the same case accepts 12000 of 12000 attempts with no refusal (MEASURED, private build). The H2 IR tabulation grid extension (advisor, this entry's neighbor above):  $n_{\text{tab}} 401 \rightarrow 496$  at the same log spacing to  $t_{\text{tab\_hi}} = 30146$  K, old nodes reproduced by the same expression, emission at 10000 K:  $\text{clamped} = \text{direct sum} = 1.57875\text{E-}18$  (was  $1.181\text{e-}18$ ); `molecular_infrared_bands.py` template carries the private `:: i` of the generated data module and regenerates it byte for byte (MEASURED).

- B3c** (`EXHALE_main.f90`, `ionization_equilibrium.f90`, `energy_semi_implicit.f90`, `molecular_reaction_heat.f90`; new suite `coupled_source_step`, 30 assertions in the advisor's run): rows 7 to 9 of the enumerated step are one local source step per cell at fixed volume, the temperature and the composition its unknowns, iterated to a fixed point ( $|dT|/T \leq 1\text{e-}8$ ,  $\max |df_{\text{sp}}| \leq 1\text{e-}8$ , cap 20 passes). The composition projection (`comp_p_from_T + W_to_U`, which rebuilt the energy row at the new particle count) is gone; the energy row is anchored on `u_th_old`, the thermal energy the cell had at its old composition; `rho` is intent(in) to the chemistry and the mass sum is a checked diagnostic (worst  $6.4\text{e-}12$ , T2.2). The two collisional deposits D0 C25 named (oxygen O1/O1r/O2/O2r/O6, associative He  $2^3\text{S}$ ) are in as differences of the one formation-energy table. RED reference kept in the suite: the removed update changed `u(3)` by  $-0.600$  with `heat = cool = 0`. **Advisor decision on the worker's deviation (accepted):** `Delta u_form` is NOT added to the energy row. In this code `heat` and `cool` are the net THERMAL sources (photoheating deposits  $h\nu - E_{\text{th}}$ , collisional ionization cooling removes  $E_{\text{th}}$ , every molecular, Penning, Lyman-Werner and oxygen reaction heat is a difference of the same table), so the formation reservoir is already accounted for and the term would double count; MEASURED with it added, `wasp_full` dies at step 0 with 368 of 504 cells at the floor. Consequence: T1.5/T1.6 as an identity on `u_th + u_form` against  $dt(\text{heat} - \text{cool})$  cannot close; the B3a instrument (`attempted_step.f90`) is to be restated as  $d u_{\text{th}} = dt(\text{heat} - \text{cool}) + \text{the transport contribution of the step}$ , or evaluated on the source substep alone (item T15-INSTRUMENT, pending). Impact (control = same snapshot with the edits reversed): `wasp_full` to its `du` stop,  $\log_{10} \text{Mdot } 13.34 \rightarrow 13.34$ ,  $T 2.5\text{e-}3$ ,  $\rho 2.4\text{e-}3$ , metals  $1.5\text{e-}2$ , one sign change of  $v$  at  $r = 1.062$  ( $2.5\text{e-}4$  of the wind); `wasp_full` 300 steps front one cell apart; molecular 12000-step snapshots up to 40 percent in  $\rho$  beyond 4  $R_p$ ,  $\text{Mdot } 10.58 \rightarrow 10.57$ ,  $10.53 \rightarrow 10.51$ ,  $10.58 \rightarrow 10.58$ ; `hydrostatic_column`  $\leq 6\text{e-}9$ . **Cost: 5 to 12 passes per step, 8.1x on wasp\_full 300, 10.3x to the du stop (18:59 -> 3:15:07).** Assigned as B3c-COST (re-sweep only the cells not yet at the fixed point; tolerance anchored on a measured convergence study, not chosen to pass). Also in and byte-identical: the carrier status pre-set removal (HYG-B3a) and the two O0FPE sites; `fpe_traps` passes at -O0 with traps. Left red on purpose: `heat_column_matches_breakdown_total`  $6.886\text{e-}3$ , because `write_heat_breakdown_eq` (`util_ion_eq.f90` 2280) is a second hand-maintained copy of the heating assembly that lacks the new deposit, and `post_process_adv.f90` ~847 a third: assigned as HEATBRK (one assembly, three callers), not the ten-line patch. Advisor verification: private build of the live tree, `coupled_source_step` 30/0, `energy_update` 48/0, `attempted_step` 31/0, `certification` 29/0, `run_mode` 30/0, `acceptance_classes` 12/0, `grid_and_gates` 92/1 (the HEATBRK red).

- **HEATBRK, first pass BLOCKED** (2026-09-06): the heating assembly the sweep uses is 210 inline lines of `ioniz_eq` (`ionization_equilibrium.f90` 2119-2331), not a routine, so the one-assembly fix has to extract it there, a file `ACCEPT-1/2` owned at the time; the worker stopped with no edit, as briefed. Re-measured RED: worst  $6.886\text{e-}3$  (cell 376, the associative  $\text{He } 2^3\text{S}$  and collisional oxygen deposits missing from the breakdown) over a  $1.24\text{e-}5$  median floor (the breakdown RE-SOLVES the field and rates on the written state while the heat column carries the rates the sweep produced it with). Interface proposed: `heating_of_composition(...)` in `utils_ion_eq` returning heat and a non-optional `heat_chan(:,17)` (channel list and header owned by the module; LW dissociation and fluorescence as two columns), called by the sweep, by `write_heat_breakdown_eq` (which then forms no rate) and by `post_process_adv` (which today omits the Balmer and associative terms from its theat). **Advisor decision:** `Heating_breakdown.txt` writes the channel array the sweep computed for the written state, not a recomputation; this is the "one state per output file" rule (rev3 section 10.2 item 2, landed) applied to this file, and the only form in which the breakdown total can equal the heat column to round-off. Resumes when `ACCEPT-1/2` releases `ionization_equilibrium.f90`.
- **ACCEPT-1/2, advisor verification** (2026-09-06): private build of the live tree, `energy_update` 53/0 (the driver was killed by error `stop 1` before), `acceptance_classes` 22/0 (did not compile before: class 6 did not exist), `attempted_step` green (10 RED before: exit status, history lines, reason breakdown), `coupled_source_step` 30/0, `certification` 29/0, `run_mode` 30/0, `carrier_retry` 97/0. Code read: the kept state of class 6 is `x_entry`, the composition the cell entered the sweep with (built once, also the solve's starting point), heat and cool of the returned composition as section 170 requires. Accepted, including the two deviations: the two-line `steady_residual.f90` change that makes `n_cells_without_chemical_root` count class 6 (without it the controller would not see a refused non-root), and `as_op_projection -> as_op_composition` (operation number 8 unchanged; B3c had removed the projection the name described). Findings taken over: cap  $1\text{e}2$  MEASURED on the whole default matrix at 400 steps (the amnesty population is a cold-start population: 111 events  $1\text{e-}6$  to 8.853 at step 0 of `oxygen_chemistry`, nothing until  $8.6\text{e}4$ , then the four cells 85-88 whose energy update failed); with the cap `oxygen_chemistry` runs its 1000 steps and exits 0 ( $\log_{10}$  Mdot 9.61), no step refused, fifteen of sixteen cases byte-identical; the comment's "res ~28 in the metals-on molecular gates" is no longer reproduced (they accept no non-root today) and was corrected in place. The energy-refusal path is exercised by the default-off hook `EXHALE_ENERGY_FAIL_AT_STEP`. Outside scope, assigned: "class 4" wording in `certification.f90/steady_newton.f90` and `steady_newton.f90`:815 `acc_resmax(4)` missing class 6 (HYG-CLASS6, folded into HEATBRK's resume); an H-only run has no acceptance classes at all (pre-existing, B6 list); `oxygen_chemistry` cannot run in phys mode (56 handoff non-roots refused by A0), a statement about that configuration, recorded.
- **HEATBRK landed, advisor verification** (2026-09-06; measured on `mol_base_handoff` 200 steps, `wasp_full` 300, `oxygen_chemistry` 1000, the `mol_metals` impact pair still running when this was written): `heating_of_composition` (`util_ion_eq.f90` 1503-1780) is the one heating assembly, 17 named channels, the total the running sum of the columns in the sweep's original order (sweep heat bit-identical); the sweep calls it once (181 inline lines of `ioniz_eq` gone), `write_heat_breakdown_eq` writes the module array

heat\_channel\_state the sweep filled and forms no rate, post\_process\_adv calls the same routine. Private build of the live tree: grid\_and\_gates 93/0 with heat\_column\_matches\_breakdown\_total 1.678e-16 (was 6.886e-3), physics\_probe 557/0 incl. heating\_channel\_closure and the scanner heating\_sum\_uniqueness (one exception recorded: the band ledger of write\_output.f90 restates the Lyman-Werner and FUV photolysis event energies, a fourth copy, assigned HYG-LEDGER-EPS). Heating\_breakdown.txt 18 -> 21 columns; \_adv profiles move by up to 1.8e-2 (the Balmer and associative deposits the third copy lacked). heat\_channel\_state is class-1 state and enters the attempted-step checkpoint through B3c-COST (owner of attempted\_step.f90), assigned. HYG-CLASS6 (class 4/6 wording, acc\_resmax over both classes) in the same landing.

- B3b-CO landed: the CO row carries two published destruction rates** (2026-09-06). src(ic\_CO) is no longer 0. It is  $-(k_{D1} n(\text{He}^+) + k_{\text{CO}}) n_{\text{CO}}$  with  $k_{D1} = 1.60\text{e-}9 \text{ cm}^3 \text{ s}^{-1}$  (UMIST RATE22 entry 4068, measured, accuracy better than 25 percent, temperature independent, READ) and  $k_{\text{CO}}$  the cell mean of  $(F_{\text{LW}}/\langle h\nu \rangle) \sigma_{\text{CO}} \Theta(\text{N}_{\text{CO}}, \text{N}_{\text{H2}}) \exp(-\tau_{\text{cont}})$  on the 912-1201 Å beam. New: co\_self\_shielding\_table.f90 (the 12CO block of Visser, van Dishoeck & Black 2009 Table 6,  $T_{\text{ex}}(\text{CO}) = 50 \text{ K}$ , with their Table 5 beside it for the spread) and co\_photodissociation.f90 (the point rate, the two-column cell mean, the fragment heat). fuv\_lw\_photon\_field gains the CO column and returns  $k_{\text{CO}}$  and  $\Theta$ ; carrier\_photolysis rebuilds  $k_{\text{CO}}$  from the current carrier columns, so the CO layer shields itself inside the relaxation. Constants, all MEASURED:  $\sigma_{\text{CO\_band}} = 1.0160\text{e-}17 \text{ cm}^2$  per beam photon (Heays et al. 2017 cross section over the 912-1118 Å window CO absorbs in),  $\langle h\nu \rangle = 12.8674 \text{ eV}$  (oscillator-strength weighted over Visser's 37 bands),  $D0(\text{CO}) = 11.1157 \text{ eV}$  and  $q(D1) = +2.2117 \text{ eV}$ , both differences of the one formation-energy table; the deposit is 1.7517 eV per dissociation. Both deposits are channels 18 and 19 of heating\_of\_composition (17 -> 19 channels) and are formed nowhere else; heat\_per\_co\_dissociation is registered in the uniqueness scanner with the band ledger of write\_output.f90 as its one recorded exception. output/FUV\_bands.txt gains  $\text{N}_{\text{CO}}$ ,  $k_{\text{CO}}$  and  $\Theta_{\text{CO}}$  and a CO row in the beam ledger (CO is printed apart from rated\_ph because beam\_loss\_ph carries no CO term: MEASURED, folding it in moved the LW budget residual from -1.8e-2 to -8.2e-2). output/Oxygen\_chemistry.txt gains the domain record  $\tau_{\text{dest}} \leq 0.1 \tau_{\text{res}}$  with the worst ratio and its radius,  $\tau_{\text{res}}/\tau_{\text{form}}$  from RATE22 8597, the count of cells above the shielding table's 512 K excitation-temperature limit and the count in which the He+ channel outruns He+ recombination. Tests: physics\_probe co\_destruction (27 assertions, RED on a mutated table entry and a mutated photon energy), fuv\_band\_ledger extended. **Impact, MEASURED on oxygen\_chemistry at its maxsteps of 1000, the only regression case that carries CO** (a control binary of the same tree with the CO source and the two deposits reverted): both run 1000 steps, exit 0 and give  $\log_{10} \text{Mdot} = 9.61$ ; the after binary costs 628 s against 11 percent less for the control at a matched 150 steps. The He+ + CO charge-transfer deposit is **65 percent of the total heating at  $r = 1.089$** , the helium ionization front where CO still survives, and under 1 percent outside 1.05-1.17; the layer's temperature moves by up to 46 percent at  $r = 1.023$  and its CO by up to 52 percent. mol\_carrier at 300 steps is byte-identical in every output file, which is the gate that the new terms are on the same thereis\_oxychem as the rest of the oxygen path; no other regression case sets Oxygen chemistry. The domain record of that run: 401374 out-of-domain cell visits, worst  $\tau_{\text{dest}}/\tau_{\text{res}} 1.34\text{e}5$  at  $r = 1.0002$ , worst  $\tau_{\text{res}}/\tau_{\text{form}} 0.219$ , 305125 cell visits above the 512 K excitation-temperature limit

and 281325 in which the He+ channel outruns He+ recombination. Goldens not refreshed. **P0 recorded:** the He+ ledger is not closed. This reaction consumes a helium ion and neither the carrier row (ion stages frozen) nor the sweep's He+ balance returns it, so where  $k_{D1} n_{CO}$  exceeds the He+ loss rate both the destruction and that heating peak are over-stated. The fix is a sink in the He+ row of `mol_heh_rows` (`System_HeH_mol.f90`), not this item's file; `dom_cells_HeP` is the observable. **Not done, carried over:** the thermodynamic CO ceiling is still in `limit_to_element_budget`; deleting it changes `EXHALE_main.f90`, `certification.f90` and `write_setup_report.f90`, none of them this item's files. The exact lines are in the item's report.

- **HYG-LEDGER-EPS closed in the same landing** (2026-09-06): the FUV band ledger of `output/FUV_bands.txt` is formed by `utils_ion_eq::fuv_band_absorption_ledger`, beside the heating assembly and reading the same imports, and `write_output.f90` writes what it returns; the `heat_LW` column of `output/Lyman_Werner.txt` is now the assembly's own channel (`heat_channel_state(:,ih_H2_LW_dissoc)`) instead of a second evaluation of  $k_{LW} n_{H2} e_{lw\_fragment\_erg}$ . The heating\_sum\_uniqueness scanner's exception list is EMPTY: every deposited energy is named only in the module that defines it and in the one assembly. RED before (the pre-move import line, two deposits reported outside the assembly), GREEN after. `FUV_bands.txt` is byte-identical across the move (MEASURED, oxygen\_chemistry at 5 steps); the `heat_LW` column moves by at most  $5.57e-4$  relative, which is the one-sweep lag between the state the file wrote and the state the energy equation deposited, and the new number is the deposited one.
- **B3c-COST, the cost of the coupled step measured and the skip specified, not implemented** (2026-09-06; `EXHALE_main.f90`, `attempted_step.f90`, the `coupled_source_step` and `attempted_step` suites). WHERE THE COST IS, MEASURED on `wasp_full` 300 steps, one thread: the run timers give `ioniz_eq` 83.6 percent and the energy solve 16.3 percent of the marching wall time, and a `gprof` build puts **78 percent of the whole marching time inside the MINPACK solve of one cell's chemistry** (one Jacobian and one QR per cell per pass; the warm start already reaches each root in one Newton step), against about 13 percent in the whole-grid rate and field work a pass repeats. So the cost is the number of CELL solves and not the field, and reusing the field between passes is worth at most that 13 percent. THE CONVERGENCE STUDY CONTRADICTS THE EXPECTATION THE ITEM WAS WRITTEN FROM: after pass 2, 308 of the 500 cells are still above the tolerances, after pass 5 still 130, and the cell that sets the pass count walks inward one cell per pass with the ionization front. Contraction 0.191 median over 2729 passes. A cell-local skip is therefore worth 2.41x on the cell solves with a one-pass rule -- but that rule is WRONG: a cell at rest moves again on 177 of the 3029 passes, 690 of the 809 wake-ups at pass 8, because the column keeps moving a cell after the cell has stopped. **Two consecutive passes of rest: zero wake-ups, 1.96x on the cell solves, 1.55x on the step.** The skip has to be a mask in the cell loop of `ioniz_eq`, which this item does not own (HEATBRK was writing to that file at 20:30 and 20:35), so it is SPECIFIED and not implemented: the optional `cell_at_rest` argument, the class-6 path (`sys_x = x_entry` straight to the extraction) as the body of the skip, and the one consequence that is not a pure skip -- the acceptance ledger describes the cells the sweep SOLVED, so a mask needs an acceptance class stored for each cell and inherited by masked cells, or `n_cells_without_chemical_root` reads a different population from the one it believes. TOLERANCE ANCHORED AND KEPT AT  $1e-8$ : the anchor the comment

gave (the 1e-6 heat-column gate) stopped depending on it when HEATBRK made the two assemblies one; the anchor that binds is `cert_tol_carrier = cert_tol_element = 1e-8` on the composition this step returns. Measured 1e-10 to 1e-5 on `wasp_full 300` and `mol_base_handoff 1500`, the adopted state moves by about the tolerance and a decade costs 1.3 passes and 15 percent of the wall time, **so the tolerance is not the cost lever**; the step's own truncation error (step-doubling, phys, 5.3e-3 relative) is five decades above the loosest value and does not bind. Newton on T with the composition's response (item 4) stays live, and the CHEAP version of it is refuted by measurement: the contraction ratio spreads by 122 percent globally and by 200 percent cell by cell over the tail of a step, so the sequence is geometric on average over a run and not within one step at one cell -- which is why B3c's Aitken failed, and why the acceleration has to come from  $dc/dT = -(dF/dc)^{-1} (dF/dT)$  against the factorization hybrid already forms, not from the iterates. ALSO FIXED, a defect of my own file found outside the task: the run's cost line divided passes accumulated over every ENTRY of the coupled loop by the number of adopted STEPS, two different populations, so it could report a mean above the worst single entry (MEASURED on the control, `wasp_full 300` in phys mode: "13.98 passes per step on average, worst 11"). It now divides by the entries and names them. ALSO: `heat_channel_state` is in the attempted-step checkpoint (save/restore/compare/perturb), the four lines HEATBRK's report asked for, with two assertions RED on the control (the moved array is not seen, and the restore leaves it 3.0 off). Tests: `coupled_source_step 36/0` (33 before), `attempted_step 97/0` (95 before). Impact, control = the same tree snapshot with this item's two production files reverted: `wasp_full 300`, `mol_base_handoff 1500` and `mol_metals 1500` all **byte-identical** in `Hydro_ioniz.txt` and `Ion_species.txt`.

- **B3c-COST** (measurement; EXHALE\_main.f90 comments and debug counters, `attempted_step.f90` checkpoint; numerically inert, `wasp_full 300`, `mol_base_handoff 1500`, `mol_metals 1500` byte-identical): the coupled source step's cost is not where the brief assumed. MEASURED on `wasp_full 300` (3029 passes): after pass 2 still 308 of 500 cells above tolerance, after pass 5 still 130; the pass count is set by the ionization front walking inward one cell per pass, not by a fixed handful of cells, so a cell-at-rest skip is worth 1.96x on the cell solves and 1.55-1.62x on the step, and the naive rule (rest on the last pass) is wrong: 177 of 3029 passes wake a resting cell, two consecutive passes of rest wake none. gprof: 78 percent of marching time is the MINPACK solve of one cell's chemistry (Jacobian + QR per cell per pass), the whole-grid field and rate work 13 percent, the energy solve 16 percent (already deactivates converged cells). Tolerance anchored and KEPT at 1e-8 (the binding anchor is `cert_tol_carrier = cert_tol_element = 1e-8` on the returned composition): a decade costs 1.3 passes and 15 percent, not the lever. Aitken and any acceleration built from the iterates fail because the contraction ratio spreads 200 percent cell by cell within a step. **The lever is a Newton on (T, composition) taken cell by cell with  $dc/dT = -(dF/dc)^{-1} dF/dT$  from the factorization hybrid already forms, inside the sweep (ionization\_equilibrium.f90), plus the two-pass rest mask with an acceptance class inherited by masked cells:** assigned as COST2, after B3b-CO releases that file. Also fixed: the cost line divided passes over loop entries by adopted steps (could report a mean above the worst); `heat_channel_state` is in the checkpoint with two assertions (`attempted_step 97/0`, `coupled_source_step 36/0`; advisor rebuild and rerun green). Outside scope recorded: EXHALE\_ERR EVERY read in both modes but the probe runs only in phys; `mol_base_handoff` in phys mode refuses 7 of its first 20 steps at the fixed point (pass cap at step 0), recovers.

- B3b-CO, advisor verification and follow-up** (2026-09-06): private build of the live tree, physics\_probe 584/0 (incl. co\_destruction 27 and the scanner heating\_sum\_uniqueness with an EMPTY exception list after HYG-LEDGER-EPS moved the band ledger into utils\_ion\_eq::fuv\_band\_absorption\_ledger), fuv\_band\_ledger 24/0, grid\_and\_gates 93/0. Accepted with the four stated departures from the design (cross section as the mean over the 912-1201 Å beam of one that ends at 1118 Å; the same flat-band cross section for a loaded spectrum, the shape-aware integral belonging to sed\_read.f90; q\_D1 as a table row because mion\_fsp is a parameter; CO on its own ledger row, co\_frac 6.39e-2 the size of decision 2's approximation). **P0 opened by the item itself and assigned at once as B3b-CO2:** the He+ + CO channel deposits its heat (65 percent of the total at the He+ front of oxygen\_chemistry,  $r = 1.089$ ) while no He+ is consumed and no C+ or O delivered anywhere; brief plan\_rev2\_brief\_B3b-CO2.md (He+ sink in both molecular systems with Jacobian, products to the free O and carbon pools, heat rate identical to the species rate from one place, element census). Also recorded: the thermodynamic CO ceiling is still present because its deletion touches EXHALE\_main.f90 34, 3641-3648, certification.f90 84, 1072-1076 (n\_active\_unvalidated\_physics loses its only producer), write\_setup\_report.f90 18, 925-934 (CEILING-DEL, after B4-1 releases main); heat is deposited for carrier reactions whose products a never-covered carrier interval does not deliver (oxygen\_chemistry in init mode never covers one), an operator-split defect that B4-1c and B5 address and that leaves the oxygen case without an end-to-end gate for its CO source; heat\_channel\_state now has 19 columns (checkpointed by B3c-COST); python/paper\_data.py HEAT\_FIXED stale.
- B3b-CO2 landed: the He+ + CO channel now consumes the helium ion** (2026-09-06). Row (2) of mol\_heh\_rows (System\_HeH\_mol.f90) loses k\_D1 n\_CO n(He+) with  $k_{D1} = rk_{D1\_Hep\_CO}() = 1.60e-9 \text{ cm}^3 \text{ s}^{-1}$ , and its turnover scale (set\_mol\_turnover\_rates) gains k\_D1 n\_CO n\_He; the helium leaves neutral, so the free neutral He that closes the helium budget receives it and no other row moves. One row serves every path: System\_HeH\_mol\_metals and the constrained-equilibrium residual both call it, and neither molecular system has an analytic Jacobian to extend (hybrd1, finite differences), so the test asserts the difference derivative the solver actually builds. CO is a background density of that row and not an unknown, because the one-sided model gives it no formation and therefore no local equilibrium; the reaction is split across the two operators, each reading the other's frozen value from the one rate, which makes the heat of channel 18 the heat of exactly the events the species ledger performs. The products needed nothing added: the free-atomic-oxygen closure of carrier\_source and  $n_{\text{free}} - n_{\text{CO}}$  in carrier\_write\_back already carry the two nuclei and element\_census already counts them in CO. Tests: new physics\_probe driver co\_helium\_ion\_sink (8 assertions, RED 4 FAIL against the pre-change module relinked into the same suite, GREEN 8 PASS; the RED reads hep\_row\_co\_sink 0.0 against -3.84e6, hep\_row\_derivative\_co\_term 0.0 against -38.4, hep\_turnover\_scale\_co\_term 0.0 against 3.84e13); physics\_probe suite PASSED after; element census with EXHALE\_ELEMENT\_ASSERT=1 on oxygen\_chemistry 150 steps exits 0 with no budget violation (a guard, not a RED/GREEN pair: the change moves no C or O nucleus). Impact, MEASURED at 1000 steps of oxygen\_chemistry: the largest share of the total heating taken by channel 18 falls from 0.672 at  $r = 1.081$  to 0.035 at  $r = 1.117$ , channel 18 at  $r = 1.0894$  from  $6.90e-7$  to  $9.75e-9 \text{ erg cm}^{-3} \text{ s}^{-1}$ ,  $n(\text{He}^+)$  there to 0.015 of its former value and to 0.0073 at  $r = 1.011$ ,  $T$  at  $r = 1.0233$  from 1156.4 K to 1088.2 K, and  $\log_{10} \text{ Mdot}$

stays 9.61 with exit 0. dom\_cells\_HeP does NOT fall (281325 -> 281382): the counter compares k\_D1 n\_CO with recombination and still says this reaction is the leading He+ loss of the layer, so what it now reports is the operator split's lag rather than a missing sink. mol\_carrier 300 steps and mol\_metals 400 steps byte-identical in every output file. Goldens not refreshed. Reported for owners elsewhere: the dom\_cells\_HeP header text in write\_output.f90 558-562 still says the frozen He+ "over-states that channel"; the mirror decomposition hydrogen\_helium\_row\_terms in constrained\_chemical\_equilibrium.f90 needs the same term after line 1859 or it stops summing to row 2; the n\_co comment in ion\_cell\_state.f90 still calls CO an equilibrium closure.

- B3b-CO2 landed, advisor verification and follow-ups** (2026-09-06): the He+ row of mol\_heh\_rows (System\_HeH\_mol.f90, the one row every caller uses) loses k\_D1 n\_CO n(He+), its turnover scale gains the same bound; CO stays a background density of the solve because the one-sided model gives it no formation channel (not a balance row). Products needed no code: the O nucleus goes to the free-oxygen closure and the C nucleus to nC\_free - n\_CO in the carrier write-back, element\_census already counts both; the carbon CHARGE is set by the sweep's stage equilibrium, the design of record ("the step moves nuclei, the sweep moves charge"), so the reaction's C+ is not imposed (recorded as a design choice, not a defect). MEASURED, oxygen\_chemistry 1000 steps: channel-18 share of the total heating 0.672 at  $r = 1.081$  -> 0.035 at  $r = 1.117$ ;  $n(\text{He}^+)$  at the front 0.015 of before (0.0073 at  $r = 1.011$ );  $T$  at  $r = 1.0233$  1156 -> 1088 K;  $\log_{10} \text{Mdot}$  9.61 both; constrained-continuation fallbacks 932 -> 1880, all accepted; mol\_carrier 300 and mol\_metals 400 byte-identical. dom\_cells\_HeP did not fall (281325 -> 281382): it measures dominance of the channel over recombination, not over-statement; its header text now says so. Advisor follow-ups applied directly: the D1 He+ + CO term added to the He+ decomposition of constrained\_chemical\_equilibrium.f90 (hydrogen\_helium\_row\_terms, whose header requires every mol\_heh\_rows reaction to appear), the FUV\_bands.txt/Oxygen\_chemistry.txt dom\_cells\_HeP header rewritten, the stale n\_co comment of ion\_cell\_state.f90 replaced. Private build: physics\_probe 592/0 (incl. co\_helium\_ion\_sink 8), fuv\_band\_ledger 24/0. Remaining: the operator split is now a LAG (the sweep destroys He+ at the entry CO while the CO row destroys CO at the previous sweep's He+) and oxygen\_chemistry never covers a carrier interval in init mode, so this row has no end-to-end gate: B4-1c/B5 territory, recorded.
- HEATBRK closed** (2026-09-06): the mol\_metals pair finished at its 12000 steps, Hydro\_ioniz.txt and Ion\_species.txt byte-identical, gate  $1.683\text{e-}16$  with 21 columns, \_adv profiles moved by at most  $8.45\text{e-}3$  (the deposits the post-process copy lacked). All three impact cases of the item therefore confirm: the sweep's heat is unchanged, only the two consumers of the assembly moved.
- B4-1** (species\_face\_flux.f90 new; Reconstruction.f90, PLM\_rec.f90, RK\_rhs.f90, EXHALE\_main.f90, binary\_element\_diffusion.f90, Makefile, new suite src/tests/species\_face\_flux/): the advective half of the ELEMENT transport now rides on the hydrodynamic face mass flux, inside the Runge-Kutta stages, on the same faces, areas, volumes and time step as the mass row. The face composition is the reconstruction of the mass fraction by the same limiter as the primitive variables, taken from the side the face mass flux selects, and the closing member of the set is one minus the others, so  $\sum_s F_s(j) = F_{\text{rho}}(j)$  holds by construction; the operator asserts it at every face of every stage (largest residual MEASURED in the unit suite:  $2.1\text{e-}16$  rel-

ative, against a tolerance of 64 ulp). The cell-velocity upwind advection of `binary_element_diffusion.f90` is gone from the marching path: the operator advects only when it is given an advecting mass flux, which the fixed-wind relaxation supplies and the marching path does not, and the trace-metal solve lost its advective term the same way. The carriers of `photochemical_transport_step` are NOT in this increment (that file was held by another item): they keep their present advective term and are B4-1c. Tests, all MEASURED with a private build: a uniform composition is preserved to 2 ulp on a compressing flow, an expanding flow and one whose face mass fluxes alternate in sign, for PLM and for WENO3; the mass fraction stays in  $[0,1]$  on a step profile carried across the domain in 300 steps; the closing member of the set differs from an independently reconstructed  $1 - Y$  by  $1.1e-16$ ; the face reconstruction converges at order 1.97 (PLM) and 1.92 (WENO3); the helium nucleus total of the domain changes only by its boundary fluxes to  $4.9e-17$ , where the cell-velocity form it replaces leaves  $1.6e-9$  on the same column and the same step (RED and GREEN in one run); the injected violation of the face identity stops the operator with the face named; the new step is refusable at its injection point and the run goes on. The element-diffusion suite `diffusion_tests` is 35 of 35 after T11 was rewritten onto the new operator (its premise, that a face-averaged upwind decouples a cell whose face velocities straddle zero, is what the face-flux form removes). Impact, MEASURED on scratch copies, `OMP_NUM_THREADS=1`, `mol_*/hp_*` at 200 steps and `mol_diffusion/lower_profile` at 1000 rather than the 12000 their maxsteps pins, `wasp_*` at 300: 14 cases byte-identical, every one of them without element transport (`wasp_full`, `wasp_he23off`, `wasp_full_newton`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_ir_bands`, `mol_sec_ion`, `mol_carrier`, `oxygen_chemistry`, `hp_zero_seed`, `hp_trace_seed`, `hp_front`, `hydrostatic_column`); `mol_diffusion` moves by at most  $7.1e-5$  in `Ion_species.txt` and  $3.6e-3$  in `Cooling_breakdown.txt`, `lower_profile` by  $1.2e-3$  and  $1.1e-2$ . The transported quantity,  $n_{\text{He}}/n_{\text{H}}$ , moves by  $1.9e-10$  at the base and  $3.5e-6$  at the molecular front of `mol_diffusion`; the base-layer mass flux  $\rho v r^2$  moves by at most  $9.2e-7$  over  $1.00 \leq r \leq 1.16$ . Those are smaller than the design's estimate because at 1000 steps neither case has separated its elements far from the reservoir ratio, and a uniform composition is what both forms carry exactly; a converged pair is what would measure the difference at the separation those cases reach, and is named as open in `docs/b4_spatial_operator_design_20260906.md`. No golden was refreshed.

- **B4-1 landed (element rows), advisor verification** (2026-09-06): new module `species_advective_transport` (`species_face_flux.f90`): the face mass fraction by the primitives' own limiter (PLM\_rec\_scalar, Reconstruct\_scalar, the MC limiter and the WENO3 geometry coefficients factored, not duplicated), upwind side = sign of  $F_{\rho}$  (coincides with the HLLC contact selection wherever both are defined, ROE/LLF offer only the sign), the closing member one minus the others so  $\sum_s F_s = F_{\rho}$  by construction and asserted at every face and stage, advanced inside the three SSP-RK3 stages with the mass row's  $dt$ ,  $dV$  and areas; `positivity_limited_fluxes` writes a replaced first-order flux back into `face_flux` so the species flux built on it is the repaired one. Transported set: helium and the hydrogen component (normalized), the trace metals riding on the same faces inside `m_1` at the reservoir abundance (trace limit stated). `binary_element_diffusion.f90` keeps the diffusive half only (`element_transport_residual` keeps its stationary advective term, equal where  $r^2 \rho v$  is constant). Carriers NOT migrated (`diffusive_photochemistry.f90` was B3b-CO's): B4-1c. Suite `species_face_flux` 13 + 2 whole-binary rows GREEN on the advisor's private build: uniform composition

preserved to 2 ulp (PLM, WENO3, three flows incl. a sign-alternating face flux), face identity 2.1e-16, bounds exact, face reconstruction order 1.97/1.92, helium nucleus total changes only by the boundary flux to 4.9e-17 where the replaced form gave 1.58e-9 (B1 AT-4d RED/GREEN in one run), identity assertion fires, element step refusable; diffusion\_tests 35/35 (T11 rewritten: its premise was the defect removed); attempted\_step, grid\_and\_gates 93/0, carrier\_retry 97/0 green. Impact (worker, control = same snapshot with the stage calls removed): 14 cases without element transport BYTE-IDENTICAL (wasp\_\* at 300, mol\_\* at 200 steps); mol\_diffusion and lower\_profile at 1000 steps move by at most 3.6e-3 / 1.1e-2 (cooling breakdown), n\_He/n\_H by  $\leq 6e-5$ , base rho  $v \cdot r^2$  by  $\leq 1e-6$ : far below the design's estimate because at 1000 steps the elements have not separated from the reservoir ratio; the converged 12000-step pair is open in the design document. Outside scope recorded: ESWENO3 measures second order on a smooth profile (1.92 against PLM 1.97, face error 8x PLM), a property of the primitive reconstruction to look at (WENO3-ORDER); hp\_\* cases need output/Hydro\_ioniz\_IC.txt in place; element\_flux\_profile.txt has a trailing text column.

- CEILING-DEL** (diffusive\_photochemistry.f90, write\_output.f90, write\_setup\_report.f90, EXHALE\_main.f90, certification.f90; tests carrier\_retry (+ scanner no\_CO\_ceiling\_identifier\_in\_src), carrier\_constraint\_attribution rewritten on the carbon clamp): the thermodynamic CO ceiling (clamp to co\_equilibrium\_density, four counters, attempted twins, three accessors, Oxygen\_chemistry.txt header line, three co\_ceiling\_\* resolved keys, the end-of-run line, validity state 4.1's only producer) is deleted per design decision 10; the domain record of B3b-CO stands in every place (seven co\_domain\_\* resolved keys under oxygen chemistry + carrier transport, the CO items under class 4.2 beside H3+ in the certification, n\_active\_unvalidated\_physics = -1 "not produced", kept because the five validity states are a fixed set). MEASURED before deletion on oxygen\_chemistry 1000 steps: co\_ceiling\_cells 0, applications 0, CO removed 0.0 (with the published destruction rates in the row and the He+ sink in the sweep the carried CO never exceeds its equilibrium), so every numerical column is byte-identical before/after; mol\_carrier 300, mol\_metals 400, wasp\_full 300 byte-identical. The design's "of order 160 cells" predates the CO rates and is retired with the clamp. Advisor verification: private build, carrier\_retry, certification 29/0, carrier\_constraint\_attribution 11/11, fuv\_band\_ledger green, no co\_ceiling identifier left in src/. Recorded for the docs sweep (DOCS-CEIL): the checkpoint enumerations and validity category 12 in a0\_run\_mode\_contract, a2\_certification\_contract, b1a\_active\_equation\_inventory, b3a\_attempted\_step\_controller\_design, b1\_target\_system (1246-1554), development\_plan\_20260905\_rev2/rev3/review3, physics\_numerics\_audit\_20260905, and b3b\_co\_destruction\_design sections 2.3/4/7 still describe the ceiling as present. The domain record is not checkpointed (a statement about the state, valid whether or not the step is accepted), unlike the ceiling record it replaces.
- DOCS-CEIL** (documentation only, 2026-09-07): sixteen documents brought level with CEILING-DEL, the current state READ from the code (destruction row  $\text{src}(\text{ic\_CO}) = -(\text{k\_D1} \cdot n(\text{He}^+) + \text{k\_CO}) \cdot n_{\text{CO}}$ , conservation-only limit\_to\_element\_budget, carrier\_co\_domain\_record, certification class 4.2 CO block, n\_active\_unvalidated\_physics = -1, seven co\_domain\_\* keys); no measurement deleted, the superseded

ones ("of order 160 cells"; "still cuts CO above 3000 K") kept with the measurement that replaced them beside them. Files: a0/a2 contracts, b1a inventory (4.1 row retired, 4.2 row the domain record, open question 3 closed), b3a design (checkpoint enumerations), b1 target system (5.1 split into the 2026-09-06 READ and the current one, AT-5 superseded, decision 12 moot), b3b\_co design (sections 2.3, 3(iv), 4, 6, 7, 7.4 as done), development plan rev2/rev3/ review3/execution, physics\_numerics\_audit (P6 resolved), code\_status, d0 (C26 closed, question 5 answered), co\_destruction\_rates\_literature (banner), To\_be\_determined\_recommend, PLAN rev2. Left as dated records: the superseded plan revisions and external reviews, and the b1 draft. Recorded: line-number citations in the 2026-09-06 design documents have drifted by up to a thousand lines (a re-anchoring sweep on routine names is a separate item, DOCS-LINES); the stale checkpoint header comment in diffusive\_photochemistry.f90 goes to B4-1c.

- WENO3-ORDER, measurement and fix** (2026-09-07): the ESWENO3 reconstruction was SECOND order on every grid the code runs on. Probe weno3\_reconstruction\_order (physics\_probe; exact volume averages by 8-point Gauss quadrature, face error against the point value,  $N = 50$  to 800): production rate 1.997 (uniform in  $r$ ), 1.988 (the define\_grid Mixed production grid), 2.940 on a grid uniform in the volume coordinate, where the defect is invisible. Cause, isolated by variants: not the ESWENO nonlinear weights ( $S = 1$  gives the same 1.997), not the floor  $\text{eps} = \text{dr}_j^2$  (rate 3.000 at kappa 0.1 to 10 once fixed), not the ideal ratios  $D1, D2$  ( $1/2 + O(h)$ , harmless); TWO of the four stencil coefficients applied the NEIGHBOR's volume share where cell  $j$ 's own belongs:  $C1(j) = c/(b+c)$  on the jump to  $j+1$  at the right face and  $C2(j-1) = a/(a+b)$  on the jump to  $j-1$  at the left face, each  $1/2 + O(h)$  where  $1/2 - O(h)$  belongs, an  $O(h)$  coefficient error on an  $O(h)$  jump. Inherited character for character from ATES v2.0 (ATES/ATES-Code-main). Advisor check of the derivation: a linear interpolant of two cell averages in the volume coordinate, evaluated at a face  $dV(j)/2$  from cell  $j$ 's center, weights the jump by  $dV(j)/(dV(j) + dV(nb))$ ; with the corrected shares a profile quadratic in the volume coordinate is reproduced to  $5.9e-16$ . **Fixed (advisor, two symbols in each of Reconstruct and Reconstruct\_scalar, Reconstruction.f90):**  $C1(j) \rightarrow C2(j)$  on  $dWp$  in WL,  $C2(j-1) \rightarrow C1(j-1)$  on  $dWm$  in WR; the probe's production mapping and header updated, the two waiting assertions enabled: weno3\_production\_third\_order rate 3.000, \_mixed 2.991 (RED before: shortfalls 0.803, 0.812); physics\_probe 603/0. **Every golden whose case reaches the WENO3 stage moves, and not by rounding** (face error down two decades at  $N = 800$ ); PLM-only cases are untouched. Consequence for the method: until now the two-stage PLM  $\rightarrow$  WENO3 recipe was a continuation between two second-order schemes with the second having the larger error constant (1.3x uniform, 3.8x on the production grid). Mdot values are to be quoted from Newton-finished runs after the gate. Attribution of the weight form and the floor to Yamaleev and Carpenter (2009, JCP 228, 3025 and 4248) stays unverified: both papers are paywalled here; requested from the user. Also recorded: stale .ipynb\_checkpoints copies of Apply\_BC.f90 and UW\_conversions.f90 under src/modules/ define the same module names as the live files (not built; to remove, HYG-CKPT).
- HYG-CKPT** (advisor, 2026-09-07): ten untracked, gitignored Jupyter checkpoint copies of production modules under src/modules/\*/ .ipynb\_checkpoints/ (Apply\_BC, Reconstruct, Source, base\_boundary, caloric\_eos, low\_mach\_dissipation, cross\_sec, UW\_conversions, h2\_photo\_channels, element\_census) re-

moved: not built, but each defined the same module name as a live file and two scanners had to skip that directory (batch 2c) to stay green.

- T15-INSTRUMENT** (attempted\_step.f90 +289/-165, suite, b3a design section 9, b1 T1.5 wording; advisor verification: private build, attempted\_step 43/0 with the two whole-binary rows skipping until the main-loop hook lands, coupled\_source\_step, certification green): the B3a instrument now states the THERMAL identity of the step,  $\sum_j dV_j [u_{th}^{n+1} - u_{th}^n] = \sum_j dV_j dt (\text{heat} - \text{cool})_j + (\text{transport contribution})$ , the transport contribution MEASURED between two marks on either side of the coupled source step, the residual the closure of the source step alone,  $\Delta u_{form}$  reported as the reservoir's change and never added (the B3c reason stated at the routine). GATE in phys mode at the energy row's own energy\_res\_tol = 1e-9 (reason as\_reject\_source\_energy = 14 at operation 9): MEASURED on a hooked scratch tree, worst lresidual/scale 7.44e-1 -> 5.40e-10 (wasp\_full 300 phys), 8.63e-1 -> 1.88e-10 (mol\_base\_handoff), zero refusals, states byte-identical in both modes. The two hook lines in EXHALE\_main.f90 (before/after the coupled\_source: loop) are assigned to COST2, the file's current owner, with the rewording of the print block. **Defect found and fixed in passing:** attempted\_step\_checkpoint\_take saved u and f\_sp through a saver that reallocates from index 1, so a reader indexing the saved grid arrays by cell compared cell j with cell j - Ng; energy\_identity was that reader, so every d\_u\_th/d\_u\_form printed so far (incl. the 28 percent reservoir share of the B3c report: with bounds fixed, same step, d\_u\_th 2.006e-5, d\_u\_form 1.298e-5) was formed against a state shifted by two cells; the state restore itself was unaffected (whole-array assignment). The second private copy of the eps\_s table in attempted\_step.f90 (no OH/H2O/CO rows) is deleted in favor of species\_formation\_energy. Recorded: n\_projection\_applied can no longer be nonzero since B3c but is still printed by the report and refused by the certification (HYG-PROJ, spans global\_parameters, certification.f90, main); design section 9 decision 8 still names the projection as a live correction (dated record).
- COST2** (measurement; the only production change is the T15 hook and print in EXHALE\_main.f90, byte-identical on wasp\_full 300 in both modes and mol\_base\_handoff 60 phys; attempted\_step whole-binary rows now run, identity residual -1.8e-10, zero refusals): the brief's premise is refuted twice. (1) MEASURED with T frozen through the coupled loop on wasp\_full: the composition sweep alone still takes 10-13 passes with the same contraction 0.15-0.20 and the same front walking one cell per pass; freezing the lagged coefficients as well collapses it to 2 passes; freezing only the radiation field leaves 3-4, only the rate coefficients 9-13. The pass count is the self-consistency of the photoionization field with the composition that attenuates it; the temperature is worth at most one pass. (2) A timer inside ioniz\_eq (60 steps, 29.95 s): PH\_heat 11.40 s (38 percent of the run), the whole cell loop 7.37 s (25 percent), pre-sweep eval\_cool 2.44 s, post-sweep assembly 2.40 s; skipping the entire cell body for resting cells saves 4.7 percent. B3c-COST's gprof attribution (78 percent in MINPACK) was inverted; the csm\_max\_pass comment still carries it (to be rewritten by COST3). (3) The two-pass rest mask was implemented, measured (6.0 percent on wasp\_full, 0 on mol\_base\_handoff, state moved 5.5e-7, step 0 of mol\_base\_handoff in phys turned into an exhausted attempt) and removed. **The lever: the field dependence is one-directional (a cell's P\_HI is set by the attenuation of the cells outside it) and the sweep already runs**

outside in, so updating the column with the composition just produced turns the Jacobi iteration into a forward substitution: same fixed point, about one pass instead of ten, and PH\_heat no longer repeated per pass. Assigned as COST3 (util\_ion\_eq.f90 PH\_heat\_HHe/PH\_heat\_H, ionization\_equilibrium.f90 sweep order; the OpenMP parallel cell loop becomes sequential in the field direction, so the design must weigh 10 parallel passes against 1-2 sequential ones and measure both at 1 and 16 threads). Also recorded: the pre-sweep eval\_cool is run for its coefficients and its cooling discarded, then a second traversal forms the cooling (16 percent together); mol\_base\_handoff in phys refuses 7 of its first 60 steps at the fixed point on the unmodified tree.

- B4-1c landed: the carriers ride on the species face flux** (2026-09-07): the carriers of photochemical\_transport\_step (H2 always, OH/H2O/CO under Oxygen chemistry, H+ under Ionization transport) join the transported set of B4-1 and are advected as  $F_{\rho} Y_c/m_c$  inside the Runge-Kutta stages. carrier\_set\_init declares them once the keys are parsed (advected\_carrier\_register in binary\_element\_diffusion.f90, which also holds the columns and hands them back through advected\_carrier\_fractions); EXHALE\_main.f90 is untouched, the existing three calls carry both sets. The carriers ride OUTSIDE the normalized set ( $n_{\text{norm}} = 0$ ): a carrier is one species inside an element, the element is already the closing member, and normalizing it beside the element would count that mass twice; what keeps carrier and element consistent is the operator's own write-back, which restores the element totals cell by cell. Their rows are advanced from cell 1 (the elements' start at 2), with the inflow composition taken from the inner ghosts: the handoff's q\_H2\_base where one is stated, the base cell's own partition where none is, one rule for both branches of decision D6 (B4-4a). diffusive\_photochemistry.f90 loses the cell-velocity advective term, carrier\_advection\_correction and the base Dirichlet branch from its MARCHING rows and their Jacobian; two paths keep the advective term, carrier\_steady\_residual (the stationary balance is the whole equation) and the fixed-wind relaxation relax\_photochemical\_composition (it takes many transport steps with no hydrodynamic stage between them, so nothing else is taking that transport), through the switch carrier\_rows\_advect. Tests: two rows added to species\_face\_flux, uniform\_carrier\_partition\_preserved 0.83 ulp (bound 8) on a sign-alternating face mass flux and h2\_nucleus\_total\_changes\_only\_by\_boundary\_flux 1.36e-15 (bound 1e-12), RED before in the strongest sense (the driver does not compile against the pre-increment module); suite 15 + 2 GREEN; carrier\_retry 101, carrier\_returned\_state\_acceptance 37, carrier\_constraint\_attribution 11, carrier\_reference\_scales 15, acceptance\_classes 22 GREEN with no reference re-derived. Impact MEASURED against a binary built from the same tree snapshot with the two files at their pre-increment text: 12 cases BYTE-IDENTICAL, every one without Molecular carrier transport (wasp\_full, wasp\_he23off, wasp\_full\_newton, mol\_base\_handoff, mol\_metals, mol\_lyman\_werner, mol\_ir\_bands, mol\_sec\_ion, mol\_diffusion, lower\_profile, oxygen\_chemistry, hydrostatic\_column); the four that do transport carriers move. mol\_carrier at 1000 steps: Ion\_species 5.0e-1 (H3+ at  $r = 1.100$ ), Cooling\_breakdown 6.2e-1, Hydro\_ioniz 2.6e-2 ( $v$  at  $r = 1.186$ ); hp\_zero\_seed/hp\_trace\_seed/hp\_front at 100 steps: 8e-3 to 1.1e-1, Hydro\_ioniz 1.3e-3 to 1.8e-3. **The front does not move; its tail does.** The  $x_{\text{H2}} = 0.5$  crossing of mol\_carrier is  $r = 1.0698$  at 1000 steps before and after and  $x_{\text{H2}}(1.05)$  is 0.7308 against 0.7310, while

$x_{H2}(1.10)$  falls from  $3.19e-3$  to  $2.02e-3$  (-37%): the molecular tail the cell-velocity form was carrying by its own numerical diffusivity, which is where the design said the difference lives, and every large entry above is a consequence of that column. Every carrier interval covered and the element limiter moved no cell, in both runs. Recorded and not fixed: oxygen\_chemistry has carrier\_transport on by default with the oxygen chemistry, but at 200 steps in initialization mode 201 of 201 intervals are uncovered before AND after, so the write-back never runs and the case is byte-identical for that reason, not for an absence of effect (its rows did change: worst CO row of step 1 |res|  $6.03e-5$  against  $5.68e-5$ ). No golden was refreshed. Also in this increment: the stale clause saying the carrier checkpoint restores "the cumulative CO record" removed from two comments in diffusive\_photochemistry.f90 (the carrier\_checkpoint type carries no such field since CEILING-DEL).

- B4-1c landed, advisor verification** (2026-09-07; the worker's own entry carries the numbers): every carrier of the photochemical transport operator ( $H_2$ ; OH/ $H_2O$ /CO under oxygen chemistry;  $H^+$  under ionization transport) rides on  $F_{rho} \ Y_c \ / \ m_c$  inside the RK stages through the registry of binary\_element\_diffusion.f90 (carriers outside the normalized set, from cell 1, base inflow composition = the inner ghosts'), species\_face\_flux.f90 unchanged (species-agnostic), EXHALE\_main.f90 untouched; the marching rows of diffusive\_photochemistry.f90 keep diffusion, drift and chemistry only; the advective term stays in carrier\_steady\_residual (the whole stationary equation) and in the fixed-wind relaxation relax\_photochemical\_composition (no hydro stage takes it there), switch carrier\_rows\_advect. Advisor: private build, species\_face\_flux 15 + 2 rows (carrier partition 0.83 ulp,  $H_2$  nucleus budget  $1.4e-15$ ), carrier\_retry, carrier\_returned\_state\_acceptance, acceptance\_classes, grid\_and\_gates green. Impact (worker, same-snapshot control): 12 cases without carrier transport BYTE-IDENTICAL; mol\_carrier 1000 steps: the  $H_2$  front does not move ( $x_{H2} = 0.5$  at  $r = 1.0698$  both), the tail the cell-velocity form carried by its numerical diffusivity does ( $x_{H2}(1.10)$   $3.19e-3 \rightarrow 2.02e-3$ , -37 percent), hence  $H_3^+$  0.50, cooling 0.62, heating 0.37 relative there;  $hp * 100$  steps  $\leq 1.7e-2$ ; base  $\rho \ v \ r^2 \leq 9.7e-4$ . Recorded: element\_advection\_\* names are now too narrow (HYG-NAME, six call sites in main, after COST3); oxygen\_chemistry in init mode covers 0 of 201 carrier intervals so it cannot judge carrier transport; a refused carrier interval discards a transport the mass row already took (B3a/B5 question); adv\_corr still checkpointed though only the residual writes it; 52 documents under docs/ carry em-dashes (DOCS-DASH, launched).
- DOCS-DASH** (documentation only, 2026-09-07): 1354 em-dashes and 237 en-dashes removed from 60 files (docs/\*.md, READMEs, test and data READMEs), each replaced by the punctuation the sentence needs (colon, comma, parentheses, or a table placeholder hyphen), en-dash ranges and joined names to the hyphen the repository already writes; 108 mechanical substitutions corrected by hand on read-back; grep over the 187-file set clean. Left: Update\_EXHALE\_stage1.md (append-only log, 319/59), and 1220 LaTeX --- in 27 .tex sources that render as the same dash (DOCS-DASH2, launched: the .md/.tex twins realign there; the built PDFs are re-generated at the gate's docs pass), plus docs/lower\_atmosphere\_figs/README.md (11) and three READMEs under backup/ (5) outside the first sweep's paths.
- DOCS-DASH2** (documentation only, 2026-09-07): 1283 LaTeX --- in 29 .tex sources (docs, paper, posters) and 30 spaced -- replaced by the punctuation each sentence needs; four exceptions kept and named (a direct

quotation of Frelikh and Murray-Clay 2026 compared letter for letter, the AASTeX \keywords separator, a literal metals.inp listing, hyphen rules); the four READMEs of the first sweep's remainder cleared. The 24 built docs/\*.pdf, literature\_survey.pdf, paper/ms.pdf and the poster PDFs are now dated and are regenerated in the gate's docs pass. Left: the append-only stage logs (Update\_EXHALE\_stage1.tex 1614, stage0.tex 37).

- COST3** (measurement; util\_ion\_eq.f90, ionization\_equilibrium.f90; default output BYTE-IDENTICAL on wasp\_full 300, mol\_base\_handoff 1500, mol\_metals 1500, hydrostatic\_column 300 at one thread; advisor rebuild: coupled\_source\_step 36, physics\_probe 862/0 incl. the new photoionization\_field\_substitution driver (259 assertions), grid\_and\_gates 93/0): the star-ward columns can now be carried forward cell block by cell block (advance\_starward\_columns, reproducing calc\_column\_dens\* bitwise; photoionization\_field\_at\_cell\*, the PH\_heat\* cell bodies extracted verbatim; xuv\_field\_block\_cells, default 0 = whole grid). MEASURED on wasp\_full 300: the substitution stops the front's walk (worst-moving cell 206 -> 177 over the passes on the lagged scheme, 202 -> 196 with the columns carried) and cuts the pass count 10.10 -> 7.13, but a pass costs the same field work either way, and the field of a whole grid is one loop sixteen threads solve at once: block width 8 buys 1.43x at one thread (136 -> 103 s) and LOSES 2.8x at sixteen (17.0 -> 42.5 s); width 64 keeps the parallel front and saves nothing. The default therefore stays the whole grid, and the width is not tied to the thread count (the widths reach states differing by  $1.5e-7$  at the  $1e-8$  tolerance, the thread count by  $7e-14$ ). **The remaining lag is the cell's OWN optical depth**, which enters its own field only on the next pass (contraction 0.17): closing it means the cell residual carries its own field, System\_HeH\*, i.e. B5. Folded: the pre-sweep eval\_cool run for its coefficients only is now chemical\_rate\_coefficients (-3.6 percent wasp\_full, -5.6 percent mol\_metals, +3.5 percent mol\_base\_handoff, the last from the 27 metal slots looped with metals off: guard on thereis\_metals, HYG-COL). Recorded: docs/openmp\_parallelization.md (2.07x at 16 threads, bit-identical across threads) is stale: measured 8.2x and round-off  $6.6e-14$  (DOCS-OMP). A cp -irp of another item blocked on an interactive prompt since 2026-09-06 was killed by PID (23818, cwd checked).
- WENO3-REF** (comments only, Reconstruction.f90 and the probe's References block; the user supplied the publisher PDFs, read with pdftotext -layout): the nonlinear weights are EXACTLY Yamaleev and Carpenter 2009, JCP 228, 3025, Eqs. (18), (20), (21), (22) (published tau already squared;  $d_0 = 2/3$ ,  $d_1 = 1/3$  entering as the ratio 1/2 that D1, D2 carry); the scheme is NOT their ESWENO: the energy-stability dissipation of Eq. (48) is absent, their analysis is a uniform-grid finite-difference flux reconstruction, and for systems only in characteristic variables (Section 5 quoted), so the code inherits the accuracy result (Eq. 24) and not the energy estimate; the geometry (C1, C2, D1, D2 on the stretched finite-volume grid) is this code's own.  $\epsilon = dr_j^2$ : a defensible stand-in, not a defect: the published  $h^2$  scaling of Eqs. (62), (64) without the solution scale of Eq. (65); MEASURED on the production grid the face error is flat to every digit for  $\kappa \geq 0.1$  over four decades (rate 2.991) and deteriorates below  $1e-2$  per Eq. (57), so in smooth regions the departure is not measurable; where it acts is the ENO biasing ( $b_r/\epsilon$  spans  $2.2e-6$  to  $8.4e3$  over the wasp\_full grid; the published floor would make the base LESS nonlinear), a candidate experiment for the base sound-wave work, not a change here. Case label and comments now say "third-order WENO"; identifiers unchanged. Advisor: paper/ms.tex

144 "the energy-stable ESWENO3" reworded to the verified attribution and the ESWENO3 name replaced by WENO3 there and in docs/code\_comparison.tex. **Recorded for the user:** every src/tests/\*/ suite of this series (19 directories) and the 2026-09-05/06 design and plan documents are untracked in git; a fresh clone of the remote carries none of them.

- **References received** (2026-09-07, from the user, publisher PDFs in references/): Zhang and Shu 2010 (JCP 229, 8918), Poinso and Lele 1992 (JCP 101, 104), Thompson 1987 (JCP 68, 1) and 1990 (JCP 89, 439), Neufeld 1990 (ApJ 350, 216), Mao and Kaastra 2016 (A&A 587, A84), Verner et al. 1996 (ApJ 465, 487), Yelle 2004 (Icarus 170, 167), Larsson et al. 2008 (CPL 462, 145), Tully 1975 (JCP 62, 1893), Slinger and Black 1982 (JCP 77, 2432). Five verification items launched against them: REF-ZHANGSHU (the positivity limiter), REF-LODI (the characteristic boundary design document), REF-NEUFELD (the Ly-alpha escape-probability closure), REF-METALS (every metal photoionization and recombination fit parameter against the published tables, a transcription error being a physics fix), REF-CHEM (the H3+ network transcriptions, the H3+ dissociative recombination coefficient, O6 and the Ly-alpha H2O yields). The series gate waits for them, since REF-METALS and REF-CHEM may move coefficients.
- **Reference data fetched** (2026-09-07, at the user's direction, each with a provenance README, third-party material unchanged): references/verner\_photo/ (D. A. Verner's distribution: phfit2.f reference routine with its fit parameters, table1.dat of Verner et al. 1996, the 1995 inner-shell fits), references/mao2016\_cds/ (CDS J/A+A/587/A84 rr\_spex3.dat, 39696 rows), references/badnell\_rr/ and badnell\_dr/ (Strathclyde TAMOC total RR and DR fit coefficients clist\_K/clist\_eV, tabulated cfout\_\*, reference routines). REF-METALS now compares every metal stage's photoionization, RR and DR coefficients against these by direct evaluation of the reference routines, and reports which source each stage uses today and any missing DR contribution.
- **HYG-MAIN** (2026-09-07; numerically inert, wasp\_full 300, mol\_carrier 200, mol\_diffusion 200, mol\_base\_handoff 300 byte-identical; advisor rebuild: certification, attempted\_step, species\_face\_flux green): HYG-NAME element\_advection\_\* -> species\_advection\_\* in binary\_element\_diffusion.f90, the main program (use list, six calls) and the two test drivers, the two design documents that still named them (advisor); HYG-PROJ n\_projection\_applied deleted from global\_parameters, the attempted\_step\_report block and both certification.f90 references (MEASURED: every reference in src/ was a read, no assignment anywhere, so the counter was provably zero since B3c), rep%n\_unbudgeted\_accepted\_correction is now n\_shapiro\_applied alone, b3a design decision 8 dated closed for row 8; HYG-COL the 27-slot metal column loop of advance\_starward\_columns guarded on thereis\_metals (inert; COST3's +3.5 percent NOT reproduced: guarded 53.7/52.8 s against unguarded 55.7/52.5 s on mol\_base\_handoff 300, within the run-to-run spread, kept as dead-work removal); DOCS-OMP docs/openmp\_parallelization.md dated and brought to COST3's measured state (8.2x at 16 threads, 6.6e-14 between thread counts). Recorded: no reduction on a physical quantity exists in src/, so the 6.6e-14 thread-count difference comes from block boundaries set by omp\_get\_max\_threads() in chemical\_rate\_coefficients/eval\_cool; which routine turns a moved boundary into a moved last bit

is not localized (THREAD-DET: until it is, the thread count is a silent input to every result at round-off); photoionization\_field\_at\_cell\_HHe makes an array temporary for nm\_j on every call (runtime warning in physics\_probe); EXHALE\_main.f90:452 is 141 characters.

- **REF-ZHANGSHU** (comments, docs/p49\_front\_energy\_mode.md 6.1, probe positivity\_limiter\_scaling 15 assertions; user-supplied PDF read): the citation was wrong as stated: positivity\_limited\_faces is not the limiter of Zhang and Shu 2010 (JCP 229, 8918) but one built on the same idea, departing on three axes (primitive, not conserved, variables; one theta per face state, not per cell, Eqs. (2.4)-(2.5), (2.10)-(2.12); a relative one-ulp floor, not their absolute  $1e-13$ ). What carries over: Lemma 2.5 (convex combination with an admissible anchor); what does not: Theorem 2.1 (the next cell average admissible under their Gauss-Lobatto set and CFL,  $N = 2$  covers  $k = 1$  only; Remark 2.6 names finite-volume WENO as open) and conservativity of the reconstruction (a theta taken face by face moves the cell's end-point mean by  $8.3e-2$  where a theta taken cell by cell gives 0). Along their conserved path  $p(t)$  is concave, so the code's theta is never larger than their  $t_{\text{eps}}$  (MEASURED 0.856 against 0.949 at Mach 60): the anticipated negative-internal-energy failure cannot occur. **Defect found and fixed (advisor applied the worker's proposal):** with eps one ulp of the average the bare update  $q_{\text{avg}} + \text{theta} (q_{\text{rec}} - q_{\text{avg}})$  is a cancellation of the size of the floor and lands on EXACTLY ZERO for 1013 of 200000 reconstructed values crossing zero (never negative), the value the floor exists to prevent (division in  $v = m/\rho$  and in the sound speed); positivity\_limited\_faces now clamps the scaled density and pressure to eps after the update (same number in exact arithmetic). Moves only runs in which the limiter fired, by one ulp except where the zero would have divided; goldens at the gate. Probe: bare\_update\_reaches\_zero (the reason), production\_limiter\_never\_delivers\_zero (the production routine over 4000 grids of zero-crossing face states) GREEN. A theta taken cell by cell (Eq. 2.11), which would restore the reconstruction average, is recorded as a change of scheme, not applied.
- **REF-LODI** (documentation, docs/phaseC\_characteristic\_base\_bc.md +282/-50; the three user-supplied journal PDFs read as page images since their text layers lose the displayed equations): the wave-amplitude normalization (Poinsot and Lele Eqs. (15)-(23); Thompson 1990 (44)-(47) with  $l_1 = (0, 1, -\rho c, 0, 0)$ ; Thompson 1987 (19)), the LODI relations ((24)-(36), equal to Thompson 1990 (48) term for term), the subsonic inflow two-condition closure (P&L section 3.1, Tables III/IV; Thompson 1990 Eq. (68)) and all four branch-table rows CONFIRMED; no constant wrong, the constraint count unchanged. Two statements of the note corrected: pair (D) is NOT singular at stagnation (Thompson 1990 Eqs. (69)-(70) have denominator  $u_1 + c$ ; the note's singularity was its own closure  $\rho_b = F_c/(r_b^2 v_b)$ ; D1 reopened with a pointer), and D6's claim that no transparent condition is expressible as a fixed point is too strong (Thompson 1990 Eq. (64)  $L_5 = \rho c g_1$  and P&L Eq. (40)  $L_1 = K (p - p_{\text{inf}})$  are algebraic in the face state; D6 stays the user's decision with a third option). One gap stated: the source projection  $l_1 C$  (gravity and the spherical divergence, Thompson 1990 footnote 1 and Eq. (52); the steady form is  $L_1 = -\rho c g_1$ , not 0) survives only because both states of (C-) sit at the same radius; any reformulation relating cell 1's center to the face must restore it. Carlson's "but not U and p" marked unverified against the primaries.
- **REF-CHEM** (comments and documents only; mol\_rates.f90 and oxygen\_rates.f90 function bodies byte-

identical; probes h3p\_recombination\_branching 12 and water\_photolysis\_lyman\_alpha\_yields 6, RED with the competing published values): every mol\_rates constant sourced to Yelle 2004 matches his Table 1 digit for digit; Yelle's R7 (H3+ + H) reference column reads "Estimated", so Frelikh's barrier-free 2.0e-9 transcribes an estimate faithfully; the code's H3+ dissociative recombination R6/R7 ( $2.16\text{e-}8 / 5.04\text{e-}8 \times (300/T)^{0.65}$ ) is already Larsson et al. 2008's 300 K total  $7.2\text{e-}8$  split by their branching 0.70, with the 0.65 index inherited from the Sundstrom/Datz fits Yelle prints (Larsson gives no temperature dependence; he states the CRYRING/ASTRID values the older  $2.9\text{e-}8$  rests on were too high from rotational excitation), so the code stays; the Slanger and Black 1982 Ly-alpha H2O yields 0.78/0.10/0.12 are exact (p. 2435) with one label corrected,  $\Phi_{12} + \Phi_{14} \rightarrow \Phi_{11} + \Phi_{12}$ , and the paper's coupling of the three yields and the 8 percent OH(A) bound recorded; Tully 1975's  $2.87\text{e-}10$  is a 300 K statistical-model value (Table II) that VULCAN flattens over 100-2100 K, and its agreeing "experiment" is a 1973 relative-rate evaluation, strengthening the audit's verdict for O6. Documentation fixed in passing: methodology\_comparison.md 6.4 listed three-body H2 formation as identical in value while the code's Cohen and Westberg 1983  $2.8\text{e-}31 T^{-0.6}$  differs from Ham et al. 1970 by 1.142 at every temperature; Baulch 1992 is in references/ (its H2 + H2 entry valid 2500-8000 K, above the whole layer); section 10 item on H2 dissociative photoionization closed (the code carries it, h2\_photo\_channels.f90 channel S).

- **REF-NEUFELD** (comments and documents; lya\_rt.f90 machine code identical before/after; physics\_probe 1118/0): the Ly-alpha escape-probability closure does NOT use Neufeld 1990 or Harrington 1973.  $\beta_{\text{esc}} = \pi^{(-1/4)} \sqrt{a/\tau}$  is the one-flight wing escape chance under complete redistribution; Ly-alpha scatters with partial redistribution and the published static-slab damping-wing solution is Harrington 1973 eq. (40),  $\langle N \rangle = 0.9093161$  B scatterings for a mid-plane source, so  $\beta = 1/\langle N \rangle$ , proportional to  $1/\tau$  and independent of  $a$ ; MEASURED on the stored wasp\_full column the code exceeds it by 234x at the base and 14x at 1.20 Rp, crossing at 1.42 Rp. Neufeld contains no expression of the code's form (his escaping fractions are escape against destruction, eq. 2.25 partitions the two faces; the code's two-face PRODUCT is unpublished); the escape frequency scales as  $(a \tau)^{(1/3)}$  (Harrington eqs. 37, 47; Neufeld eq. 4.32), not the code's  $(a \tau)^{(1/2)}$ ; Neufeld's validity  $(a \tau)^{(1/3)} > 10$  holds in 320 of 504 cells (inside 1.156 Rp); Neufeld is static, the Sobolev factor is Sobolev 1960 radial-ray. The header's J\_int formula lacked the  $(1 - \beta)$  factor the code applies (fixed). **Advisor decision: adopt the published solution (LYA-BETA, launched):**  $\beta = 1/$  of eq. (40) joined to the thin limit, the face partition of eq. 2.25 in place of the product, the  $(a \tau)^{(1/3)}$  escape frequency, a validity record, A\_2p1s  $6.3\text{e}8 \rightarrow 6.2648\text{e}8$  (spectroscopic), every case with Jlya escape-prob: True moves (expected at frozen state: J\_int/n\_2p up to 90x higher inside 1.02 Rp, 14x at 1.20 Rp, up to 7x lower beyond 1.47 Rp; Mdot to be measured to the du stop). The same question for the metal lines' line\_escape\_probability\_one\_face (Cool\_coeff.f90) is LYA-BETA-METALS, after REF-METALS.
- **REF-METALS** (cross\_sec.f90, Cool\_coeff.f90; probes metal\_photoionization\_fits, recombination\_coefficient\_fits, 255 assertions; advisor rebuild green apart from LYA-BETA's in-progress probe): all 132 fit parameters of the seventeen metal photoionization cross sections match Verner et al. 1996 Table 1 (verner\_photo/photo.dat) digit for digit and metal\_photoion\_sigma reproduces Verner's own phfit2 exactly wherever the fit is the one phfit2 uses; 220 Badnell RR and DR parameters (fourteen

daughters; cfout\_K witness within 0.045 dex), the H/He rows and the Mao and Kaastra  $n = 1$  rows match. **Two fixes:** the Mg II turn-on gated at Verner's rounded 15.04 eV while the grid edge is the NIST 15.035 (e\_th\_MgII now; byte-identical on wasp\_full, the band geometric means miss the sliver); alpha\_rr\_metal('CaI') used Verner rrfit.f's rrec(:,20,19), the row that FORMS Ca II (second index = the recombined ion's electron count, fixed by rnew(:,1,1) = H+ + e and rnew(:,2,2) = He+ + e), 6.05x the Ca I row ( $1.120 \times 10^{-13}$ , 0.900) at  $1 \times 10^4$  K; corrected. Impact (worker, scratch): wasp\_full 300 Ca I 0.87 relative at its 1.196 Rp maximum, Ca II +2.8 percent at the base,  $T$   $4.3 \times 10^{-3}$ ,  $v$   $9.9 \times 10^{-3}$ ; mol\_metals 400 Ca II/III  $\times 2.5$  at 1.38 Rp, cool  $2.9 \times 10^{-2}$ ; wasp\_full, mol\_metals, lower\_profile move at the gate through calcium. **Two physics items opened:** (a) the outer-shell fit is evaluated on the whole photon grid to 1240 eV while every row carries an E\_max (34-558 eV) above which phfit2 hands over to the 1995 subshell fit and adds the inner shells: the code carries 0.1-12 percent of the true metal photoabsorption there, and Fe I's row ( $Q = 1.54$ ) RISES with energy to 49x the truth at 1240 eV; the handover of phfit2 itself is adopted (METALS-OUTER, launched); (b) the inner-shell absorption and its Auger cascade (an atom two or more stages up) cannot be represented by the three-stage metal ladder: To\_be\_determined\_by\_user\_20260906.md item 7. Also: verner\_photo/README.md corrected (photo.dat is the 1996 Table 1, table1.dat the 1995 fits); rev3 line 944 closed; e\_th\_MgI/e\_th\_MgII provenance lines in parameters.f90 to be annotated (METALS-OUTER, after LYA-BETA releases the file).

- **METALS-OUTER** (cross\_sec.f90; generator src/utls/metal\_photoion\_table.py from verner\_photo/photo.dat + table1.dat; probes metal\_photoionization\_fits 188 and metal\_photoion\_table\_transcription 15; docs/photoion\_cross\_sections.tex + PDF; advisor rebuild: physics\_probe green): the seventeen sigma\_XX routines are one parameter block, and metal\_photoion\_sigma makes phfit2's handover: the 1996 outer-shell fit below E\_max, the 1995 fit of the same shell at and above it (MEASURED: einn of phfit2 equals photo.dat's E\_max for all sixteen metal rows; the production routine reproduces phfit2's outer shell to  $1 \times 10^{-10}$  at 17 ions  $\times$  12 energies where 110 of 204 rows disagreed before, worst Fe I at 1240 eV 24.9 Mb against 0.00172 Mb, a factor 14464). Inner shells still absent by decision item 7 (option (a) recommended; the measured sizes now in that item). Impact (scratch, metals-off mol\_base\_handoff byte-identical): wasp\_full 300  $v$  0.15,  $T$   $4.7 \times 10^{-2}$ , heat  $7.8 \times 10^{-2}$ , Fe III 0.99, C III 0.60 relative (hard photons now reach the base, the top stages fall at small radius; metal share of the heating at the base 17.4  $\rightarrow$  8.5 percent); mol\_metals 400  $v$  0.56, heat 0.42; lower\_profile 200 heat  $8.3 \times 10^{-3}$ ; log10 Mdot unchanged to 0.01 dex in the bounded snapshots. Both Mg thresholds already agree with NIST to their digits (item 4 needed no move; provenance note pending parameters.f90).
- **LYA-BETA-METALS** (measurement by the worker, fix applied by the advisor; Cool\_coeff.f90, probe fine\_structure\_escape\_probability, docs/resonance\_line\_trapping.md sections 6.1 and 10; publisher scans of de Jong, Boland and Dalgarno 1980 and Osterbrock 1962 fetched into references/): line\_escape\_probability\_one\_face is de Jong et al. 1980 eq. (B-7), both branches to round-off, applied to the eight ground-term fine-structure lines only; NO metal resonance line uses an escape probability (Mg II h&k, Ca II H&K, Na I D, Fe I/II at beta = 1, justified in that memo's sections 1-5 by  $n_{\text{crit,eff}} = \beta A/q \sim 1 \times 10^{10} - 1 \times 10^{15}$  against  $n_e \leq 4.3 \times 10^9$ ), and the transit post-process uses none. The fine-structure lines have a  $\sim 1 \times 10^{-8}$

and a  $\tau \leq 1e-9$ , so the Ly-alpha correction does not transfer. **Defect:** (B-7) is written in the frequency-integrated depth  $\sqrt{\pi} \tau_0$  (normalized profile; equal to Hollenbach and McKee 1979 eq. (5.10) once the convention is matched, 3.4 percent at worst over  $\tau = 3$  to  $1e6$ ) while the caller passed the line-center column: beta too large by 1 (thin) to 2.11 (branch point) to 1.77 (asymptotic). **Fixed:** each cell's depth is now  $\sqrt{\pi}$  times the line-center depth in `fine_structure_line_transfer`. MEASURED regime on the shipped columns: largest line-center depth 0.14 ([O I] 63  $\mu$ m at the hot-Uranus base), so the factor there is 1.12 and the change of the total radiative losses is below 0.1 percent (`mol_metals` 0.096 percent), but a factor of two in any base thick in these lines and in the Base IR field incident term the same beta scales. `physics_probe` 1291/0 after the fix. Metals cases move at the gate at that level.

- **Decision item 7, literature check** (2026-09-07; the user supplied Locci 2022 PSJ 3, 1; Locci 2024 PSJ 5, 58; Cecchi-Pestellini 2006 A&A 458, L13 and 2009 A&A 496, 863; Fox 2008 SSRv 139, 3; all in references/): the published exoplanet X-ray photochemistry models take the inner-shell absorption as opacity summed over all ionization channels (Cecchi-Pestellini 2009 Eq. 1 with Verner 1993 cross sections), put the photoelectron and the Auger electrons into the secondary-electron cascade (Locci 2022 Eq. 3), and carry singly charged ions only, i.e. option (b) of the item with the charge under-count accepted; no escape hydrodynamics code includes inner shells. Recommendation revised from (a) to (b) in `To_be_determined_by_user_20260906.md`; the user decides.
- **METALS-INNER** (decision 7 option (b), user 2026-09-07; `cross_sec.f90` subshell block from `phfit2.f`'s PH1 data (its thresholds differ from `table1.dat` in six rows, `phfit2` being the reference), `set_energy_vectors.f90` (every subshell turn-on of an active metal ion a bin edge; bin count unchanged, 17 -> 60 threshold cuts on `wasp_full`), `util_ion_eq.f90` comment only (the existing rate  $\sigma/e_v$  and heat photoelectron\_share( $E_{th}$ )  $\sigma$  integrals already implement "one stage, one electron of  $h\nu - E_{th}(\text{outer})$  to the SvS85/Dalgarno cascade" once the cross section is the total); generator `metal_photoion_table.py` --check; probes 392 metal assertions, `physics_probe` 1479/0; docs `photoion_cross_sections.tex` (+PDF), `code_status` new L row): the metal photoabsorption is the SHELL SUM `phfit2` returns, reproduced to  $1e-10$  at 17 ions x 12 energies (130 of 204 rows disagreed before). It also closed a cliff METALS-OUTER's handover had left at  $E_{max}$  where the 1996 fit stood for a valence complex the 1995 outer row does not cover (Fe I x0.012 and Fe II x0.0066 steps, now 1.23 and 1.13; genuine K edges C/N/O x19-21), and put  $E_{max}$ , a step of the cross section, on a bin edge (it was not). Approximation stated at the code site: ion advanced one stage, one electron of  $h\nu - E_{th}(\text{outer})$  to the cascade (MEASURED with the degradation routine: 0.95-1.09x the heat and 1.10-1.37x the secondary H I ionizations of the true two-electron event; the un-spent  $I_2$  is 24-35 eV of a 300-550 eV photon), fluorescence neglected (with  $e_{top} = 1240$  eV only C, N, O, Na have K holes; the high-Z ions contribute L shells only). Impact (worker, same-snapshot control; metals-off `mol_base_handoff` byte-identical): `wasp_full` 300 the wind 1-3 percent hotter and faster, base heating +26 percent, T at most +9.5 percent, Fe II-I/Mg III/N III at the base x74-160 from vanishing values, **Mg II falls 84 percent at  $r = 1.178$**  (the h&k transit line); `mol_metals` base heating +26 percent, T  $1.4e-3$ ; `lower_profile` heating +21 percent at the base, T  $2e-5$ ; metals' share of the 100-1240 eV absorption 2.2 -> 3.5 percent (`wasp_full`), the band 0.38 percent of the whole;

log10 Mdot unchanged to 0.01 dex in the snapshots. Krause 1979 (JPCRD 8, 307) K/L fluorescence yields could not be obtained (NIST 503, AIP 403): requested from the user; no yield is quoted anywhere. Advisor verification: physics\_probe 1479/0, grid\_and\_gates 93/0; spectrum\_type segfaulted in loaded\_sed\_threshold\_edges (its threshold array was sized  $n_{\text{mion}} + 16 = 43$  while the active list now reaches 126), fixed by sizing it with the module's  $n_{\text{thr\_max}}$ ; suite green after. ( $n_{\text{thr\_max}}$  was a local parameter of set\_energy\_vectors; hoisted to module scope, public, so every caller of ionization\_thresholds\_active sizes its array by the one bound; spectrum\_type 154/0, physics\_probe 1479/0, grid\_and\_gates 93/0 after.)

- **METALS-INNER addendum, Krause 1979 yields** (comments and documents only; user-supplied Krause\_1979\_JPCRD\_8\_307.pdf, read from page images where the OCR is noisy, alignment checked on  $\omega_K(\text{Fe}) = 0.340$ ): Table 3  $\omega_K$  for C/N/O/Na  $2.8\text{e-}3/5.2\text{e-}3/8.3\text{e-}3/0.023$ , Mg/Si/S/K/Ca/Fe  $0.030/0.050/0.078/0.140/0.163/0.340$ ; Table 5 effective L1/L2 yields (Coster-Kronig included per the text,  $\nu_3 = \omega_3$ ) for Mg to Fe all below  $6.5\text{e-}3$ ; Table 2 uncertainties ( $\omega_K$  40-10 percent at Z 5-10, footnote (a) on molecules and solids). With  $e_{\text{top}} = 1240$  eV only C, N, O and Na I can hold a K hole, so the neglected radiative branch is at most 2.3 percent of a K-hole event (under 1 percent for the three coolants) and under 0.7 percent of any L-shell event; raising  $e_{\text{top}}$  past 1.3 keV (Mg, Si K) or 4 keV (Ca, Fe K) is the condition under which the approximation must be revisited, stated at the code site, in photoion\_cross\_sections.tex (new tab:kedge), code\_status and decision item 7.
- **LYA-BETA landed, advisor verification** (lya\_rt.f90, hydrogen\_n2\_rates.f90 ( $A_{2p1s} 6.3\text{e}8 \rightarrow 6.2649\text{e}8$ , the nonrelativistic dipole value times the reduced-mass factor, = NIST ASD's Wiese and Fuhr 2009 value), parameters.f90 comments; probe lya\_escape\_probability 24 assertions; advisor rebuild physics\_probe 1479/0, certification 29/0): the escape probability is the published static-slab solution,  $\langle N \rangle = \text{Neufeld 1990 eq. (3.27) at zero destruction} = \text{Harrington 1973 eq. (40)} (4 \sqrt{6}/\pi^2) u_2 B = 0.909316 B$  for a mid-plane source (the 0.9093161 of the REF-NEUFELD entry was a transcription slip),  $\beta = 1/(1 + \langle N \rangle)$  (exact at both limits: a generated photon is emitted  $1 + \langle N \rangle$  times, and the budget  $n_{2p} A \beta = P$  wants the escape chance per emission), the two faces sharing one escaping population by Neufeld eq. (2.25) instead of the former product,  $\Phi(\xi)$  summed in closed form; the Voigt parameter has left the escape probability. a still enters only lya\_wing\_penetration\_coefficient (where an incident stellar photon crosses in one flight, a different question, judged at the code site). Defect found in passing:  $\beta_{\text{tot}} = 1 - (1 - \beta_{\text{esc}})(1 - \beta_{\text{sob}})$  lost digits at the base ( $4.8\text{e-}9$  relative), now  $\beta_{\text{esc}} + \beta_{\text{sob}} (1 - \beta_{\text{esc}})$  (closure  $2.2\text{e-}16$ ). RED before  $4.79\text{e}2$  / GREEN  $2.19\text{e-}16$  on the end-to-end identity. **Impact, wasp\_full to its du stop (2 h 59 m): log10 Mdot 13.34  $\rightarrow$  13.34 (+1.6 percent, 0.0067 dex); Jint/n\_2p x84 at the base, x73 at 1.05, x37 at 1.10, x10 at 1.20, x0.75 at 1.50 R\_p; T within 2 percent; the old min(1, .) cap had zeroed the internal field from 1.15 R\_p outward. H(n=2) chord column x31.5 at  $b = 1.00$ , x2.8 at 1.20, x1.005 at 1.45: every H-alpha/H-beta transit number in this repository from a Jlya escape-prob: True run is superseded (He 10830 not affected). Validity  $(a \tau)^{1/3} > 10$  holds in 349 of 504 cells, out to  $1.193 R_p$ , exactly where the change is an order of magnitude. Advisor follow-ups applied: the second  $A_{2p1s}$  in exhale\_transit\_lib.py and docs/transmission\_spectrum.tex set to  $6.2649\text{e}8$  (one value with the wind model), the stale  $E = \min(\text{boost},$**

1/beta) comment in parameters.f90 replaced by the implemented  $1 + (\text{boost}-1)(1-\text{beta})$  T\_star. Pending: the B6-style print of the Ly-alpha domain record in certification.f90 (the record and accessor exist; LYA-DOM, with the gate's follow-ups).

- **LYA-DOM** (advisor; certification.f90): the Ly-alpha damping-wing domain record (lya\_wing\_domain\_record: cell visits below the  $(a \tau)^{1/3}$  limit, visits seen, the smallest value, the limit) joins the out-of-domain closure count and is printed beside the H3+ and CO records, informational like them. certification, run\_mode, attempted\_step green. **Every worker item of the series has landed; the series gate begins (single build, make test, the 16-case matrix, movement table, one golden refresh, fcheck, docs pass).**
- **Series gate, 2026-09-07** (PLAN rev2 steps A to C and the items above, one series): single build of the shared tree (make -q up to date, EXHALE.x md5 ba3d34e6eae); make test green apart from the two steady suites, which then passed on the wasp\_full\_newton run log (EXHALE\_STEADY\_RUNLOGS); the 16 default cases run against that binary (the harness holds a single-instance lock, so wasp\_full ran through run\_check.sh check and the other fifteen through a runner replicating its steps for each case, all OMP\_NUM\_THREADS=1); every case moves against the previous goldens, as every item of the series said it would. Movement of the physical columns over the cells where the golden value is at least  $1e-3$  of the column maximum (max relative; T median; the change of  $\log_{10} \text{Mdot}$  from the mean of  $r^2 \rho v$  over 1.20-1.45  $R_p$ ): | case | rho max | v max | p max | T max | T median | dlog10 Mdot | log10 Mdot new | |---|---|---|---|---|---|---| | wasp\_full | 2.97e-01 | 1.28e+00 | 1.65e-01 | 9.18e-02 | 3.8e-02 | +0.0095 | 13.35 | | wasp\_he23off | 2.95e-01 | 1.30e+00 | 1.62e-01 | 9.18e-02 | 3.9e-02 | +0.0084 | 13.35 | | wasp\_full\_newton | 6.96e-01 | 3.51e-01 | 4.30e-01 | 2.35e-01 | 8.7e-02 | +0.0414 | 13.30 | | mol\_base\_handoff | 1.60e-01 | 2.50e-01 | 1.52e-01 | 2.33e-01 | 2.2e-02 | -0.0440 | 10.57 | | mol\_metals | 1.51e-01 | 2.25e-01 | 1.53e-01 | 2.07e-01 | 2.3e-02 | -0.0476 | 10.58 | | mol\_lyman\_werner | 2.12e-01 | 4.87e-01 | 1.99e-01 | 2.55e-01 | 2.6e-02 | -0.0011 | 10.54 | | mol\_diffusion | 1.59e-01 | 2.50e-01 | 1.52e-01 | 2.33e-01 | 2.2e-02 | -0.0440 | 10.57 | | mol\_ir\_bands | 1.51e-01 | 2.26e-01 | 1.53e-01 | 2.07e-01 | 2.3e-02 | -0.0475 | 10.58 | | mol\_sec\_ion | 2.16e-01 | 4.03e-01 | 1.96e-01 | 2.53e-01 | 3.0e-02 | +0.0149 | 10.51 | | mol\_carrier | 1.54e-01 | 6.37e-01 | 1.44e-01 | 2.24e-01 | 1.9e-02 | +0.0106 | 10.56 | | lower\_profile | 9.40e-03 | 3.22e+01 | 7.50e-03 | 9.77e-01 | 8.8e-03 | -0.0335 | 7.29 | | hydrostatic\_column | 2.09e-12 | 2.81e-03 | 3.54e-12 | 1.40e-03 | 2.5e-14 | -0.0000 | 8.27 | | oxygen\_chemistry | 8.62e-01 | 1.29e+02 | 9.76e-01 | 7.61e-01 | 3.7e-01 | +nan | 9.61 | | hp\_zero\_seed | 2.25e-05 | 1.81e-04 | 1.61e-04 | 3.20e-03 | 4.4e-05 | +0.0000 | 10.56 | | hp\_trace\_seed | 2.24e-05 | 1.81e-04 | 1.61e-04 | 3.20e-03 | 4.4e-05 | +0.0000 | 10.56 | | hp\_front | 2.07e-05 | 5.77e-04 | 3.66e-04 | 1.00e-02 | 6.7e-05 | +0.0000 | 10.56 | The maxima sit at the fronts, which move by a cell; the T medians (2-9 percent on the wind cases) and the Mdot changes (within  $\pm 0.05$  dex everywhere it is defined; oxygen\_chemistry has a reversed flow in the window) are the state of the physics after B3c, B4-1/1c, LYA-BETA, the WENO3 coefficient fix, the metals (REF-METALS, METALS-OUTER/INNER), the LW band, the CO channels, ACCEPT-1/2 and the rest. wasp\_he23off exits 2 (its stationary claim refused by four inventory entries, as A2tol2 measured at the default tolerance; outputs written in full); wasp\_full\_newton exits 0, certified, log10 Mdot 13.30. PDFs of the 25 dated documents, the paper and both posters rebuilt. **Golden refresh pending the user's hand:** the archive

copy golden -> golden\_pre\_series\_20260907 and run\_check.sh golden were refused by the session's permission classifier; the commands are in the report to the user. The final confirmation (make check against the refreshed goldens, serial) follows the refresh. **Golden refresh done by the user (2026-09-07 13:19):** golden/archived as golden\_pre\_series\_20260907/ (17 cases) and re-snapshotted from the case outputs of this gate (checked: golden equals output on wasp\_full, mol\_metals, oxygen\_chemistry, lower\_profile). The serial make check against the refreshed goldens runs next as the bitwise confirmation, then run\_fcheck.sh.

- **User manual, environment-variable table** (user report, 2026-09-07): the fixed table float of the environment hooks (67 rows) overflowed page 11 into the footer; it is now a longtable with the caption at its head, breaking across pages 10-11 with a "continued" foot. PDF rebuilt. Pages 20-21 (user report): the keyword table's long file paths were unbreakable \texttt tokens and ran past the right margin; 18 such paths are now \path{} (url package, breaks at / and \_), one inside a caption kept as \texttt (a moving argument). Rebuilt with latexmk, so the longtable column widths and cross-references are resolved (a single pdf<sub>l</sub>atex pass leaves both unresolved, which is what produced the "Table ??" and the first overflow the user saw).
- **Update log TeX twin** (user request, 2026-09-07): docs/Update\_EXHALE\_stage2.tex and its PDF now exist, GENERATED from this Markdown by src/utis/update\_log\_to\_tex.py (pandoc body, the memo-class preamble of the stage logs); regenerate after appending here. 56 pages at this entry.
- **Decision item 4 closed** (user, 2026-09-07): candidate C, wasp121b\_composite\_huang2023.txt (solar XUV shape below 1700 Å rescaled to Huang et al. 2023's F\_XUV and Ly-alpha, the MUSCLES WASP-17 photosphere above), is the WASP-121 b spectrum of record, as both input.inp files already state; A and B remain sensitivity inputs. No decision item of To\_be\_determined\_by\_user\_20260906.md is open. **make check against the refreshed goldens: REGRESSION PASS, all 16 cases byte-identical (64 files), 13:21-16:25 KST 2026-09-07** (the harness runs the cases eight at a time, one thread each). run\_fcheck.sh follows. **run\_fcheck.sh: CLEAN** (1500 bounded steps of the HD 209458 b folder case under -fcheck=bounds,do,mem,log10 Mdot 9.56) after one harness fix: the scratch copy of the planet input named its spectrum by the folder's relative path, which the script now rewrites to the repository's absolute path for the Spectrum file and Lower atmosphere profile keys (the first run failed on "Cannot open file ../inputdata/sed/...", not on a runtime check). Production EXHALE.x restored, make -q up to date, md5 ba3d34e6eae. **The series is closed:** goldens refreshed once, make check byte-identical, fcheck clean, docs and PDFs rebuilt. What follows is in docs/session\_handoff\_20260907.md.
- **HYG-PY** (advisor, 2026-09-07; python/paper\_data.py, python/make\_struct\_figures.py): HEAT\_FIXED (the headerless-file fallback) and HEAT\_CH\_LABEL/HEAT\_CH\_ORDER (the heating-budget panel) now carry all 19 deposit channels of heat\_channel\_name with the descriptive brackets stripped the way \_clean\_name strips them (heat\_He23S\_assoc, the two Lyman-Werner halves, heat\_mol\_chem, heat\_FUV\_photolysis, heat\_oxygen\_collisional, the two CO channels); the panel loop already skips a channel a run's file lacks. Both scripts parse; the loader reads the file header, so the fallback list was never what a real file was read by.
- **THREAD-DET** (advisor, 2026-09-07; util\_ion\_eq.f90): the thread count is no longer an input to any result. Mechanism, MEASURED: Cool\_coeff.o imports glibc libmvec's two-lane \_ZGVbN2v\_exp, \_ZGVbN2v\_log,

\_ZGVbN2vv\_pow (gfortran -O3 vectorizes the rate-coefficient loops), and the vector variants agree with the scalar libm to one ulp, not bitwise; inside a block of cells the loop takes lane pairs from `j_lo` and an odd remainder by the scalar call, so where a block STARTS decides which cell meets which variant. `chemical_rate_coefficients` and `eval_cool` cut the grid into `2*omp_get_max_threads()` blocks, so the block boundaries, and with them the last bit of every rate, moved with the thread count (the 6.6e-14 of COST3/HYG-MAIN). Fixed: both sweeps use blocks of the fixed even length `xuv_rate_block = 32` (16 blocks on the 504-cell grids); the threads decide only who takes which block. MEASURED on `wasp_full 300`: old binary at 1 thread, new at 1 thread and new at 16 threads are byte-identical in `Hydro_ioniz`, `Ion_species`, `Heating_breakdown`, `Cooling_breakdown` (the old single-thread partition happened to start both its blocks at even offsets, so the refreshed goldens are untouched). Shared EXHALE.x rebuilt from the tree (make -q up to date).

- **DOCS-LINES** (documentation only, 2026-09-07): every `file.f90`:NNN citation in the nine design and status documents (d0 922 tokens on 309 lines, b1a 127, b1 64, b3b 24, a2 7, a0 5, b3a 5, b4 2) re-anchored to file + routine, declaration or labeled comment block, read against the live tree; the step tables of d0 6.1 and b1 7.1 index by the `as_op_*` operation tag of `attempted_step.f90`; the convention is stated in each document. Citations of removed code (the composition projection, the two-iteration energy update, the CO ceiling, the cell-velocity carrier advection, the element advective term, DB96 eq. (39), the H3+ clamp, the second H2 partition function) kept as dated records naming the removing item. Stale values corrected in passing: `e_lw_photon_erg` 1.88021e-11 erg (11.7354 eV) over 912-1201 Å, `binary_element_diffusion.f90` 3037 lines, a2 section 1's chemical count now passed by the marching caller. All 190 names introduced grepped present in `src/`. Recorded: pre-existing markdown table pipe misalignments (unescaped | in math) in a2, b4, `code_status`, d0; two markdown line citations of `grid_and_gates/README.md` left.
- **B5, carrier rows** (`steady_newton.f90`, `diffusive_photochemistry.f90`, `input_read.f90`; new suite `steady_species_rows` 41 + 8 rows; advisor rebuild: it and certification, `run_mode`, `carrier_retry`, `coupled_source_step`, `species_face_flux` green): the Newton unknown vector carries one row and one unknown per solved carrier (H2, OH, H2O, CO, H+) through a row registry in place of the H2-only `nvar = 4` special case; band geometry derived, `kl = ku = 3 nvar - 1`, `ncolor = 6 nvar - 1` (reproduces the 8/17 and 11/23 the code carried); the completion flag reads the species rows of the system it solved; EXHALE\_SPECIES\_JAC\_TEST asserts `Jr` inside the solve (the run-level test builds its Jacobian before the registry exists). Ionization transport + Solver: Newton supported by derivation when the proton is a Newton unknown (`x_hp_fix` handed to the sweep), refused with the reason otherwise; MEASURED `nvar = 5`, Jacobian action 7.2e-7 (all rows), 4.6e-5 (species rows), 0 unresolved colors. Two pre-existing defects fixed: `frozen_residual` left the species slots of an `intent(out)` array unwritten (directional check relative error 1.0 from stale memory) and `n_headroom_iterate` was never assigned (the trial-admissibility threshold stood at 0; with the four oxygen balances as rows all 41 colors came back unresolved). `wasp_full_newton` stays certified (13.30, byte-identical); H2-only control byte-identical; marching path byte-identical (`wasp_full`, `mol_base_handoff 300`). **Findings:** no regression case enters the steady solver, so every golden is a marching snapshot and `cert_tol_carrier/cert_tol_element`

still have no anchor (contract section 11 records two rows from solves that did not meet their target,  $7.46e-3$  and  $5.82e-3$ , not anchors). **Open, and a physical-consistency defect:** the stationary carrier row still uses the cell-velocity advective form while the marching rows ride on  $F_{\rho} Y_s^{\text{face}} / m_s$  since B4-1c (two operators, two fixed points): B5b. Also open: the element rows (element\_transport\_residual must return a residual per element, not the worst at each cell), the cell's own field in the cell residual (interface written: FIELD-SELF), reload consistency, three EXHALE\_main.f90 sites (the registry name, EXHALE\_JAC\_TEST coverage, the PTC refusal text).

- **B5b** (species\_face\_flux.f90, steady\_residual.f90, steady\_newton.f90, certification.f90, binary\_element\_diffusion.f90, diffusive\_photochemistry.f90, three sites of EXHALE\_main.f90; suite steady\_species\_rows 67 rows; advisor rebuild: it and species\_face\_flux, certification, carrier\_retry, coupled\_source\_step, run\_mode, acceptance\_classes green): ONE discretization of the material advective term exists in the code: the RK stages, the stationary carrier rows, the fixed-wind carrier relaxation and the stationary element rows all form it as the divergence of the same face species fluxes (species\_face\_fraction, species\_face\_flux, new species\_flux\_divergence) on the mass row's faces, areas and volumes; the cell-velocity upwind form, its deferred van Leer correction (carrier\_advection\_correction, adv\_corr, carrier\_slope), the base Dirichlet record it needed and the advective coefficient ntv are gone from diffusive\_photochemistry.f90 (RED 76.7 relative against a stage-1 update on a sign-changing face flux, GREEN 7.5e-14). element\_transport\_residual returns one residual per element (the certification prints one entry per element: lower\_profile C 0.977, O 0.921, N 0.968, seven elements "not applicable"); rows srow\_element\_he and one srow\_element\_trace per metal element registered before the carriers, project\_element\_mass\_fractions the write-back. mol\_diffusion reloaded with Coupled carrier solve: nvar = 4, one element row, Jacobian action  $4.9e-3 / 1.3e-3$ , 0 unresolved colors, the solve refused at the He/H row (0.9987) with no descent direction: the element system is expressible and its Jacobian right, not yet solved from that start (the element unknown moves the whole column's opacity, which the banded model does not carry; the "all rows" bound of the suite widened  $1e-4 \rightarrow 1e-2$  with the measurement at the site). Reload consistency asserted on the binary. EXHALE\_main.f90: set\_transportated\_species\_rows (not carrier-specific), the two EXHALE\_JAC\_TEST inserts, the PTC text; at that site two pre-existing defects fixed: the species slots of Yvec were read uninitialized ( $2.4e2$  against 1.85) and the probe direction was unscaled ( $3.3e1$  at nvar = 4); now  $3.2e-6 / 4.5e-6$ . **By construction the stationary species rows now inherit the mass row's imbalance** ( $\text{div}(F_{\rho} Y) = Y \text{ div } F_{\rho} + F_{\rho} \text{ grad } Y$ ; the old form was the second term alone), so on marching snapshots whose mass rows read 1.1-1.9 of their terms the rows read  $O(1)$  where they read  $4e-4$  to 0.4 before. Marching path byte-identical on wasp\_full, mol\_base\_handoff, mol\_carrier, mol\_diffusion, lower\_profile at 300 steps; wasp\_full\_newton reloaded certifies (13.30,  $2.6e-9$  of the stored state). Anchors still UNCONFIRMED (contract 11.3). Recorded: element\_diffusion\_step / relax\_element\_composition still advect on the smoothed steady mass flux, so the elemental Picard alternation and the elemental row are two operators (B5c); carrier\_rows\_advect kept as the on/off switch of one operator; Coupled carrier solve now also governs the element rows (key not renamed); ionization\_equilibrium.f90 cites carrier\_base\_state, which no longer exists (to FIELD-SELF). THREAD-DET confirmed by the official check: make check on the rebuilt

binary, REGRESSION PASS, every case byte-identical (16:58-20:04 KST 2026-09-07; the binary of that check also carried whatever of B5 had landed by 16:58, and B5's marching path was measured byte-identical anyway). A further make check follows FIELD-SELF and B5c.

- **B5c** (binary\_element\_diffusion.f90; suite steady\_species\_rows 60/0; advisor rebuild: it and species\_face\_flux, certification, carrier\_retry green): the elemental relaxation and the stationary elemental row are one operator. element\_diffusion\_step takes the FACE MASS FLUX and forms its advective term as the divergence of the face element fluxes through the three shared routines (element\_advective\_divergence, one place); relax\_element\_composition reads that flux from the mass row of the state (face\_mass\_flux\_of\_state) and no longer smooths a steady  $\dot{m}/(4\pi r^2)$ ; element\_transport\_residual takes both halves of the helium row from composition\_residual and the trace rows from the new trace\_composition\_residual, the routines the step itself solves, so the fixed point is the row's zero by construction; solve\_trace\_element\_in\_hydrogen is a deferred-correction loop that takes one pass, term for term the old direct solve, when the advection is off (the marching path). MEASURED on a synthetic column relaxed to its own convergence: the row reads 8.65e-13 (He alone), 8.45e-13 / 5.38e-13 (He + ten trace elements) of its terms, the advection carrying 0.993 of them; the replaced form leaves 6.4e-3 on the same state. The smoothed-flux rationale does not survive: it repaired a form (the element equation minus  $X$  times the mass equation) that is gone, and its consequence is now a measured property of the equation: on a face mass flux that does not conserve mass the conservative balance has no bounded steady composition ( $X$  driven onto 0 and 1). Marching path byte-identical (mol\_diffusion, lower\_profile 300 steps); the direct-steady reload of mol\_diffusion byte-identical, and its outer pass never reaches the relaxation (the solve ends info = 2, B5b 6.3), so the run-level RED/GREEN does not exist. **Found: diffusion\_tests.x has not linked since B5b** (binary\_element\_diffusion now uses steady\_residual\_mod, whose chain the test's DIFT\_SRC does not carry): a functions/ module depending on time\_step/steady\_residual is a layering inversion; the flux goes by argument (DIFT-LINK, launched), and make test's diffusion\_tests row is red until it lands. Also recorded: the relaxation's convergence measure is helium-only (DIFT-LINK folds the trace drift in); the vanished-element write-back branch leaves a trace row at 1e-4 to 2e-2 where an element is driven to exactly zero (deliberate guard, measured, documented at the test site); element\_transport\_residual's dummy  $v$  is unread.
- **FIELD-SELF** (ionization\_equilibrium.f90 self-field loop, default one pass; util\_ion\_eq.f90 and System\_HeH\* comments; probe photoionization\_field\_self\_consistency 12; advisor rebuild: coupled\_source\_step 36/0, physics\_probe 1491/0, certification, acceptance\_classes green): the cell's own optical depth CAN now be closed inside the cell solve (one pass leaves a 4.06e-2 self-consistency defect in the ionized fraction of a cell one optical depth thick, the loop closes it to 2.8e-15 in 13 passes, contraction 0.080; the field routines already formed the B4-6 cell mean from the cell's own densities, so no field code changed), **and doing so buys nothing: COST3's attribution is refuted**. MEASURED on wasp\_full 300: coupled passes 10.09 -> 10.01 for 1.65x the wall time (2 self-field passes) and 1.87x (8); mol\_base\_handoff 1500: 7.19 -> 7.24 for 2.0x; the decisive binary with BOTH levers (carried columns at block 8 AND four self-field passes): 7.20 passes, 190 s, against COST3's 7.13 / 103 s for the columns alone. (**Superseded by**

**COST4 below:** with both stellar lags closed at the settings that close them the count is 8.18, not seven, and the two couplings named here are worth 0.08 of it; the count is the slowest rung of a ladder of modes.) Same fixed point: state movement  $\leq 1.4e-7$ , at the  $1e-8$  tolerance's level. Default output byte-identical on `wasp_full 300, mol_base_handoff 1500, mol_metals 1500, hydrostatic_column; wasp_full_newton` uncapped CERTIFIED, both files byte-identical to its golden (that binary carried B5b too). Kept in the tree at its inert default on COST3's precedent (the switch is the record of a measured negative result). Comment fixes in passing: `System_HeH_metals.f90` said the sweep is serial (untrue since the parallelization; its arrays are threadprivate); the `carrier_base_state` citation of `ionization_equilibrium.f90` rewritten to `carrier_base_composition_imposed`. Recorded: `he_ground_singlet_density's` clamp counter needs a lock when the self-field loop runs (a slot per cell would be free, `composition.f90`); `f_vib_quench/e_vib_bound` stay at the entry composition inside the loop; the COST3 attribution in that report, its log entry above and the `csm_max_pass` comment of `EXHALE_main.f90` are superseded by this measurement (comment to be corrected when DIFT-LINK releases main).

- **DIFT-LINK** (`binary_element_diffusion.f90` and its three call sites in `EXHALE_main.f90`, `steady_newton.f90`, `certification.f90`; suite `steady_species_rows` 62 rows; advisor rebuild: it and `species_face_flux`, `certification`, `coupled_source_step` green): the element module no longer uses `steady_residual_mod`; the face mass flux is a required argument of `relax_element_composition` and `element_transport_residual`, obtained by each caller with `face_mass_flux_of_state`. The layering inversion is gone and **make diffusion\_tests links again with the Makefile and DIFT\_SRC untouched: 35 passed, 0 failed** (MEASURED by the advisor on the live tree; it failed at the link since B5b). The relaxation's convergence measure now runs over every element it relaxed, each against its own reservoir (`element_composition_distance`): RED 5.99 / GREEN 0.0 on a column where Fe moves 1.78 while helium moves 0.25, so the helium-only measure returned a number 7x too small to bound the element that moved. Byte-identical on `mol_diffusion`, `lower_profile 300` steps and the `mol_diffusion` direct-steady reload. Dead code removed (`helium_mass_fraction`, an unread `element_mass_fractions` call). **Advisor fixes on its findings:** the limiter of `species_face_fraction` divided by  $Y_{av} - Y_f(j)$ , which vanishes when the donor cell's average is itself negative (a trace species handed back at a small negative density; MEASURED as a SIGFPE under `-ffpe-trap=zero`): the anchor is now the average clamped into  $[0,1]$ , which makes both denominators strictly positive and is what the comment already claimed; `mol_diffusion 300` steps byte-identical, since at production flags the infinity was clamped to the same 0. The `csm_max_pass` comment of `EXHALE_main.f90` and `docs/steady_solver_design.md` section 15 corrected to the FIELD-SELF and B5c measurements. Recorded: `diffusion_tests T9` reads  $1.098e-12$  against its  $1e-12$  bound at `-O1 -fcheck` (an optimization-level rounding sensitivity of `relative_settling_mass`, 35/0 at `-O3`); the outer loop's under-relaxation damps helium only while the drift it tests can now be set by a trace element.
- **Batch gate after B5/B5b/B5c/FIELD-SELF/DIFT-LINK/THREAD-DET** (2026-09-07/08): shared tree rebuilt (`EXHALE.x md5 eba2d7f2260f`, `make -q` up to date); `make test` green, the two steady suites passing on the `wasp_full_newton` run log and the `diffusion_tests` row restored; `make check` **REGRESSION PASS, all 16**

**cases byte-identical** (20:54 to 00:05 KST). Every item of this batch was designed to leave the marching path alone and the check confirms it: the goldens refreshed at the 2026-09-07 series gate still stand, and no golden was refreshed here.

- **HYG-T9** (advisor, 2026-09-08; `src/tests/diffusion_tests.f90`): T9's ambipolar-limit bound sat exactly on the arithmetic's noise floor, so the suite passed at -O3 and failed at -O1 with checks (DIFT-LINK's finding). MEASURED at both levels: the neutral limit is exact (no electron term), the H+ plasma reads 8.207e-13 (-O3) and 7.740e-13 (-O1), the He++ plasma 7.083e-13 and 1.098e-12. Mechanism, written at the site: the limits are exact algebra in the masses, so the deviation is arithmetic, and the ambipolar term of the plasma limits comes from a centered difference of  $\ln \rho$  over  $dr$  whose logarithm is  $O(b_0/r)$ , so each ulp reaches  $dm_{eff}$  amplified by  $1/dr$ . The bound is now one decade above the largest measured ( $1e-11$ ), the anchoring rule the certification tolerances use, and the test still catches a wrong ambipolar limit by ten decades (a wrong limit misses its target by 0.1 to 1) with the neutral limit still asserted exact. 35/0 at -O3 and at -O1 -fcheck.
- **B5d** (`steady_newton.f90` only; suite `steady_species_rows` 77 rows; advisor rebuild: it and certification, `species_face_flux`, `coupled_source_step`, `carrier_retry` green): the item's own brief expected the element rows to fail because an element unknown moves the whole column's opacity, too non-local for the banded model. **MEASURED and REFUTED**: the scaled Jacobian column of an element unknown carries at most 6.3 percent of its 2-norm outside the band, the same order as a hydrodynamic column (mass 0.0018, momentum and energy 0.0000), so the diagonal correction, the rank-one column-integral update and the identity block the brief offered would each have repaired a defect that is not there; none was written. What does break the model, located by the same measurements and fixed: (1) the base element row was structurally EMPTY, cell 1 being the Dirichlet reservoir of the element operator, so the banded model was singular by construction (its LU pivot grew to 8.413e19 against a largest band entry of 5.229e3; now 1.433e6) and it now carries the boundary condition the operator states; (2) the element ROW SCALE collapses on the abandoned iterate (helium rows spanning 1.277e-29 to 2.360, so an ordinary  $dF/dY = 46.7$  became a scaled band entry of 4.890e22 and  $||grad\ merit|| = 1.104e44$ ), and the row scale is now floored at the flow-time rate of the quantity the row transports, the hydrodynamic convention with no chosen constant. Result: the same case reaches  $||R|| = 1.723$  against 1.974 at outer iteration 18 instead of 8 with  $||grad\ merit|| = 4.813e3$ , 40 decades lower and a number the model's own entries can produce. **The element solve still does not converge and the item does not claim it does**; `cert_tol_element` and `cert_tol_carrier` are unmoved and UNCONFIRMED. New diagnostics: `banded_model_conditioning` (empty rows and columns stated once per registry, pivots and scales under EXHALE\_BAND\_REPORT), `jacobian_column_reach_beyond_the_band` (EXHALE\_JAC\_COLUMN\_REACH). `mol_diffusion` 300 steps byte-identical; `wasp_full_newton` byte-identical, `info = 0`, CERTIFIED, 13.30, same 6 outer iterations. **Named as what is left: the globalization** (the non-monotone search accepts steps that raise  $||R||$  from 1.7 to 25; the trust-region route gives up at radius 3.8e-16 after 12 ray-test refusals while the merit is still falling): item B5e. Also recorded: `carrier_row_term_scale` has the same collapsing structure and was not measured (no carrier row in that case); the trace-metal case of the floor is covered by inspection only.

- **COST4** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to the snapshot it took; every experiment a private binary in a scratch tree whose no-switch build is byte-identical to the control): **the coupled loop's pass count is a ladder of nested modes and equals the SLOWEST of them, not their sum.** MEASURED on wasp\_full 300, one thread, removing one composition coupling at a time: the production loop 10.09 passes; the stellar field's dependence on the composition removed, 7.16; the temperature removed as well (the energy solve replaced by its own converged answer), 4.94; the He recombination rates too, **2.00** (which reproduces, with a different probe, the 2 passes COST2 measured with every composition-dependent input frozen). Both hypotheses of the brief are refuted: the two diffuse couplings are worth at most 0.08 of the 10.09 (Balmer off 10.09, He rec off 10.01, both off 10.01, He rec frozen after pass 1 10.01, and on mol\_base\_handoff all three read the control's 7.19 to the last digit), and **the Balmer continuum is not a lag at all**: excited\_H\_update fills its arrays once BEFORE the loop, so they are constants of it, and refreshing it every pass costs 1.5 percent of wall for +0.01 passes. The stellar lag closed IN FULL for the first time (a one-cell block and eight self-field passes): 8.18 passes for 218 s against 10.09 for 139 s, the next rung, with freezing the field entirely giving 7.16. The temperature is not the residual either: the composition sweep alone, at the temperature the step converged to, takes 10.94 passes (6.32 against 7.19 on mol\_base\_handoff). **Decision: change nothing** (no loop-internal Balmer refresh, no self-consistent He recombination coupling, no default moved); to go below seven passes every rung must come out at once. Advisor: the three statements this refutes are corrected in place, the csm\_max\_pass comment of EXHALE\_main.f90, the FIELD-SELF entry above (marked superseded) and open item 1 of the 2026-09-08 handoff. **The one remainder, now open item 1**: on mol\_base\_handoff the ladder is flat (freezing the field RAISES the count 7.19 -> 7.41) and everything the item could reach accounts for 0.6 of its 7.19 passes; the named candidate, READ and not measured, is n\_tot, the third-body density of the three-body molecular reactions, built by calc\_ntot from the ENTRY composition, and with it the H2 vibrational quench fraction and bound energy. Also recorded: xuv\_field\_block\_cells and xuv\_self\_field\_passes are compile-time parameters, so measuring the two together needs a rebuild; heating\_of\_composition calls he\_rec\_coupling a second time whole-grid every pass for its heating alone (both calls together bounded at 1.7 percent of wall); on mol\_base\_handoff the He recombination coupling is what keeps step 0 inside the pass cap.
- **COST5** (measurement; NOTHING in the repository changed by the item, its four candidate files byte-identical to its own snapshot; probe binaries byte-identical to the control at their default): the molecular remainder COST4 left is named, quantified, and the ladder reaches its bottom rung. **The count is arithmetic, and the law is measured and then tested**:  $\text{passes} = 2 + \log_{10}(\text{the step's own excursion} / \text{the stopping tolerance}) / (\text{decades per pass})$ . On mol\_base\_handoff the composition contracts by 0.171 a pass and the temperature by 0.072 (0.77 decades a pass) from an excursion of  $7.0e-5$  to a tolerance of  $1e-8$ : 7.00 predicted, 6.93 MEASURED; on wasp\_full 0.222 and 0.199 from an excursion seven times larger, 9.2 predicted, 10.09 measured (so the molecular cases are the cheap ones, not the pathological ones). Tested, not fitted: the law says the count depends on the stopping tolerance through one logarithm at 1.30 passes per decade, and moving both tolerances together gives 8.06 / 6.96 / 5.68 / 4.04 at  $1e-9$  /  $1e-8$  /  $1e-7$  /  $1e-6$ , i.e. 1.34 over three decades; and it is not the cell solve, six decades of hybrd1 tolerance leaving the count at 6.97 for up to 1.99x the wall. Of the four candidates only

the named one is an iterate: `n_tot`, the third body of R12/R13/R15 (the ONLY composition dependence of the molecular network outside the field: of seventeen hoisted coefficients exactly two see it, both linearly). Holding it: `mol_base_handoff` 7.19 -> 6.03 at 1500 steps, `mol_metals` 6.99 -> 6.01; it limits the composition mode's RATE (0.171 -> 0.073) but the count falls only 6.93 -> 6.11 because the temperature mode then binds. The other three are entry state through the operator-split field (vibrational pair 6.96, electron density 6.94, Lyman-Werner 6.96, against 6.96), and the metal element totals are invariant across a sweep by construction (6.99 against 6.99, exact). With every input AND the temperature held the sweep needs **2.01 passes**, reproducing COST2's and COST4's atomic bottom rung on the molecular path. **Decision: nothing closed.** The mode that sets the count is the temperature/composition alternation (`EXHALE_main.f90`, `energy_semi_implicit.f90`, not that item's files); closing the `n_tot` lag exactly needs two sites in `System_HeH_mol.f90` (also not its), and the in-file alternative costs a second cell solve per pass, which the cell-tolerance scan bounds at more than the 0.80 passes it returns. The production treatment is not wrong: at the fixed point the entry and returned compositions are one state, so the third body the network is solved with is the third body of the accepted state, to `csm_comp_tol`. Advisor: the law is now in the `csm_max_pass` comment; the misleading indentation of `composition.f90`'s `calc_ntot` branches fixed. Recorded: the loop's stopping test compares an ABSOLUTE mass-fraction change against `csm_comp_tol`, so it is a relative  $1e-8$  on H2 and no constraint on a  $1e-12$  species. **The one lever that keeps both the fixed point and the tolerance is sequence extrapolation** on the (T, c) pair after the third pass (the decay is geometric with a ratio stable from pass 2 and close across cases): ESTIMATED at three passes of seven on the molecular cases and four of ten on `wasp_full`, item COST6.

- **HYG-4** (advisor, direct; the four hygiene items of the 2026-09-08 handoff): (i) `he_ground_singlet_density`'s run-wide clamp counter is incremented with `!$omp atomic update` inside the function, so the named critical section around its one call inside a parallel region (`ionization_equilibrium.f90`, self-consistent field loop) is gone and the three comments that justified serial column formation by the counter are reworded (the columns are still formed before the blocks are solved, for the ordering, not for the counter); (ii) `element_transport_residual` no longer takes the cell velocity it never read (four call sites: `certification.f90`, `steady_newton.f90`, two test rows), the balance being the divergence of the FACE mass flux handed in; (iii) the outer Picard loop of `steady_wind_with_element_diffusion` halved the element under-relaxation  $\omega$  whenever `comp_drift` failed to fall, but `comp_drift` is the worse of the element drift and the carrier drift while  $\omega$  damps the element relaxation alone (the carrier pass is bounded by `carrier_trust`), so a carrier drift that failed to fall slowed the element alternation for nothing; the damping test now reads the element relaxation's own drift (`elem_drift`) and the exit test keeps the worst of the two. No regression case runs both `He_diffusion` and the carrier transport, so the matrix cannot move. (iv) `f_vib_quench/e_vib_bound` staying at the entry composition inside the self-field loop is CLOSED BY MEASUREMENT (COST5): they are entry state through the operator-split field, and holding them at the first-pass value is worth 0.00 passes (6.96 against 6.96). Suites: `steady_species_rows` 74/0, `certification` 29/0, `diffusion_tests` 35/0.
- **B5e** (steady solve globalization; `steady_newton.f90`, the `steady_species_rows` suite, design doc section 17, contract section 11.5): **the first stationary solve with a species row that meets its own target.** MEASURED on

mol\_diffusion reloaded with Coupled carrier solve: True (one element row, nvar = 4): the scaled trust region takes  $||R||$  from 1.986 to **4.345e-09** against a target of 1e-08 in 66 outer iterations (54 accepted), where the pseudo-transient line search aborts at iteration 18 on "no descent direction exists for the banded model" at 1.702. Two defects were named by instrumenting the eleven exits of trust\_region\_step and repaired where the measurement pointed: (i) the radius collapse was the CAUCHY LEG ASCENDING the model, because its gradient is the transpose of the BANDED A and has no sign (MEASURED  $r_0 \cdot A g = -2.1e13, -2.2e19$  at the collapsing iterates, pred negative and exactly linear in the radius); where  $r_0 \cdot A g \leq 0$  the leg is now dropped and the dogleg reduces to the Krylov leg cut to the ball, which always predicts a reduction (GMRES gives  $r_0 \cdot A s_N < -||A s_N||^2/2$ ); and pred is formed as  $-(r_0 \cdot A s) - ||A s||^2/2$  instead of the cancelling difference of two squared norms, whose sign on a short step was rounding; (ii) the best-iterate ledger ranked on  $||R||$  (three hydrodynamic rows) while the state is judged on those AND every species row, so it could hand back the iterate with the worse element balance; it now ranks on distance\_from\_certification, the largest of the judged row measures over their tolerances, formed once inside eval\_residual under an optional argument; the three-unknown branch keeps its comparison character for character. **Refused by measurement:** bounding the STEP by the judged rows (implemented in all three step controls) stalled the element solve at  $||R|| = 1.92$  on 13 refusals; and the banded-gradient Cauchy length  $(r_0 \cdot A g)/||A g||^2$ , more defensible on paper, left the same solve crawling at 1.8e-4 after 81 iterations against 4.3e-9 in 66 with the original length, so the original stays, recorded at the site with both numbers. **Default moved:** use\_tr = (nspec\_row > 0), the trust region is the step control of the coupled route (EXHALE\_TRUST\_REGION=0 restores the line search for measurement); no regression case sets Coupled carrier solve: True, so no golden can move. **Anchor:** the elemental transport He/H row of the returned state reads **4.633e-11 against cert\_tol\_element = 1e-8** (215x inside), the first measurement under that tolerance; it is NOT moved, because contract sections 9 and 10 want a ladder (the same case at Resid tol 1e-9 and 1e-10, not run) to tell a target-limited row from one at its floor. The state is not certified: its cell-1 hydrodynamic mass row reads 6.2e-11 against 3e-12, a base-cell matter and not an element-row one. **The carrier-row solve is not rescued:** on mol\_carrier the trust region stagnates at 1.873 (line search 1.855) and 18 of its 49 iterations end on "no Krylov direction could be sampled", so cert\_tol\_carrier stays UNCONFIRMED, and the measurement says it is not a step-length problem. Worker MEASURED wasp\_full\_newton byte-identical (it 6,  $||R|| = 8.131e-09$ , info 0, CERTIFIED, log10 Mdot 13.30) and the mol\_diffusion trust-region run identical to the pre-item text under EXHALE\_TRUST\_REGION=1 (every landed change inert on a path that already worked). Recorded, not fixed: EXHALE\_SPECIES\_JAC\_TEST=1 perturbs the run it measures (iterations 1-5 agree to every printed digit, iteration 6 differs, the two runs then diverge; some module state survives a probe eval\_residual; the same root writes n\_eq\_sweeps\_model and the frozen background); and the ray test can refuse a step that reduced the merit far MORE than the model promised (ratio = 7187), which Algorithm 4.1 of Nocedal and Wright never does, the obvious amendment (gate it on ratio <= eta\_accept) left unmade for want of a control measurement. The worker was stopped by a rate limit after writing its report; advisor verified the tree: build clean; suites steady\_species\_rows 74/0 (six new driver rows), certification 29/0, species\_face\_flux 16/0, carrier\_retry 101/0, coupled\_source\_step 30/0; one "per-iteration" phrase in run.sh reworded; wasp\_full\_newton re-run by the advisor against its golden with

the B5e binary: all four files byte-identical, info 0, CERTIFIED, log10 Mdot 13.30.

- Batch gate 2 (2026-09-08, 05:04 to 08:08):** make check on the tree as of B5e + HYG-4 (before COST6 landed; the harness built at 05:04, COST6's first edit is 05:17) **PASS, all 65 files of the matrix byte-identical** to the goldens refreshed by the user on 2026-09-07. make test in the same script failed 7 suites on their build-staleness precheck only (the shared build/ had not been remade after HYG-4 when the suites ran, the harness remade it afterwards); the suites themselves were run on the advisor's private build of the same tree: steady\_species\_rows 74/0, certification 29/0, species\_face\_flux 16/0, carrier\_retry 101/0, coupled\_source\_step 30/0 then 39/0 with COST6, diffusion\_tests 35/0, physics\_probe 1491/0, and energy\_update, grid\_and\_gates, spectrum\_type 53/0, 93/0, 154/0. Every suite of make test is therefore green on this tree. No golden touched.
- COST6** (extrapolation of the coupled loop's geometric decay; EXHALE\_main.f90 +532 lines, coupled\_source\_step +3 whole-binary rows; **default OFF**, EXHALE\_CSM\_EXTRAP=1 turns it on): implemented, guarded, measured. The object is the GLOBAL pair (T, f\_sp), which is exactly the loop's iterate (the energy solve rewrites u(3,:) and W(3,:) in full from T and the entry anchor each pass, so an undo of the pair is exact, and the guard-forced-refusal row reproduces the off run byte for byte); one Rayleigh-quotient ratio per block over all cells and species, the projection onto the dominant mode COST5 measured stable, and one damped ray, so no cell can move in a direction the global mode lacks. That is how it differs from COST2's Aitken taken cell by cell (12.00 against 11.05 passes): a ratio taken cell by cell spreads 200 percent across the grid and cannot be tested for validity; the global one is tested ( $||d_k - \theta d_{k-1}|| \leq 0.5 ||d_k||$ ) before use. Admissibility is by construction (fractions in [0,1], He 2<sup>3</sup>S inside its He I column, T inside the energy solve's own bracket, all linear in the step, so the largest admissible multiple is taken); MEASURED, the worst violation is identically zero on every case at 300 and 1500 steps, 1 and 16 threads, and the step was never damped below 0.95. The stopping test and its tolerances are untouched: a candidate is kept only if the pass taken FROM it meets them. Three things the measurement taught: the mode ALTERNATES (cosine of consecutive increments -0.85 to -1.00 on mol\_base\_handoff, +0.71 to +0.98 on wasp\_full; the first implementation demanded  $\theta > 0$  and refused 1188 of 1188 candidates); the one-mode model does not reach back into early passes (the ratio drifts 0.77, 0.47, 0.31, 0.22, 0.17, 0.14, 0.11 over one wasp\_full step), hence a REACH guard, take the limit only where the modelled remaining error is already inside the tolerance ball (reach scan 6.61 / 6.15 / 6.36 / 7.81 / 7.74 / 7.75 at 1/2/3/10/30/100, control 6.96); and undoing a mediocre candidate costs more than it saves (7.29 strict against 6.94 lenient, control 6.99 on mol\_metals). MEASURED, sweeps control -> on: mol\_base\_handoff 1500 10780 -> 10102 (**-6.3 percent**, matched pair one binary 285.4 -> 268.0 s), mol\_carrier 300 -2.3, mol\_metals 400 -0.75, wasp\_full 300 -0.63, hydrostatic\_column 0; state movement 8.3e-8 to 5.0e-7 relative, log10 Mdot unchanged. **The brief's estimate of three passes of seven is NOT reachable, and the reason is the item's real finding:** the stopping test measures the INCREMENT, and the error is  $|\theta/(1-\theta)|$  times it, one to two decades smaller, so the loop is one to two decades stricter than its tolerances say and spends 1.0 to 1.6 passes on that; the extrapolation's ceiling is the last factor  $1/|\theta|$ . Redefining the test on the estimated error is a change of what the loop returns and is put to the user as decision item 8 of

docs/To\_be\_determined\_by\_user\_20260906.md (recommendation (b)). Tests: coupled\_source\_step RED 2 rows against the control, GREEN 39/0 (extrapolation\_lowers\_the\_pass\_count 6.09 against 6.96 at 300 steps; refused\_extrapolation\_costs\_nothing identical at 6.80; mean\_passes\_within\_bound re-measured 6.80, unchanged); physics\_probe 1491/0. Worker-introduced defect fixed inside the item: a ledger write with six integer descriptors for seven items. Recorded, not fixed: write\_setup\_report.f90 does not echo the switch; n\_csm\_x\_kept can only count candidates the stopping test did not meet and reads 0 everywhere (the number that matters, met on the very next pass, is printed beside it); the csm\_max\_pass comment block is now some 60 lines of measured tables and belongs in a docs/ memo. Advisor: the measured tables and the item history that had grown to some 230 comment lines around the coupled-step constants are moved to docs/coupled\_source\_loop\_cost.md (sections 1-7, decision item 8 linked); the code site keeps each constant's conclusion and a section pointer. The setup report does not echo environment switches (none of them, by design), so the switch is reported by the run counter line alone. Advisor verification: coupled\_source\_step 39/0 and physics\_probe 1491/0 on the advisor build; default-off byte-identity against the GOLDENS at matrix length (which checks B5e, HYG-4 and COST6-off together): mol\_base\_handoff 12000 steps and wasp\_full to its own stop, both Hydro\_ioniz.txt and Ion\_species.txt IDENTICAL to the goldens (both runs end with status 2, "not certified", exactly as the golden runs do).

- **COST7** (decision item 8, option (b), user 2026-09-08; EXHALE\_main.f90, coupled\_source\_step suite, docs/coupled\_source\_loop\_cost.md sections 7-8): **the coupled source loop stops on the ESTIMATED REMAINING ERROR of the pair (T, composition), not on the last increment.** The estimate is  $l_{\theta}/(1 - \theta)$  times this pass's increment in each of the two norms the test uses,  $\theta$  the global Rayleigh-quotient ratio, formed in ONE routine (coupled\_pair\_geometric\_error\_estimate) that the COST6 extrapolation now reads too (its reach guard is literally the stopping test at a loosened tolerance; shared objects renamed csm\_x\_\* -> csm\_geom\_\*, env hooks EXHALE\_CSM\_GEOM\_\*). Where the sequence gives no estimate (passes 1-2, a guard failing) the increment test stands, and that fallback (EXHALE\_CSM\_GEOM\_REFUSE=1) is byte-identical to the pre-COST7 binary on all 10 written files of wasp\_full 300 and all 7 of mol\_base\_handoff 1500. **The item's finding:** the estimate is an L2 quotient against two max-norm tolerances, and MEASURED with a probe that runs the loop on to  $1e-12$  and rolls back, the unfactored estimate accepts states OUTSIDE csm\_comp\_tol on 100 of 299 steps of mol\_base\_handoff (1.8x the tolerance at worst); a factor 2 leaves 1 of 299; **a measured safety factor csm\_err\_safety = 3 leaves none**, the accepted state's distance to its own fixed point reading  $7.4e-9$  and below (composition) and  $2.1e-9$  and below (temperature) on every one of the 2496 accepted states probed. The price is that almost none of the passes COST6 estimated survive: wasp\_full 300 10.09 -> 10.01 passes (sweeps -0.9 percent), mol\_base\_handoff 1500 7.19 -> 6.96 (-3.1), mol\_metals 400 6.99 -> 6.93 (-0.9), **mol\_carrier 300 6.01 -> 5.01 (-16.6)**, hydrostatic\_column 1.01 -> 1.01 (+1 sweep of 302: on its single three-pass step the estimate EXCEEDS the increment, so the new test is the stricter one there; it is not uniformly looser). log10 Mdot unchanged on every case; state movement  $4.3e-10$  to  $1.1e-7$  relative, at the tolerance level. What the change buys is the MEANING of the two tolerances, the decision's own first reason, not passes. The COST6 extrapolation under the new test is worth nothing (6.79 against 6.82 passes) and stays default off. Tests: coupled\_source\_step RED 41/3 against the control, GREEN 44/0; mean\_passes\_within\_bound re-measured 6.23 (was 6.80), both

at the site; extrapolation\_lowers\_the\_pass\_count rewritten as extrapolation\_does\_not\_cost\_passes; the fallback-is-one-path row replaces the brief's byte-identity row (a suite cannot hold a golden of the previous binary; the identity is measured in the report). physics\_probe 1491/0. Recorded, not fixed: the composition increment of mol\_base\_handoff has a floor near  $5e-9$  (127 of 1500 probes cannot reach  $1e-12$  in 59 passes), so  $\text{csm\_comp\_tol} = 1e-8$  is within a factor of two of what that case can reach at all; COST6's  $\text{n\_csm\_x\_kept}$  still reads 0. **Every golden of the matrix moves at the 1e-8 level:** gate 3 below.

- Gate 3 (2026-09-08, 09:29 to 12:00)**, tree after COST7 (with COST6 default off, HYG-4, B5e): shared build remade, **make test all suites green**; make check verdict FAIL against the  $1e-3$  relative tolerance, as expected for a change of the stopping test, and every movement examined.  $\log_{10}$  Mdot unchanged on all 14 cases that print one (13.35, 13.35, 13.30, 10.57, 10.58, 10.54, 10.57, 10.58, 10.51, 10.56, 7.29, 8.27, 9.61, 7.98). Movement table (largest relative change of any column, new against golden): wasp\_full 6.6e-5 (v at the base, cell 136), wasp\_he23off 3.8e-4 (v at cell 146), wasp\_full\_newton 1.1e-10, mol\_base\_handoff 2.8e-6, mol\_metals 2.4e-6, mol\_lyman\_werner 4.1e-7, mol\_diffusion 2.8e-6 (adv file 2.2e-3 at the outer boundary, cell 504), mol\_ir\_bands 2.4e-6, mol\_sec\_ion 4.5e-3 (v = 0.091 cm/s at cell 177, a near-zero base velocity, adv 1.5e-3), mol\_carrier 5.2e-7, lower\_profile 9.3e-6, hp\_zero\_seed / hp\_trace\_seed / hp\_front identical, roundtrip within. Two cases move by order one and both are pre-existing base artefacts, not COST7's: **oxygen\_chemistry** (1000-step snapshot) reads 42x in  $n(\text{H II})$  at cell 22 because the GOLDEN carries an odd-even sawtooth there (cells 21, 24, 25, 27 at  $7e3\text{--}9e3$  against  $2e5$  in their neighbors) and the new run is smooth ( $1.6e5$  to  $3.9e5$  across cells 18-26;  $\rho$ , p, T within  $1e-6$ ); **hydrostatic\_column** reads 0.999 in the adv file's T at cell 350 because the adv post-process gives  $T_{\text{adv}} = 0.78$  K inside  $1.40 R_p$  in this static column (the original T is 1124 K throughout) and the edge of that region moved by one cell; the non-adv files move by  $4.6e-8$  ( $\text{Ion\_species}$  by 0.16 at cell 2 on values of  $1e-182$ , denormal noise). The adv defect is recorded as an open item. **Goldens to be refreshed by the user** (archive first), then make check re-run.
- ADV-STATIC** (post\_process\_adv.f90, new suite adv\_static\_limit, docs/postprocess\_advection\_validity.md): **the brief's premise is refuted and the 0.78 K of hydrostatic\_column's \_adv file is named.** MEASURED: the flow where it sits is Mach 0.23 to 0.82 (the case is a 300-step relaxation snapshot, not a static column:  $\rho v^2$  spans seven decades); the IONIZATION correction is held at equilibrium in 503 of 503 cells ( $\text{Ion\_species\_adv.txt}$  equals  $\text{Ion\_species.txt}$  to the bit) and its Damkohler condition already gives the continuous  $v \rightarrow 0$  limit the brief asked for, so nothing there was touched. The number is the exact solution of the equation the \_adv ENERGY solve poses: the steady internal-energy equation with the enthalpy flux of the mass-flux divergence DROPPED,  $\rho v \frac{de}{dr} - p v \frac{d \ln(\rho)}{dr} = \text{heat} - \text{cool}$ , which equals the exact  $\dots + h \text{div}(\rho v) = \text{heat} - \text{cool}$  only where  $\rho v^2$  is stationary; with the radiation off it reduces to the discrete adiabat  $T_j/T_{j-1} = \rho_j/[\rho_j - (\gamma-1)(\rho_j - \rho_{j-1})]$ , and integrating that on the run's density reproduces the reported temperature to  $6e-11$  over the 347 cells before the first fallback (1138 K  $\rightarrow$  0.78 K at  $1.396 R_p$ ; the dropped term exceeds the kept ones in 480 of 502 cells). With heat/cool negligible the equation is homogeneous in v, so no  $v \rightarrow 0$  condition can change its answer (99.96 percent departure at  $|v| \leq 1.4e-3 c_s$ ). Landed: a

validity condition of the energy solve's own, the thermal Damkohler number  $Da = t_{\text{cross}} | \text{heat} - \text{cool} | / u_{\text{th}}$  (new pure thermal\_damkohler\_number; above one the cell keeps the run's own  $T$ ), the counterpart of ionization condition (ii) that the bare  $v \leq 0$  sign test lacked; the stationary-mass-flux assumption stated at the site with its exact dropped term and MEASURED at run time (largest ratio, where, how many cells exceed one; nothing acts on it yet); the first eval\_cool writes a named tcool\_in instead of a discard slot; the stale "is not evaluated" sentence of the validity memo rewritten with both tables. Impact: byte-identical everywhere measured (hydrostatic\_column full case all four files; wasp\_full, mol\_base\_handoff, lower\_profile post-processed from the same golden state by both binaries, all four files; max thermal  $Da$  over tested cells 0.951 / 0.534 / 0.055 /  $3e-19$ , 0 cells above one anywhere). Suite RED/GREEN (rows 1-7 cannot link before; row D measured=silent before), advisor 10/0. Also MEASURED: no matrix case has a stationary mass flux by the cell-to-cell measure ( $\rho v r^2$  over outflowing cells spans 1.27x its median in wasp\_full, 2.2x in mol\_base\_handoff, 83x in lower\_profile); the du stop is taken on a restricted window and does not bound this. **Follow-up decided by the advisor under the physical-correctness rule: restore the dropped term (item ADV-ENERGY)**, since an energy equation that is exact only for a stationary mass flux is wrong on every snapshot and at the  $1e-3$  level on every converged wind; MEASURED worth on the static column 569 K at 1.40  $R_p$  and 121 K at 2.96  $R_p$  in place of 0.78 K and 6.0 K. The alternative, refusing the column where the assumption fails, would rewrite 162 of 502\_adv temperatures of wasp\_full (median 5 percent) and is decision 14 (T8.1) of the target-system document, left to the user. Advisor fixed the stale provenance paragraph of the case README.

- **ADV-ENERGY** (T\_equation.f90, ion\_cell\_state.f90, post\_process\_adv.f90, adv\_static\_limit +7 rows, docs/postprocess\_advection\_validity.md): the advection post-process's energy equation is now the EXACT steady internal-energy equation. The enthalpy flux of the mass-flux divergence,  $h \text{ div}(\rho v)$  with  $h = e + p/\rho$ , enters T\_equation through a new zero-default field of the cell record, teq\_cell%div\_rho\_v, as a separate additive term in both branches (caloric  $\mu_m \text{ dr div\_rho\_v } (E(x_{H2}, T) + T)$ , monatomic  $\gamma \mu_m \text{ dr div\_rho\_v } T$ ;  $\mu_m, \mu_p$ , coeff untouched; zero field an exact zero, tested to the bit).  $\text{div}(\rho v)$  is the mass row's own operator  $(A_p F_p - A_m F_m)/dV$  over the control volume between the two points the upwind difference uses. MEASURED corrections to the briefs: T\_equation has no caller outside post\_process\_adv (no marching solve to protect; the bitwise property holds anyway); and ADV-STATIC's 569 K prediction had put back  $p \text{ div}(v)$  alone, not  $e \text{ div}(\rho v)$ ; the whole term gives **1.7e7 K at 1.40  $R_p$  and 1.6e9 K at 2.96  $R_p$**  on the static-column snapshot, the closed form  $w \sim \rho^{(\gamma-1)} F^{(-\gamma)}$  with  $F = \rho v r^2$  falling 8 percent from cell to cell. Tests RED (nine "not a member" compile errors before) / GREEN (closed-form solution  $T \sim r^{(-\gamma)}$  on a diverging column to  $6.9e-4$  at 400 cells, first-order convergence  $5.59e-3 / 2.79e-3 / 1.39e-3 / 6.95e-4$ , the adiabat departing by 2.17); advisor adv\_static\_limit 17/0, physics\_probe 1491/0. **Impact, MEASURED, same golden state post-processed by both binaries:** marching files bitwise in every case; T\_adv movement wasp\_full\_newton 1.4 percent (the only Newton-converged state, He 2<sup>3</sup>S 1.9 percent), wasp\_full (du stop) 16 percent at 1.088  $R_p$  (He 2<sup>3</sup>S 16 percent), mol\_base\_handoff snapshot 301 percent, lower\_profile snapshot 456 percent (He 2<sup>3</sup>S 238x), hydrostatic\_column 4.2e9 (reaching 2.6e10 K). Restored term against the kept ones: wasp\_full 67x at 1.033  $R_p$  in 161 of 502 cells, lower\_profile 25x in

299 cells, wasp\_full\_newton 24x in 2 cells. **The \_adv temperature of a state that violates continuity is now unbounded above where it was bounded below, and the transit tools read those columns:** on such a state NO steady post-process has a meaning, and what to output there is decision 14 of the target-system document, now put to the user as decision item 9 with a recommendation. Goldens: every \_adv file moves by design; refresh deferred by the user to this item's gate.

- B5g** (steady\_newton.f90 +467/-35, steady\_species\_rows suite, design doc sections 18, 18.5): **a residual evaluation is now a function of its arguments.** Three class-(b) caches survived a probe eval\_residual and were read back by the solve: the elemental rows and scales (erow\_he/escale\_he/erow\_tr/escale\_tr), the carrier measures (carrier\_relnorm\_last...), and the carrier arrays of diffusive\_photochemistry (col\_scale\_car, row\_scale\_car, row\_terms, headroom\_car, the frozen backgrounds); the Newton ROW SCALE, the acceptance gate, the best-iterate ledger and the choice of the state handed back read them. Fixed on the reader side: eval\_residual gained state\_is\_discarded and a probe holds/puts back that set (reusing the save\_/restore\_carrier\_module\_state pair certification.f90 already uses); the loop-top row scale is taken AFTER the iterate's own evaluation; the gates read a new carrier\_relnorm\_state. The two self-tests also tag their sweeps as candidate sweeps (under the iterate tag they were writing the run's acceptance ledger and ieq\_nonroot\_streak, whose limit stops the run). PROVEN: EXHALE\_SPECIES\_JAC\_TEST=1 and the untested run are the same run, every (JFNK) it line identical over 17 iterations, Ion\_species.txt byte-identical (RED: 82 differing lines, first at iteration 6). wasp\_full\_newton byte-identical (info 0, 6 iterations, CERTIFIED, 13.30). **What the fix uncovered is bigger than the item:** the mol\_diffusion element solve that B5e brought to 4.3e-9 in 66 iterations now STAGNATES at 1.957 after 17; the runs agree for five iterations and part on a 3 percent difference of Drow between two evaluations of the SAME state, because the residual's seed dependence (resid\_sc\_probe) is **0.96 in units of Resid tol** against  $||R|| = 1.986$ , and tightening the sweep tolerance (1e-12, 1e-14 with pass caps 200, 500) never satisfies the increment test from either seed: **the eliminated composition residual of that state has no fixed point it reaches**, and B5e's convergence was a property of one sequence of evaluations. The change is kept (a row scale must be the row's own terms of the same evaluation). Part 2, the carrier rows: all 18 "no Krylov direction" exits and all 514 refused samples were ONE thing, the H2 carrier of cell 500 at -7.4e-19 (a species unknown on its lower bound), not the element budget (the counter's label named one screen of several; corrected, census added); jv\_product now takes the backward difference  $[F(Y) - F(Y - \epsilon v)]/\epsilon$  only after every forward halving is refused: 49/0/0 accepted/rejected/ refused where 31/18/18, GMRES truncations 18 -> 0, radius 8e-3 -> 2.05;  $||R||$  unchanged at 1.873 because the dogleg cuts steps by  $2^{-25}..2^{-49}$  against the same bound from the trial side, a bound-constrained unknown space (decision, not repair). cert\_tol\_carrier UNCONFIRMED. Suites (advisor build): steady\_species\_rows 74/0 (88/0 with the self-test pair logs), certification 29/0, coupled\_source\_step 44/0, carrier\_retry 100/0, species\_face\_flux green; advisor re-ran wasp\_full\_newton in full with the B5g binary: Hydro\_ioniz.txt and Ion\_species.txt IDENTICAL to the gate-3 output, CERTIFIED, log10 Mdot 13.30.
- B5f** (docs/a2\_certification\_contract\_20260906.md section 11.6; no constant moved): the cert\_tol\_element ladder on mol\_diffusion at Resid tol 1e-8 / 1e-9 / 1e-10 gives ONE state: the

1e-8 solve meets its target in 66 iterations (element row 4.633e-11, reproducing B5e digit for digit, so COST7 leaves the stationary solve untouched); the 1e-9 and 1e-10 solves abort after 88 iterations with the radius at 9.3e-23 (9 of 31 refusals on "the ray difference is below its cancellation floor") and hand back the SAME state, bitwise. The row is floor-limited at 4.63e-11 (knee at  $||R||$  about 1e-7, 1.5 percent movement over the last factor 26); proposed value by the contract's rule 5e-10, NOT applied (two rungs are not solves that met their target; a decade above one configuration's floor is not a decade above the row's floor, which the operator's header puts between 1e-12 and 1e-5 across configurations; no second configuration exists). **Read together with B5g:** B5f ran on the pre-B5g text, and B5g shows the state it anchored on is reached by one sequence of evaluations only (with the probe caches removed the same solve stagnates at 1.957), so this rung stands as a measurement of that state's row, not as evidence that the solve converges. B5f also MEASURED that the coupled route's convergence depends on eq\_sweep\_reltol (1e-10 loses the 1e-9 solve at iteration 17), the same fact from the other side. Recorded: the JFNK of a reloaded 1e-8 state starts from  $||R|| = 1.226$  because the marching loop takes two CFL steps first (the O(dt) kick, eight decades, seen for the first time from a stationary state); the certification prints no cell-resolved record of the species rows, which blocks any floor attribution.

- **ADV-REFUSE** (decision item 9, option (a), user 2026-09-08; post\_process\_adv.f90 +227/-43, write\_output.f90 +94/-10, examples/exhale\_io.py, EXHALE\_transit.py, README.md, README\_HOWTO.md, user manual section 4 (PDF rebuilt, 76 pages), docs/postprocess\_advection\_validity.md, adv\_static\_limit +13 rows): **the advection post-process refuses the correction in every cell whose own energy-equation term ratio says the flow is not stationary; that cell keeps the run's temperature and the ionization equilibrium at it; a new integer column adv\_status at the end of Hydro\_ioniz\_adv.txt records the reason row by row** (0 corrected; 1 thermal Damkohler; 2 not stationary; 3 ionization conditions (i)-(iii); 4 cell solve failed; five named module parameters in write\_output, the # columns line extended, a # adv\_status\_counts header line counted from the array written). The criterion is enthalpy\_flux\_term\_ratio (new pure function beside thermal\_damkohler\_number), the restored term over the LARGER kept term, refused on a STRICT > 1 so a stationary flux with round-off is never refused; formed ONCE on the state handed in over every cell, so the refused set cannot move with the iterate and the converged product does not depend on the iteration history; applied to temperature AND composition (a non-stationary cell has no steady advective ionization correction either). write\_output gains an optional adv\_status dummy (the four 'eq' calls untouched, Hydro\_ioniz.txt keeps seven columns); exhale\_io.load\_hydro now takes names from the header (seven historical keys kept, Run.adv\_status exposed, None for an equilibrium file); EXHALE\_transit.py prints how many rows inside  $1 \leq b \leq \text{Rib}$  are refused, by reason; EXHALE\_plots.py and the four other header parsers adapt unchanged. MEASURED (same golden state, "Do only PP"; marching files bitwise everywhere): hydrostatic\_column's corrected temperature falls from 2.6e10 K to **1182.32 K, exactly the run's own maximum, no row above it** (435 before); status counts 0/1/2/3/4: wasp\_full\_newton 501/0/2/1/0, mol\_base\_handoff 470/0/33/1/0, wasp\_full 205/0/297/2/0 (160 of the 297 beyond conditions (i)-(iii), matching ADV-STATIC's route-2 estimate of 162; the larger total is the reference point, the state handed in over every cell including the inflow cells), lower\_profile 182/0/321/1/0, hydrostatic\_column 0/0/481/23/0. Every refused row carries the run's T to the BIT and the equilibrium density of every species exactly (one

1.8e-16 reassociation in lower\_profile's He I, pre-existing). Movement against the pre-ADV-ENERGY goldens, T\_adv: wasp\_full\_newton 3.3 percent, wasp\_full 8.0, mol\_base\_handoff 64, lower\_profile 1.8x, hydrostatic\_column 1525x (0.78 -> 1182 K); species: He III 5.4 percent, He 2^3S 29 percent, He 2^3S 110x (at the base of mol\_base\_handoff), H II 11x (lower\_profile base), He III 16 percent. **Transit effect on wasp\_full**: 295 of 500 rows inside the impact-parameter range are refused (printed), and the metrics move at 1e-4 relative or less (He 10830 red 4.102 percent both, blue 2.843 -> 2.844, FWHM 0.899 A both; largest line movement O I 1302 +0.55 percent relative), because the refused rows sit in the base under the extended wind. Tests RED (seven "not found in module" errors; measured=7 columns on the old binary) / GREEN; advisor adv\_static\_limit 30/0, physics\_probe 1491/0. Worker also fixed a dangling \ref{sec:retiredbase} in the manual (the four retired base keys named instead). Recorded: two documents describe the \_adv validity conditions (manual, HOWTO) and had both drifted; four # columns parsers exist in python; dV\_cv division now appears twice unguarded (a zero is a broken grid).

- B5h** (ionization\_equilibrium.f90 +68/-2 diagnostics, steady\_newton.f90 +211/-2 probe only, steady\_selfconsistent\_residual assertion 4; NO behavior changed, wasp\_full\_newton and the element run byte-identical to a control built from the same source with the additions removed): **why one state has two residuals**. MEASURED on mol\_diffusion's element configuration with a lockstep two-seed trace of the composition elimination (EXHALE\_RESID\_SC\_TRACE), a nine-seed basin ladder (EXHALE\_RESID\_SC\_BASIN) and a cell acceptance report (EXHALE\_IEQ\_REPORT\_CELL): the amnesty is never used (classes 4 and 6 zero at every pass, every cell class 1 with reaction residual 1e-25 against 1e-6), the self-field lag is not it (xuv\_self\_field\_passes 1 -> 20, identical to every digit), the He recombination coupling is not it (off, identical to four digits), the stopping test is not it (a fixed count of one pass per call, chained, the difference frozen from pass 2 to 60). **Two things are**. (1) Two separated ROOTS of the network at the same incident field: from a seed displaced five cells the sweep reaches a composition 8.9x apart in x(H2) at the H2 front (cell 214) and 19 percent apart in x(H I) at the ionization front (cell 192), bitwise identical from cell 216 outward, both exact roots (class 1, 1e-25). (2) **A marginal direction: the local-equilibrium elimination does not determine the shielded layer's H2 CONTENT**. Retention of a seed perturbation of the H2 partition is 0.77 at the base rising to 0.84 at  $r = 1.064$  and falling to 0 outside cell 216, the SAME curve to three digits at perturbations of 1e-6 and 1e-2, so the sweep's map has eigenvalue 1 in that direction and the seed dependence is a derivative, linear over four decades with gain 3.2e3 (the base cell's energy row amplifies the ghost composition by ~4e3), reaching Resid tol at a seed change of 3e-12. READ from the network: the fast H2 -> H2+ -> H3+ -> H2 cycle returns every nucleus, so the H2 row as scaled by its turnover pins the PARTITION and leaves the total to  $k_{15} n_{HI}^2$  against  $k_{12} n_{tot} n_{H2}$ , a detailed-balance pair that in the shielded layer is far below the turnover; and pinning it would pin the fully molecular limit (0.9997) where the state carries 0.376: the content is upstream and transport data, exactly why it is imposed at the ghosts. **The repair the code already owns, MEASURED**: with Molecular carrier transport: True (n(H2) a Newton unknown) the same ladder falls from 3.155e-3 to **9.736e-10** at delta 1e-6 and x(H2) of the front cell is bit-identical from all nine seeds; the residual is a function of its arguments below Resid tol. What remains at large seeds (0.98) is the ionization front's two roots, a physics decision (which root a stationary residual may land on, or a statement that the front is unresolved on this grid), reported. The element solve with H2 carried

then meets B5g's carrier obstruction ( $|R|$  flat at 1.986 over 19 iterations, carrier row 0.91, run stopped by PID at 4 min an iteration). Also MEASURED: eq\_sweep\_reltol below  $\sim 1e-10$  is unreachable by construction, the increment measure  $\max|df|/\max(|f|, 1e-12)$  reporting round-off of He I and H2+ fractions  $\sim 1e-20$  in the ionized wind (plateau 6e-11 for sixty passes), which is what B5f's and B5g's "never met at 1e-12, 1e-14" was. Tests: assertion 4 residual\_is\_one\_number\_for\_basin RED 3.155e-3 with H2 eliminated, GREEN 9.736e-10 with H2 carried (the tolerance read from the run's own log); steady\_species\_rows 86/0 with the self-test pair; B5g's part 1 reproduced as a by-product. A defect of the item's own fixed inside it (a cell-report default of -1 was the lower ghost). **Decision put to the user as item 10:** refuse Coupled carrier solve: True on a molecular configuration without carrier transport, or turn the transport on for it (input\_read.f90, not the item's).

- B5i** (decision item 10, option (a), user 2026-09-08; input\_read.f90 +51/-2, grid\_and\_gates new row coupled\_carrier\_h2 (+5 assertions), user manual (a Coupled carrier solve entry, which the manual had NOT had; the Molecular carrier transport entry corrected from "inert without oxygen chemistry" to "needs molecular chemistry"), README.md, README\_HOWTO.md (new block; the oxygen paragraph's dangling "It" corrected), docs/steady\_solver\_design.md section 19): **Coupled carrier solve: True with Molecular chemistry: True and Molecular carrier transport: False is refused at startup**, after every key is resolved (so the oxygen default of the transport key is seen: MEASURED, Oxygen chemistry: True plus the coupled key is accepted, the same with the transport stated False is refused), with a message naming what was asked, why there is no root (B5h's numbers) and the remedy key; error stop 1, exit status 1 asserted (2 is the certification's refusal of a complete run). Scope exactly carrier\_in\_newton .and. thereis\_mol .and. .not. carrier\_transport, not gated on the Newton key (the direct steady route EXHALE\_PTC=1 runs a coupled solve too, and it is the route B5b to B5h used); the atomic coupled solve with an element row is accepted (asserted). No shipped configuration is refused (MEASURED by grep: no \*.inp in the tree sets the key; 62 of 81 B5-series scratch inputs would be, as the decision says). Two molecular cases byte-identical before/after at 200 steps. The parse-site comment that claimed "checked below" for a check that did not exist is corrected. A refused run leaves a zero-byte EXHALE\_setup.out and no EXHALE\_resolved.out, like every other input\_read refusal. Tests RED (the eliminated-H2 coupled run completed and printed a Mdot) / GREEN; advisor grid\_and\_gates 98/0, physics\_probe 1491/0. Advisor corrected docs/input\_schema.md K15e ("needs K15d" now qualified to the molecular configuration). Recorded: EXHALE\_main.f90 opens EXHALE\_setup.out (truncating) before input\_read, so any refusal erases the previous run's setup report (fix: open it after the input is accepted); the other "checked below" comments of input\_read.f90 deserve a sweep against the checks that exist.
- HYG-SETUP** (advisor, direct; EXHALE\_main.f90): the setup report EXHALE\_setup.out is opened AFTER input\_read returns instead of before it, so a configuration input\_read refuses no longer truncates the report a previous accepted run left in the same directory (B5i's finding). input\_read writes nothing to that unit. grid\_and\_gates green on the advisor build; the marching path is untouched.
- HYG-CHECKED** (input\_read.f90 comments plus one new report block, docs/input\_schema.md, README\_HOWTO.md, user manual (PDF 77 pages), grid\_and\_gates new row carrier\_transport\_inert +4

assertions): every requirement statement of `input_read.f90` and every "Needs K.." / "refused with" clause of `docs/input_schema.md` checked against the code that enforces it. 40 enforced as stated. ONE not enforced (Molecular carrier transport: True with the molecular network off) is now a startup WARNING, not a refusal, by the code's own convention for a present-and-inert key (every consumer is guarded by `thereis_mol`, MEASURED inert; eight shipped inputs all carry the network). THREE enforced differently from the statement, one with the sense INVERTED: the schema and the HOWTO said Ionization transport is refused with Solver: Newton and with Coupled carrier solve, where the code REQUIRES the second whenever the first is set (K15e does carry a proton row; what is refused outright is the EXHALE\_PTC=1 route); "Solver: PTC refuses it" named a solver value that does not exist; the base-grid resolution check lives in `write_setup_report` (warns below 10 cells per scale height). SIX stale statements corrected: `known_keys` also feeds the duplicate-key REFUSAL (MEASURED, all 99 entries used, no hole); the sub-Lyman absorbers are not mutually exclusive; `Kzz_base/He_Kzz` are NOT inert with `He_diffusion` off (the carrier transport reads `kzz_cell`), corrected in four places; Molecular reaction heat default True is guarded by `with_molecules` (a refusal would refuse every atomic run); the coronal cutoff window 0.08-0.13 superseded by `coronal_cutoff_width.md` 7.2; four K15b cross-references that pointed at the Stellar LW flux row corrected to K15. MEASURED byte-identical on `wasp_full` 100 and `mol_base_handoff` 200 (17 of 17 files). Advisor `grid_and_gates` 102/0, `physics_probe` 1491/0. Recorded: all 63 "F" line citations of the schema are stale by 300-1200 lines (marked stale at the convention's definition; recommend deleting the tokens); Coupled carrier solve: True is silently inert in an atomic run without `He_diffusion` (a report belongs where the row registry is); the user manual has no Ionization transport entry.

- Gate 4 (2026-09-08, 15:39 to 18:45)**, tree after COST7, COST6 (off), B5e, B5g, B5h, HYG-4, ADV-STATIC, ADV-ENERGY, ADV-REFUSE (B5i, HYG-SETUP and HYG-CHECKED landed during the matrix; they touch startup checks and a file-open order only, and were verified on the advisor build with the full `grid_and_gates` suite, 102/0): shared build remade, **make test all suites green**; `make check` FAIL against 1e-3 as expected. **The marching files are unchanged from gate 3 to the digit**: `Hydro_ioniz.txt` and `Ion_species.txt` carry exactly COST7's movement on every case (identical max-rel values in both logs, the three `hp_*` cases still identical), so B5g, B5h, the three ADV items, B5i and the hygiene items moved nothing on the marching path, and `log10 Mdot` is unchanged on all 14 cases. The `_adv` files move by design: `Hydro_ioniz_adv.txt` now has eight columns (`adv_status`, the harness reports SHAPE 8 vs 7), and `Ion_species_adv.txt` moves by 0.05 (`wasp_full_newton`, He I at the outer edge) to order one (the molecular cases and the `hp_*` cases, whose refused base cells now carry the equilibrium densities; the relative measure is element-wise, so a species the old correction had moved off its equilibrium value reads 1). **Goldens to be refreshed by the user, once, for COST7 + ADV-ENERGY + ADV-REFUSE together** (archive first), then `make check` re-run by the advisor.
- Golden refresh (2026-09-08 19:05, run by the advisor on the user's instruction, the user being away from the machine)**: the previous set archived to `backup/regression/golden_pre_cost7_20260908/` (verified identical to `golden/` before the refresh), then `run_check.sh` golden snapshotted the gate-4 outputs of all 16 cases. The refreshed goldens carry COST7 (the stopping test on the estimated error), ADV-ENERGY and ADV-

REFUSE (decisions 8b, 9a). Gate 5, a make check on the current tree (B5i, HYG-SETUP, HYG-CHECKED on top, all startup-only), follows.

- **HYG-DOCS** (documents only: docs/input\_schema.md, user manual (PDF 82 pages), README\_HOWTO.md): the 91 stale "F" line tokens (77 lines) and the 37 GUI<n> numbers (MEASURED stale by a uniform 13 lines) are gone; a key is located by its label and the header says so; nine Notes cells that held only a citation got real notes READ from the parse sites. The manual gained the missing Ionization transport entry (in the sense HYG-CHECKED corrected) and the four other keys the schema documented and the manual did not (Reconstruction continuation, Atomic rate set, Caloric EOS, Photoelectron heating); MEASURED afterwards, all 97 schema keys have a manual entry. Three stale manual entries corrected (Resid norm is retired; Resid tol's norm is the row scaling of sections 143/145, not  $|R|/\max|u|$ ; Flux spread tol default  $2e-5$ , not  $5e-3$ ); the validity paragraph's "one of three" made four. The \_adv validity conditions live in ONE place (manual section 4, \label{sec:advvalidity}), the HOWTO points there; the HOWTO names exhale\_io.py as the loader every new script should use (four # columns parsers exist). Advisor: the same three stale statements in parameters.f90's comments corrected (default  $1e-5$ ; the row scaling; the retired base valve), and the four schema rows whose code spans carried a | (a strict renderer split them) rewritten; table widths verified.
- **B5j** (steady\_newton.f90 +1061/-41 only; steady\_residual.f90 and diffusive\_photochemistry.f90 bit-wise unchanged; steady\_species\_rows +14 rows, steady\_selfconsistent\_residual assertion 5): **the carrier row's lower bound, MEASURED as two mechanisms in sequence.** The outermost carrier row is linear in its own unknown: at the hand-off its root is  $+2.1e-11$  with the iterate 4 percent above (positivity-preserving, the Newton right to reduce it); twenty iterations later the row's constant term has changed sign and the root sits at  $-5.5e-13$ , below the bound, so the row has no admissible root and the unknown is pinned; and at the cell that binds the solve (215) the root is  $+2.75e-6$  while the Newton has driven the unknown to exactly zero, eleven decades below it, the row reading -1.00000 of its own scale. A species unknown at zero whose row wants it positive cannot be sampled at all (a two-sided difference leaves the box on both sides). Landed: the carrier mass fraction clamped onto [0,1] in write\_species\_rows\_into\_composition (the rule the element fractions had); the trial projected onto the faces of the species box with the predicted reduction recomputed for the step taken; Bertsekas (1982) epsilon-active set held for the step and applied to every sampled direction; the probe step cut to the box with the side chosen by the room (subsuming B5g's backward sample); the Cauchy point as the step when the Krylov leg cannot be sampled; a defect of the item's own found and fixed (the accepted trial re-evaluated as  $Y + D$  s rather than as the vector evaluated, whose round trip does not close for a projected component: a carrier at  $-4.8e-35$ ). MEASURED on mol\_carrier, three configurations, none converging: baseline 49 iterations ||R|| 1.929, 527 refused samples, cut-backs to  $2^{-49}$ ; (a) bound-aware region 63 it, 1.415, 0 refused,  $2^{-4}$ ; (b) carrier as  $\ln n$  (EXHALE\_CARRIER\_LOG\_UNKNOWN=1 (the spelling at B5j; since B5k the log unknown is the default and  $=0$  restores the density)) 65 it, **0.9972**, merit 17.6  $\rightarrow$  0.924, smallest adopted unknown  $2.2e-11$ . (b) is better by every measure and is NOT the default (it changes the unknown space of a route a shipped key selects): decision item 11. **Part 2:** the elimination's stopping test is now the change of the three quantities the residual reads the composition FOR (particle count, H2 caloric share, the radiative pair), EXHALE\_EQ\_MEASURE=species restor-

ing the old one: at eq\_sweep\_reltol 1e-12 wasp\_full\_newton reaches it in 33/49 passes where the species measure never does (plateau 2e-11); the element case's remaining plateau (1.85e-11) is the radiative channel at cells 441-459 wandering, that case's own non-convergence (B5h), not the measure's floor. Seed dependence unchanged to four digits (9.736e-10). **wasp\_full\_newton moves 1.1e-9 relative**, isolated to the new measure by EXHALE\_EQ\_MEASURE=species reproducing the control byte for byte: the solve converges FURTHER ( $|R|$  8.1e-9  $\rightarrow$  1.7e-9 in 8 iterations instead of 6, info 0, CERTIFIED, 13.30); below the 0.1 percent rule, the golden is not touched (the harness will read "within"). Suites (advisor build): steady\_species\_rows 93/0, steady\_selfconsistent\_residual 3/0, certification 29/0, coupled\_source\_step 44/0, physics\_probe 1491/0; one "per-iteration" phrase in run.sh reworded. Recorded: a carrier row at an active bound is an error to the certification (row 1.00000 at cell 215) and a constrained optimum to the solver, so a bound-constrained stationary state has a multiplier there, not a zero row (decision item 12); 28 of 64 steps of configuration (a) are the Cauchy point alone (Bertsekas' two-metric projection not taken); load\_IC refuses a molecular restart written at the input's own He/H by comparing printed digits for equality (7.9307E-02 against 0.0793). Advisor: wasp\_full\_newton re-run on the matrix path (marching then Newton) with the B5j binary against the refreshed golden: movement 5.9e-9 (Hydro\_ioniz.txt) and 9.6e-9 (Ion\_species.txt), 13 outer iterations to  $|R|$  = 4.3e-9, CERTIFIED, log10 Mdot 13.30; below the 0.1 percent rule, golden untouched.

- **Gate 5 (2026-09-08, 19:03 to 22:02):** make check on the tree as of B5i, HYG-SETUP, HYG-CHECKED (built 19:03, before B5j landed) against the refreshed goldens: **REGRESSION PASS, all 64 files of the 16 cases byte-identical**. The refreshed goldens are therefore the current tree's own outputs, and the three startup-only items moved nothing. B5j's 1.1e-9 movement of wasp\_full\_newton (the Newton converging further under the new increment measure) is not in this gate's binary and will read "within" at the next one; below the 0.1 percent rule it refreshes nothing.
- **B5k** (decision item 11, option (a), user 2026-09-08; steady\_newton.f90 +346/-85, steady\_species\_rows +137/-6 (114/0), design doc section 20, schema K15e, manual entry, PDF 82 pages): **the molecular carrier densities of the coupled steady solve are carried as  $\ln n$  by default** (EXHALE\_CARRIER\_LOG\_UNKNOWN=0 restores the density unknown; the element fractions untouched; no certification change, decision 12 (c)). The floor under  $\ln n$  is the transport operator's own admissibility floor, 1e-20 of the carrier's element budget in that cell (carrier\_headroom), the fraction carrier\_residual already puts under its row scale; MEASURED 5.6e-28 to 4.6e-21 across mol\_carrier's column (B5j's log unknown had  $\log = -\text{huge}$ , unsound: exp underflows to an exact 0). Stated at the site: the column factor  $n$  of the Jacobian (both the banded model and the Krylov action probe a RELATIVE change), the region asymmetric in the density ( $n \rightarrow n \exp(+\Delta)$ ), and  $\ln 0$ . **A real finding at the arming point:** carrier\_headroom does not exist when the solve reads its controls (allocated inside carrier\_steady\_residual, first reached by the opening eval\_residual), so the first implementation silently fell back to the density unknown on every coupled run; the opening state is now packed as a density and rewritten as logarithms after that evaluation, sound because the residual is a function of the state and not of its coordinates. Floor array made rank 2 (cell AND carrier row; two carriers of one cell have different elements). MEASURED, one binary, the two runs one variable apart: mol\_carrier density 63 it  $|R|$  1.415, 28

Cauchy-only steps, 10 unknowns on a bound, 716 adopted carriers at the floor;  $\ln n$  66 it ||R|| 0.9957, merit 0.960, 0 Cauchy-only, 0 on a bound, 0 backward products, 0 at the floor, smallest adopted carrier  $2.2e-11 = 2.5e15$  times its floor; mol\_diffusion (H2 carried + element row) density unknown 0 accepted steps of 12, log unknown 50 accepted, ||R|| 1.986 -> 1.982. **Neither converges, and the next obstruction is named and is NOT the density unknown's:** the log unknown's steps are model-accurate (ratio 0.39-1.09) and SHORT (||s|| 0.09-0.17 against a radius of 2.5 sitting at its ceiling for 30 of 68 iterations), cut by 3-5 halvings before the trial is describable, and the refusing screen is the ELEMENT BUDGET (157 samples, against 0 for a negative carrier): the carrier CEILING, which species\_unknowns\_outside\_their\_bounds deliberately does not treat as a face of the box. Bertsekas' two-metric projection is not what the log unknown needs (it holds no bound). Gates: wasp\_full\_newton byte-identical twice (full case control vs new, and the reload identical to B5j's binary including every JFNK line); marching bitwise on mol\_carrier 200 steps. New rows: default is  $\ln n$ , switch reaches density, the floor identical under both unknowns, carrier\_is\_above\_its\_own\_floor (RED on the density unknown: 0, 716 at the floor). Advisor: steady\_species\_rows 93/0, steady\_selfconsistent\_residual 3/0, certification 29/0, physics\_probe 1491/0; the two documents that named the old =1 spelling annotated. Recorded: gm\_m = 40 is a literal at the call site in EXHALE\_main.f90 and the log-unknown solve sits at it in 58 of 66 iterations (unmeasured whether raising it helps, the cycle reports iterations used, not the residual reached); carrier\_headroom returns huge as an availability sentinel. **Next item B5l:** the element budget as a face of the unknown box.

- **B5l** (steady\_newton.f90 +392/-82; diffusive\_photochemistry.f90 +51/-6, the headroom accessor only; steady\_species\_rows suite +161 and +108; docs/steady\_solver\_design.md section 21; certification.f90, EXHALE\_main.f90, steady\_residual.f90, ionization\_equilibrium.f90, input\_read.f90 untouched): **the element budget is a face of the unknown box and the projection treats it exactly.** freeze\_species\_unknown\_box is now the one place the admissible set of a species unknown is written down (one lower and one upper bound per unknown of the flat vector, formed once per outer iteration from the iterate and from the element budget the carrier operator froze at that iterate); the three routines that had each computed their own bounds (species\_unknowns\_outside\_their\_bounds, fix\_active\_species\_bounds, largest\_step\_inside\_the\_species\_box) read it. **A drift between the three was found and fixed on the way:** the projection had read the carrier ceiling from the trial's own density slot while the other two read it from the iterate, so the set a trial was written onto and the set the Krylov cycle was sampled under were not the same set. Faces of a carrier unknown: below, zero (carrier\_log\_floor in  $\ln n$ ); above, the cell density and the element budget (carrier\_headroom, for H2 half the hydrogen the carriers may take), the tighter of the two. The budget is frozen for the step, not evaluated at the trial, which is the carrier operator's own choice (headroom\_car assigned under weno\_mode == 1, the one evaluation that IS the iterate; refreshing it per trial was tried there and is worse), so box and screen describe one state. The budget face is raised to the iterate where the iterate stands outside it (counted, n\_budget\_face\_at\_the\_iterate; 0 on both coupled cases with the default step control); it is stored one floating-point step inside the budget (budget\_face\_inside = 1 - 8 eps, representability in  $\ln n$ , not a margin). An unknown whose box is a point (zero budget) is held without asking the gradient. EXHALE\_SPECIES\_BUDGET\_FACE=0 restores the entry text's box (measurement switch,

default on) [retired by N4b: the budget became a constraint row, EXHALE\_ELEMENT\_CONSTRAINT\_ROWS=0 is the measurement switch now]. pgmres reports the relative residual it REACHED against the tolerance asked; EXHALE\_GM\_M overrides gm\_m (the caller's 40 stays the default and the literal at EXHALE\_main.f90 line 5343 is untouched). **MEASURED on mol\_carrier (coupled, ln n), face off vs on: 66 vs 52 outer iterations, both ended by the stall detector; ||R|| handed back 0.9957 vs 1.174; trials halved 3 or more times 38 of 68 vs 0 of 53; residual samples refused by the element budget 157 vs 0; trials written onto a face 0 vs 29; adopted carriers above their own budget 0 vs 0 (largest ratio 0.9886 / 0.9878); iterations ending "the model promises no reduction" 0 vs 19; worst Krylov relative residual reached 0.938 vs 0.569 where 0.1 was asked; log10 Mdot 10.63 both.** The halvings and refusals against the budget are gone exactly; ||R|| does not improve; the limit moved to a named place, the Krylov leg of the dogleg. **The Krylov budget is not the limit:** gm\_m 40 / 80 / 160 give ||R|| 1.174 / 1.103 / 1.271, worst relative residual reached 0.569 / 0.630 / 0.614, ||s|| of the last accepted steps 0.04-0.24 in all three, log10 Mdot 10.63-10.64; the cycle spends every product and stops at the same 0.6 of its right-hand side, so it stagnates on this operator (preconditioner or operator, not subspace). On mol\_diffusion (H2 carried) the face does the same to the step control (halved 6-11 times in 50 of 50 vs 0 of 41; budget refusals 388 vs 0) and the solve stalls where it always has (H2 front, cell 212, element row at 1.000 of its scale; ||R|| 1.982 vs 1.943; log10 Mdot 7.97 both), with the probe showing instead 419 backward-side products, 26 unknowns held on a bound, 12 Cauchy-only steps and 18 truncated cycles. A defect in the item's own first measurement was found and fixed (the adopted carrier was recorded against the PREVIOUS iterate's budget, which had reported 67 adopted carriers above budget, worst 1.058; the call now follows the iterate's own residual evaluation, arithmetic unchanged iteration for iteration). **Gates:** wasp\_full\_newton bitwise against the control binary (Ion\_species.txt bitwise, Hydro\_ioniz.txt identical but for the provenance line, every (JFNK) line identical, ||R|| = 4.299E-09, log10 Mdot 13.30, 13 iterations); marching bitwise on mol\_carrier 200 steps; EXHALE\_SPECIES\_BUDGET\_FACE=0 reproduces the entry-text binary exactly; steady\_species\_rows 118/0 with the new rows RED without the face and GREEN with it; physics\_probe PASSED. **Advisor:** the five diffs match the tree against the worker's entry snapshots; forbidden-word scan clean; the current tree builds warning-free (build\_lwv); the worker's wasp\_full\_newton output against the current golden moves 5.87e-9 / 9.58e-9, exactly the B5j movement already on record, so B5l itself moves nothing on the matrix path. No golden refreshed. **Recorded, not repaired (outside this item's files):** carrier\_element\_headroom states a box side per carrier but no joint bound for carriers sharing an element (the physical constraint  $2 n(\text{H}_2) + n(\text{OH}) + 2 n(\text{H}_2\text{O}) + n(\text{H}^+) \leq n\text{H}_{\text{avail}}$  is a hyperplane the code does not state), and OH/H2O hold hydrogen nuclei their own headroom does not mention, so a state can pass every headroom and still leave the write-back's hydrogen remainder negative (the branch that zeroes H I, H II, H2+, H3+); neither reaches the {H2} carrier set both measured cases use. Also: the epsilon-active band (one finite-difference step) never engages on mol\_carrier with the face (29 trials projected, 0 held; no switch, a wider band unmeasured); largest\_step\_inside\_the\_species\_box returns zero for the whole direction when one component has no room (18 truncated cycles on mol\_diffusion); note\_adopted\_species\_unknowns is not called for a converged state (pre-existing, cannot affect a stall). **Next obstruction on the carrier-row solve: the Krylov leg (banded preconditioner or operator), not bounds and not subspace.** Per the user's instruction

of 2026-09-09 ("leave running things alone, stop code modification for now") no further item is started; B5m (the atomic element-row solve with no Krylov direction) stays planned.

- Preparation of PLAN\_20260909\_rev1 (items N-1 and N17; advisor, documentation and file moves only, no source change; user's go-ahead 2026-09-09 with the instruction not to rerun regression checks after simple changes).** docs/PLAN\_20260909.md was written from the review of the issues inventory (docs/ISSUES\_20260909\_review.md), reviewed (docs/PLAN\_20260909\_review.md) and revised to docs/PLAN\_20260909\_rev1.md, the plan of record (items N-1 to N17, stages A to E, decisions 13 to 19). Before writing rev 1 the advisor re-read the review's new source claims: the sampled integration-error estimate is printed and dropped and no production caller issues `as_reject_int_error` (EXHALE\_main.f90 2806-2821); after a retry inside the full pass the half passes start from  $0.5 \cdot dt\_step$  and the clock advances by the saved `dt_step` (1706-1734, 2797-2799, 2820, 2836); the audit harness asserts the OLD failure (`if (ieee_is_finite(x(1))) error stop 4`); `use_tr = (nspec_row > 0)` (steady\_newton.f90 5763) so `wasp_full_newton` never enters the trust region, which corrects a statement of the first plan. All confirmed. **N-1:** the B5m reproducer moved from the session scratchpad into the tree as `backup/regression/atomic_elem_newton/` (input, metals, setup and resolved reports, `run.log`, the two handed-back state files, `run.sh`, README; 6.8 MB, of which `run.log` 6.2 MB; not in DEFAULT\_CASES); the common worker brief became `docs/worker_rules.md` with the session paths generalized and three rules added (one owner per source file per increment; affected-path validation only; named outcomes in reports); the review harness `docs/audit_20260905/run_issues_review_20260909.sh` was run once on the current tree, its output kept as `docs/audit_20260905/run_issues_review_20260909.out` (zero-radius step norm 0; Cauchy merit 0.5 to 40.5, line minimizer 0; identity GMRES 1; zero-operator GMRES non-finite with reported residual 0; enthalpy ratio 0.5 on a nonzero mass divergence; He/H pair  $8.83e-5$ ; hydrogen inventory 2 at the individual ceilings; outer-shell chord 11.83; "defects reproduced, not repaired"), the harness itself untouched. **N17:** ISSUES\_20260909.md corrected per rev 1 section 3 (seed retention no longer stated as an eigenvalue of one; B5f marked a withdrawn, uncertified reference; 3.5 scoped to the present fixed-point map; the restart row now states the `heh_dev > 1e-6` branch and the  $8.83e-5$  discrepancy; `carrier_headroom_known` noted; `hydro_row_max` and `use_tr` bullets; the `_adv` statement in section 5 made conditional; new section 3.8 on the physical step never being refused for its integration error). `code_status_20260905.md`: the Huang et al. 2023 shell-opacity note added to the photoionization row; four pre-existing spelling/forbidden-phrase slips fixed on the way (per-cent to percent twice, two per-molecule phrases reworded). Verification: forbidden-word scans of the four documents; the reproducer's `run.sh` resolves its default binary path; nothing built or rerun.
- N16a** (PLAN\_20260909\_rev1 Stage A, the physical step can be refused and the clock is right; EXHALE\_main.f90, attempted\_step.f90, `src/tests/attempted_step/{run.sh,attempted_step_tests.f90}`; user 2026-09-09 "N16a 병행"): **the sampled step-doubling comparison is one transaction over one macro-interval H:** the full pass covers H, the two half passes  $0.5 \cdot H$  each and end at  $t + H$  by construction (pass 3 no longer inherits whatever `dt` held); a refusal inside any pass, a finite estimate above one, or a nonfinite estimate rejects the WHOLE macrostep (`as_reject_int_error` now issued by the production caller), restores `as_entry`, shortens H by

the controller and recomputes both trajectories; exhaustion of the transaction's retry budget (`n_macro_retry`, cap 8) is a named failure that stops with status 2, names the pass and prints the checksum of the restored state (`stop_on_exhausted_retry_budget`, one report for both exits). The clock, the accepted count and the physical accumulations move only at the adoption boundary below the macrostep loop. **The order of the complete split update is MEASURED as  $p = 1$ :** at one fixed state the estimate scales as  $H^2$  over five decades ( $p + 1 = 1.964, 1.993, 1.9985, 1.9997, 1.9999, 2.0004, 2.0000$  for  $H$  from  $2e-2$  s to  $2.6e-7$  s; the coarsest  $1e-1$  s is outside that regime and excluded), so `attempted_step_error_estimate` now returns the retained full step's error  $\Delta/(1 - 2^{-p}) = 2 \Delta$  (`err_order_p`, `err_full_step_factor`, convention at the declaration) [superseded by N16d/N16e:  $p = 1$  was the base-face leak; the conservative and electron/T rows are  $p = 3$ , factor 8/7, the species rows  $p = 1$ ] and the reduction is  $0.9 e^{-(1/(p+1))}$ . The brief's first procedure (one interval by 1, 2, 4, 8 prescribed steps, ratios of the output rows) was INCONCLUSIVE,  $p$  0.4 to 4.0, because the third difference ( $2.2e-8$  relative) sits at the coupled source step's fixed-point tolerance: the inner error is not below the temporal error there (review F3's condition, not yet enforced). Test hooks, default off: `EXHALE_AS_ERR_INJECT[_COUNT]`, `EXHALE_AS_REJECT_IN_PASS`, `EXHALE_AS_FIXED_DT` (seconds; in initialization mode with local stepping it flattens the local steps, stated at the hook). `EXHALE_STEP_CLOCK=1` now also prints an `err-pass`: line per pass and `state=` (checkpoint checksum) per accepted step. **Tests:** suite RED on the entry binary 117/16 (the sixteen new rows: passes cover one interval, forced error rejects, clock unchanged, shorter retry accepted, refusal aimed at pass 1/2/3 keeps the endpoint and names the pass, exhaustion in each pass and on the error is named and restores the accepted state by checksum), GREEN on the final binary 133/0; advisor reran the suite with the worker's final binary (md5 5ea530d3251a8b8ce77e72415ddef0ab): 133/0. **Runs:** `mol_carrier` marching 200 steps byte-identical apart from the provenance line (initialization mode never sets `n_err_passes = 3`; the guard lines quoted in the report); the suite's case in physical mode, 60 steps: identical through step 19, at step 20  $e = 1.215$  (raw 0.607, the number the old code printed and dropped) rejects,  $H$  1.118  $\rightarrow$  0.913 ( $e$  1.77)  $\rightarrow$  0.618 s ( $e$  0.89) accepted,  $t_{\text{phys}}$  66.40  $\rightarrow$  65.92 s; two rejections in 60 steps, all by the integration error. Not run: the matrix, `wasp_full_newton`, `run_fcheck.sh` (out of the affected path; the module compiles clean under the suite's `-fcheck=all`). No manual entry (the manual carries no run-mode or clock text; nothing in it became false). **Advisor:** file scope as briefed; six British per cent (two new, four pre-existing) changed to percent. **Reported, not fixed (outside the item):** (1) the accepted state of a retried macrostep depends on which pass was rejected, up to  $9.1e-5$  relative in the base cell's velocity ( $5e-6$  elsewhere), with the checkpoint round trip exact, so some state read by the coupled source step is neither saved nor rebuilt (seeds or caches of `ionization_equilibrium`, `constrained_chemical_equilibrium`, `diffusive_photochemistry`): a rejected attempt is not yet without consequence; (2) one interval of the order sequence ( $H$   $6.48e-5$  s from the 0.405 controller reduction) sits 2.3x off the  $H^2$  law the other seven follow, unexplained, for N16b; (3) `err_rtol = 1e-3`, `err_atol = 1e-12` are flat, unvalidated and cover no species (N16b); the gate now shortens physical-mode steps by about two at the sampled steps on this case.

- **N0, N3, N1, N2** (PLAN\_20260909\_rev1 Stage A, one worker, four diffs; `steady_newton.f90` +738/-143 over the four, new suite `src/tests/krylov_and_dogleg/`; no other file). **N0:** 14 rows calling the PRODUCTION `pgmres`, `dogleg_step` and the factored step-length and predicted-drop code (a test-operator

hook in pgmres in the style of carrier\_headroom\_set\_for\_test); RED on the entry text 10/7, among them zero\_operator\_step\_is\_finite = F, approximate\_gradient\_length\_decreases\_the\_model = 40.5. **N3**: seven named linear outcomes (gm\_tolerance\_reached, gm\_subspace\_exhausted, gm\_no\_direction\_sampled, gm\_jacobian\_action\_unusable, gm\_preconditioner\_failed, gm\_reduced\_system\_singular, gm\_nonfinite); a zero right-hand side returns before any factorization; hypot rotations; rank tested before the triangular division and the leading nonsingular block solved; LAPACK status and finiteness checked after each preconditioner apply and on the returned step; the TRUE scaled residual verified with one product at a breakdown; the cycle exits at a breakdown. Reorthogonalization not added: worst max  $|V_i \cdot V_j|$  MEASURED  $6.9e-13$  on the carrier fixture. **N1**: absent elements (reservoir abundance zero, the melem\_ab  $\leq 0$  test set\_energy\_vectors and load\_IC use) are no longer registered as rows; initial\_trust\_region\_radius with named statuses, holding the active bounds before its own cycle; a positive floor tr\_delta\_min =  $1e-6 \sqrt{\text{neq}}$  in scaled coordinates (257x and 1157x below the first radius the two fixtures take); tr\_dmax follows the radius; "no feasible descent" is info = 3. **N2**: the length along the direction used,  $\tau = (r \cdot A g) / \|A g\|^2$ , the point named an approximate descent point, the paragraph defending the old length by B5e removed, a dropped Cauchy leg with a Krylov residual  $\geq 1$  exits as tr\_no\_direction\_reduces\_model. Suite after diff 4: 31/0, also under -fcheck=bounds,do,mem.

**Root cause of B5m (R1's mechanism, its cause)**: the 1500 unknowns on a bound of the atomic element reload were the three elements metals.inp does not carry (Si, K, S x 500 cells), registered with an identity row and sitting exactly on 0, so the fraction-to-the-boundary rule zeroed ANY direction with a component there and the first cycle took 0 products; the radius became 0 and stayed so. MEASURED after (worker and advisor agree): unknowns on a bound 1500  $\rightarrow$  0; initial radius 0  $\rightarrow$   $1.909e-2$  from a Krylov step that reached tolerance (3 products,  $6.1e-2$  against 0.1); first step accepted, ratio 0.995;  $\|R\|$  1.888 constant  $\rightarrow$   $8.3e-2$  at iteration 50, then an oscillation between  $6.9e-2$  and  $1.1e-1$  with the merit creeping  $7.29e-3 \rightarrow 7.23e-3$  over 258 iterations (advisor run stopped by its 20-minute cap; 178 accepted, 80 refused; every cycle spends all 40 products). Not converged: the next obstruction is the model quality of the element rows near  $\|R\|$   $1e-1$ . **Carrier reload** (control reproduces B5l's run exactly): after, 54/1 accepted,  $\|R\|$  handed back  $1.174 \rightarrow 1.151$ , worst achieved Krylov residual  $0.569 \rightarrow 0.575$ , radius range unchanged, log10 Mdot 10.63 both, outcomes now named (8 tolerance reached, 47 subspace exhausted). **The concurrent N16a edits are inert on every fixture** (paired controls bitwise). **wasp\_full\_newton** (once, final binary): not bitwise, largest relative movement  $2.1e-10$  (Ion\_species.txt, Ca I) and  $8.8e-11$  (Hydro\_ioniz.txt), info = 0, log10 Mdot 13.30; by code path the hypot rotation of N3 (use\_tr is false there); below the 0.1% rule, identical, no refresh. lower\_profile (the only default case with He\_metal\_diffusion and a partial reservoir) bitwise to its golden in both builds (it stops at its marching cap). Other suites (worker): steady\_species\_rows 106/0, certification 29/0, steady\_selfconsistent\_residual 3/0; physics\_probe does not include steady\_newton (source\_closure.py). **Reported, not fixed**: largest\_step\_inside\_the\_species\_box still annihilates a whole direction when any unheld unknown has no room on the side the direction points (an element physically zero in one cell can trigger it; the outcome is now named and the radius survives, the direction is lost); an active-set rule decision, for N4b. Two dead locals removed from pgmres.

- **Stage A gate (advisor, 2026-09-09).** Tree after N16a, N0, N3, N1, N2. make OBJDIR=build\_lwv EXE=EXHALE\_lwv.x warning-free, md5 dd233d6a (identical to the worker's final binary, so the whole tree is what both workers measured). Suites on that build: krylov\_and\_dogleg 31/0, steady\_species\_rows 106/0, steady\_selfconsistent\_residual 3/0, certification 29/0, attempted\_step 133/0 (with the N16a binary, md5 5ea530d3, same source). N7's first two iterations on the atomic reload logged above with named outcomes. Not run, by the affected-path rule: the matrix, physics\_probe, run\_fcheck.sh; the shared build/ and EXHALE.x remain behind the sources. No golden refreshed. Gate PASSED; Stage B (N4a, N5, N8a) and the N7 diagnostic list are next.
- **N13** (PLAN\_20260909\_rev1 Stage D, transit refusal census; EXHALE\_transit.py, exhale\_transit\_lib.py, new Python suite src/tests/transit\_census/; no Fortran, no output format change, so decision 17 is untouched): the census mask ( $r \geq 1$ ) & ( $r \leq \text{Rib}$ ) was geometrically wrong (the ray integral uses every shell with  $|r| \geq b$ ); one function chord\_shell\_indices now serves the census, resonance\_depth and resonance\_spectrum, and the census reports, line by line at eight impact parameters and at its maximum, the share of the line-center optical depth carried by rows whose correction was refused (refused\_line\_center\_tau\_share, the trapezoid rule rewritten as a sum over rows so that shares of disjoint subsets add to one), labeled a CONTRIBUTION, not an uncertainty. MEASURED: every tpm\_\*.txt of wasp\_full (10) and mol\_base\_handoff (5) bitwise before/after; the defect is latent when  $\text{Rib} = r[-1]$  (both cases as they stand) and real when the domain passes the stellar radius (wasp\_full read at EXHALE\_TRANSIT\_RSTAR\_RSUN=0.30: 419 -> 500 sampled rows, 81 above the cap); on wasp\_full the rays near  $b = 1$  take 99 percent of their Ly-alpha line-center depth from uncorrected rows. Suite RED on the pre-change copies (11 rows, the mask present and no shared selection), GREEN 11/0 on the tree; advisor rerun 11/0; +0.3 s on the whole script. Advisor: one per-line in the library header reworded. Reported, not fixed: the wavelength-resolved share and the file metadata wait for decision 17; np.loadtxt(..., max\_rows=1) at line 896 warns on every run and duplicates the column count of line 325.
- **N12** (PLAN\_20260909\_rev1 Stage D, the upstream caloric energy at the upstream composition; T\_equation.f90, post\_process\_adv.f90, ion\_cell\_state.f90 (two fields x\_h2\_up, e\_up), src/tests/adv\_static\_limit/ (new driver upstream\_caloric\_energy.f90, rows U0-U4)): the upwind term of the \_adv energy residual now receives the upstream cell's SPECIFIC internal energy  $e_{\text{up}} = E(x_{\text{H2},\text{up}}, T_{\text{up}})/\mu_{\text{up}}$  formed by the caller, and the caloric branch is taken when EITHER side carries H2 (an atomic cell below a molecular neighbor no longer loses the incoming rovibrational energy); the monatomic branch is verbatim when both sides are atomic. **A second, larger defect found and fixed in the same path:** x\_h2\_cell was assigned only inside the  $k == 1$  stationarity loop (old 650) and read by the temperature loop for EVERY cell (old 1316), so the whole molecular layer was solved at the H2 fraction of the outermost ghost ( $1.5\text{e-}7$  on mol\_base\_handoff,  $2.3\text{e-}5$  on mol\_carrier, against 0.84 at the base); advisor confirmed both lines in the pre-change source. The stationarity diagnostic's lower energy e\_lo is likewise evaluated at the upstream composition (the ratio's definition, threshold and operator unchanged). Energy convention unchanged: the caloric EOS carries the bound rovibrational ladder only, the 4.478 eV bond lives in the formation reservoir

(molecular\_reaction\_heat.f90), no term is counted twice. **Tests** RED/GREEN (MEASURED): isothermal H2 gradient residual 1.508 -> 0; atomic cell below molecular upstream 1.006 -> 0; dissociation-layer march against the analytic discrete solution, largest departure 0.155 -> 2.4e-11 (RED marched 1000, 892, 782, 615, 560 K against the discrete 1000, 975, 863, 728, 596; GREEN both 1000, 974.76, 943.62, 880.65, 853.52); controls unchanged. Suite 36/0; advisor rerun on build\_lwv (md5 8d027f8b, the tree with N12 and N13) 36/0 (a first advisor run reported one no\_output FAIL from a relative EXHALE\_EXE path, the advisor's mistake, not the suite's). **Impact** (post-process restart Do only PP: True of each case's own state; the before binary is the Stage A binary): wasp\_full\_adv files bitwise (atomic, both x\_h2 zero, monatomic branch verbatim); mol\_base\_handoff and mol\_carrier\_adv move in 469 and 467 of 504 rows, T\_adv by up to 94.2 percent (218 -> 424 K at r 1.0941) and 103.2 percent (169 -> 343 K), split about 35/45 and 41/46 percent between the stale-composition and the upstream-composition defects; both a warming of the molecular layer (5/2 kT per H2 instead of 3/2); p\_adv, cool\_adv follow, heat\_adv at most 4.8 percent; adv\_status changes in one cell of mol\_carrier (row 26, 0 -> 2, from the e\_lo composition in the ratio); run states (Hydro\_ioniz.txt) bitwise in all three. **The molecular\_adv goldens of the matrix will move; refresh at the Stage D gate, reported then, not now.** The writer's warning that molecular and oxygen columns are not advection-corrected stays; the product is still not a molecular stationary solution. Advisor: one per-thread phrase in ion\_cell\_state.f90 reworted; the worker corrected four British spellings in comments. Reported, not fixed: a stale scalar formed in a k == 1 block and read later is a class worth a sweep of the file; post\_process\_adv.f90 is named by both N11 and N12 (one owner at a time; N11 waits for decision 16).

- **N9** (PLAN\_20260909\_rev1 Stage D, the restart He/H discrepancy; load\_IC.f90 +84/-12, new grid\_and\_gates row restart\_round\_trip (9 assertions); write\_output.f90 untouched, no key, no tolerance change): **the reported failure was a message defect**: the refusal printed the He/H of the TOP physical cell while the branch tests the largest departure over all physical cells; on the failing state (B5j/scan500\_stag, a coupled-carrier JFNK iterate of the hot-Uranus gate) the top cell agreed with the input to 17 digits and cell 199 was off by 3.96 percent, so the message read "7.9307E-02 against 7.9307E-02 ... cannot be rescaled". Review R5's 8.83e-5 is the distance between the INPUT values of two different runs, not a property of any state. Fixed: the comparison records the cell it attained and report\_deviating\_heh\_cell prints cell, radius, file and input ratios at ES16.9, the departure and the threshold (heh\_dev\_tol = 1e-6, now named, value unchanged); RED/GREEN on a state with helium raised by half at cell 199. **The round trip is clean** (MEASURED on the 12000-step mol\_base\_handoff state with EXHALE\_DUMP\_IC=1): r, v, p bitwise; every species column and every inventory (H, He, O, C nuclei, charge, He/H, H2 ratio) within 4.2e-16; T 5.6e-16; the rho column 4.9e-12 because the loader never reads it and rebuilds mass from the species (perturbing the column by 1e-15 changes nothing, bitwise), so 4.9e-12 is the writer state's own mass-closure defect (4.1e-14 on a 40-step state); a second trip 5.0e-16, no drift; no clock advanced. **The producing path moved the ratio, not the loader** (MEASURED): He/H departs from the input by 1.0e-11 and 8.9e-12 after 12000 marching steps with carrier transport off and on, but by 3.9 to 7.9 percent on every coupled-carrier density-unknown JFNK state and 41 to 43 percent on the ln n states (B5j, B5k): the carrier column is written after the element projection precisely so the Newton's carrier is not rescaled (write\_species\_rows\_into\_composition 1439-1469), the sweep reads the nuclei totals from

the state (hydrogen\_helium\_nuclei\_density), and with helium diffusion off no row re-imposes He/H. This is the element constraint N4a/N4b are about, seen from the restart side; reported, not touched. Also fixed on the way: parse\_labels had no bound on 80-long label arrays (a file with more than 80 columns wrote past them); it now stops with a message (present schema at most 42). Atomic restart path MEASURED unchanged: wasp\_full\_newton's state reloaded before/after bitwise, no He/H line printed. Advisor: restart\_round\_trip 9/0 on build\_lwv (conda gfortran; the worker's private build used /usr/bin gfortran 13, against the standing toolchain rule, so the advisor's run on the standard toolchain is the one on record). **Reported, not fixed:** the window integrals of the assembled residual are not Lipschitz in the state in the odd-even band above the base (one round trip at  $4.9\text{e-}16$  moves the 1.03-1.10 mass window by 16 percent and flips cell residual signs at cells 183-187; one ulp on every species column reproduces it), so a round-trip residual budget must be stated on the row maxima (N5); restart\_grid\_guard.sh carries a stale header describing a fixed defect; nothing checks the writer's rho column against its species (decision 15's contract); EXHALE\_RESIDUAL=1 runs one sweep unconditionally, so a true "no chemistry on load" evaluation has no entry point yet (N10).

- **N7 (PLAN\_20260909\_rev1 Stage C, the atomic element-row diagnostic after Stage A; steady\_newton.f90 +477/-0, diagnostic prints under EXHALE\_ELEM\_DIAG=1 (iterations 1, 2, 60; =n moves the third), one extra Jacobian product per selected iteration with the refusal statistics held aside; no arithmetic of the step changed, MEASURED: every (JFNK) line identical hook off/on over 2027 lines, across four builds). Verdict, MEASURED on the reload fixture:** the oscillation at  $||R||$   $7\text{e-}2$  to  $1.1\text{e-}1$  is NOT a model failure (predicted/actual ratio 0.9946, 1.0001, 1.0003 at iterations 1, 2, 60; the refused iteration's model is right to  $3\text{e-}4$ ), NOT a bound effect (0 unknowns held, 0 samples refused in 293 iterations), and NOT limited by the linear solve although the solve fails its tolerance (40 of 40 products, achieved 0.6685 against 0.1, TRUE residual 0.6685 by one extra product, Arnoldi gap  $2.3\text{e-}4$ ; the dogleg takes  $2.1\text{e-}4$  of the Newton leg so that inaccuracy is second order). **Proximate cause: the ray test** (the finite-difference check of the model slope) refuses the step: at the refusing iteration model slope  $-9.13\text{e-}9$  against ray slope  $-1.48\text{e-}9$  (factor 6.2 against the allowed 4) with  $|f_{2p} - f_{2m}| = 3.0\text{e-}12$  only 11x above the floor  $\text{eq\_sweep\_reltol} * |f_2| = 2.7\text{e-}13$  (at iteration 1 the same test has a factor 1371 of room and the slopes agree to 0.5 percent); 82 of the 91 refusals in 293 iterations are this test, 3 its sign form, 6 a real over-prediction. Limit cycle read off the radius:  $9.78\text{e-}4$  accepted  $\rightarrow 1.96\text{e-}3$  accepted  $\rightarrow 3.91\text{e-}3$  refused by the ray test  $\rightarrow$  cut to  $9.78\text{e-}4$ , period three; the merit falls  $1\text{e-}4$  of itself per accepted step, so Resid tol  $1\text{e-}8$  is seven decades away. **Underlying cause: the merit and the gate are different functionals.** At iteration 60 the eight element rows hold 0.437 of the squared merit (He 0.305) and NOTHING of  $||R||$  (energy row holds 1.000 of  $||R||$ , 0.560 of the merit); their own certification measures sit at  $1.2\text{e-}3$  to  $7.4\text{e-}3$  against  $\text{cert\_tol\_element} = 1\text{e-}8$ ; accepted steps swing  $||R||$  by 40 percent ( $7.8\text{e-}2$  to  $1.09\text{e-}1$ ) while the merit stands at  $7.291\text{e-}3$  to four digits. At iteration 20 the brief's option (c) is met outright:  $||R||$  1.400 set entirely by the mass row, the energy row 0.810 of the merit, seven of eight elements worst at cells 12 or 17. **The Cauchy leg is worth nothing here:**  $||sU||$   $6.6\text{e-}9$  against a radius  $3.9\text{e-}3$  ( $g = B^T r_0$  from the band,  $||A g||$   $1\text{e}4$  times  $r_0$  .  $A g$ ), predicted decrease  $1.3\text{e-}9$  against the dogleg's  $7.6\text{e-}9$ ; one product per iteration for  $1.7\text{e-}6$  of the step. **The band is not the defect** (0 zero, nonfinite or round-off columns, dgbtrf info 0), though it conditions worse along the solve (dgbcon  $7.3\text{e-}8 \rightarrow 9.4\text{e-}10$ , smallest pivot always a trace element of cell 500). **gm\_m probe:**

unlike the carrier-row solve, the larger subspace changes the achieved residual on this operator (first divergence at iteration 17: 40 of 40 at 0.1035 against 48 of 80 at 0.09999; deep in the stall 0.67 against 0.29) and the solve stalls WORSE ( $m = 80$  plateau  $||R||$  0.27 to 0.34, merit  $1.57e-2$ , against  $7.7e-2$  to  $1.1e-1$  and  $7.3e-3$  at  $m = 40$ ): longer steps lift the ray probe off its floor and the refusal becomes an honest over-prediction, and the solve still goes nowhere. **Proposals, not implemented:** P1 measure the ray test's floor from two evaluations of one state instead of assuming  $eq\_sweep\_reltol * |f2|$ ; P2 do not let the slope comparison overrule a reduction ratio within  $[0.998, 1.001]$  (every ray-size refusal inspected had one; the over-prediction guard stays live); P3 drop or rescale the banded-gradient leg where  $||sU||$  is below a small fraction of the radius (one product saved); P4 a GMRES restart buys  $m = 80$ 's reduction at  $m = 40$ 's memory but does not fix the stall; **P5 the acceptance functional and the merit must be reconciled** (a merit that reads the element rows on their certification scales, or a separate control for them): a design decision, decision item 20 (section below). Also fixed in the same file: the `tr_exit_text/gm_outcome_text` buffers (48 and 64 characters) truncated two 67-character messages; all three are 80 now. **Reported, not fixed:** the worst  $j, k$  on the (JFNK) it line is the merit's largest scaled row while  $||R||$  on the same line is a certification maximum, and the two need not be the same row (iteration 20:  $||R||$  is the mass row's, worst is an energy cell); the brief's belief that Solver: Newton 100.0 caps the iterations is wrong (the third word is `newton_du_switch`; the JFNK cap is the literal 500 at `EXHALE_main.f90` 3509 and 3000 on the direct route, no key or hook; runs were capped by wall clock); `atomic_elem_newton/README.md` is right but incomplete (the refused trials have a correct model). Advisor: rebuild warning-free, `krylov_and_dogleg` 31/0 on the tree with N7 and N9.

- **N4a** (PLAN\_20260909\_rev1 Stage B, one elemental constraint map with three contexts; new `src/modules/lower_atmosphere/element_inventory.f90` (745 lines, one SRC line in the Makefile), `diffusive_photochemistry.f90` (its three element densities, hydrogen budget and five carrier box sides now derived from the map with unchanged arithmetic), `steady_species_rows` suite, `docs/element_inventory_contexts_20260909.md` corrected; `steady_newton.f90` untouched): **the map reports, nothing enforces it yet**, so the Stage B gate may not claim from this item that trial construction, chemistry and certification enforce one constraint (N4b). `nuclei_per_particle` is the one place a stoichiometric coefficient is read (`bsp_nH/nHe/nO/nC`, `mion_elem`); `element_inventory_of_cell` gives the full inventory grouped carried/locked/reserved/closure with the budgets and the shared-constraint breach as a MAGNITUDE (against the budget the sides spend, and against the budget the chemistry closes with, separately); `element_inventory_of_candidate` evaluates a trial's carriers against the frozen element densities; `EXHALE_INVENTORY_REPORT=1` prints it at the operator's entry. **Physical judgment:** the separate box sides are a wrong feasible set by stoichiometry (two carriers of one element spend one budget); a cell at both ceilings breaches by exactly 1.0 of the available hydrogen (MEASURED). **Seven corrections to the advisor's draft**, each checked in the source: `nO_free/nC_free` include the neutral stage while `cbg_nOion` excludes it (the F7 gap as a number: the oxygen box side spends the element total while `carrier_source` closes against the total less the ion stages, so the box admits states whose source rows clamp free O at zero; N4b's feasible set); `nH_avail` is floored; H I and He I are the closure, not rows; CO is fixed data of the local solve (its destruction row is the carrier operator's); **the shared sums ARE enforced on the marching path by `limit_to_element_budget`**

(what has no shared statement is the coupled stationary solve, whose `eval_residual` compares a COUNT of cells); **max(rest, 0) in the write-back CREATES nuclei** (with `rest < 0` the carriers keep what they hold and the closure cannot go negative, so the written element total exceeds the handed one by `|rest|`; only `element_census_verify` at EXHALE\_ELEMENT\_ASSERT 1 or 2 sees it, off by default); absent elements are no longer registered (N1). **Tests** 149/0 (advisor on `build_lwv`: 145/0, the four coupled-log rows unset), RED shown by EXHALE\_INVENTORY\_SHARED=0 (the three infeasibility rows fail with that setting); all nine molecular goldens classified: worst carrier breach exactly 0.0, tightest hydrogen occupancy 0.96 to 0.99 (oxygen\_chemistry occupancy 1.0 is carbon's, 175 of 500 cells with all C in CO). **The N9 finding caught live**: inside a coupled carrier JFNK the He/H departure from the reservoir grows  $1.4\text{e-}11 \rightarrow 4.9\text{e-}5$  (third iterate, the first with a Newton  $n(\text{H}_2)$ )  $\rightarrow 1.1\text{e-}1$  (thirteenth) while every carrier budget stays at breach 0.0, so the fix belongs at the write-back of each iterate (the element row or a projection there), not at the end of the solve; the two saved coupled states have carrier breach 0.0 and He/H departures  $4.0\text{e-}2$  and  $6.7\text{e-}1$ . **Marching**: `mol_carrier` 200 steps byte-identical (two independent before/after pairs); `mol_diffusion` 12000 steps byte-identical over the 7511 steps both builds had covered at report time (step trace character for character, the step-7001 snapshot files byte-identical apart from provenance); the pair finished later in the session scratch and the advisor compared its end: `Hydro_ioniz.txt` and `Ion_species.txt` at 12000 steps IDENTICAL apart from the provenance line, `log10 Mdot` 10.57 in both builds. Reported, not fixed: `element_ratio_gate = 1e-9` now exists in two modules because `element_census_tolerance()` is private. Rule breach reported by the worker: one `pkill -f` on its own shell pattern (it matched only its own shell; the other workers' runs were verified alive). Advisor: five British percent in the two files fixed; the advisor's earlier `pgrep` for the worker's binary name missed its renamed scratch copies, a false "finished" the worker corrected.

- **N7b** (N7 proposals P1 and P2, and the JFNK iteration hook; `steady_newton.f90` +249/-40, `krylov_and_dogleg` +12 rows): **P2 opens the cycle; P1 is implemented and MEASURED inactive, and the measurement refutes N7's reading of the refusals.** The ray test's decision is extracted into `ray_slope_verdict` (public, tested), its floor is  $\max(\text{eq\_sweep\_reltol} * |f_2|, \text{ray\_floor\_factor} * \text{measured\_reproducibility})$  with the reproducibility measured once per outer iteration by two evaluations of the merit at one state (cost one residual evaluation per outer iteration, +1.5 percent on the carrier reload): MEASURED  $1.8\text{e-}15$  at a merit of 14.95 and 0 to  $4\text{e-}19$  at  $2.7\text{e-}5$ , four to seven decades BELOW the assumed part ( $2.6\text{e-}13$ ), which therefore governs everywhere; the refused trials'  $|f_{2p} - f_{2m}|$  ( $1\text{e-}10$  to  $3\text{e-}10$ ) stand two to three decades above the floor, so they are real differences and NOT the probe near its cancellation floor as N7 and ISSUES 3.1 (d) said (corrected there). What they are: at the refusing iterate the ratio is 0.999 while the model slope ( $-5.5\text{e-}7$ ) and the ray slope ( $+1.7\text{e-}7$ ) have OPPOSITE signs; the model slope is set by the banded-gradient leg, which is not in the step ( $|s|$   $6.6\text{e-}9$  against a radius  $3.9\text{e-}3$ ) but carries  $|A_g|$   $3.1\text{e}3$  into  $r_0$ . As, so the slope test is right that the slope is wrong and the leg is what is wrong (N7 P3). `ratio_sound_band = 1e-2`: within it the ray verdict is formed and printed but not acted on; `tr_ratio_below_acceptance` untouched. EXHALE\_JFNK\_MAXIT read once per solve, resolved by `jfnk_outer_iteration_cap`, printed with its origin. **Atomic reload, 150 iterations, before -> after**: accepted 106 -> 129, ray-size refusals 35 -> 1, ratio-below-eta refusals 6 -> 15 (the guard takes over), radius  $9.8\text{e-}4 \dots 9.8 \rightarrow 3.8\text{e-}2 \dots 2.7\text{e}8$  (no longer binds; the accepted step

is the interior Krylov point at  $||s||$  1.0e-5, ratio 0.72),  $||R||$  7.27e-2  $\rightarrow$  6.92e-2, merit 7.26e-3  $\rightarrow$  7.08e-3, achieved linear residual still 0.69 against 0.1 with 40 of 40 products: **the limiter is now the linear solve and the model behind it** (N7 items 4 and 7, P3/P4), not the step length. No convergence claimed. **Carrier reload:** every numeric (JFNK) line identical; 20 of 55 iterations had carried a STALE refusal label ("the model promises no reduction" printed on ACCEPTED iterations because refuse was reset only inside .not. model\_ok; pre-existing, fixed): earlier reports quoting "iterations that ended on N: the model promises no reduction" counted that defect wherever N exceeds the refused count (B5I's 19 and 20 included). wasp\_full\_newton BITWISE on all ten output files (three log lines differ: two wall clocks, one new cap line). Tests RED on the extraction stage (4 of 43 fail: the measured floor rows and the sound-ratio rows), GREEN 43/0; advisor rebuild warning-free, krylov\_and\_dogleg 43/0, binary md5 6d4bf63e identical to the worker's final. Two dead locals removed (md5 unchanged). Advisor: four British neighbor spellings in comments fixed. Advisor decision: P1's measured floor stays (correct in form, one evaluation per iteration). **Reported, not fixed:** the ray test runs before the acceptance test, so a catastrophic over-prediction (ratio -533, -1493 at iterations 47, 50) is named a slope disagreement; ray\_h = 1e-3 is not a measured interval; tr\_dmax doubles without an absolute cap and with P2 runs to 2.7e8 (harmless, the interior point is taken, but delta stops being a diagnostic); n\_tr\_exit has no index guard.

- **N5** (PLAN\_20260909\_rev1 Stage B, the residual contract; steady\_newton.f90 +877, ionization\_equilibrium.f90 +274/-6, EXHALE\_main.f90 replay driver +134/-60, src/tests/residual\_determinism/ rewritten with a suite run.sh (cases: the two in the tree plus EXHALE\_RESID\_RELOADS=<dir>[:<dir>], binary EXHALE\_RESID\_EXE)): **the old determinism suite had a hole:** it replayed the THREE-unknown residual in every configuration (0 of 1500 entries, the species-row registry is empty before a solve); the replay now registers the rows and returns 0 of 1500 / 2000 / 2500 / 5500 entries differing (3, 4, 5, 11 unknowns per cell) after three trials, a discarded trust-region trial and a jv\_product probe with the active bounds held (**contract 1 GREEN on four configurations;** a rejected macrostep is not reachable from the driver, the one part unmeasured). **Contract 2, conditional closure convergence, MEASURED and it exceeds the certification tolerances:** the judged row maximum moves, per unit relative change of the eliminated partition's seed, by 4.2e2 cert\_tol\_carrier on the carrier reload, 3.1e1 cert\_tol\_element and 1.7e1 Resid\_tol on the atomic element reload, 6e-8 on the element + carrier reload; a seed above 2.4e-3 (carrier) or 3.3e-2 (atomic) moves the judged row past its tolerance; the retention of a scaled seed at the fronts is ABOVE one (1.39, 3.69, 3.46) where B5h had 0.77 at the base. Two suite rows stay FAIL by design as the standing measurement (closure\_spread\_within\_the\_row\_tolerance\_{atomic\_elem\_newton, det\_car}, 61.5 and 8.5 against < 1). **Contract 3, branch discovery** (EXHALE\_RESID\_BRANCH\_REPORT=1, roots grouped by a branch-datum distance 1e-1 anchored on B5h's two roots): with H2 transported B5h's two-root H2 front does NOT reproduce (x(H2) at the front bitwise identical over 10 seeds, the transported value replaces the seed in the H2 row); the H I partition shows one root on the carrier reload and TWO on the element + carrier reload (9 and 1 seeds, 1.05e-1 apart); whether the second is a distinct branch or a slow iteration is N6's question. **Chemical decay against transport** on the resolved gradient length: H2 t\_adv \* lambda 3.68 (base), 4.0e-3 (front), 0.82 (wind); H2+, H3+, HeH+ 3 to 12 decades faster than transport everywhere, so the transported-H2 / eliminated-ion split stands and no promotion is proposed (conservation modes are removed

by the parameterization itself: the neutral stage closes each element). **Rounding floor of the species rows in the judged measure is exactly epsilon** ( $2.2\text{e-}16$ ) against tolerances  $1\text{e-}8$ , 7.65 decades apart, condition numbers 2 to  $1\text{e}3$ : compensated summation (the Al-Refaie et al. 2024 reading) is not a lever. Validation: krylov\_and\_dogleg 43/0, steady\_species\_rows 145/0, mol\_carrier 200 steps bitwise, wasp\_full\_newton bitwise on all ten files (one new log line, N7b's cap). Advisor: rebuild warning-free, md5 913e6579 identical to the worker's final; krylov\_and\_dogleg 43/0; residual\_determinism 5 PASS / 1 FAIL, the FAIL being the atomic closure-spread row by design (a first advisor run with the wrong variable name EXHALE\_EXE tested the stale shared EXHALE.x, the advisor's mistake); two British percent in ionization\_equilibrium.f90 fixed, the worker fixed five neighbor. The worker's session was cut once by the Opus limit (09:50 KST) and resumed; nothing was lost. **Reported, not fixed:** a discarded evaluation restores n\_eq\_sweeps\_last, so a caller reads the previous state's pass count; report\_accepted\_cell\_state writes inside the parallel region without a lock; carrier\_diffusion\_coefficient reads the marching operator's pct\_Dco, which the stationary residual never fills (a pure stationary path would read zero; diffusive\_photochemistry.f90); the global count shadows the intrinsic.

- **N16b** (PLAN\_20260909\_rev1 Stage E brought forward, species-aware integration error; attempted\_step.f90 +546/-28, EXHALE\_main.f90 macrostep code +125/-7, attempted\_step suite +289): the step-doubling estimate reads four row classes on their own scales (conservative rows; transported species fractions, the f\_sp columns carrier\_state builds fc from: H+, H2, OH, H2O, CO; element mass fractions under He\_diffusion; electron fraction and T through the caloric EOS), the species scale being the species' own value with an absolute floor of element\_ratio\_gate ( $1\text{e-}9$ ) of the largest density the species can reach in the cell, read from the N4a map's stoichiometry; the flat  $1\text{e-}12$  no longer judges any species row (a carrier at  $1\text{e-}20$  changed by 1 percent:  $e = 20$  on its own scale,  $2\text{e-}10$  on the flat floor). **The inner nonlinear error is collected cell by cell** (the coupled pair's safety-factored geometric factor on the cell's own increment, and the carrier substep's row residual) and the estimate carries the interval e\_lower .. e\_upper the three inner errors leave it in: reject when above one at the bottom, accept when below one at the top, UNRESOLVED in between (named outcome, counted in the end-of-run block); the brief's fraction rule for each class (1/10) is kept as the report, not as the verdict, because applied literally it called EVERY estimate of a healthy run unresolved (species difference  $6\text{e-}17$  five decades below the pair's own error) and would have made F1's rejection inert. **p = 1 confirmed on every class** (atomic case, eight levels over five decades: species p + 1 1.9987, 1.9799; electron/T 1.9943 to 1.9997; conservative 1.9809 to 2.0004; molecular case coupled ladder H = 5, 2.5, 1.25 s: species 1.949, 2.299, electron/T 1.912, 1.983), so the retained-step factor 2 holds for all rows. On the molecular case at  $H \leq 0.4$  s the species class reports itself unresolved (inner over difference  $2.7\text{e}5$ ): review F3's condition, MEASURED and detected. **The 60-step physical run is unchanged to 17 digits** (t\_phys, checksum, both files): the conservative rows still decide it. mol\_carrier marching bitwise. **N16a's 9.1e-5 state leak IDENTIFIED** (not fixed, files not owned): Apply\_BC.f90 base\_face\_W and base\_face\_lower\_W (lines 48, 50; written only by Apply\_BC, read by Rec\_BC in the first Reconstruct of a retaken pass BEFORE the pass's first Apply\_BC, so the base face's left state is the refused attempt's) and caloric\_eos.f90 caloric\_mixture\_active, nk\_per\_mass, x\_h2, molecular\_cell (125, 145-147; written only inside get\_species\_densities, which rebuild\_state\_from\_checkpoint calls

AFTER `U_to_W`, so the restored `p` and `T` use the refused attempt's caloric composition); the two reinforce each other at the base cell, which is the measured signature (`v`  $9.1\text{e-}5$ , `T`  $6.8\text{e-}8$ , cooling  $1.7\text{e-}5$ ). Row `rejected_pass_leaves_no_state_behind` stays RED at  $9.1162\text{e-}5$ , labeled with the files and lines. N16a's 2.3x outlier reproduced and bounded (only at controller-formula levels, conservative and charge classes, always the base cell; 0.665 and 0.53 of the  $\text{H}^2$  prediction), not explained; two candidates named (the unbounded inner error of the mass/momentum rows through the second half's coupling; the base-face leak). `err_rtol`, `err_atol` unchanged (the CFL step 20 would need  $1.21\text{e-}3$ ; the estimate RISES at the first shortened interval, a second reason not to anchor on this case). Tests 155/1 (the one FAIL by design), RED on the entry text 150/6 (an estimate that was the solvers' own error rejected ten macrosteps there; now 0, named). Advisor: suite rerun with the worker's final binary (md5 7bd041dd) 155/1; four modelled in `EXHALE_main.f90` comments changed to modeled. **Reported, not fixed:** the mass and momentum rows carry no inner error; the carrier substep's number is a row residual, not a solution error (conditioning unmeasured); `NCO_col`, `k_co_diss`, `theta_co_shield` are outside the checkpoint by an unstated exception (read only after their write within a sweep).

- **N16d** (the checkpoint's restore is complete; `EXHALE_main.f90` `rebuild_state_from_checkpoint` 4 code lines, comments in `Apply_BC.f90`, `caloric_eos.f90`, `attempted_step.f90`, the suite row extended to both files): **closed by rebuilding, not saving.** The five leaking quantities (`base_face_W`, `base_face_lower_W`, `caloric_mixture_active`, `nk_per_mass`, `x_h2`, `molecular_cell`) are functions of (`u`, `f_sp`), so the rebuild now runs `rho = u(1,:)`, `get_species_densities`, `Apply_BC`, then `U_to_W` and `comp_T_from_p` (composition first because the pressure map and the boundary read it; boundary second so the face states are the restored state's). Every one of the six call sites follows a restore, so no restore-free path can move. **MEASURED:** `rejected_pass_leaves_no_state_behind`  $9.1162\text{e-}5 \rightarrow 0.0$  bitwise on both files; suite 156/0 (advisor rerun with the worker's final binary md5 abdf17f7: 156/0). **The fix changes the measured order of the conservative rows:** the full-versus-two-half difference was 40x the real splitting difference (the leaked base face carried an  $\text{O}(\text{H})$  error into one face of one RK stage of an  $\text{H}/2$  step, an  $\text{O}(\text{H}^2)$  contribution) and now falls as  $\text{H}^4$  (`p` + 1 3.9966 to 4.0104 over the ladder), the local-error order of the three-stage Runge-Kutta; the electron/T rows follow it; the transported species stay  $\text{H}^2$  (unchanged to all printed digits). N16a's 2.3x outlier is EXPLAINED: the two controller-formula levels now sit on the  $\text{H}^4$  law to 0.1 percent; the leaked base face carried the difference of two intervals whose ratio was not the ladder's. **Consequence not yet fixed (N16e, launched):** `err_order_p = 1` was calibrated on the leak; the conservative and electron/T estimates carry a factor 2 where 8/7 is right (inflated 1.75x) and the reduction exponent is -1/2 where -1/4 is; the species class is genuinely `p = 1`, so the order is a property of each class. `mol_carrier` marching 200 steps bitwise (0 refused attempts, no restore); with one injected refusal the initialization-mode state moves  $2.1\text{e-}5$  (`v` at `r` 1.0185) and  $1.2\text{e-}5$  (`He`  $2^3\text{S}$ ), the fix of the same defect with both channels live (atomic hydrostatic\_column had the base-face channel alone). `wasp_full_newton` not run (no restore unless an attempt is refused; whether it has zero refusals was not measured). The reload route does not share the rebuild (checked). Advisor: two Per-cell phrases in `caloric_eos.f90` reworded. **Reported, not fixed:** the same composition-before-pressure ordering stands at three other sites of `EXHALE_main.f90` (1075-1078, 5162-5166, 5312-5326) and `upmap_bc_roundtrip` measures an identity its comment says is not one (N16e); the added `Apply_BC` adds one boundary evaluation per restore to

the run-total diagnostic counters.

- **N20** (decision 20 a with N7's P3; steady\_newton.f90 +314/-84, krylov\_and\_dogleg +13 rows, steady\_species\_rows +1 row; certification.f90 unchanged, cert\_scale\_floor was already public): **one functional, implemented exactly, with a measured consequence that is a decision, not hidden.** merit\_row\_scale\_from\_certification is the one denominator a species-row-route row is read through; cell\_row\_scales builds the hydrodynamic rows through residual\_row\_scale(k,j,u), the element rows through their certification scale, the carrier rows through the term sum the certification divides by; base cells of element rows keep the unknown's own scale (Dirichlet reservoir); the three-unknown route is untouched by construction ( $D_r = D$  when  $n_{\text{spec\_row}} \leq 0$ ). MEASURED: on both reloads every element row's largest scaled merit row EQUALS its certification measure at the same cell and the merit's largest scaled row IS  $||R||$ 's row (before: factor 2.5 apart at iteration 1, 112 at iteration 60). **P3**: the banded-gradient leg is dropped from the step AND its image when shorter than  $\text{cauchy\_leg\_min\_fraction} = 1e-3$  of the radius (measured  $1.4e-5$  to  $8.7e-7$  on both fixtures, so everywhere), so the slope handed to ray\_slope\_verdict is the step's own; products saved: zero, because tau\_c needs A g (a lazy leg would save one per iteration, P-N20-1, reported). A quotient verdict at  $\text{ratio} \leq \text{eta\_accept}$  is named tr\_ratio\_below\_acceptance. tr\_dmax capped at  $1e3 \sqrt{\text{neq}}$  ( $7.4e4$  on the atomic fixture; binds on neither fixture now, would have bound the  $2.7e8$  of N7b). **The consequence**: the carrier reload improves 2.6x (best  $||R||$   $1.151 \rightarrow 0.446$ , both builds on the stagnation detector, +16 percent evaluations, dbgcon  $5.4e-10 \rightarrow 1.9e-11$ ) while **the atomic element reload no longer solves**: the certification row scales span 11.4 decades ( $3.7e-10$  ..  $92.7$ ) against 8.3 before, dbgcon  $7.3e-8 \rightarrow 8.9e-10$  at iteration 1, the Krylov point collapses to  $||s||$   $2.5e-15$  inside a radius 19.6 at iteration 11 and the solve aborts on the no-descent watchdog at  $||R||$  1.777 (before:  $7.38e-2$  at 150 iterations). A third scaling choice (element/carrier rows on certification scales, hydrodynamic rows on the state scales) does not collapse but is worse than the entry text on both fixtures (0.741, 1.026): the hydrodynamic reconciliation is what helps the carrier reload and kills the atomic one. The merit's row scaling IS the operator's ( $r_0 = F/D_{\text{row}}$ ,  $A = D_{\text{row}}^{-1} J D$ ), so the remaining freedom is the column scaling D and the preconditioner: N8a's, launched with the atomic collapse as its RED. Tests RED on the entry objects (the element-row scale row 4.0 against 1.0; the ratio-naming row), GREEN krylov\_and\_dogleg 56/0, steady\_species\_rows 145/0, certification 29/0; wasp\_full\_newton BITWISE on all ten files (two wall-clock lines). Advisor: rebuild warning-free, steady\_newton.o md5 6f3aef50 identical to the worker's, the three suites 56/0, 145/0, 29/0. Two pre-existing defects fixed (a dead local; [diag 7] printed at cell 0 for the carrier). **Advisor decision**: (a) stays in the tree; the atomic reload's collapse is a conditioning problem of the scaled operator and is the first deliverable of N8a, not a reason to give the step control a functional other than the gate's; the state of the atomic reload is reported as WORSE than before N20 until N8a lands. Reported, not fixed: EXHALE\_main.f90 1182 should pass u to cell\_row\_scales; docs/p54\_base\_layer\_mass\_flux.md 10.4 now has a second measurement behind it.
- **N16e** (the order of each row class is its own integrator's; attempted\_step.f90 +106/-50, EXHALE\_main.f90 +60/-21, attempted\_step suite +75/-15): err\_order\_p(n\_err\_class) = [3, 1, 1, 3] (hydro, species, element, charge/T) with the measured ladders at the declaration (conservative 3.9966 to 4.0104; electron/T

3.9962 to 3.9867; species 1.9987, 1.9799; the element class is assigned the species' order by construction, said so); `err_full_step_factor` is a class's own (8/7 for the RK3-carried rows, 2 for the first-order split ones), the estimate records the deciding class (the one whose lower bound reached the gate) and `attempted_step_reduced_dt` uses that class's  $-1/(p+1)$ . MEASURED: the conservative and electron/T estimates fall by exactly 4/7 (inflation 1.7500004 removed), the 60-step physical run's trajectory does not move (N16d had already removed the step-20 rejection, so the brief's expected "1.21 rejected -> 0.69 accepted" belonged to N16b's entry text, not to this item's; stated); on the forced-error ladder the controller with exponent -1/4 asks for 0.846 and 0.226 of the interval where -1/2 asked 0.601 and hit the 0.2 floor, so the floor no longer does the sizing. Stale sites: EXHALE\_RESIDUAL=1 and `physical_handoff_check` NOT reordered, the composition cannot have moved there (`init` and `equilibrate_loaded_composition` both end with `get_species_densities`), the invariant is now stated at both; `update_map_begin_step`'s install branch reordered (composition before U\_to\_W; EXHALE\_UPDATE\_MAP on `mol_carrier` moves 0.0); `upmap_bc_roundtrip` renamed `upmap_bc_interior`, required exactly zero (`Apply_BC` writes only the ghosts), MEASURED 0. Tests 163/0 (156 before; the unit rows cannot be run RED against a scalar `err_order_p`, the RED is the entry suite's own contradiction,  $p = 1$  asserted and  $p + 1 = 4$  measured in one run; the whole-binary row 0 of 2 -> 2 of 2). `mol_carrier` marching bitwise. Advisor: suite rerun with the worker's final binary (md5 0ae8d8d9) 163/0. **Reported, not fixed:** `update_map_begin_step`'s install branch owes the next Reconstruct an `Apply_BC` (same family as N16d, a diagnostic route); `err_atol = 1e-12` still floors the conservative rows on a code-unit density; the mass/momentum inner error is still unbounded (the deciding row's interval is a point); no fixture has the species class deciding a step, so the exponent switch is covered by unit rows only.

- **N8a** (PLAN\_20260909\_rev1 Stage B, scales, boundary equations, derivatives and the inner budget, with N20's atomic collapse as its RED; `steady_newton.f90`, `krylov_and_dogleg` +17 rows, `steady_species_rows` +2 rows; `certification.f90` and `steady_residual.f90` unchanged, no tolerance moved): **the collapse was one exactly empty Jacobian column, not the spread of the row scales.** At iteration 11 of the atomic reload the carbon unknown of cell 500 had been driven to zero, its column scale sat on the absolute floor  $1e-20$ , the probe step  $\sqrt{\text{eps}} * 1e-20$  underflowed the residual difference, the column was exactly 0 and the banded preconditioner singular (`dgbcon` 3.98e-27, smallest pivot 3.7e-19); the  $2.5e-15$  Krylov point is what a triangular solve with that factor returns. Iteration 1's  $8.9e-10$  was only the 7.3-decade spread of the certification row scaling (no empty column). **Two repairs, the model untouched to the last bit** ( $r_0 = F/D_{row}$  and A unchanged, MEASURED by the scaled column norms): (a) `element_unknown_column_scale = max(|x|, 1e-4 |reservoir|, 1e-20)`, the arithmetic at the function (the column is resolvable while the probe step stands far above  $\text{eps} * X_{\text{reservoir}}$ ); (b) `equilibrate_the_preconditioner`: two-sided equilibration of the band that is FACTORIZED only, powers of two, `banded_preconditioner_solve` applying C `dgbtrs(R v)`. Attribution (first solve, accepted/refused; empty columns at it 11; `dgbcon` at it 11): entry 14/16, 1,  $4e-27$ ; floor alone 69/16, 0,  $4.9e-11$ ; both 114/36, 0,  $1.1e-9$ . **Atomic reload (EXHALE\_JFNK\_MAXIT=150):** entry abort at  $||R||$  1.777 (no descent, 30 iterations); floor alone 1.189 (stagnation, 86); both  **$3.800e-2$  at the 150 cap, still descending** ( $2.5e-1$  to  $3.8e-2$  over the last thirteen iterations), `dgbcon` at iteration 1  $8.9e-10$  ->  $1.65e-6$ , the brief's target (the pre-N20  $7.4e-2$ ) passed by 1.94x with one functional kept; refusals by name shift from ray-floor to ratio-below-eta (35)

and one slope-size. **Carrier reload** 0.446 -> **0.1513** (2.95x), `dbgcon`  $1.9\text{e-}11$  ->  $8.7\text{e-}6$ , +30 percent evaluations. The model's own column equilibration (`EXHALE_COLUMN_EQUIL=1`, default off) was built, MEASURED and NOT adopted: it moves the trust-region coordinates, the radius runs to the cap and trials ascend by  $4\text{e}4$  times the predicted decrease. **First cell rows declared once** (`species_row_base_equation`: element row = Dirichlet reservoir, READ from `element_transport_residual`; carrier row = control-volume balance, READ from `carrier_residual`'s `j == 1` branch; no equation changed). **Base derivatives**: the directional-derivative sweep at the base is a clean V with its minimum  $4.8\text{e-}8$  at `sqrt(eps)`, the step `build_banded_jac_full` takes; whole-system rows  $8.2\text{e-}8$ . **Closure map at the base** (`measure_the_closure_map_at_the_base`, four evaluations once per solve): the sweep exits after ONE pass at  $9\text{e-}12$ , four decades inside the  $1\text{e-}8$  asked, so the reference is a pass count; quotients  $2.1\text{e-}4$  (atomic, attenuation) and 5.34 (carrier); `outer_residual_closure_target` (what the outer residual may keep, what the ray floor reads) is now separate from `eq_sweep_reltol` (what the sweep is asked = target / measured amplification;  $1.87\text{e-}9$  on the carrier reload). **B5h's ~4e3 base amplification is not reproduced on either fixture** and is not claimed. Manufactured states in `krylov_and_dogleg` (columns spanning eight decades, pivot ratio  $5\text{e-}18$  -> 0.54 after equilibration; exactness rows: the map and the unknown-space step unchanged to the last bit). Tests `krylov_and_dogleg` 73/0, `steady_species_rows` 147/0, certification 29/0; the new binary with both changes off reproduces the entry text to every digit; `wasp_full_newton` BITWISE (ten files; `nspec_row = 0` short-circuits every new path, the hydrodynamic column scales untouched). **Advisor**: rebuild warning-free, `steady_newton.o` md5 `c7621824` identical to the worker's, the three suites 73/0, 147/0, 29/0; the atomic reload rerun on the advisor binary reproduces the first solve to the digit (`||R||`  $3.800\text{e-}2$  at 150, `info = 1`), and the run's SECOND stationary solve (the code re-enters the Newton from the handed-back state, another 150 iterations) reaches `||R||` **5.273e-4, flux spread 2.3e-6**, still not certified (`log10 Mdot` 10.10; exit 2), 32 min at 16 threads: the reload now descends four decades in 300 iterations where it stood still since B5m. Not converged. **Reported, not fixed**: `n_eq_sweeps_model = 1` so the pass count, not `eq_sweep_reltol`, controls the closure on both fixtures; `EXHALE_JFNK_MAXIT` caps each solve, not the run; `fix_active_species_bounds` lost its D argument (the epsilon-active width now on `state_column_scale`); the [diag 4] true-residual row is not valid for long steps (its finite-difference product is out of range). The fixture README updated by the advisor.

- **Advisor measurement after N8a (atomic reload, EXHALE\_JFNK\_MAXIT=600, 16 threads, 18 min).** One solve with the cap at 600 stops on the stagnation detector at iteration 224 with `||R||`  $2.671\text{e-}2$  (`info = 2`; 170 accepted, 55 refused, all but three by "the reduction ratio is below eta"; `log10 Mdot` 10.06), WORSE than the same binary's 150 + 150 route, where the run's re-entry into a second solve from the handed-back state reached  $5.273\text{e-}4$ . [The advisor's reading here was WRONG and N21 corrected it: the run does not re-enter the Newton from the handed-back state; `info = 1` sends it back to MARCHING for 2000 steps (the log's Newton finish at step 2002), and the second solve starts from the marched state at `||R||`  $6.06\text{e-}2$ . The two decades are gained by the marching, not by a fresh trust region; N21 measured every reset and none helps.] Not converged either way.
- **N4b** (decision 14 route (i), a step that moves along the element-budget surface; `steady_newton.f90`, `element_inventory.f90` +94 (the derivatives of the map, `element_constraint_derivatives`),

steady\_species\_rows +35 rows; diffusive\_photochemistry.f90 untouched, the frozen budgets recovered exactly through the public carrier\_headroom): **the acceptance row is met by ten decades**. On the carrier reload the He/H nucleus ratio against its reservoir stays at  $1.39\text{e-}11$  at every one of 68 iterates where the entry text drifted from  $1.39\text{e-}11$  to  $2.016\text{e-}1$  (20 percent of the ratio the run was given) over 79 iterates: the write-back of each iterate now returns the element totals it was handed (the N9 / N4a finding closed where those reports said it belonged). Best  $\|R\|$  0.1513 -> **0.0899** at 14 percent fewer residual evaluations (5690 -> 4891). The shared constraint  $2\text{ n(H2)} + \text{n(H+)} + \text{n(OH)}$

- $2\text{ n(H2O)} \leq \text{nH}_{\text{avail}}$  (and the O and C rows) replaces the five separate corners as the step limit (each corner is implied by the row, so nothing is lost and the direction along the surface is gained); a nonlinear feasibility screen by magnitude at every candidate; a restoration phase for an infeasible start (on the element-and-carrier reload 52 trials restored over 1165 cell-steps, worst violation  $1.630 \rightarrow 2\text{e-}16$ ; 6 candidates refused by magnitude; 167 write-back cells whose closure could not return an element total, counted). **\*\*Stated, not claimed\*\***: on both reload fixtures the shared hydrogen row is never within a probe step of being tight (occupancy 0.985 to 0.989, active band  $1\text{e-}8$ ), so the active-set PROJECTION onto the constraint hyperplane, the part route (i) is named for, fired on ZERO iterations of either run; its correctness is stated by the suite rows, and the measured improvement comes from the write-back, the budget leaving the coordinate box and the shared step limit. The atomic element reload (element rows, no carrier row) is identical line for line over its 150-iteration first solve ( $\|R\| 3.800\text{e-}2$  both). Tests steady\_species\_rows 182/0 (RED with EXHALE\_ELEMENT\_CONSTRAINT\_ROWS=0, 20 rows fail), krylov\_and\_dogleg 73/0, certification 29/0; mol\_carrier marching 200 steps bitwise; wasp\_full\_newton bitwise ( $\|R\| 4.213\text{e-}9$ , info = 0, log10 Mdot 13.30). **\*\*Advisor\*\***: rebuild warning-free, steady\_newton. omd542fcf223 and binary md59c9ba732 identical to the worker's; the three suites 73/0, 182/0, 29/0; wasp\_full\_newton's two matrix files compared by the advisor against the Stage A reference: IDENTICAL. Two earlier builds of the item were discarded and their runs rerun (report section 8). Certification unchanged (decision 12 c). **\*\*Stage B gate\*\***: N4a map, N4b constrained step and element-conserving write-back, N5's three contracts measured (contract 2 fails by design on two reloads), N8a scales and closure map: the gate's rows are in; what is NOT met is convergence of any species-row solve, which is Stage C's criterion, not Stage B's. N4b's user-visible switches: EXHALE\_SPECIES\_BUDGET\_FACE is retired and replaced by EXHALE\_ELEMENT\_CONSTRAINT\_ROWS (=0 restores the five corners), EXHALE\_ELEMENT\_WRITE\_BACK added (the element-conserving write-back); no input key, output column or unknown layout changed. Also found by N4b:  $\text{dc/d(nd)} = \text{c/nd}$  exactly (the constraint is homogeneous of degree one in the density, so an active row cannot push the mass unknown); two earlier builds of the item were discarded on the

atomic fixture's line-for-line check (a blocked probe component zeroed unconditionally made the sampled operator nonlinear, factor 33 regression; the probe sides compared before capping). The advisor annotated the retired switch in the B5l entry and indocs/steady\_solver\_design.md'.

- **N21** (what limits the species-row solves now; steady\_newton.f90 +417, krylov\_and\_dogleg +31 rows): **no reset of the trust-region state helps, and the advisor's premise was wrong.** The run's second stationary solve does not start from the handed-back state: info = 1 returns the run to MARCHING for about 2000 steps and the second solve enters at  $||R||$  6.06e-2 (advisor's own log: Newton finish at step 2002, then step 4002, the third solve reaching 4.26e-4 on the stagnation detector), so the two decades the 150 + 150 route gains are gained by the marching between the solves. MEASURED on the atomic reload from the SAME 150-iteration prefix, EXHALE\_JFNK\_MAXIT=300: continue (control) 227 iterations, stagnation,  $||R||$  2.671e-2 (170/55, reproducing the 600-cap run); resetting the pseudo-transient shift, the acceptance memory or the closure map: bitwise identical logs (three exact no-ops: dtau already sits on its floor dtau0, the Grippo window is reported and never acted on on the coupled route, the closure map re-measures to amplification 1.000); resetting the radius and ceiling 2.851e-2; returning to the best iterate 2.921e-2; all five together (what a re-entry does) 3.161e-2; a second Krylov cycle of 40 2.927e-2; a restart every 10 iterations without improvement runs to the 300 cap at the same 2.671e-2. Carrier reload: control reproduces N4b (8.993e-2), periodic restart 1.153e-1, second Krylov cycle 1.728e-1, both worse. **The periodic restart trigger is delivered default OFF** (tr\_restart\_stall\_default = 0) by the brief's own condition. **The linear solve is not the limiter:** after N8a the atomic cycle reaches a median 0.157 of its right-hand side over iterations 1 to 150 (N7: 0.57 to 0.69) but 0.74 to 0.82 over 151 to 225; the carrier cycle 0.963; doubling the products halves the linear residual and makes the outer solve WORSE on both fixtures; in the second atomic solve a cycle reaching 0.98 coexists with a two-decade fall of  $||R||$ . The radius reset is worse for a stated reason: the carried radius 9.85e-9 accepts steps at ratio 1.000 while a re-initialized 1.15e2 over-predicts by 39x then 3392x and costs four refused iterations. READ: the band is refactored every outer iteration (a re-entry changes the preconditioner only through the shift) and the Krylov cycle starts from zero every iteration. Tests RED (13 symbols absent from the entry module), GREEN krylov\_and\_dogleg 104/0, steady\_species\_rows 182/0; with everything disarmed the delivered binary reproduces the entry text's 20-iteration solve line for line; wasp\_full\_newton BITWISE (ten files). Advisor: rebuild warning-free, steady\_newton.o md5 cdbbf535 and binary e140284a identical to the worker's, the two suites 104/0, 182/0; the advisor's own n8a\_reload log confirms the marching between the solves. **The finding that names the next item (N22):** the predicted decrease stops scaling with the step below about 1e-7 (from iteration 140: the step falls 4x and pred stays 7.099e-4; every longer step of the resulting period-two cycle is an ascent of 3e-3 against a promise of 3e-4, ratios -4, -8, -13, refused, quartered, then accepted at ratio exactly 1.000); the production dogleg geometry is cleared by two new rows (quartered radius quarters the step and the predicted decrease to 2.5e-1 and 2.50009e-1), so what remains is trust\_region\_step's bookkeeping at those lengths or a precision floor in  $r_0 \cdot A_s$  and  $||A_s||^2$ ; the separating measurement is one diagnostic line ( $||A_s||$ ,  $r_0 \cdot A_s$  beside  $||s||$  at two radii of one iterate). README of the fixture updated by the advisor.

- **N13b** (decision 17 option a, transit spectra carry their validity; EXHALE\_transit.py, exhale\_transit\_lib.py, transit\_census +8 rows; Python only): one reader read\_adv\_validity locates the status columns by name in the # columns line and distinguishes three cases: schema 2 (N11's two fields: a row is refused for the census when adv\_T\_status is not 0, the composition field reported beside it), schema 1 (today's single column), legacy (no schema: validity UNKNOWN, and both the console and the saved metadata say UNKNOWN rather than 0 refused). Every tpm\_\*.txt gains a comment metadata block (transit\_schema 1, the input profile and its schema, its adv\_input\_certified / stationarity operator / model restrictions lines copied, its provenance lines, the tool's own identity by git hash and file mtime, the census for that line with the eight-point refused-share ladder and its maximum, the CONTRIBUTION-not-uncertainty sentence, the EXHALE\_TRANSIT\_\* overrides in effect); the two hand-written products get the same block. MEASURED: numerical columns bitwise on all 10 wasp\_full and 5 mol\_base\_handoff products; the schema-2 read tested on synthetic files and on a relabeled wasp\_full profile (the relabeling is the worker's, not N11's mapping; the production file comes with N11); every consumer in the tree (python/make\_transit\_figures.py, the WASP-52b and LHS 1140 b notebooks, two LHS scripts) reads through np.loadtxt or skips #, none edited; the planet folders' tpm\_\*.txt untouched (mtime 2026-08-19). Suite RED 8 rows (no reader existed), GREEN 19/0; advisor rerun 19/0. Fixed on the way: the second np.loadtxt(..., max\_rows=1) column count (NumPy warning on every run) replaced by one rule. Reported: the refused share is taken over the temperature field only (a companion share over the composition field would be the honest partner); still line center only; README\_HOWTO.md should mention the metadata block (advisor added the sentence).
- **N11** (decision 16 option a; post\_process\_adv.f90, the \_adv writer of write\_output.f90, examples/exhale\_io.py, EXHALE\_plots.py, adv\_static\_limit +driver M1-M5 and rows R8-R9): **stationarity of the input state is judged by the certification's own mass operator** (assemble\_residual, residual\_row\_scale, the mass row |R<sub>1</sub>|/s<sub>1</sub> against adv\_mass\_row\_tol = cert\_tol\_mass), the enthalpy term ratio kept as a printed sensitivity diagnostic that refuses nothing; **two status fields** adv\_T\_status, adv\_comp\_status (0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not\_evaluated) in BOTH \_adv files, columns 8 and 9, with write\_adv\_validity\_header writing # adv\_schema 2, the two legends, # adv\_status\_counts T <5> comp <5>, # adv\_input\_certified <T|F> <reason> (the certification's verdict on this very state), # adv\_stationarity\_operator face\_flux\_mass, # adv\_product (a ONE-WAY correction on the input's density and velocity field; a converged scalar temperature solve does not make a row self-consistent), # adv\_model\_restrictions; adv\_unsupported where the omitted molecular and oxygen species carry more of the cell's particle count than the carried ones (structural, no threshold); exhale\_io exposes both fields and the header (load\_adv\_header, load\_adv\_status), a legacy file reads schema 1 with both fields ADV\_UNKNOWN, never a translation of the old column; EXHALE\_plots.py shades the refused rows. **MEASURED, the physics of the item:** the ratio screen and the mass row are not interchangeable and the ratio is the weaker: on the analytic outflow fixture the ratio is 1.1e-13 everywhere while the mass row stands at 2.1e-7 (RED of the old criterion); on wasp\_full the ratio refused 297 of 503 rows, the mass row refuses 503 of 503 (largest 1.04 at r 1.032); mol\_base\_handoff 33 vs 503 (0.245 at r 4.21); wasp\_full\_newton (the one certified state) 2 vs 3. So on the two relaxation snapshots every row is retained and the \_adv product IS the run's own state (T\_adv equal

to T in 504 of 504 rows; the old product differed by up to 6.4 and 207 percent); on the certified state the change is confined to the base (p, T 7e-5; species 7.7e-2 at r 1.0002). Run states bitwise in all three; rho, v in the \_adv files moved by exactly zero. Tests 48/0 (advisor rerun on build\_lwv, md5 4320f5b7 identical to the worker's: 48/0). No golden refreshed; every \_adv golden will move. **Advisor decision (N11b, launched)**: the criterion is the right OPERATOR but the wrong TOLERANCE for a conditional correction.  $\text{cert\_tol\_mass} = 3\text{e-}12$  is the tolerance of a certified stationary state; the advective correction is a first-order local correction whose own error is of the order of the mass row's measure (the fractional change of the face mass flux across the cell), so a du-stopped or snapshot state with a mass row of  $1\text{e-}3$  is exactly what the tool was built for and is what review R6 calls the "approximate local use" with "a tolerance tied to the required error". N11b writes the mass row measure itself as a column of Hydro\_ioniz\_adv.txt, sets  $\text{adv\_conditional\_tol} = 1\text{e-}2$  (the correction is described as accurate to that fraction, stated in the header beside adv\_input\_certified), and keeps the certification verdict as the separate statement it is; the product is then a CONDITIONAL correction with its condition written in every row. Advisor: two per-field and two percent (one in a printed message) fixed. **Reported, not fixed**: assemble\_residual leaves  $R(:, 1-\text{Ng})$  undefined (RK\_rhs assigns  $2-\text{Ng}..N+\text{Ng}$  only; no caller reads it today; RK\_rhs.f90/steady\_residual.f90); README.md 154, README\_HOWTO.md 959 and the manual section 4 still describe the single adv\_status column (N11b updates them with the final schema); a suite row for corrected needs a converged fixture.

- **N11b** (the \_adv product is a CONDITIONAL correction with its condition in every row; post\_process\_adv.f90, the \_adv writer of write\_output.f90, exhale\_io.py, EXHALE\_plots.py, EXHALE\_transit.py, exhale\_transit\_lib.py, adv\_static\_limit (53 rows), transit\_census (21 rows), README.md, README\_HOWTO.md, manual section 4): the row decision is taken against  $\text{adv\_conditional\_tol} = 1\text{e-}2$  (the fraction a corrected row is accurate to in the mass flux, the first-order argument at its declaration), the measure  $m_j = |R_1|/s_1$  is written as column 10 adv\_mass\_row of Hydro\_ioniz\_adv.txt and named in # columns, the header carries # adv\_conditional\_tol and # adv\_mass\_row, and # adv\_input\_certified stays the certification's verdict with cert\_tol\_mass; schema stays 2 (readers are header-driven; the one place that counted columns from the end of the row was a test row, fixed to locate by name). MEASURED against the N11 binary: corrected rows (T) wasp\_full 0 -> 215 of 504, mol\_base\_handoff 0 -> 341, wasp\_full\_newton 500 -> 502 (the two others are ghost rows below the energy loop's first cell); the column equals the certification's own printed measure on the same state to every digit (1.942392 at cell 432); run states and rho, v of the \_adv files bitwise. exhale\_io returns adv\_mass\_row and the tolerance; the transit metadata block copies the tolerance line and the largest measure; the census still refuses on adv\_T\_status. Tests 53/0 and 21/0 (advisor rerun on build\_lwv, md5 3131cab0 identical to the worker's: 53/0, 21/0). **Every \_adv golden moves, far on the two snapshot cases: refreshed at the Stage D gate.** For the record: the first-order argument bounds the dropped term; in some corrected cells the KEPT enthalpy-flux term of the mass divergence still dominates the energy equation (452 of 503 cells of hydrostatic\_column against 390 retained), and whether one percent of the mass flux bounds the temperature error of such a cell is not measured; a corrected row is not close to the run's own row and is not expected to be (the certified state moves T by  $6.7\text{e-}2$  at  $m_j$   $3\text{e-}16$ ). Advisor: one percent in README.md fixed; docs/postprocess\_advection\_validity.md marked stale with a pointer.

- **N10** (decision 15 option a, the restart contract; `input_read.f90` +137, `load_IC.f90` +551, `write_output.f90` +13, `EXHALE_main.f90` +166, `grid_and_gates` +1 row script of 20 assertions, `docs/input_schema.md` K43 and appendix D.2, manual entry; PDF not rebuilt): the key Restart intent: trajectory | relaxation | stationary [evaluate|equilibrate] with the three consistency refusals (default trajectory in phys, relaxation otherwise, so no existing input changes behavior); both state files carry the eight-line block (`restart_schema` 1, `reservoir`, `species_columns`, `grid`, `constants`, `options` with 20 switches, `t_phys[s]`, `source`), built and parsed in `load_IC.f90` (one definition), every number ES23.16; no block = legacy, loaded as before and marked provenance unknown in the log and in every file the run writes (the mark is inherited); a disagreeing reservoir (within `heh_dev_tol`), `grid`, `constants` or `options` is a refusal naming the field, `species_columns` and `source` are reported; a pair whose halves carry different blocks is refused. `stationary_state_of_the_loaded_restart`: WENO3 fixed, the N16d rebuild order, one equilibrium sweep (part of the residual's definition), the loaded composition's departure from that sweep's root printed, the residual and the certification held to the FILE's own certified=, then evaluate writes back and stops or (default) enters `steady_wind_with_element_diffusion` at once; no CFL step. **MEASURED**: the atomic element fixture AS LOADED stands at `||R||` 1.8924 (the marched hand-off read 1.888: the two CFL steps); the certified `wasp_full_newton` state is NOT the fixed point of its own sweep (particle count moves  $2.1\text{e-}11$ , species  $5.0\text{e-}10$ , cooling  $3.6\text{e-}10$ : the cell solve's `xtol = sqrt(eps)` and the file's digits), reported at every evaluation rather than equilibrated away; `stationary evaluate` takes 0 steps and returns the certification unchanged; `stationary` on that state enters the JFNK at  $7.133\text{e-}9$  and exits `info = 0` with no step. Tests 20 rows GREEN (RED on the entry binary: the key warned and ignored, one step marched, no block, no refusal); `grid_and_gates` 130/0 (advisor rerun 131/0 on `build_lwv`). `mol_carrier` 200 steps numeric-line bitwise (eight new # lines); `run_check.sh` line 297 excludes # lines, so the block cannot change a matrix verdict; `wasp_full_newton` cold numeric-line bitwise against a control binary of the same tree without N10 (its  $5.8\text{e-}9$  /  $9.4\text{e-}9$  movement against the golden is B5j's recorded movement, not N22's as the worker supposed). **Fixed on the way**: the header loop matched columns as a SUBSTRING (`index(line, 'columns') > 0`), so the `_adv` files' NOTE prose or the new `species_columns` field would have replaced the species labels silently; now the first token (`comment_field_is`); the same hazard in `restart_round_trip.sh`'s Python readers fixed. Deviations from the design, stated: no EXHALE\_STATIONARY\_EVALUATE\_ONLY env hook (the second word is the one answer); the provenance unknown mark is not yet in `EXHALE_setup.out` (`write_setup_report.f90` not owned; one line); `species_columns` reported not refused (the by-label species map is a documented path); `source` is the git revision, not an executable md5. **Consequence for the user (decision item 21)**: the exact options comparison forbids the option-ladder workflow (converge without an option, restart with it on); legacy pairs are unaffected, every new rung would be refused; the disciplined fix is a key naming the tokens allowed to differ. Also corrected: `docs/input_schema.md` appendix D listed the retired `valve/fluxconst` fields and the wrong `base_bc` values. Reported: Run mode is absent from `input_schema.md`; the EXHALE\_PTC=1 route writes no setup report; a re-evaluated state writes `sec_ion_step=0`.
- **N22** (the predicted decrease stops scaling below `||s||`  $1\text{e-}7$ ; `steady_newton.f90` +195/-10, `krylov_and_dogleg` +16 rows): **all three candidates of the brief refuted by measurement, and the**

**cause named:** the trials that break the scaling are exactly the trials whose species unknown was written onto a face of its box (nout = 1), and that write is a move of the state no model of the step contains (11 of 11 such trials refused with a fixed ascent of  $3e-3$  over five decades of step; 10 of 10 others accepted at ratio 1.000). When the radius is cut the SCALED step quarters but the step in the unknowns the residual reads,  $||D \ s||$ , moves 0.4 percent, so the model image and pred are unchanged to seven digits;  $r0 \cdot A \ s = -||A \ s||^2$  to every digit (pred =  $0.5 \ (r0 \cdot A \ g)^2 / ||A \ g||^2$ , radius-free, the interior minimizer along the leg: not a defect); a fresh Jacobian action at the same step reproduces pred to every digit (no stale cache); the probe displacement is  $\sqrt{\text{eps}}(1 + ||Y||) = 1.8e-7$  whatever the direction, nine decades above the residual's reproducibility (no floor; the reverse: the stalling step is  $2.7e-3$  of the probe arc). **The fix:** hold\_the\_step\_where\_it\_leaves\_the\_species\_box zeroes the step at an unknown it would carry out of its box (the iterate is feasible, so the trial is, and the step is one the operator can be sampled along); probe\_length\_of\_the\_jacobian\_action is the one definition of the probe arc; a diagnostic line for each trial under EXHALE\_ELEM\_DIAG. **Atomic reload, cap 300, before -> after:** 227 -> 113 iterations (both stagnation), accepted/refused 170/55 -> 91/19, negative ratios 49 -> 14, best  $||R|| \ 2.353e-2 \rightarrow \mathbf{1.333e-2}$ , handed back  $2.671e-2 \rightarrow \mathbf{1.356e-2}$ ; the period-two cycle is gone (iterations 95-98: pred doubles with the step, all accepted at 0.998 to 1.001, steps  $1e-2$  to  $1e-1$  instead of  $1e-9$ ); the closing refusals are ray/floor verdicts at POSITIVE ratios. Carrier reload BITWISE (the branch never fires); wasp\_full\_newton BITWISE (ten files; the probe-arc refactor is shared through jv\_product and this settles it). Tests: the twelve-rung ladder rows are GREEN at entry (the geometry was never the cause), the new rows RED by four absent symbols, GREEN 120/0; steady\_species\_rows 182/0. Advisor: rebuild warning-free, steady\_newton.o md5 d70a078b identical to the worker's delivered object, the two suites 120/0, 182/0. Not converged. **Reported, the next item (N23):** the approximate-gradient leg is 0.7 percent of the step and supplies ALL of its model image ( $||sU|| / \delta = 6.9e-3$  passes  $\text{cauchy\_leg\_min\_fraction} = 1e-3$  while  $||A \ s|| = \tau_c \ ||A \ g||$  to four digits); the criterion must be the share of the IMAGE, not the length ratio (N7b's finding at a length the threshold admits).

- **N10b** (decision 21 option a; load\_IC.f90 +319/-20, input\_read.f90 +145/-1, write\_setup\_report.f90 +62/-6, grid\_and\_gates +1 script of 11 rows, docs/input\_schema.md K44 and D.2, manual entry): the key Restart option change: <token>[, ...]; the options field is compared TOKEN BY TOKEN from one table (opt\_name(20), opt\_value, the emitted text bitwise unchanged), only the named tokens may differ, every other difference refuses the load naming the token with its from -> to; six layout tokens (metals, mol, oxychem, carrier, carrier\_newton, iontrans) may never be named (they decide the rows the state carries: a cold start, not a restart); an unknown token, a grid/reservoir/constant word, the key with no token or without Load IC? True are input errors; the change is written into both halves of the new state as # option\_change from -> to at restart of <source>, inherited (history of 32 lines); EXHALE\_setup.out gains Restart provenance: UNKNOWN | UNKNOWN by descent | states its configuration and the permitted changes (N10's deviation 2 closed; the "by descent" wording was added because the first wording was measured false on a ladder's second rung). **Correction of the advisor's brief:** LW is a FLUX (Stellar LW flux), not an option token, so the armA\_noLW -> armA\_LW rung never needed this key; and every state written before the Jupiter-radius unification (those two included) is refused by the grid guard at  $2.25e-2$  in the outermost cell, so the armA\_\* direc-

tories are configurations, not restart sources. The ladder step was measured on the `armA_noLW` configuration with `base_ir`, `mol_ir` flipped: refused by token without the key, accepted with it, the provenance line in both halves. Tests RED 11/11 at entry (the key was an unrecognized line), GREEN 11/11; `grid_and_gates` 141/1 with the one FAIL attributed (below); `mol_carrier` 200 steps numeric-line bitwise (the rebuilt # options line bitwise). Also fixed: N10's test stage A read the LIVE `wasp_full_newton/output/`, which the matrix rewrites (a false failure of the legacy-mark row as soon as the pair carried a block); it now strips the block from its own copy. **Advisor:** rebuild warning-free, md5 `b10641e5` identical to the worker's; `grid_and_gates` 141/1, the FAIL being `stationary_evaluate_rebuilt_within_sweep_budget` reading the live `output/` of `wasp_full_newton`, which the running Stage D matrix had left as a mid-run relaxation snapshot ( $3.2e-3$  against  $1e-8$ ; the same pair evaluated by the entry and the N10b binary is bitwise, so not N10b's); the advisor pinned the certified state as `backup/regression/wasp_full_newton/IC/` (copied from the golden, `certified=T`, README there) and pointed the row at it. Reported: `restart_grid_guard.sh`'s stale header (third time); `sec_ion_step=0` on a re-evaluated state; whether `molbase`, `he_diff`, `he_metal_diff` belong with the layout tokens is a reviewer's question (the worker's reading: they add operators, not rows).

- **N23** (the approximate-gradient leg admitted by its share of the model image; `steady_newton.f90` +220/-5, `krylov_and_dogleg` +16 rows): two constants `cauchy_leg_image_share_max = 0.5`, `cauchy_leg_step_share_max = 0.1`, two pure routines (`cauchy_leg_shares_of_step_and_image`, `cauchy_leg_is_image_without_step`: image share above the first AND step share below the second, so a leg that IS the step keeps its image), and a branch in the `dogleg`'s trial loop that DROPS such a leg from the step and its image and rebuilds the `dogleg` at no extra product (drop, not rescale: a rescale would either shorten A SU without SU, which is not a step of the operator, or move the step off the `dogleg` path); the model slope handed to `ray_slope_verdict` is the slope of the step taken in both branches. **MEASURED, the reading ([diag 9]):** at the reload's second solve quartering the radius moves the promise by 0.9959 with the leg admitted and by 0.2501 with it dropped (six iterates: 0.9959 to 0.8938 against 0.2501 to 0.2517), so the promise follows the radius again where it did not. **Honest impact: the new rule fires ZERO times on both fixtures and all three runs are BITWISE** (atomic 113 iterations,  $1.356e-2$ ; carrier  $8.993e-2$ ; `wasp_full_newton` 13.30): on the post-N22 trajectories the length test reaches every image-dominant leg first (117 length drops, 0 image drops); the shares printed at every trial say the image test is the right criterion nonetheless (0.9986 of the image at  $3.4e-5$  of the step; eleven of seventeen read trials satisfy the predicate on their own), and the leg N22 saw at iteration 145 sat on a trajectory the N22 hold removed. Tests RED (four symbols absent), GREEN 136/0; `steady_species_rows` 182/0; -Wall warnings 6 before and after. **Advisor:** rebuild warning-free, the two suites 136/0 and 182/0 on `build_lvw` (the object md5 differs from the worker's delivered one by a post-measurement comment edit that moved line numbers, which the I/O statements carry; the worker's line-for-line reproduction of the solve covers it). Not converged; the rule is not claimed to improve either fixture. Reported: `cauchy_leg_min_fraction` is now a proxy for a test the code can make directly (letting the image test alone decide is the next increment, not taken because no fixture exercises the difference); the trial diagnostic's evaluations enter the cost counters.

- **N24** (what refuses the remaining trials of the atomic element reload; `steady_newton.f90` +51/-2,

krylov\_and\_dogleg +13 rows): **the obstruction is the linear model, and the row is named.** Atomic reload, cap 300: 110 trust-region iterations (the "113" of earlier reports counted three summary lines), 91 accepted / 19 refused. The 19 are two populations, neither a step-control defect: 14 ascents at LONG steps ( $||s||$  1.6 to 51, five with zero species unknowns held, so N22's mechanism is not what is left) and five closing refusals at positive ratios ( $||s||$   $2.3e-4$  to  $8.9e-7$  with the promise quartering exactly while the true decrease keeps a step-independent floor of  $3e-11$  to  $1.6e-10$ ). One accepted step (outer 59, ratio 0.948,  $||R||$   $7.37e-2 \rightarrow 2.12e-2$ ) moves the binding row from the energy equation to **the sodium element row of the outermost cell (cell 500, r 4.06 Rp)** and the solve never leaves it: from outer 60 to 110 that row holds 0.435 of the squared merit and the whole judged distance ( $5.3e-2$  of its own scale,  $5.36e6$  at `cert_tol_element`  $1e-8$ ) while  $||R||$  stands at  $1.333e-2$ . At outer 105 everything else is sound (0 held, 0 refused samples, 0 empty columns, `dgbcon`  $1.25e-6$ , step ratio 0.9998); what fails is the linear model: 40 of 40 products, true relative residual 0.58 against 0.1, **the Arnoldi image 21 percent away from the operator** ( $1.6e-6$  at iteration 1,  $2.3e-4$  at N7's iteration 60),  $||sN||$   $8.6e2$  against a radius  $5.9e-2$ , so the region uses  $7e-5$  of an inaccurate direction; the stuck row is at once the smallest scaled column and the smallest scaled row of the band. The stagnation detector read the judged distance, which IS that row;  $||R||$  was not falling ( $1.337e-2 \rightarrow 1.333e-2$  over 32 iterations). **The image-only leg test** (EXHALE\_CAUCHY\_LEG\_BY\_IMAGE=1, default off) measured and NOT adopted: worse on both fixtures (atomic  $2.30e-2$  against  $1.36e-2$ ; carrier 0.324 against 0.090, to the cap), the two runs parting at outer 7, so N23's "the rules agree" was that trajectory's property. Deliverable 3 STOPPED as briefed (linear solve / model / one row's physics, not the step control). Tests RED (two symbols absent), GREEN 149/0; `steady_species_rows` 182/0; the delivered binary reproduces the entry solve line for line over 925 lines; `wasp_full_newton` BITWISE. Advisor: rebuild warning-free, 149/0, 182/0 on `build_lwv` (the advisor's binary of N23 had been built from an earlier revision of `steady_newton.f90`, as the worker showed; the suites and fixtures were the check). **Proposal (N25):** (1) explain the 21 percent Arnoldi gap (EXHALE\_GM\_ORTHO; Jv reproducibility by two products); (2) ask the cycle for the direction ON THE BALL (Steihaug-Toint truncation) instead of solving to 0.1 a leg four decades beyond the radius; (3) the sodium row's own terms at cells 490 to 500: the element operator gives cell 500 a zero-gradient outer ghost, so nothing outward constrains it (a boundary question of the element operator, `binary_element_diffusion.f90`); (4) `cert_tol_element`  $1e-8$  is what makes this row the whole judged distance (UNCONFIRMED, ISSUES 3.1 e; not moved). Reported: the (JFNK) it line prints  $k=*$  for a species row without naming the element (in this case the identity was the whole diagnosis); the stagnation messages quote  $||R||$  where the detector read the judged distance.

- **Stage D gate and golden refresh (advisor, 2026-09-09 21:14 to 2026-09-10 00:27, make check on the shared build/; refresh 00:40).** Tree after every item to N24 (N25 was running on a private build). Verdict against the goldens of 2026-09-08 19:05: on all 16 default cases the two marching files are **data identical** (15 cases bitwise; `wasp_full_newton` WITHIN at  $5.78e-9$  /  $9.41e-9$ , B5j's recorded movement, which the 0.1 percent rule counts as identical) and all 32 `_adv` files FAIL by SHAPE (10 against 8 and 40/36/43 against 38/34/41 columns: the two status fields and the measure column of N11/N11b), so make check exits 2 as expected. The `_adv` movement is the physics of N12 (stale composition, upstream energy), N11 (mass-operator stationarity) and N11b (conditional tolerance): `T_adv` changes in 477 of 504 rows on `mol_base_handoff` (max 61.6 percent,

364 -> 588 K at r 1.093), 479 of 504 on mol\_carrier (max 138 percent, 379 -> 901 K at r 1.098), 215 of 504 on wasp\_full (max 2.3 percent, 5637 -> 5505 K at r 1.131). **Goldens refreshed:** the previous set archived as backup/regression/golden\_pre\_stageD\_20260910/, run\_check.sh golden snapshot the outputs the gate had just produced (17 cases, including the three hp\_\* and oxygen\_chemistry extras), and the advisor compared the 68 snapshot files against the case outputs with the comparator's own #-filter: 68 identical, 0 differ (this comparison, not a three-hour rerun, is the "check again" of the standing rule, since a snapshot is a copy; the user asked that regression checks not be repeated). The shared build/ and EXHALE.x are now current with the tree (md5 f6ccb498 at the gate's build; N23/N24 built after it touched steady\_newton.f90 only, so the marching path of the matrix is the gate's). State files now carry the N10 metadata block and the \_adv files the N11/N11b header; both are # lines the comparator ignores.

- **N25** (the Arnoldi gap, a direction on the ball, the sodium row at the outer boundary; steady\_newton.f90, krylov\_and\_dogleg +25 rows, binary\_element\_diffusion.f90 a default-off print only, 33 lines, no arithmetic): **(1) the 21 percent Arnoldi gap is the NONLINEARITY of the finite-difference action across directions**, shown positively: every one of the 40 Arnoldi columns matches its own product to 1e-16 to 3e-16, two products of one direction agree exactly, the basis's orthogonality loss is 4.4e-13, yet the action of  $V_1 + V_2$  differs from the sum of the two actions by 1.1e-4 and the assembled gap tracks the step length (8.5e-5 and 2.5e-4 per unit of  $||sN||$  at outer 1 and 105;  $||sN||$  12 -> 865), the signature of a curvature term over the fixed probe arc  $\sqrt{\text{eps}}(1 + ||Y||) = 1.8e-7$ . Reorthogonalization NOT adopted (loss 1e-13 to 3e-12 against the 1e-8 level; as a measured option EXHALE\_GM\_REORTH0=1 it is worse on both fixtures: 1.487 and 0.43). **(2) Steihaug-Toint truncation** is in as EXHALE\_KRYLOV\_ON\_THE\_BALL=1, default off: products per iteration 33 -> 7 (atomic), 40 -> 1.5 (carrier); atomic better on the judged distance (1.69e6 against 5.31e6, lowest  $||R||$  9.4e-3 against 1.33e-2) but worse on  $||R||$  handed back (1.69e-2 against 1.36e-2); carrier worse (0.486 against 0.090); the brief's condition fails, default stays off. **(3) The sodium row of cell 500 is not a well-posed outflow condition, and the reason is a DEFECT of the stationary residual evaluation, not of the element operator:** the outer ghost of the element mixing ratio is FROZEN at the pre-solve composition (1.73111e-6, the metals.inp reservoir the handed-in state carried) throughout the solve while cell 500 falls to 1.0e-11, so at the stagnating iterate the ghost is 1.7e5 times the cell it "continues"; the marching path refreshes that ghost (the second solve, entered at step 2002, has ghost = cell 500 again); the accepted step of outer 59 did not move  $fX(500)$  at all (it moved the outer face mass flux); the scaled Jacobian row is dominated by the momentum of cells 499/500 and the sodium of 499 while its own diagonal is 1.3e-4 of the largest entry (the row is nearly independent of its own unknown). The physically stated condition (zero gradient at the top, no diffusive flux) requires the ghost to follow the ITERATE; the repair is in the composition write-back / ghost policy of eval\_residual, which is N26 (launched). **(4)** the (JFNK) it line names the element (worst row: element Na of cell 500) and the stagnation messages quote the judged distance and dj\_best. Tests RED (eight symbols absent), GREEN 174/0; steady\_species\_rows 182/0; default path unchanged (atomic 109 iteration lines identical to the digit; carrier identical); wasp\_full\_newton BITWISE. Advisor: rebuild warning-free, 174/0, 182/0 on build\_lwv; three British neighbor in the element operator's comments fixed. Not converged; the on-the-ball run's best iterate is chosen by the judged distance while the control is  $||R||$  (the two rank iterates differently, which limits the

comparisons).

- **N26** (the upper ghost of a transported element or carrier follows the iterate; steady\_newton.f90 +135/-20, steady\_species\_rows +7 rows): in write\_species\_rows\_into\_composition (the ONE place a trial composition is assembled for the residual: eval\_residual and the three best-iterate restores) the transported columns of the upper ghosts take the outermost cell's value by the rule the marching path uses (READ:  $Y_{\text{etr}}(N+1:N+Ng, q) = Y_{\text{etr}}(N, q)$  in species\_advection\_stage;  $f_{\text{try}}/f_{\text{c}}(N+1:N+Ng, ic) = \dots(N, ic)$  in the carrier operator; Apply\_BC never touches f\_sp), the element totals of a ghost restored after the carrier copy, a ghost deficit clamped as the operator clamps its own; lower ghosts untouched (Dirichlet reservoir); the three-unknown route returns before reading anything. The mixing ratio follows the mass fraction exactly (equal He mass fraction at N and its ghost gives equal  $n_{\text{He}}/n_{\text{H}}$ , hence equal  $n_{\text{X}}/n_{\text{H}}$ ; MEASURED 1.4e-16). **MEASURED, atomic element reload (EXHALE\_JFNK\_MAXIT=300)**: ||R|| handed back 1.356e-2 -> **3.719e-4** (36x), judged distance 5.36e6 -> 6.21e4 (86x), flux spread 6.6e-4 -> 5.5e-7, 91/19 -> 235/18 accepted/refused, the sodium row of cell 500 binds on **0 of 252** iterations (was 51 of 109) and no row of cell 500 is ever the worst; the solve now stagnates on the INTERIOR helium row of cell 246 (r 1.155), info = 2, uncertified. **Carrier reload**: binding cell (H2 of cell 205, interior) and judged distance (4.807e7) unchanged to four digits, ||R|| handed back 8.993e-2 -> **2.519e-1** (worse by 2.8x, a path difference of a stagnating solve, reported not judged), flux spread down 12 percent. mol\_carrier marching bitwise (no caller outside solve\_steady\_jfnk); wasp\_full\_newton BITWISE. Tests RED on the entry text with only the export added (4 rows: ghost mass fraction 6.7e4 off, mixing ratio 1.0e5 off, the outermost element row of a flat column 1.19e-2 instead of 0, the carrier ghost 0.80 off), GREEN 189/0 (a non-vacuous row keeps the stale ghost and reads 1.19e-2 in both builds); krylov\_and\_dogleg 174/0. Advisor: rebuild warning-free, steady\_newton.o md5 4bd3138a and binary e56609ff identical to the worker's, 174/0 and 189/0 on build\_lwv; one neighbors fixed. **Reported, the next small item (N26b)**: the relaxation solvers solve\_trace\_element\_in\_hydrogen and element\_diffusion\_step refill their working arrays' upper ghosts before every pass while element\_transport\_residual, documented as the same operator, reads the ghost from f\_sp with no refill, so the operator's boundary depends on its caller (binary\_element\_diffusion.f90, two lines). Not claimed: convergence; an explanation of the collapse of the outermost cells to 1e-11 of the reservoir.
- **N26b** (the element operator refills its own upper ghosts; binary\_element\_diffusion.f90 +26/-0, new suite src/tests/element\_operator/ with 5 rows): element\_transport\_residual now writes  $X_{\text{he}}(N+1:N+Ng) = X_{\text{he}}(N)$ ,  $f_{\text{X}}(N+1:N+Ng) = f_{\text{X}}(N)$  and  $w_{\text{Ytr}}(N+1:N+Ng) = w_{\text{Ytr}}(N)$  (the third line the relaxation twin gets for free through project\_elements) so the operator poses ONE outer boundary (zero gradient, no diffusive flux at faces 0 and N) whatever the caller left in the ghosts; the lower ghosts stay the Dirichlet reservoir; the header states the boundary. Both outermost rows (N-1 and N, WENO3 reaches cell N+1 at face N-1) are covered. MEASURED: the relaxation's own fixed point handed to the residual with stale ghosts read 1.19e-2 of its scale in the outermost row before and 1.5e-13 after (the interior level is 1.3e-13); the same column with stale and with zero-gradient ghosts now gives the two outermost rows bitwise. Suite RED 3 of 5 on the entry text, GREEN 5/0; certification 29/0; the atomic reload's first two JFNK iterations identical but for the merit's reproducibility

going  $2.8\text{e-}14 \rightarrow 0$  exactly (the residual no longer reads a ghost the sweep leaves at a different last bit), the outputs moving at  $7.9\text{e-}12$  (T) and  $1.6\text{e-}10$  (an `_adv` status/measure column) because N26's write-back hands over the ghost to  $2\text{e-}16$  and the refill to the bit; `mol_diffusion` marching 200 steps BITWISE (the marching path reaches the relaxation twin, never the residual). Built and measured in an isolated source tree with `steady_newton.f90` pinned, because N27 was editing it. Advisor: rebuild warning-free, `element_operator` 5/0, certification 29/0 on `build_lvw`; one percent fixed. Reported: the write-back's ghost is the cell's to round-off, not exactly (any other consumer of those ghosts reads a  $2\text{e-}16$  gradient).

- N27** (the helium row of cell 246 and the H2 row of cell 205 read side by side; `steady_newton.f90` +191/-3 diagnostic only, `steady_species_rows` one row restated): **the helium row of cell 246 is held by the LINEAR SOLVE, not by its physics, its boundary, a bound, the closure or the tolerance.** At outer 245 the row is  $+2.43\text{e-}22 = \text{diffusive (settling)} + 2.40\text{e-}22$  plus advective  $+3.3\text{e-}24$ , same sign (they add); the Newton removed the advective term (a factor 7200 since outer 1) and barely touched the diffusive one ( $\times 1.28$ ); the measure  $6.17\text{e-}4$  is the maximum of a flat plateau ( $5.9\text{e-}4$  at 240 to  $5.4\text{e-}4$  at 253), not a bad cell; 0 held, 0 refused samples in 246 iterations, band healthy (`dgbcon`  $7.5\text{e-}7$ ), merit reproducibility exactly 0, the trust region inert (ratio 0.933, 0 cuts, `aN` 1.0) and the Krylov cycle spends 40 of 40 products on a direction whose true relative residual is **0.9917** against 0.1 with the Arnoldi image 14.7 percent off the operator; the step buys  $4.7\text{e-}5$  of the merit per iteration. The helium Jacobian triple is a proper tridiagonal stencil with the diagonal largest (the unknown controls its row; the wind dominates the row's entries). The element rows hold 0.98 of the merit (He 0.63, Fe 0.36);  $||R||$  is the energy row at  $1.37\text{e-}4$ ; the stagnation detector is honest (merit and  $||R||$  fall while the judged distance rises  $6.12\text{e}4 \rightarrow 6.21\text{e}4$ ). **A number that does not fit N25's reading:** the Arnoldi gap is 0.147 at  $||sN||$   $1.8\text{e-}2$ , 8 per unit against N25's  $2.5\text{e-}4$  per unit, so the gap is not a function of the assembled length alone. **The H2 row of cell 205** is the same shape on a coarser front: measure 0.480 (not a cancellation, half its own terms), `f(H2)` falling  $1.4\text{e-}1 \rightarrow 1.1\text{e-}3 \rightarrow 4.9\text{e-}4$  over cells 195-207 (logarithmic slope -757), `f(H I)` flat (1.7 percent), so cell 205 is NOT a two-root H I front (consistent with N5); linear 40/40 at 0.97, gap  $3.1\text{e-}2$ . Deliverable 3 STOPPED (nothing in the step control to change at the binding iterate). Deliverable 4: `cert_tol_element` would have to be  $6.2\text{e-}4$  (4.7 decades; far above the  $1\text{e-}5$  the operator's header gives as its loosest floor); the closure's reproducibility ( $3.1\text{e-}7$  in the row's measure per unit seed, N5) is not what holds it. Proposals: re-measure the Arnoldi gap at three iterates (it no longer tracks  $||sN||$ ); the untried freedom is the COLUMN preconditioner (row scaling is fixed by decision 20 a); whether the detector should stop a solve whose merit still descends while the judged row does not is an acceptance decision; the carrier front is a grid statement. Tests 189/0, 174/0; advisor rerun 189/0, 174/0. **Incident, found by N27 and CONFIRMED by the advisor:** with N26b's ghost refill in `element_transport_residual` the atomic element reload parts from N26's run at outer 10 and stands at  $||R||$  1.199 at iteration 40 where N26's run was at  $3.21\text{e-}2$  (advisor run, cap 40, current tree: 1.888, 1.891, 1.677, 1.484, 1.518, 1.582, 1.199 at iterations 1, 5, 10, 15, 20, 30, 40 against N26's 1.888, 1.891, 1.678, 1.955, 1.758, 0.912, 0.0321); N27's own diagnostic is inert (hook on = hook off). N26b's validation (the first two iterations, the certification suite, `mol_diffusion` marching) did not reach it, and the advisor's verification of N26b did not rerun the reload: the advisor's mistake. N26b's refill is therefore NOT accepted as it stands; N26c (launched) finds why an operator that poses its own boundary breaks a solve whose caller already hands it that boundary to  $2\text{e-}16$ , or reverts the refill. N27 restated

the row `a_stale_upper_ghost_is_read_` by the outermost row to its negation because N26b made the original false; N26c reconciles the two rows with whichever text stands.

- N26c** (why N26b's ghost refill broke the atomic solve; `binary_element_diffusion.f90` refill REVERTED with the header restated, `element_operator` rows restated (6), N26's `steady_species_rows` row restored): **the refill changes the residual by one element's outermost row in its twelfth digit and nothing else, and the atomic element reload does not survive a last-bit change of the upper ghosts, whatever makes it.** MEASURED: the caller's ghosts already equal the cell to 1 to 4 units in the last place (Xhe 9e-16, fX 1.9e-15, wYtr 6e-16 over the whole solve); the two texts' outermost rows agree to 17 digits on every element but nitrogen's, which differs by 9.8e-13 relative (6.8e-17 in its own measure); no other ghost of the routine is read (face coefficients stop at N-1); the band's stencil already carries the dependence of rows N-1, N on cell N through the write-back. **A control that multiplies the same three ghost quantities by  $1 + 2e-16$  and poses no boundary at all breaks the reload as badly** (`|R|`) at iteration 40: caller's ghosts  $3.212e-2 = N26$  to the digit; refill 1.199; one-ulp control 1.125; the three part at outer 10-11 and from each other). Where it amplifies: the Krylov leg at its one-digit tolerance (relative residual reached  $8.86e-2 / 8.89e-2 / 8.93e-2$  at outer 10, pred 0.4 percent apart, then 2.7x apart at outer 11, then different branches of the acceptance at outer 12: accepted 0.874 / REFUSED -1.98 / accepted 85); the solve's own merit reproducibility prints  $1.4e-14$ , a hundred times the ghost movement. **The refill is physically right and is reverted anyway:** no version of it can be bit-inert on the production caller, and the reload's acceptance is a trajectory no last-bit change preserves. Delivered text reproduces N26's atomic reload line for line over 252 records (`|R|`)  $3.719e-4$ , judged  $6.22e4$ , He of cell 246, 27543 samples, exit 2) and N26's carrier reload over 63 (0.2519,  $4.806e7$ , H2 of cell 205); `mol_diffusion` marching bitwise; suites certification 29/0, `element_operator` 6/0 (restated to ASSERT the measured caller dependence,  $1.194e-2$ , so the suite exits 0; the advisor accepts that choice), `steady_species_rows` 189/0, `krylov_and_dogleg` 174/0; advisor rerun of the four on `build_lwv` (md5 3201310a, the worker's delivered binary): 6/0, 29/0, 189/0, 174/0. **The finding that matters beyond this item:** the atomic element reload is NOT a reproducible quantity; its `|R|` at a fixed iteration count is set by which Arnoldi vector crosses a one-digit tolerance at the last bits of the residual, so "reproduces the previous run" rejects correct changes as readily as wrong ones, and what the reload can carry is named outcomes (binding rows, refusal counts, flux spread), not `|R|` at iteration N. Recorded in ISSUES 3.1 (d) and in the fixture's README. The asymmetry of the element operator (the relaxation twins write their ghosts, the residual takes the caller's) stays in the tree, documented and measured ( $1.19e-2$  on a stale caller). Advisor: a `cp -irp` of the N4b worker hung on its interactive prompt since 2026-09-09 (PID 61163, cwd the repository root, copying from the session scratch) was killed by PID. CORRECTION within the hour: killing the `cp` let the worker's shell run its NEXT statement, which re-applied N4b's first (discarded) patch onto the delivered `element_inventory.f90` (832 -> 926 lines, a duplicate routine, would not compile); the N4b worker, woken by the stale task, rebuilt the file from its saved entry copy plus the delivered patch and it reproduces `diff_element_inventory.diff` line for line; the advisor confirmed 832 lines, one routine, a warning-free build (md5 3201310a, N26c's delivered binary) and `element_operator` 6/0, `steady_species_rows` 189/0. Lesson written into `docs/worker_rules.md`: `\cp -f / \mv -f` always, and a stale command is canceled as a task, not killed as a process.

- **Direction after N26c (user, 2026-09-10: "권고안대로 진행")**. The advisor's recommendation adopted: (b) N8b, anchoring the species-row certification tolerances on the two candidate states by the review's five measurements (reproducibility floor, derivative convergence, a manufactured steady solution of the element operator, grid refinement of the candidate state, conserved-flux error), reported per row class and regime with a PROPOSAL for the user (decision item 22) and no value moved; and in parallel (c) N28, the code items of ISSUES 3.7/3.8 (run-wide JFNK cap, one element\_ratio\_gate, the update map's install branch owing an Apply\_BC, the undefined innermost ghost column of assemble\_residual, the PTC route's setup report, sec\_ion\_step on a re-evaluated state, the stale test header, Run mode in the input schema). (a), the column preconditioner, waits for N8b's numbers. The user manual PDF was rebuilt by the advisor (85 pages; it was two keys behind its source: Restart intent, Restart option change).
- **N8b** (anchoring the species-row certification tolerances; certification.f90 +173 additive, no value moved; steady\_newton.f90 one diagnostic; certification suite 29 -> 42 rows; memo docs/certification\_tolerance\_anchoring\_20260910.md): **1e-8 is refused by no floor of the arithmetic, the closure or the representation (five to eleven decades below it); what refuses it is the DISCRETIZATION and the ELEMENT FLUX CONSERVATION**. MEASURED on the two candidates: reproducibility exactly 0; closure-seed floor  $5.3\text{e-}15$  (the state's own sweep seed  $1.7\text{e-}10$ ); state-perturbation amplification 0.23 to 0.30 in the wind but  $1.1\text{e}3$  to  $2.0\text{e}3$  at  $r < 1.01$  (floors  $5\text{e-}17$  and  $4.4\text{e-}13$ ); derivative: a clean V at the base ( $4.5\text{e-}7$ ), a PLATEAU at  $6.4\text{e-}5$  over four decades of probe length at the binding cell 245 (the part of the derivative outside the band) so the smallest row a Newton step resolves there is  $3.8\text{e-}8$ ; manufactured relaxed column at 801/401/201 cells  $5.4\text{e-}13$  /  $2.47\text{e-}6$  /  $7.42\text{e-}6$ , order 1.59 (the flat column  $2.7\text{e-}14$ ); grid refinement of the candidate at  $r$  1.1515: the He/H row  $5.97\text{e-}4$  ( $N = 500$ ) ->  $2.48\text{e-}4$  ( $N = 1000$ ), remap control  $1.6\text{e-}5$ , so **about 78 percent of the  $6.0\text{e-}4$  is discretization** (a C2 interpolant is needed; below  $r$  1.03 the remap dominates); element flux spread 39.3 in the layer,  $2.6\text{e-}2$  in the wind,  $2.2\text{e-}3$  over the escape window against the  $5.5\text{e-}7$  mass-flux spread the candidate is quoted with. Memo section 7 is the proposal (decision item 22, added by the advisor to To\_be\_determined): (a)  $1\text{e-}5$  in the wind with the layer reported, (b) keep  $1\text{e-}8$  and state the fixtures cannot certify, (c) two regimes gating; no proposal certifies either candidate. **Two defects found outside the item's files, reported:** (1) **the candidate's species do not carry the run's own density**: mass closure  $6.68\text{e-}3$  at the top against  $1.9\text{e-}9$  on the three-unknown control; reproduced in the suite on a column closed to  $6\text{e-}16$  that reads  $3.37\text{e-}3$  after ONE relax\_element\_composition, so project\_elements does not return to hydrogen the mass it takes from helium (N29, launched); (2) the writer-to-loader round trip does not return the state: the atomic candidate reloads at  $||R||$  0.486 instead of  $3.7\text{e-}4$  because load\_IC rebuilds rho from the species (a consequence of 1). Tests 42/0 (RED at the build level: the suite does not compile against the entry objects); the default certification report byte-identical (anchoring under EXHALE\_CERT\_ANCHOR=1). Advisor: rebuild warning-free, certification 42/0; one neighborhood and two percent fixed.
- **N28** (the code items of ISSUES 3.7; EXHALE\_main.f90, RK\_rhs.f90, steady\_residual.f90, element\_census.f90, element\_inventory.f90 and attempted\_step.f90 (the duplicate constant), grid\_and\_gates +2 rows and headers, docs/input\_schema.md K45): (1) EXHALE\_JFNK\_MAXIT caps

the RUN: the three call sites of `steady_wind_with_element_diffusion` pass one named `default_jfnk_outer_iterations_default = 3000` (the marching hand-off's literal was 500, the one default that moved; `wasp_full_newton` converges in 13 and cannot reach either), and under the variable the run takes its FIRST stationary solve and refuses later entries by name (MEASURED: three solves of 2 iterations -> one); (2) `element_ratio_gate` has one owner, `element_census.f90` (public, `element_census_tolerance()` public with its override), read by `element_inventory.f90` and `attempted_step.f90`; (3) `update_map_begin_step's` install branch calls `Apply_BC` in the N16d order (the EXHALE\_UPDATE\_MAP diagnostic's second block on `mol_carrier` moves 5 percent in the mass and energy rows: the fix of a defect, a boundary of one state under an interior of another); (4) `assemble_residual` defines the innermost ghost column (dF, S, R zero at 1-Ng; two callers read it in whole-array expressions); `-fcheck clean` on the certification driver and on a 20-step `mol_carrier` (the run that reaches it); (5) the EXHALE\_PTC=1 route writes its setup report (was 0 lines; new row RED/GREEN); (6) a Restart intent: stationary evaluate keeps the file's `sec_ion_step` (2402 -> 2402, was 0; new row RED/GREEN); (7) `restart_grid_guard.sh's` stale header rewritten and the script made to honor EXHALE\_TEST\_OUT; (8) Run mode documented as K45. Item 9 (`n_eq_sweeps_last` on a discarded evaluation) is `steady_newton.f90's`, reported. Validation: `mol_carrier` 200 steps and `wasp_full_newton` cold BITWISE on every output file (the one log difference: cap 500 -> 3000, from the caller); `grid_and_gates` 146/0; `attempted_step` 162/1 with the FAIL proven pre-existing (rebuilt from entry sources: same row); advisor rerun on `build_lwv: grid_and_gates` 146/0, `attempted_step` **163/0** (the N28b fix below had reached the tree by then). **Found, reported, N28b launched:** `bg_cell` (`type(ion_rates)`) is allocated without initialization and the type has no default values, so before the first sweep the checkpoint checksum reads 8.7e244 and the row `a_perturbed_state_is_not_the_checkpoint` could never fire (a perturbation of 1e2 lost in the round-off of 1e244). Advisor: three forbidden phrases fixed (per-element in a printed line, neighboring, per-region). One deviation stated by the worker: an extra private `-fcheck build` (the shared `run_fcheck.sh` runs `make distclean`).

- **N28b** (`bg_cell` allocated and never initialized; `ion_cell_state.f90` the type only, `ionization_equilibrium.f90` the allocation site, `attempted_step` suite +1 row and a deterministic dirty-heap setup): **the defect is real, wider than `bg_cell`, and includes a LOGICAL.** Three grid-sized objects of `ionization_equilibrium` came out of `allocate` holding the recycled block: `bg_cell` and `ieq_rate_cell` (both `type(ion_rates)`, no default values) and the real array `heat_chem` (the one array left out of the routine's zero list). MEASURED with a probe that frees a 16 kB block before the allocation: T\_K, P\_HI 8.7e244 and **x\_h2\_fixed TRUE** (an imposed H2 partition the caller never asked for, read from garbage, standing in the array the carrier transport reads), on a fresh heap all zero (which is why N28 saw the dead row fail and the advisor's rerun saw it pass: the defect appears only when a freed block is recycled). Fix: `ion_rates` carries its zero in every one of its 37 fields (34 reals, 3 logicals) as default initializers, stated as the invariant "a cell not yet swept has no gas, no rates and no imposed partition", which reaches `bg_cell`, `ieq_rate_cell`, `bg_cell_adopted`, `bg_cell_best`, the closure scratch and the four checkpoint copies at once; `heat_chem = 0` at the allocation. Two comments that claimed a default initializer would not reach threadprivate copies were corrected (MEASURED with a four-thread program: it does, as OpenMP specifies).

Tests: RED on the entry text with the new driver 68/2 (the\_unswept\_background\_carries\_no\_rates -3.4e250; a\_perturbed\_state\_is\_not\_the\_checkpoint h1 == h0), GREEN 164/0 (the intermediate build located heat\_chem at 8.7e244 by a field-by-field peek). mol\_carrier 200 steps bitwise (every reader of the three objects is gated on bg\_ready/ieq\_rates\_ready, set after the loops that write them; heat\_chem has no physics reader, only the checkpoint); wasp\_full\_newton not run (atomic: the arrays never enter a physics path). Advisor: rebuild warning-free, attempted\_step 164/0. Reported: three test fixtures enumerate all 37 fields of bg\_cell by hand (now redundant) and element\_census\_tests.f90 set two fields and left the rest to the allocator (now covered by the type).

- **N29** (the element projection does not conserve mass; binary\_element\_diffusion.f90 +258/-113, element\_operator suite 6 -> 14 rows): **the leak is the mass the write-back is told to hand back, not the write-back.** project\_elements took the mixture mass as  $m_{sum} = m_1 n_H + m_{He} n_{He}$  with  $m_1 = \text{mass\_per\_H\_nucleus\_without\_He}()$ , the RESERVOIR mass per hydrogen nucleus (the trace metals at melem\_ab), while the cell carries the metals at whatever the trace transport left; the projection then wrote the difference into hydrogen at every call. MEASURED at one cell of the certification column (metal/H 0.024401 against the reservoir's 0.028227):  $\sum m_i f_i$  1.0000000 -> 1.0000636 after one step, helium's gain exceeding the hydrogen and metal loss by 6.4e-5: mass created. A second leak of the same kind: the trace-metal diffusion wrote metal densities with no counter-flux in the hydrogen it diffuses through (3.0e-3 on the suite column with the metals frozen). **Fix:** mixture\_mass\_split reads the mass the cell's species actually carry (bsp\_mass and melem\_A, no species weight restated), project\_elements(f\_sp, Xhe, msum, scale\_metals) scales component 1 as a whole, every helium-free species included (CO too), the metals written first and the hydrogen group closing the mixture; one projection after the metal loop puts the metals' mass change back on hydrogen; settling\_coefficient takes its component-1 mass from the same split. Metals off is bitwise by construction ( $m_1 = m_H = 1$  exactly). **MEASURED:** certification column closure after the relaxation 3.37e-3 -> 2.3e-15; the handed-back atomic candidate 6.44e-3 -> 1.94e-9 (the file's own precision, the three-unknown control's 1.93e-9); the writer-to-loader round trip of that state now returns it (writer ||R|| 1.112 and reload 1.1123 on the SAME row, against 3.2e-2 vs 0.4385 on different rows before: memo section 6 closed); mol\_diffusion and wasp\_full\_newton BITWISE (metals off / no element row); **lower\_profile MOVES and should:** its stored golden is 3.6e-4 out of closure (O/H 4.61e-4 against the reservoir 4.90e-4), a 1500-step cold start goes 6.77e-3 -> 3.14e-9 in closure with T at 1.6 Rp moving 1.3 percent and H I 6.3 percent: **a golden that moves for a physics fix, refreshed at the next gate.** Atomic reload cap 40 by named outcomes: closure 6.44e-3 -> 1.94e-9, 32/8 -> 31/9, refusals 0 in both, binding row energy of cell 2 -> cell 4, flux spread 2.3e-3 -> 1.4e-2, same stop. Tests RED 4 rows on the entry text (3.44e-3, 4.78e-4, 3.44e-3, 3.82e-4), GREEN 14/0; certification 42/0, diffusion\_tests 35/0, krylov\_and\_dogleg 174/0. **The worker's control discipline:** the advisor's EXHALE\_lwv.x is rebuilt at every verification and is a build of the live tree, so the worker built its own control (the delivered objects with only the entry-text binary\_element\_diffusion.o) and discarded three comparisons that had used the advisor's binary as "before"; workers are to build their own controls (added to the rules below). **Advisor:** steady\_species\_rows\_tests.f90 referenced element\_ratio\_gate through element\_inventory, which since N28 imports it with use, only

and does not re-export it, so the suite would not build (N28's validation did not include this suite; the advisor's did not either): one use `element_census`, only: `element_ratio_gate` line added by the advisor; rebuild warning-free (md5 a9627a14), `element_operator` 14/0, `certification` 42/0, `steady_species_rows` 189/0, `krylov_and_dogleg` 174/0; one percent fixed. Reported: the metal diffusion's counter-flux is closed on hydrogen but not on momentum or internal energy (a trace approximation, named); `element_census_verify` gates the element ratios and only reports the mass closure (the 6.4e-3 state passed every gate it has): whether it should gate is a decision.

- **N8c** (the five anchors re-measured on the states after N29; the memo docs/certification\_tolerance\_anchoring\_20260910.md only, 344 -> 524 lines, no source file changed): **the reading is unchanged and is now taken on states worth measuring.** Post-N29 candidates (advisor-verified: no src file newer than the brief): atomic reload 275 iterations, 254/22, ||R|| 2.271e-4, binding row He/H of cell 245 (r 1.153), flux spread 9.8e-7, **mass closure 1.95e-9** (was 6.68e-3), 11 of 13 entries refuse; carrier reload byte-identical to N26's (no metals, no element row; the corrected projection is never reached), closure 1.93e-9. The anchors: two evaluations of one state 0 exactly; the state's own sweep seed 1.7e-10 -> 1.6e-14 (the loader no longer repairs a composition whose mass missed the density), closure floor 5e-19; state-perturbation amplification 1.2e3 to 2.0e3 at  $r < 1.01$  and 0.4 to 0.7 elsewhere; derivative: a clean V at the base (7.6e-8), the PLATEAU at 6.6e-5 at the binding cell stays (smallest resolvable row 3.9e-8 there, 6.5e-9 in the wind; both taken at the first JFNK iterate of a relaxation restart, two CFL steps from the loaded state, stated); manufactured column order 1.589, closure 2.3e-15 (was 3.4e-3), extrapolated 1.52e-6 at the production spacing; grid refinement of the candidate 5.93e-4 (N = 500) -> 2.47e-4 (N = 1000), ratio 2.40, remap control 1.5e-5, Richardson limit 1.31e-4: **78 percent of the row is discretization**; element flux spread in the layer **39.3 -> 9.79** (He), 37.3 -> 9.09 (H), the wind 2.6e-2 -> 2.7e-2, the escape window 2.0e-3. Memo: sections 3 (mass closure) and 6 (the reload) marked CLOSED by N29 with the readings; section 5 rewritten; section 7's proposal re-stated with the new floors and **option (a) supported as written** (1e-5 gating at  $r \geq \text{cert\_regime\_wind\_r}$ , the layer reported and not gating); two of its three prerequisites discharged, the layer's flux conservation (9.8) standing. No proposal certifies either candidate (wind element row 3.05e-4 and 7.2e-2 against a proposed 1e-5). `certification` 42/0. Reported: the iron wind row of the atomic candidate rose 18x between the two chaotic states (a size, not a trend); the derivative anchor is not measured at the loaded state itself (needs the probe under stationary `evaluate`, `steady_newton.f90`).
- **Decision 22 (user, 2026-09-10): option (a).** `cert_tol_element` and `cert_tol_carrier` become 1e-5, gating, for cells at  $r \geq \text{cert\_regime\_wind\_r}$  (1.20); the layer's element and carrier rows ( $r < \text{cert\_regime\_layer\_r}$ , 1.10) are reported and do not gate until the layer's element flux conservation (9.8 after N29) is repaired; the hydrodynamic tolerances do not change; a wind-certified state says so in one line. Anchor: the operator's stated floor and a decade above the manufactured column's discretization at the production spacing (1.52e-6); no existing state is admitted by it (3.0e-4, 7.2e-2). Implemented as N30 (launched).
- **Gate after N29 and the lower\_profile refresh (advisor, make check 2026-09-10 07:58 to 11:03; refresh 11:10).** Tree after N28, N28b, N29 (N30 running on a private build). Against the 2026-09-10 00:40 goldens: **15**

of 16 default cases data identical on all four files (wasp\_full\_ newton now identical too, its golden being the 00:40 snapshot); **lower\_profile FAILS on all four, by design** (N29's mass fix: its golden was  $3.6\text{e-}4$  out of closure;  $v$  moves up to 4.8 percent at  $r$  3.18 in 85 of 504 rows,  $\rho$  1.0 percent in 4 rows,  $T$  0.5 percent in 2 rows; the `_adv` measure column 57 percent at one row;  $\log_{10}$   $\dot{M}$  7.29). The failing case was identified by its own section of the log (the advisor's first parse attributed the summary line to `hp_front`, which PASSES). Refreshed: golden/lower\_profile archived as `golden_pre_n29_20260910/lower_profile`, `run_check.sh` golden lower\_profile snapshotted the gate's output, the four files compared identical to the case output with the comparator's filter. No other golden touched.

- N30** (decision 22 a; `certification.f90` +296/-54, `steady_newton.f90` +50/-30, `certification` suite 42 -> 57 rows, `steady_species_rows` 189 -> 194, `docs/input_schema.md` the `cert_reason` row): `cert_tol_element` and `cert_tol_carrier` are gone as single numbers; `cert_tol_element_wind` = `cert_tol_carrier_wind` =  $1\text{e-}5$  with one accessor each (`cert_tol_*_at(r)`, a step at `cert_regime_wind_r` = 1.20; `cert_tol_reported_only` = huge below), the anchoring written at the declaration with the memo cited; **one reduction** `certification_species_row_gate` decides a species row over the GATED cells (no gated cell = refused, never a satisfied row); the entries carry gated and reported maxima with their cells; the report prints gated at  $r \geq 1.2$  against  $1\text{e-}5$ : `<measure>` at cell `<j>` and, for a certified state with species rows, the one line certified IN THE WIND ... the rows below it are reported and do not gate: worst `<measure>` at cell `<j>`; the state file carries `cert_reason=certified_in_wind` (the field already existed). **The band  $1.10 \leq r < 1.20$  is reported and does not gate** (the decision was silent; the band holds the candidates' binding cell at  $r$  1.153 where the discretization is  $4.6\text{e-}4$ , so a  $1\text{e-}5$  gate would ask the residual to fall 1.5 decades below the error of its own discrete equation; no anchor for the band exists). `distance_from_certification` and `certified_row_measures` read the gated measures and the accessors, so **the trust region's ranking changes and both reloads move** (that is the decision's intended reach): atomic reload 275 -> 166 iterations at the same  $\log_{10}$   $\dot{M}$  10.10, binding row He/H  $2.896\text{e-}4$  at cell 288 (a WIND cell) against  $1\text{e-}5$ , NOT certified; carrier reload 63 -> 76 iterations, H2  $7.284\text{e-}2$  at cell 285 against  $1\text{e-}5$ , NOT certified; the hydrodynamic tolerances, the H(n=2) level balance ( $4.8\text{e-}10$  against  $1\text{e-}6$ ) and `steady_gates_met`'s own carrier gate (`carrier_resid_th`, a volume ratio, not this tolerance) untouched. `wasp_full_newton` bitwise, still certified with no reason. Tests RED (the superseded rule refused a layer- $1\text{e-}2$  / wind- $5\text{e-}6$  row and a wind- $5\text{e-}5$  row alike, on a layer cell; four judged-distance rows moved onto the new divisor), GREEN 57/0, 194/0, `krylov_and_dogleg` 174/0. No key added (the radii are properties of the flow, constants of `certification.f90`). Advisor: rebuild warning-free, 57/0, 194/0, 174/0 on `build_lwv`; the stale comment in `EXHALE_main.f90` (580) rewritten; `write_output.f90`'s `# adv_input_certified` T branch now carries the reason (a wind-certified state keeps its qualification in the `_adv` header); superseded notes appended to `a2_certification_contract`, `steady_solver_design` and `element_inventory_contexts`; one neighborhood fixed. Reported: whether `steady_gates_met` should read the anchored tolerance is a separate decision on a shared interface.
- DOC1 to DOC3** (the documentation brought to the stage-2 state, 2026-09-10, three concurrent documentation workers, no source touched, no binary run): **DOC1** `EXHALE_BC_and_IC.tex` (29 pages): the whole lower-

boundary chapter described the retired component-wise ghost closure (Dirichlet density, one-way valve, isothermal ghost pressure, four keys that `input_read.f90` now refuses with error `stop`); rewritten as the characteristic  $(p, s)$  face condition of `base_boundary.f90` (branches by face Mach number, the smoothstep of width `base_face_mach_blend`, the Gauss-Legendre ghost averages, the three surviving controls), the upper boundary as `free_outflow_ghost`'s isothermal continuation, a new section on the transported columns' boundaries (N26, N26c, N8a, N29), the checkpoint rebuild order (N16d), and the old test matrix retitled as the retired closure's, not reproducible by today's binary. `composition_restart_and_base_handoff.tex` (18): thirteen base species in the nuclei count (OH, H<sub>2</sub>O, CO added), every `load_IC.f90` line citation re-derived, a new section for the restart contract (intent, the eight-line metadata block field by field with refuse/report on each, provenance unknown, the grid guard, `Restart option change`), the round-trip and projection evidence of N9 and N29 labeled as read from this log. `transmission_spectrum.tex` (11): a new section for `read_adv_validity` (schema 2, the five statuses, `adv_conditional_tol`), the `chord_shell_indices` census with the CONTRIBUTION statement, and the `transit_schema 1` block. **DOC2** `steady_solver_memo.tex` (16): a new section 7 for the species-row solve as it stands (merit on certification scales, the trust region with named outcomes and the leg rules, the constraint rows, the empty-column repair and the equilibrated preconditioner, the three options measured and disarmed, the linear solve as the limiter with the additivity finding, the ulp-level chaos, the residual contract, the certification by regime), eleven hooks added to its table. `molecular_hydrogen_treatment.tex` (23): the `_adv` row corrected (the energy residual reads the upstream caloric energy since N12), the layer table RE-MEASURED on the current `mol_lyman_werner` output (144/184/476 cells, H<sub>2</sub> peak 5.15e12, 1 percent edge 4.48e9 against the stale 227/251, 4.63e13, 4.42e9), thirteen of sixteen cases pinned, a new section on the element budget, the H<sub>3</sub><sup>+</sup> departure factors marked STALE in place, the `mol_ir_bands` peak re-measured (2.148e-8 at  $r$  1.0059 against the stale 2.934e-7). `lower_atmosphere_coupling.tex` (28): a new section 7.4 on the element operator (N29's conservation, N26b/N26c's boundary, N8c's flux spread table as the open item); two pre-existing overfull boxes fixed. `EXHALE_user_manual.tex` (88): certification by regime in section 2.7, the Run mode key entry, eleven hooks, the coupling header example replaced by the line the code writes (`cert_reason`), `Cooling_breakdown.txt` columns 12 to 14 (H<sub>2</sub>, H<sub>2</sub>O, CO infrared) restored to the column list. **DOC3** `README.md` (sixteen-case matrix, 25 suites, the restart contract, the transit metadata, certification stated, nineteen examples), `README_HOWTO.md` (`Resid tol` default 1e-5 not 1e-3, `Flux spread tol` default 2e-5 on the face flux not 5e-3, a certification table, a restart section, an eleven-row hook table, the sixteen cases and 25 suites), the `NEW docs/code_status_20260910.md` (418 lines, replaces the 2026-09-05 file as the read-first document; section 3.3 states what is and is not certified and what limits the rest), the project `CLAUDE.md` pointer line, `TO_BE_DONE.md` (an index of what is open, three headings corrected, 35 em-dashes removed), `steady_solver_design.md` section 22. All eleven PDFs rebuilt, `grep -c '^!' 0` on every log. Advisor: every default and constant quoted checked against the source (`flux_spread_th_default` 2.0e-5, `resid_max` 1e-5, `cert_tol*_wind` 1e-5, `cert_regime_wind_r` 1.20, `ratio_sound_band` 1e-2, `tr_delta_min_dimensionless` 1e-6, the `Cooling_breakdown` header); one constant name corrected in the memo; pre-existing British spellings fixed in the workers' files, in `ISSUES_20260909.md` and in the handoff; the 51 em-dashes of the workspace `CLAUDE.md` replaced. **Reported by the workers and CONFIRMED by the advisor** (ISSUES 3.7): the base

reservoir is stated once at startup and the refreshes of `dp_bc` and `ntot_bc` after every sweep are dead writes (the boundary itself is right: the reservoir equals the user's base pressure by construction); `golden/mol_ir_bands/` carries no `Cooling_breakdown.txt`; `examples/README.md` covers part of nineteen folders under the wrong planet; the `.md` twins of two `.tex` documents have parted from them; the `max(rest, 0)` write-back clamp (N4b) restated. ISSUES section 5 rewritten to the state after N30; the handoff opens with a dated 2026-09-10 state block. The carrier reload is still a scratch recipe (`B5k/mkrun.sh carrier`), not a pinned fixture: to pin with N31.

- **N31** (why the finite-difference Jacobian action is not additive across directions, and two probe options; `steady_newton.f90` +563/-3, of which the arithmetic is two probe rules and the rest is a default-off hook, `krylov_and_dogleg` 174 -> 190 rows, new fixture `backup/regression/carrier_elem_newton`): **the action is not additive because the residual carries a NON-SMOOTHNESS FLOOR five to nine decades above the double-precision rounding of its own rows, and the difference quotient along the PRECONDITIONED Krylov directions divides that floor by an increment five decades smaller than a generic direction produces.** N25's reading (curvature over the fixed arc) is refuted at the iterates that matter and confirmed at one that does not. MEASURED on the atomic element reload with the new hook `EXHALE_JV_ADDITIVITY=1` ([diag 13], at outer 1, 20 and the last of a capped run, on the first two Arnoldi directions and on two directions off the basis, at a tenth, one and ten times the probe arc): the defect of the Arnoldi pair scales as the arc to the power **-1.026** at outer 1 and **-0.999** at outer 40 (rounding), the off-basis pair as **+1.125** and **+2.109** (curvature, and  $3.2\text{e-}8$  and  $2.1\text{e-}8$ , negligible), and at outer 20 both are curvature at cells 1 and 2. The second difference of the residual on the defect-holding rows confirms it directly: ratios 0.774 and 1.676 at outer 1 where a smooth residual gives 4 and 4, and the floor stands **9.5e5** (outer 1) to **2.0e9** (outer 40) times epsilon times the residual on those same rows, so it is not the arithmetic of the assembly. READ, the only tolerance in the atomic path that can set a floor of that size: `ionization_equilibrium.f90` line 969 hands `hybrd1` an `xtol` of `sqrt(dpmpar(1)) = 1.49e-8` (lines 1432, 2110, 2114), a statement about the step and not the residual, so the composition is a piecewise map of the state; `eval_residual` fixes the number of composition passes but not `hybrd1`'s own iteration count. **A decision of the feasible set is ruled out:** on all six probes of the three iterates, components blocked 0, forward side, and the step taken equal to the nominal step. **The localization gives no unknown class to fix:** at outer 1 the squared defect is 0.31 mass, 0.64 energy, 0.03 momentum, 0.02 helium, under  $1\text{e-}4$  in the eight trace elements, and nine tenths of it sits in 35 of the 500 cells between 1 and 75. The scale reading is a uniform shortfall: the largest unknown of the state is displaced by  $2.2\text{e-}8$  of itself (about `sqrt(epsilon)`) and almost every other by  $1\text{e-}10$  to  $6\text{e-}9$ , one to two decades short, with no unknown near a bound (nearest face  $2.9\text{e-}5$  against a displacement of  $4.5\text{e-}14$ ). The model that follows, `defect ~ floor / (eps || Jv ||)`, predicts both pairs at outer 1 within a factor three. **Two options, both default off.** `EXHALE_JV_COLUMN_SCALE=1` (`probe_step_on_the_column_scales`, `eps0 = sqrt(eps)(1 + ||Dc-1 Y||) / ||Dc-1 v||`) is the brief's remedy and it is WORSE, for the reason the diagnosis gives: the Krylov direction's weight sits on unknowns of small column scale, so the rule SHORTENS the arc by 42 and the defect rises  $8.430\text{e-}6$  ->  $1.466\text{e-}4$  at exponent -0.965, the 1/arc law reproduced by a change of the arc made for another reason. `EXHALE_JV_PROBE_ARC=<x>` (`jv_probe_arc_scale`, default 1) is the lever the measurement points at, and it wins the linear question and loses the Newton by the same change: at `x = 100` the defect of the Arnoldi

pair falls  $8.430\text{e-}6 \rightarrow 2.240\text{e-}6$ , the exponent turns  $+1.139$  and the second differences become  $4.065$  and  $4.022$ , so the rounding regime is gone and the quotient is truncation-limited, while the atomic reload then ABORTS at outer 33 (info = 2,  $||R||$  1.504, flux spread  $5.902\text{e-}2$ , 12 steps refused by the model with the actual reduction 26.4 times the predicted, radius  $2.028\text{e-}7$ ) where the control runs 144 iterations to  $3.641\text{e-}4$  and  $5.557\text{e-}6$  with no model refusal. At  $x = 20$  it aborts at outer 43 at 1.524 with 21 model refusals and a radius of  $2.076\text{e-}13$ ; the carrier reload runs its 40 iterations at both values and hands back 1.006 at  $x = 100$  and  $9.076\text{e-}1$  at  $x = 20$ , each with 2 model refusals, against the control's  $9.572\text{e-}2$  with none. **At  $x = 5$ , which does NOT stall**, the binding regime is reached (outer 58,  $||R||$  3.087e-3, element Fe of cell 146, against the control's  $3.386\text{e-}3$  and element Fe of cell 147 at outer 60) and THERE the Krylov cycle still spends 40 of 40 products at a relative residual of 0.898 to 0.980 against the 0.1 asked, indistinguishable from the control's 0.935 to 0.959. **The substance of the item:** the additivity of the linear operator wants a LONGER arc (defect ~ floor / arc) and the trust region's model wants a SHORTER one (truncation ~ arc, and the ray test refuses a step whose model and true slope differ in size), the arc already sits at the Newton's balance point, and every direction was measured to lose, so NO probe rule fixes this. What can is lowering the residual's own floor, which is not double precision. Column-scale named outcomes at cap 40, option against control: atomic  $||R||$  at 40 1.674 against  $7.607\text{e-}2$ , handed back 1.633 against  $5.377\text{e-}2$ , flux spread  $4.985\text{e-}1$  against  $3.555\text{e-}3$ , 30/10 against 32/8; carrier  $||R||$  at 40  $1.357\text{e-}1$  against  $1.070\text{e-}1$ , handed back  $1.263\text{e-}1$  against  $9.572\text{e-}2$ , flux spread  $7.718\text{e-}2$  against  $4.635\text{e-}2$ , 31/9 against 33/7, SAME binding row (carrier H2 of cell 205), no new gm\_no\_direction\_sampled and no refusal of any class in either. The one thing the column scaling improves is the count of Krylov cycles reaching the tolerance asked (40 of 40 against 26 of 40), which is the REDUCED problem's residual on an operator that is not the Jacobian and is precisely why that count must not be read as progress. NEITHER OPTION IS ADOPTED: both fail the first clause of the acceptance rule outright. **Default path unchanged:** with every hook off the atomic reload's 40 iteration lines are identical to the control's to the digit, the carrier reload's binding row and  $||R||$  at 40 are the control's, and wasp\_full\_newton is BITWISE but for the provenance timestamp; with EXHALE\_JV\_ADDITIVITY=1 on, the 40 iteration lines are still identical, so the hook is inert armed as well as disarmed. Tests: krylov\_and\_dogleg RED on the control objects (seven symbols absent), GREEN 190/0; steady\_species\_rows 192/0, certification 57/0, element\_operator 14/0, residual\_determinism 5/1 with the same row failing by design in the control build. **Fixture pinned (deliverable 0):** backup/regression/carrier\_elem\_newton now holds the carrier reload that existed only as a scratch recipe, with run.sh, a README and the control of the unmodified tree at cap 40 ( $||R||$  1.969, 1.984, 1.921,  $6.800\text{e-}1$ ,  $1.070\text{e-}1$  at 1, 5, 10, 20, 40; handed back  $9.572\text{e-}2$ ; flux spread  $4.635\text{e-}2$ ; info = 1; 33 accepted, 7 rejected; 40 of 40 products every cycle; binding row carrier H2 of cell 205; 2754 samples admitted, none refused). A capped run of that fixture never reaches a mass-loss rate. **Defect fixed on the way:** the new probe rule was first written with a local integer  $n$ , which is the grid size  $N$  of global\_parameters under Fortran's case-insensitive rules, so its own size guard fell through on every call; the new suite rows caught it at once and the local is now nunk. Not claimed: that the floor is hybrid1's xtol -- that is READ from the source and not measured, and the experiment that closes it (tighten that xtol, re-measure the floor) belongs to ionization\_equilibrium.f90, which this item does not own. The comment of the suite row the\_probe\_arc\_stands\_above\_the\_residual\_noise was restated: the arc IS nine decades above the

- residual's reproducibility as a length of the whole vector, but the conclusion drawn from it about the quotient does not follow. Advisor: rebuild on build\_lwv (one linker warning, pre-existing), krylov\_and\_dogleg 190/0, steady\_species\_rows 194/0, residual\_determinism one row FAIL by design; atomic reload at cap 40 with every hook off, 8 threads: 1.888, 1.891, 1.819, 1.583, 1.861, 8.964e-1, 7.607e-2 at 1, 5, 10, 15, 20, 30, 40, handed back 5.377e-2, flux spread 3.555e-3, identical to the worker's control; the carrier fixture's README control checked against its own run.log; the fixture README is the reference for every later carrier-reload number (the 0.25 quoted after N26 was a scratch-recipe run, not this fixture). Accepted. **Next item N32:** the residual's floor (the inner solves of the composition and of the temperature stop on step or bracket tolerances, newton\_dense, hybrd1 xtol sqrt(eps), brent\_root 1e-10); measure whether the additivity floor follows them.
- **N32** (which inner tolerance sets the residual's non-smoothness floor; two default-off tolerance hooks and a widened [diag 13], ionization\_equilibrium.f90 +47/-2, T\_equation.f90 +42/-3, steady\_newton.f90 about +130 inside the additivity hook, krylov\_and\_dogleg 190 -> 203 rows): **NEITHER. The floor follows neither the composition solve's stopping tolerance nor the temperature bracket's, and it is not in the chemistry at all: it sits in the HYDRODYNAMIC rows of the residual.** New hooks EXHALE\_IEQ\_TOL=<x> (replaces tol = sqrt(dpmpar(1)) for the run; new function composition\_solve\_tolerance, cached once by ioniz\_eq, announced only when armed) and EXHALE\_TEQ\_TOL=<x> (the bracket of brent\_root, now a module variable at its old 1e-10 through temperature\_bracket\_tolerance). MEASURED on the atomic element reload at outer 1 (cap 40, 8 threads, one entry state for every run): the additivity defect of the Arnoldi pair is 8.174e-6 at exponent -1.063 (default 1.49e-8), 7.452e-6 at -0.995 (1e-10), 9.348e-6 at -1.059 (1e-12) and 9.546e-6 at -1.064 (1e-14), and the second difference on the defect rows stays on its floor at 5.6e-12 to 1.6e-11 with no trend in the tolerance: **the law is FLAT**, over six decades. The measurement is exactly reproducible (three control runs, and 1 thread against 8, give 8.174e-6 and 1.362e-11 to every printed digit), so the +-17 percent between runs is a real movement of the state and not a fall. EXHALE\_TEQ\_TOL=1e-13 is inert to the bit through all 40 iterations, which confirms as MEASURED what was READ, that brent\_root is reached only from post\_process\_adv and never from eval\_residual. **Why the composition cannot be the floor:** quadratic convergence pins its root at the SQUARE of the step tolerance. On the reported cell the accepted  $x(H^+)$  is 2.037266074305163e-12 at the default and 2.037266074305300e-12 at 1e-14, agreeing to 13 significant digits (7e-14 relative), and the pinned fixture's own run.log records an accepted reaction residual of 2.91e-16 at maximum for the steady sweeps; the floor implies a jitter of about 1e-9 relative, five decades above that. The tightest tolerance costs 30 percent more wall time per residual evaluation (0.090 s against 0.066 s at 4 threads, 17 evaluations each) and buys nothing. **Where the floor IS**, from the widened hook: the energy SOURCE, which is where excited\_hydrogen, lya\_rt, the cooling table interpolation, the write-back clamps and comp\_T\_from\_p all end up, is smooth on the arc and five decades below the floor (second difference of heat - cool on the defect cells 3.039e-17, 2.474e-16, 1.046e-15 at half the arc, the arc and twice it, ratios 8.1 and 4.2, textbook h squared, against the residual's own 1.362e-11 on the same rows); by row class over the whole column the floor is mass 1.801e-11, energy 3.090e-11, momentum 5.946e-12 against element He 4.269e-12 and every trace element at 1e-16 to 2e-13; nine tenths of it sits in 29 of 500 cells between cell 1 and cell 58. The mass row carries no source at all, so no part of the chemistry can be its cause. The competing explanation, that the row is a small difference of large terms and

its floor is their rounding, is REFUTED: the largest of heat and cool on those cells is  $3.424\text{e-}3$  against  $||F||$   $5.507\text{e-}3$ , a ratio of 0.62, and epsilon times it is  $7.6\text{e-}19$ . **Deliverable 2 (the stationary-solve-only polish) was NOT BUILT:** the brief conditions it on deliverable 1 naming a tolerance, and none is named. **A correction to the brief and to N31:** on this fixture `solve_ieq` and `newton_dense` never run. The helium triplet is on (the fixture's `EXHALE_setup.out`) and `metals.inp` is present, so the sweep calls `hybrd1(ion_system_HeH_TR_metals, ...)` directly; MEASURED `nt_calls` 0 over the whole hook and no "ioniz-eq solver" line in the fixture's reference run.log. `hybrd1` returns `info = 1` at the default, at  $1\text{e-}10$  and at  $1\text{e-}14$  alike, so there is no `info = 2/3` and no fallback to count, and `newton_dense`'s hardcoded step test  $1\text{e-}11 \times \text{xscale}$  is not in this path. **Default path unchanged:** `wasp_full_newton BITWISE` in `Ion_species.txt` and identical but for the provenance timestamp in `Hydro_ioniz.txt`, every (JFNK) line identical; the atomic reload's 40 iteration lines identical to the control's with every hook off. Tests: `krylov_and_dogleg` RED on the control objects (both new functions absent), GREEN 203/0 with 13 rows added; `steady_species_rows` 194/0, certification 57/0, `element_operator` 14/0, `adv_static_limit` 53/0, `coupled_source_step` 30/0, `physics_probe` 1491/0, `residual_determinism` 5/1 with the same row failing by design as in the control. Noticed, not fixed: the fixture README does not say the helium triplet is on, which is what selects the direct `hybrd1` branch and is what sent this item's brief to the wrong routine; and `newton_dense` applies the caller's tolerance to `fnorm` while its step test uses a hardcoded  $1\text{e-}11 \times \text{xscale}$  no caller can set, which would be the real floor on a fixture that does reach it. **Next item:** the floor is in the flux assembly, not in `ionization_equilibrium.f90` -- the candidates left are the reconstruction where the limiter is not the frozen WENO3 weight, the Riemann wave-speed estimate, `Apply_BC`, and the transport coefficients of the diffusion operator, which are the one remaining way the composition reaches a hydrodynamic row. Advisor: rebuild on `build_lwv`; `krylov_and_dogleg` 203/0, `steady_species_rows` 194/0, `adv_static_limit` 53/0, `residual_determinism` the same single row FAIL by design; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit (1.888, 1.891, 1.819, 1.583, 1.861,  $8.964\text{e-}1$ ,  $7.607\text{e-}2$ ; handed back  $5.377\text{e-}2$ , spread  $3.555\text{e-}3$ ), so the worker's 1.112 at iteration 40 is the thread-count sensitivity of N26c, not a change of the path. The atomic fixture README now states the triplet is on and which branch that selects; `newton_dense`'s hardcoded step test recorded in ISSUES 3.7. Accepted; deliverable 2 correctly not built. **Next item N33:** the jump itself, scanned along the Arnoldi direction across four arcs and split by term and operator (base boundary, reconstruction, Riemann flux, caloric inversion, composition read, transport sources).

- **N33** (where the residual's non-smoothness floor is: the term, the operator and the branch; a default-off multi-scale jump scan [`diag 14`], `steady_newton.f90` +710, `krylov_and_dogleg` 203 -> 220 rows): **THERE IS NO BRANCH. The floor is the ROUNDING OF THE FLUX ASSEMBLY, raised above the last bit of the row by two cancellations that compound.** The Roe flux of a face is built from the JUMPS of the two reconstructed states, and in the nearly hydrostatic base layer those jumps are  $3.2\text{e}5$  to  $5.9\text{e}6$  times smaller than the states themselves, so the interface flux carries the last bit of  $O(1)$  quantities and not of its own value; the row is then the flux difference divided by the cell volume,  $r^2/dV = 5.12\text{e}3$ , one over the  $1.955\text{e-}4$  width of a base cell. The bound `epsilon x (face state) x r^2/dV` is what a correctly rounded assembly cannot go below, and MEASURED it IS the floor:  $1.675\text{e-}12$  against  $1.240\text{e-}11$  at cell 1,  $2.520\text{e-}13$  against  $1.644\text{e-}12$  at cell 84,  $4.326\text{e-}15$  against  $2.366\text{e-}14$  at cell 167,  $4.241\text{e-}17$  against  $4.409\text{e-}16$  at cell 251 -- the bound falling

4.6 decades with the measured floor following it, ratio 5.5 to 12 throughout, and the ten cells holding the largest second difference are cells 1 to 23, every one a base cell. Beyond cell about 250 the bound falls below the residual's own curvature and stops binding, which is what a floor does. New hook EXHALE\_RESID\_JUMP\_SCAN=1 samples F at 65 points of  $Y + t \cdot v$  on ten windows, each a quarter of the one before, from two probe arcs down to two arcs over  $4^9$ ; one window is not enough, because at the standard arc every interval already holds thousands of steps and none stands above its neighbors. **The measurement:** on the atomic element reload at outer 1 the tracked row is the energy row of cell 1, its median first difference falls by exactly four whenever the spacing does (the smooth part is a straight line), and its second difference does not fall at all --  $1.0e-11$  to  $1.6e-11$  over 5.4 decades of spacing,  $1.8e-5$  down to  $6.9e-11$  -- which is a step of F and not a kink or curvature. Isolated steps appear once the smooth increment per interval falls below the floor: 28 of 64 intervals at  $3.05e-5$  arcs and 61 of 64 at  $7.63e-6$  arcs, at least  $4.0e6$  stepping intervals per probe arc, the largest moving its row by  $2.625e-12$ ,  $2.749e-9$  of the row. **The term and the operator,** at the largest step (cell 9, mass row): S of the mass row is identically zero and the whole step is in dF; every quantity upstream of the flux -- the ghosts, T,  $n_{tot} + n_e$  and the four reconstructed face states -- moves by one to eight units in the last place of an O(1) number with no outlier, in exact multiples of  $2.220e-16$ ; the first quantity that steps is the interface flux, by  $4.803e-16$  where its neighbors move by  $3.2e-17$ , and  $4.803e-16 \times 5116.3 = 2.457e-12$  against the row's measured  $2.625e-12$ . **No branch fires anywhere near it:** faces falling back from Roe to HLLE 0 and unchanged, face states the positivity limiter scaled 0 and unchanged, so the base boundary's Newton, characteristic\_branch\_weight, the Mach cap, the ghost quadrature, the stencil selection, the wave-speed branch and comp\_T\_from\_p are all continuous here. The off-basis direction carries the SAME floor,  $9.1e-12$  to  $1.4e-11$  flat over 5.4 decades, and isolates no step at any window, because its smooth increment never falls below it: the floor is a property of the residual and not of the direction, and what the preconditioned direction does is shrink the numerator of the difference quotient by five decades. The carrier reload reproduces the law (floor  $1.532e-11$ , bound  $1.382e-12$ , ratio 11.09), so it is the discretization at the base and not one fixture. **N31 corrected:** the floor's size stands, but "five to nine decades above the rounding of its own rows" compared it against  $\epsilon ||F||$  on the row, which is the wrong reference for a row built as a flux difference over a cell width; against the right one it is a factor of ten. N32's localization is confirmed and made quantitative. **Deliverable 2 NOT BUILT:** the brief conditions the deliverable on a branch being named and there is none. The treatment the measurement does name, for the advisor, is a WELL-BALANCED flux difference at the base -- assemble the rows from the departure of the face states from the local hydrostatic isentrope, which base\_boundary.f90 already integrates, so the balance cancels analytically instead of in floating point -- in Num\_Fluxes.f90, RK\_rhs.f90 and Source.f90, none of them this item's files; the cheaper partial lever is the base cell width, on which the floor depends as one over it. **Default path unchanged:** the atomic reload at cap 40 gives all 338 (JFNK) lines identical to the control's through done info=1  $||R|| = 1.112E+00$ ; with the scan armed the outer-1 line is the control's to every digit; wasp\_full\_newton BITWISE in Ion\_species.txt, Ion\_species\_adv.txt and the three breakdown files, differing only in the provenance timestamp in the two hydro files, log10 Mdot 13.30 on both. Tests: krylov\_and\_dogleg RED on the control objects (resid\_jump\_scan\_on absent), GREEN 220/0 with 17 rows added; steady\_species\_rows 194/0, element\_operator 14/0, certification 57/0, residual\_determinism

5/1 with the same row failing by design and the same measured  $3.354e+01$  on the control binary. Noticed, not fixed: `global_parameters` declares `integer :: count`, which shadows the intrinsic `COUNT` in every scope that uses the module, so `count(mask)` written anywhere fails to compile -- the same class as the `t0` and `g2s` incidents and worse in that the shadowed name is an intrinsic; renaming a public global was out of scope. Advisor: rebuild on `build_lvw`; `krylov_and_dogleg 220/0`, `steady_species_rows 194/0`; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit ( $7.607e-2$  at 40, handed back  $5.377e-2$ , spread  $3.555e-3$ ), no [diag 14] line; two British spellings in the new comments fixed. Accepted; deliverable 2 correctly not built (there is no branch to smooth). The finding is structural and reaches the marching discretization, so it is put to the user as DECISION 23 (well-balanced flux differencing recommended; quad accumulation of the flux difference as the control experiment; coarser base cells; accept the floor). The count global shadowing the intrinsic recorded in ISSUES 3.7 (HYG). **Next item N34** (runs without a decision, changes no default): the control experiment, quad accumulation of the face fluxes and their difference inside the stationary residual only, as a measured option.

- **N34** (the control experiment of decision 23: the hydrodynamic rows of the stationary residual in quadruple precision). `EXHALE_RESID_QUAD=1` sends the STATIONARY evaluations of `assemble_residual` -- the tag `ieq_sweep_state_kind` puts on an iterate or a probe; the marching stages reach `RK_rhs` directly and are untouched -- through a quadruple-precision instantiation of the whole flux assembly, from the conserved state through the caloric EOS, the PLM limiter and the WENO3 weights, the Roe flux with its entropy fix and HLLE fallback, the geometric and gravitational sources and the division by the cell volume, rounded to double once, at the row. One source text (`hydrodynamic_rows_body.inc`, 1111 lines, kind-generic) instantiated twice by `hydrodynamic_rows.f90`; the composition, the heating and the cooling stay double and enter converted exactly. **The double instantiation is the production operator BIT FOR BIT**, asserted three ways: a suite row on a stated grid (largest difference exactly zero in `dF`, `S` and the four face states, tolerance 0), and two whole 40-iteration reloads under `EXHALE_RESID_QUAD=2` -- the atomic (all 355 (JFNK) lines identical) and the carrier, which is the molecular one and so exercises the generic caloric EOS (all 364 identical). The quadruple instantiation reaches the analytic mass row of a uniform state to  $1e-30$  before its rounding, where the double one stands at its own. **Verdict: the floor is confirmed and is NOT removed.** On the atomic reload at outer 1 the tracked row's non-smoothness floor falls only  $1.240e-11$  to  $5.917e-12$ , and cell by cell it lands on `epsilon_DOUBLE`  $\times$  face state  $\times r^2/dV$  -- ratio 0.93 to 1.4 over the base cells against 5.7 to 8.9 in the control -- and not on the quadruple bound, which is  $1.224e-30$ ,  $4.8e18$  below what is measured. So N33's amplifier (the face cancellation  $3.2e5$  times the  $r^2/dV$   $5.1e3$ ) is confirmed a second way and its attribution completed: four fifths of the floor is the arithmetic INSIDE the assembly, the rest is the double representation of what the assembly is handed and hands back -- the ghosts `Apply_BC` writes, the sweep's temperature and particle count, and the double row itself -- which precision inside one operator cannot reach. The detector isolates no step at any of the ten windows (control: 28 of 64 and 61 of 64 at the two narrowest), but the second difference is still flat over 5.4 decades and the row still changes sign 39 times in 64 intervals: a staircase with a smaller rise. **What the quadruple-precision accumulation does buy is the fidelity of the Krylov model:** the additivity defect falls  $8.174e-6$  to  $1.009e-6$  with the exponent in the arc still negative ( $-1.063$  to  $-0.901$ , so still rounding and not curvature), and at the binding iterate of the atomic reload at cap 40 the relative gap of the Arnoldi image falls  $1.188e-1$  to  $3.813e-3$ , a factor 31, the returned relative residual

then being the true one to four digits (0.2114 against 0.2114; the control returned 0.1122 for a true 0.1617). The element rows, which do not pass through the assembly, are unmoved to four digits (C 1.367e-13 to 1.366e-13), which is the control inside the control. **The Newton does not converge and the Krylov cycle still stalls.** Atomic cap 40, quad accumulation on: 40 of 40 products at 0.21 against the requested 0.10,  $||R||$  8.413e-1 handed back against the control's 1.112 with trajectories parting at iteration 9 (the solve is chaotic at the ulp level, so that is not by itself a result). Atomic cap 250: both stop on the stagnation detector, control at iteration 167 at 3.047e-4, quad at 136 at 4.096e-4, both with 40 of 40 at 0.983 to 0.989, both binding on the element Fe rows of cells 132 to 137, both NOT CERTIFIED with 11 entries of the inventory refusing. Carrier cap 40: unchanged in every named outcome -- binding row the H2 carrier row of cell 205 throughout, 40 of 40 at 0.9886 against 0.9729,  $||R||$  9.336e-2 against 9.427e-2 -- as it must be, since that row is assembled by carrier\_steady\_residual and never sees the flux. Cost MEASURED: the flux assembly alone 46x at 8 threads and 74x at one (500 cells, 3.628e-5 s to 1.681e-3 s), the whole atomic reload at cap 40 1.11x (271 s to 302 s), because a stationary residual is mostly the ionization sweep. **Nothing adopted, no default changed:** wasp\_full\_newton bitwise but the provenance timestamp, log10 Mdot 13.30 on both; the atomic and carrier reloads identical line for line with the option off. Tests: krylov\_and\_dogleg RED on the control objects, GREEN 232/0 with 12 rows added; steady\_species\_rows 194/0, element\_operator 14/0, certification 57/0, energy\_update 53/0, acceptance\_classes 22/0, adv\_static\_limit 53/0, species\_face\_flux 1/0, spectrum\_type 154/0, grid\_and\_gates 146/0, attempted\_step 70/0, coupled\_source\_step 30/0, residual\_determinism 5/1 with the same row failing by design and the same 3.354e+01 on the control binary. One file outside the brief's list: caloric\_eos.f90 gained three read-only accessors (the mixture ratios and the H2 rovibrational table nodes) with no arithmetic and no behavior change, because the generic pipeline cannot evaluate the molecular caloric EOS without them; the molecular whole-run bitwise comparison is the evidence that they changed nothing. Noticed, not fixed: backup/regression/atomic\_elem\_newton/README.md still names the helium row of cell 246 as the binding row at cap 250 from N26/N27, where the measurement on this tree is the element Fe rows of cells 132 to 137 at iteration 167 and  $||R||$  3.047e-4. **For decision 23:** extra precision confirms the diagnosis and does not cure it; the amplifier has to go (well-balanced flux differencing, or a coarser base cell, which buys one factor of the width), and even with the hydrodynamic rows made model-faithful the two rows that actually bind -- the element rows of the atomic reload and the H2 carrier row of the carrier reload -- are not hydrodynamic rows and are untouched by any of it. Advisor: rebuild on build\_lwv (the new hydrodynamic\_rows.f90 in SRC); krylov\_and\_dogleg 232/0, steady\_species\_rows 194/0, energy\_update 53/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit (7.607e-2 at 40, handed back 5.377e-2). The caloric\_eos.f90 accessors (read-only, outside the brief's list) accepted on the molecular reload's bitwise evidence. Accepted; nothing adopted. Decision 23 restated with this result: option (b) retired as a remedy, (a) still the right discretization but not by itself the cure of the species-row solve; the atomic fixture README gained a 2026-09-10 state section (binding rows Fe of cells 132 to 137 at cap 250 on this tree). **Next item N35** (runs without a decision): the conditioning of the preconditioned operator on the element and carrier rows at the binding iterate, and what the banded preconditioner misses there.

- **N35** (what holds the Krylov cycle on the species rows once the rounding floor is removed: the conditioning of the

preconditioned operator and what the banded preconditioner misses). Three default-off measurement hooks on the linear system of one outer iteration, adopting nothing: EXHALE\_KRYLOV\_SIZE\_SCAN=1 ([diag 16]) reruns the SAME cycle at 40, 80, 160 and 320 products and the last two again with the basis orthogonalized twice, with EXHALE\_GM\_HISTORY=1 printing the reduced residual after every product and the true one every twenty; EXHALE\_PRECOND\_SPECTRUM=1 ([diag 15]) takes 200 Arnoldi products of  $A_z M^{-1}$  from a deterministic start, diagonalizes the Hessenberg (dgeev) and reports the Ritz values of the whole operator and of its compressions onto the species and the hydrodynamic rows, with the localization of the three smallest Ritz vectors; EXHALE\_BAND\_DIFFERENCE=1 ([diag 17]) measures  $(A - A_{\text{band}}) v$  on those Ritz vectors and on the first two Arnoldi directions, splits the columns they live on into the part inside the band, the band's disagreement there and the part outside, with the radiation live and frozen, and reads the binding row's band row by class as a fraction of its diagonal. **Verdict: the band misses nothing, and the operator itself is near-singular on the species rows.** Its action agrees with the full finite-difference one to  $4.0\text{e-}5$  to  $1.5\text{e-}3$  relative on every direction tested on both fixtures; the columns hold 99.4 to 100.0 percent of their norm inside the band (the largest outside part is 0.6 percent, element He of cell 308); and what remains is not on the rows that bind at all, but 0.999 in the momentum row of the LAST cell (carrier) and 0.83 to 0.89 momentum plus 0.11 to 0.17 energy in five to seven cells (atomic), the element and carrier rows taking 0.0000 to 0.003 of it. So the two remedies the brief conditioned on -- a rank-few radiation correction by the Woodbury identity, or a wider  $kl\_jac$  -- would each repair something already right, and **nothing was built. What IS wrong, MEASURED at the binding iterate of both fixtures.** (1) The preconditioned operator is well conditioned on the hydrodynamic rows and near-singular on the species rows: the Ritz spectrum of the compression onto the hydrodynamic rows has a magnitude ratio 43.4 (carrier, carrier H2 of cell 205 binding) and 158 (atomic, energy of cell 52 binding), none and three of 200 below  $1\text{e-}2$ ; onto the species rows,  $6.56\text{e}5$  (carrier, smallest  $1.493\text{e-}6$ , nineteen below  $1\text{e-}2$ , one below  $1\text{e-}4$ ) and  $1.06\text{e}4$  (atomic, smallest  $9.366\text{e-}5$ ). The smallest Ritz vector is 0.71 to 0.91 carrier rows on the front cells 199 to 205 and 458 to 466 (carrier) and 0.9955 element rows over 145 cells (atomic), and the carrier fixture's near-zero part is a complex pair, not a real null direction. The attribution is there from outer 1 ( $2.17\text{e}5$  against 39.6). And the discretization statement behind it: the band row of carrier H2 of cell 205 has diagonal  $1.2396\text{e-}1$  against off-diagonal magnitudes  $4.1026\text{e}2$ , **3.304e3 times the diagonal to the hydrodynamic unknowns of its own stencil and 5.65 times to the neighboring carriers**, so the binding row's own unknown is three decades below its coupling to another class and there is no diagonal for a preconditioner to stand on. (2) **A larger subspace makes the step worse, not better**, which is neither of the brief's two alternatives. The reduced least-squares residual falls monotonically with the size while the TRUE residual of the returned step stops falling at 60 to 80 products and rises: atomic with quad accumulation on, reduced  $2.658\text{e-}1$ ,  $1.781\text{e-}1$ ,  $1.144\text{e-}1$ ,  $9.958\text{e-}2$  at 40, 80, 160 and 182 products against TRUE  $2.851\text{e-}1$ ,  $2.575\text{e-}1$ ,  $3.006\text{e-}1$ ,  $3.203\text{e-}1$  -- the cycle announces its tolerance reached at 182 and hands back a step leaving 0.32, three times what was asked; carrier, reduced  $9.677\text{e-}1$ ,  $8.001\text{e-}1$ ,  $4.639\text{e-}1$ ,  $1.434\text{e-}1$  against TRUE  $9.731\text{e-}1$ ,  $8.638\text{e-}1$ ,  $9.668\text{e-}1$ , **1.472e+00**. With quad accumulation off the true residuals are a factor 3 to 6 worse at every size (atomic  $7.856\text{e-}1$ ,  $3.905\text{e-}1$ ,  $7.580\text{e-}1$ ,  $2.004\text{e+}00$ ) and the shape is the same. Reorthogonalizing the basis changes nothing at any size, the measured loss of orthogonality being  $0.000\text{e+}00$ , and the Arnoldi image of each column at the same iterates is

exact to 3.30e-16 and 3.77e-16: the recursion is not the defect. What fails is additivity -- the action of the sum of the first two directions departs from the sum of their actions by 1.357e-3 (atomic) and 7.601e-5 (carrier), the assembled 40-product step's image by 1.068e-1 and 2.256e-1 -- because the matrix-free action is the secant of a nonlinear residual over a fixed arc and the step it is extrapolated to grows from 0.31 to 11.4 while the arc stays 5.2e-6. Tests: krylov\_and\_dogleg 251/0 with 19 rows added (RED on the control objects, which do not carry the routines), among them the Ritz values of a stated upper triangular 6 by 6 spanning six decades reproduced to 1.171e-12 with the recursion carried to the dimension of the space, the column split summing to the whole difference with an empty band and with an exact one, and the four hooks off by default; steady\_species\_rows 194/0, element\_operator 14/0 (the three suites whose driver links steady\_newton). Default path: the atomic reload at cap 40 with every hook off, new binary against my control, all 354 (JFNK) lines IDENTICAL through ||R|| 5.377E-02, flux spread 3.555E-03, the N31 ladder to the digit; wasp\_full\_newton bitwise but the provenance timestamp, log10 Mdot 13.30 on both. Cost MEASURED: the size ladder is 1082 products per diagnostic iteration (the carrier reload at cap 40 from about 3.5 to about 9 minutes at 8 threads), the spectrum 600 and the band difference about 15. Noticed, not fixed: the frozen-radiation column of element He of cell 410 carries 7.267e-4 of its 2.816e-1 norm outside the band where the WENO3 stencil says it should carry none, 0.26 percent and two decades below anything in this item's conclusions; the likely route is the base boundary state recomputed from the iterate, not confirmed. Also corrected in the reading: the preconditioner is build\_banded\_jac\_full, a coloring of the FULL residual, not of the frozen one -- build\_banded\_jac on frozen\_residual is called only by the standalone check in EXHALE\_main.f90 -- so "inside the band" carries the coloring's contamination by the non-local response, and the frozen column is a separate control rather than the band's own approximation. **For the advisor:** the next question is not the preconditioner. The species rows' row scaling (the binding row's diagonal 3.3e3 below its hydrodynamic coupling is set by the certification row scale of decision 20 a divided by the column scale, and using one scale for the merit and another for the model is a decision); the cycle has an optimal length around 60 to 80 products on both fixtures and the step control asks for a tolerance the operator does not honor; and gm\_resid\_rel is verified against the operator only at a breakdown or a rank loss, where the measurement says it needs verifying whenever the cycle ran more than a few tens of products. Advisor: rebuild on build\_lwv; krylov\_and\_dogleg 251/0, steady\_species\_rows 194/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit (7.607e-2 at 40, handed back 5.377e-2). Accepted; deliverable 4 correctly not built. The two levers named are the LINEAR model's row scaling (a freedom decision 20 a did not fix: the merit and gate keep the certification scales, the linear inner product may differ) and a Krylov cycle that returns the step its TRUE residual chooses (the optimum lies at 60 to 80 products; the reduced residual lies past it). **Next item N36:** both as measured options, default off.

- **N36** (the two levers N35 named: the row scaling of the LINEAR model, separate from the merit's, and a Krylov cycle that returns the step its TRUE residual chooses). Two default-off options on the linear solve, adopting nothing. EXHALE\_MODEL\_ROW\_EQUIL=1 (unit\_infinity\_norm\_row\_scaling\_of\_the\_band) forms the diagonal E that brings every row of the banded model to unit infinity norm as a power of two, applies it to the band that is factored and, inside pgmres, to the right-hand side and to every action of the operator, and maps the image and the relative residual BACK to the certification scales at every exit, so that the merit, the gate, the trust region's

predicted decrease and the certification keep the scales decision 20 a fixed and only the linear model's inner product changes; ab itself is untouched, so the merit's gradient is where it was. EXHALE\_GM\_TRUE\_RESIDUAL=1 makes the cycle measure its candidate step against the operator from 20 products on, every 10 and at every point where it would otherwise end, keep the best, and stop on two consecutive rises or on the tolerance IN THE TRUE NORM; what comes back is that step, its own image from the product that measured it, and the true residual as gm\_resid\_rel, under a new eighth outcome gm\_true\_residual\_turned. **Verdict: both options work as specified, both fail the acceptance, both stay off.** (1) **The row scaling is a real freedom and taking it is worse.** The cycle then minimizes a functional the merit does not read, and the two disagree by a factor 1.2 to 673 MEASURED ([row equil], equilibrated against certification: atomic 9.956e-2 / 4.111e-1 at outer 2 and 9.937e-2 / 1.218e-1 at outer 40; carrier 2.353e-1 / 5.730e+1 and 1.286e-1 / 8.657e+1), so the cycle announces the 1e-1 it was asked for while handing back a step that leaves 86.6. It cannot touch what binds: the preconditioned operator  $A_{eq} M_{eq}^{-1} = E (A B^{-1}) E^{-1}$  is SIMILAR to the unequilibrated one, and the measurement at the same iterate agrees (hydrodynamic Ritz ratio 39.62 against 39.60 carrier, 57.02 against 56.36 atomic; the count of species-row Ritz values below 1e-2 unchanged at 18 and 2), while the banded factorization is bit for bit the same matrix because equilibrate\_the\_preconditioner already equilibrates both sides; and a row scaling leaves every ratio WITHIN a row where it was, so the binding carrier row's diagonal 1.2396e-1 against 4.1026e2 of coupling, 3.304e3 times the diagonal to the hydrodynamic unknowns of its own stencil, is exactly N35's number still. What it costs: the carrier reload at cap 40 goes from  $\|R\|$  9.572e-2, info=1, binding carrier H2 of cell 205 to 1.955, **info=2 with four entries of the inventory refusing**, and the true residual of the returned step at 40 products from 0.973 to 1.050e2. (2) **The true-residual cycle is the right instrument and it does not turn the solve.** At the binding iterate it caps the cycle where the step stops improving: on the ladder 40/80/160/320 the control's returned step gets WORSE with size (atomic TRUE 2.851e-1, 2.575e-1, 3.006e-1, 3.203e-1; carrier 9.731e-1, 8.638e-1, 9.668e-1, **1.472e+00**) while the true-residual cycle stops at 80 and at 120 products and returns 2.660e-1 and 8.244e-1, never worse than the control's best at any size and at fewer products. Two cycles of the atomic reload at cap 40 name the new outcome. But the best true residual any subspace reaches is 0.27 and 0.82, the Newton does not descend past its plateau, and  $\|R\|$  handed back at cap 40 is 1.060 against the control's 5.377e-2 (atomic; the trajectory parts at iteration 15, the first cycle to pass 20 products) and IDENTICAL on the carrier, where a subspace of 40 always makes the last candidate the best and the option changes only the number reported (true 9.870e-1 against reduced 9.839e-1). **Acceptance, by the named clauses:** the true residual below 0.1 within 80 products on both fixtures fails for both options (option 1 atomic 8.79e-2 but carrier 7.51e+1; option 2 atomic 2.66e-1, carrier 8.64e-1), and the refusal count at cap 40 does not fall by half (atomic 9 to 10 for both options; carrier 8 to 7 and 8 to 8), with no new refusal class and no new gm\_no\_direction\_sampled anywhere. Corrected mid-item and re-measured: the first true-residual cycle left the reduced tolerance in place as a stop and so discarded a step it never measured (carrier outer 1, the 80-product cycle ended at product 49 and handed back product 40's step at TRUE 1.259e-1 where product 49 reaches 9.724e-2); with the reduced residual no longer ending a cycle, the numbers above are the corrected cycle's. Tests: krylov\_and\_dogleg RED on the control objects (nine symbols absent), GREEN 264/0 with 13 rows added, among them the equilibrated solve returning the same step as the unequilibrated one to 9.90e-16

on a stated dense system solved over the whole space, the equilibrated cycle reporting the CERTIFICATION residual of a truncated solve to eight digits, and a stated operator whose action stops being one linear map after 40 products, on which the cycle names `gm_true_residual_turned` and stops before its subspace is spent; `steady_species_rows` 194/0, `element_operator` 14/0, `residual_determinism` 5/1 with the same row failing by design and the same measured  $3.354e+01$ . **Default path:** the atomic reload at cap 40 with every hook off, all 354 (JFNK) lines IDENTICAL to my control through  $||R||$   $5.377E-02$ , `flux` spread  $3.555E-03$ ; the carrier reload all 364 identical; `wasp_full_newton` bitwise but the provenance timestamp, `log10` `Mdot` 13.30. With the options ON `wasp_full_newton` stays CERTIFIED: option 1  $||R||$   $3.043e-9$  against  $1.667e-9$  with a largest relative movement  $5.2e-10$  in the output, option 2 bitwise in `Ion_species.txt` with 8 extra residual evaluations, its cycles never reaching the 20 products the needs. **For the advisor:** what the two options rule out is the linear algebra. The norm the cycle minimizes is free and wrong to change; the length of the cycle is free and already at its optimum; the operator's species block is near-singular at every scaling and the step it can produce leaves 0.27 to 0.82 of the residual whatever the subspace. The remaining lever is the one N35 named and N34 confirmed the arithmetic cannot reach: the DISCRETIZATION of the rows that bind. Advisor: rebuild on `build_lwv`; `krylov_and_dogleg` 264/0, `steady_species_rows` 194/0; atomic reload at cap 40 with every hook off, 8 threads: the N31 ladder to the digit ( $7.607e-2$  at 40, handed back  $5.377e-2$ ). Accepted; both options stay off. **The linear algebra is exhausted as a lever** (N31 to N36: probe rule, inner tolerances, extended precision, band coverage, row scaling of the linear model, cycle length). STOPPED here for the user: decision 23, and the next question is the discretization of the species rows that bind at the front (the carrier row of cell 205 whose own diagonal is  $3.3e3$  below its hydrodynamic coupling). `code_status_20260910.md` section 5 task 1, ISSUES 3.1 and 5, the handoff and decision 23 restated accordingly.

- **N37** (decision 23 (a): a well-balanced flux difference for the near-hydrostatic layer, as the default-off option `Well balanced:`). With the key on the reconstruction, the Riemann jumps and the pressure force carry the DEPARTURE from the cell's own local hydrostatic equilibrium instead of the state: within cell  $j$  that equilibrium is the constant-density one through its own  $(\rho_j, p_j)$ ,  $p_{eq,j}(r) = p_j - \rho_j (\phi(r) - \phi(r_j))$ , which assumes no thermal, entropy or compositional stratification and so preserves ours (Kaeppli and Mishra 2016, A&A 587, A94, sections 2.1.1 and 2.1.3; the momentum source as the face-pressure difference of the cell's own equilibrium is their 2014 paper, J. Comput. Phys. 259, 199, eq. 2.26). The pressure stencil acts on the neighbor's pressure measured against that equilibrium continued THROUGH the face (this cell's density up to the face, the neighbor's beyond it), which is what vanishes on the discrete equilibrium; the two cell pressures are differenced first, while that is exact, and the hydrostatic terms added to the difference. The momentum row is then assembled from the face pressure measured against the same equilibrium and source returns  $S(2) = 0$ : with  $A+ P_{up} - A- P_{dn} - (A+ - A-) p_j = -\rho_j [A+ (\phi_i(j) - \phi_c(j)) + A- (\phi_c(j) - \phi_i(j-1))]$  identically, the equilibrium's flux difference and its gravitational source cancel in the ALGEBRA and the cell's own  $O(1)$  pressure never enters the row. Mirrored in the kind-generic copy, so EXHALE\_RESID\_QUAD=2 is the production operator bit for bit under both key values (suite row, largest difference exactly zero, tolerance 0). **Exact preservation, MEASURED:** on the scheme's own discrete equilibrium (the isothermal solution of the face-matching condition, the spherical form of their eq. 42) the momentum row falls from  $7.0e-5$  (PLM) and

2.5e-3 (WENO3) of the local weight  $\rho_g$  at  $N = 250$  to 4.1e-14 and 3.9e-14, and stays at 4.5e-14 to 5.5e-14 at  $N = 500, 1000, 2000$  while the base scheme falls at its design order; the mass and energy rows reach 1e-16 against 1e-8. Twelve new assertions in `hydrostatic_residual`, plus eleven rows in `krylov_and_dogleg` (with no gravity the well-balanced option is the base scheme to 8e-16 on a uniform state and to 1.4e-16 on a Sod tube, for both reconstructions; and the same exactness on a second grid and geometry). **What it does NOT reach, and this was written into the design before the code:** N33's non-smoothness floor does not move (energy row of cell 1, 1.009e-11 with the option against 1.046e-11 without; cells 84, 167, 251 the same to a factor 1.3; six to ten times  $\epsilon \times \text{face state} \times r^2/dV$  either way), because a cell pressure is itself a rounded  $O(1)$  double and every jump built from two of them steps by its last bit whatever the grouping. What does move is the additivity defect at ten times the probe arc, 2.410e-7 with the option against 3.151e-5 without. **The stationary solves are worse with the key on, and the reason is not the option:** `store_row_terms` scales the momentum row by  $\max(|dF(2)|, |S(2)|, |Smom|)$ , the option returns  $S(2) = 0$ , so the row IS its own scale and every normalized momentum residual is exactly 1 (MEASURED on the carrier reload as loaded:  $k=2$  is 1.000000E+00 with the option against 1.954 without, while the mass row is the same number to seven digits and the energy row to four). The merit then carries no momentum information:  $||Fs||_2$  rises from 16.5 to 28.1 at the same state, and  $28.1^2 - 16.5^2 = 517$  is the 500 momentum rows pinned at unity. The atomic reload at cap 40 reaches 1.415 against the control's 1.112 and the carrier reload 1.732 against 9.34e-2; at cap 250 the option's solve aborts at iteration 61 on "no descent step in 12 consecutive iterations" with  $||R||$  and the worst scaled row both exactly 1.000 and that row the MOMENTUM of cell 1, which is what a merit whose momentum rows are all pinned at one looks like from inside the line search. `wasp_full_newton` with the key on says the same from the other side: it is NOT CERTIFIED on exactly one entry, the momentum row at 1.000E+00, while its flux spread is 1.539e-15 against the control's 6.233e-14, its energy row 9.911e-12 against 1.667e-9, its mass row 1.228e-13 against 1.453e-13 and  $\log_{10} Mdot$  13.30 either way, the profiles moving by at most 1.2e-4 (velocity). **The mechanical column drains by the same amount with the option as without** (`hydrostatic_column`,  $\max |v|/c_s$  0.796 against 0.792 at 300 steps and 2.776 against 2.777 at 3000): that case's state is not the scheme's equilibrium and its drain is set by the outer boundary, as B4 recorded. **Default path:** with the key absent or False every path is bit for bit my control (`wasp_full_newton` reload, the atomic reload's 196 (JFNK) lines, `hydrostatic_column` at 300 steps, `mol_carrier` and `mol_diffusion` at their `maxsteps`); the one line that moves in every output file is the # options field, which now carries the 21st token `wellbal`, nameable in `Restart` option change. `EXHALE_resolved.out` and the parse dump each carry one new line. Key documented as K46. **For the advisor:** the option is kept in the tree and stays off; before it can ever be default on, the momentum row's reference scale has to become the pressure force the option cancels analytically,  $|\rho_j (A^+ (\phi_i(j) - \phi_c(j)) + A^- (\phi_c(j) - \phi_i(j-1)))|/dV$ , which is two multiplications from what `RK_rhs` already holds but sits in an acceptance interface this item does not own. Advisor: rebuild on `build_lwv`; `krylov_and_dogleg` 279/0, `grid_and_gates` 162/0 (four rows FAIL when the shell rows are pointed at the stale shared `EXHALE.x`: pass `EXHALE_EXE` with `EXHALE_OBJDIR`), `steady_species_rows` 194/0, `certification` 57/0, `element_operator` 14/0, `residual_determinism` the same single row FAIL by design; atomic reload at cap 40 with every hook off and the key absent, 8 threads: the N31 ladder to the digit (7.607e-2 at 40, handed back 5.377e-2). The regression

comparison strips comment lines, so the new wellbal token in the # options header is invisible to make check. Accepted as a default-off option; the momentum-row reference scale in store\_row\_terms (the option's S(2) = 0 makes the row its own scale) is recorded in ISSUES 3.7 as the prerequisite of any default-on decision; the publisher PDFs of the two Kappeli and Mishra papers remain wanted (the institutional versions were read in full).

- **N38** (the discretized element and carrier rows at the front: why the row that binds is nearly independent of its own unknown). One default-off measurement hook, adopting nothing. EXHALE\_FRONT\_ROW=1 ([diag 18]) takes the species row that carries the largest scaled residual and its two neighbors and prints, at the selected outer iteration: every TERM of the row on its certification scale, with the advective term split per FACE by masking the face mass flux (the divergence of a face species flux is exactly zero where the mass flux it multiplies is, so the masked call leaves one face standing alone and the SAME operator is used, never a copy); the reconstructed face fraction read back out of that masked divergence beside the donor cell's own; the chemical source of a carrier from the sweep's own rates; the band's entries for the row, to its own unknown, to the same unknown in the neighbors and to every hydrodynamic unknown of the three cells, each as a number and as a multiple of the diagonal; those same entries SPLIT BY TERM, by one-sided differences of the row's own terms along the same column with the same step build\_banded\_jac\_full uses; the cell Peclet number, the logarithmic gradient and the chemical against the advective time scale; and the column scale of every unknown of the binding cell beside the unknown itself. **Verdict: the binding row's diagonal is NOT small -- it is the size of the row -- and the 3.3e3 of N35 is a CHOICE OF COLUMN SCALE that the Krylov cycle cannot see.** (1) **Every hydrodynamic entry of the row is its ADVECTIVE term**, on all three fronts measured: carrier H2 of cell 205 (carrier reload, outer 40) momentum of cell 205 -7.40478e+1 of which the diffusive part is 2.0e-5 and the chemical -2.3e-5; element He of cell 191 (atomic, outer 1) momentum of cell 190 -1.86758e+4 with a diffusive part 1.9e-6; element Fe of cell 137 (atomic cap 250, outer 140, one of the rows N34 named) momentum of cell 136 -1.25980e+4 with 2.0e-6. Summed over the reported stencil the hydrodynamic columns carry **3389, 4176 and 2424 diagonals** and the neighboring species unknowns 5.855, 1.052 and 1.005, which is N35's 3.304e3 and 5.646 reproduced. Each species sub-block is meanwhile a clean diffusive tridiagonal (15.385/-7.772/-7.612 helium, 17.432/-8.861/-8.564 iron): N27's reading of an interior element row holds at the front too, and it is the wind columns that bury it. (2) **The size of that coupling is one over the Mach number**, because the momentum column scale is  $|\rho v| + \rho c_s$ , the SOUND speed in a deeply subsonic layer: MEASURED scale over unknown **3.839e+2** (carrier front, momentum 6.83274e-5 against scale 2.62332e-2), **1.361e+5** (helium front) and **5.149e+4** (iron front), against 1.000 for the mass, the energy and every species unknown. The arithmetic closes to a factor 1.4 to 1.6: one face carries -0.5322 of the carrier row and  $383.9 \times 0.5322 = 204$  against the measured 129; one face carries -0.1969 of the helium row and  $1.361e5 \times 0.1969 = 2.68e4$  against the measured 1.87e4. (3) **The three other candidates are measured and are not the cause.** The species column scale's reservoir floor is INACTIVE everywhere it was looked at (the carrier unknown is  $\ln n$  with column scale exactly 1.0, the default; the helium and iron unknowns' scales are their own magnitudes to five digits), so raising or lowering element\_column\_scale\_floor by a decade changes the diagonal by nothing. A stiff advective/chemical cancellation in the diagonal is a factor 1.8 and not decades (carrier diagonal

$1.19228e-1$  = diffusive  $+2.17367e-1$  and advective  $-9.89614e-2$ , with the chemical derivative  $+9.44223e-4$ , 0.8 percent, although the chemical time scale is 220 times the advective one), and there is no cancellation at all in the helium diagonal. The limiter is a live switch (the reconstructed face fractions stand at 0.587 and 0.657 of the donor at the carrier front, and frozen at first order the advective part of the diagonal changes SIGN,  $-9.89614e-2$  to  $+2.50409e-1$ ) but it moves the ratio by less than 2, because freezing it raises the momentum coupling by the same kind of factor. N5's two-root front was not re-run: cell 205 is not one. (4) **The test that settles it:** the existing default-off column equilibration (EXHALE\_COLUMN\_EQUIL=1) is exactly the remedy branch (i) names, and MEASURED at outer 1, where both runs stand at the same iterate, it brings the binding row's hydrodynamic coupling from 3389 to **0.993** (carrier) and 4176 to **2.926** (atomic) while the Krylov cycle returns the SAME step at the SAME residual to every printed digit (carrier 40 of 40, reduced  $1.257e-1$ , TRUE  $1.259e-1$ ; atomic 8 of 40,  $7.821e-2$ , TRUE  $7.824e-2$ ) and the species-row Ritz count below  $1e-2$  is unchanged at 18 and 2. It cannot be otherwise: a column scaling E is applied to the operator and to the band built FROM it, and  $(A E)(\text{band}(A) E)^{-1} = A \text{band}(A)^{-1}$  IDENTICALLY. N36 showed a row scaling is a similarity; this shows a column scaling is an identity, so **both sides of the scaling are now exhausted**. What the column equilibration costs, since it also reshapes the trust-region ball and the probe steps: cap 40 ends at  $||R||$  1.822 (carrier) and 1.592 (atomic) against  $9.572e-2$  and  $5.377e-2$ . Acceptance by the named clauses: the Ritz ratio below  $1e3$  FAILS ( $2.07e5$ ,  $1.45e3$ ), the cycle reaching 0.1 within 40 products FAILS on the carrier, named outcomes are lost on both; it stays off. **Deliverable 3 built nothing**, by the branch N35's deliverable 4 took: the remedy branch (i) names is either already the default (ln n) or already measured to be invisible. Tests: krylov\_and\_dogleg RED on the control objects (attributed\_jacobian\_entry absent), GREEN **279/0** with 6 rows added (the hook off by default; a stated term set composing its own row and its two faces composing its advective term through species\_row\_from\_its\_terms, the expression the measurement itself uses; the attributed entry of a term linear in the unknown equal to its coefficient times the scaling, with a zero probe step and a row with no scale attributing nothing) -- 279 rather than 270 because N37 added nine rows to the same file, a brief overlap reported and not resolved; steady\_species\_rows 194/0, element\_operator 14/0, steady\_selfconsistent\_residual 3/0 on the wasp\_full\_newton log, residual\_determinism 5/1 with the same row failing by design and the same  $3.354e+01$ . Default path: the atomic reload at cap 40 with every hook off, all 354 (JFNK) lines IDENTICAL to my control through  $||R||$   $5.377E-02$ , flux spread  $3.555E-03$ ; the carrier reload all 364 identical at  $9.572e-2$ ; wasp\_full\_newton done info=0  $||R||$   $1.667E-09$  on both binaries with all 28 lines the same; and with the hook ON both reloads still end at the control's values, so the measurement is not an operation on the state. Cost MEASURED: one residual evaluation for each of twenty columns at each of three outer iterations, about four minutes on the carrier reload at cap 40 and 8 threads. Also corrected in the reading: N35's "there is no diagonal for a preconditioner to stand on" does not follow from the band row -- the diagonal is the size of the row, the disparity beside it is a column scale, and the near-singularity of the species-row compression of  $A^{-1}$  is a separate fact that the band-row ratio is not evidence for. Advisor: verified on the same final build (the suites above); atomic reload identical. Accepted; deliverable 3 correctly not built. N35's sentence "no diagonal for a preconditioner to stand on" is withdrawn: the diagonal is the size of the row, the  $3.3e3$  is the momentum column scale (the sound speed) against a layer Mach number of  $1e-3$  to  $1e-5$ , and column scalings are invisible to the

cycle, so the near-singularity of the species-row compression is a fact of its own.

- **Close of the species-row solver program** (2026-09-10 evening, user's instruction: "어떤 것을 해도, 끝이 없어 보이니, 엄청난 차이가 나오지 않는 한 이번 건만 마무리 하고 끝내기 바래", then "있는 그대로 정리하고, md tex 파일들 update 후 commit/push"): N37 and N38 were finished as briefed and no further item is started. State of the tree at the close, verified on the advisor's rebuilt build\_lvw: krylov\_and\_dogleg 279/0 (174 at the morning commit), grid\_and\_gates 162/0, steady\_species\_rows 194/0, certification 57/0, element\_operator 14/0, residual\_determinism one row FAIL by design; the atomic reload at cap 40 with every hook off and every new key absent gives the N31 ladder to the digit (1.888, 1.891, 1.819, 1.583, 1.861, 8.964e-1, 7.607e-2; handed back 5.377e-2, flux spread 3.555e-3); the wasp\_full\_newton reload ends info = 0, ||R|| 1.667e-9, the number every worker's paired control gave (its output is not compared with the cold-start golden, which is a different path: sec\_ion\_step 2402 against 0). Everything N31 to N38 added is a default-off hook, option or key: EXHALE\_JV\_ADDITIVITY, EXHALE\_JV\_COLUMN\_SCALE, EXHALE\_JV\_PROBE\_ARC, EXHALE\_IEQ\_TOL, EXHALE\_TEQ\_TOL, EXHALE\_RESID\_JUMP\_SCAN, EXHALE\_RESID\_QUAD, EXHALE\_KRYLOV\_SIZE\_SCAN, EXHALE\_GM\_HISTORY, EXHALE\_PRECOND\_SPECTRUM, EXHALE\_BAND\_DIFFERENCE, EXHALE\_MODEL\_ROW\_EQUIL, EXHALE\_GM\_TRUE\_RESIDUAL, EXHALE\_FRONT\_ROW, and the input key Well balanced: (K46); the marching path and the goldens are untouched (the # options header gains the wellbal token, which the regression comparison strips with every comment line). New fixture backup/regression/carrier\_elem\_newton. Documents closed to this state: ISSUES\_20260909.md 3.1, 3.7 and 5; code\_status\_20260910.md 3.3 and task 1; session\_handoff\_20260910.md (new, the handoff of record); To\_be\_determined\_by\_user\_20260906.md decision 23 with its outcome; steady\_solver\_design.md 22.4; input\_schema.md K46 and the manual's key table (PDF rebuilt, 90 pages); README.md (the key). The analysis the user asked for, of why the coupled solve fails and what a different approach would be (a segregated solve, the CETIMB way read from Koskinen et al. 2013 section 2.1.3 and 2022 Appendix B, a different treatment of the layer), is docs/solver\_approach\_analysis\_20260910.md. Reported, not fixed (ISSUES 3.7): the count global shadowing the intrinsic; newton\_dense's hardcoded step test; the dead dp\_bc/ntot\_bc refreshes; store\_row\_terms's momentum scale under the new key; fortdep.py blind to include; golden/mol\_ir\_bands/ without Cooling\_breakdown.txt; examples/README.md; the two .md twins.

## 8. PLAN\_20260911: the partitioned stationary route (2026-09-11)

Plan of record: docs/PLAN\_20260911\_partitioned\_solver.md. User instruction of 2026-09-11: read docs/solver\_approach\_analysis\_20260910.md, its review docs/solver\_approach\_analysis\_20260910\_review.md and docs/solver\_partition\_experiment\_20260911.md, correct the errors they establish, run the solver experiments, and adopt the partitioned route in the production solver if it measures better than the coupled species-row solve. This reopens the stage-2 solver program that the 2026-09-10 instruction had closed after N37 and N38. Items P1 to P9, then P10 to P12 on the same day's further instruction to fix the open items the plan had left to the user, then P14 and P15 on the two findings of that afternoon's Codex adversarial

review and P16 to P18 on the user's two decisions of the same afternoon, then P20 and P21 in the evening and the terminology sweep A1 to A4 on the user's instruction of that evening (P13, P19 and P22 are the record itself); the rules every worker on this tree follows are docs/worker\_rules.md.

### The items

- P1 (defect D1): a failed element composition solve is no longer handed back as an update** (binary\_element\_diffusion.f90 +311/-23, src/tests/element\_operator/element\_operator\_tests.f90 +190/-13; every line count of this section is MEASURED from `git diff --numstat` on the tree at the close). `solve_mass_fraction` returns an explicit success result and no longer accepts the last halved trial of a line search that never descended; `element_diffusion_step` takes an optional status and, when asked, restores the composition it was handed unless the candidate is finite, inside  $[0,1]$  to  $1e-12$  before the range clip, and no farther from the density the step held fixed than the entry composition was, to  $1e-10$ ; `relax_element_composition` steps on a copy, discards a refused step, retries at half the length within a bounded budget and reports three outcomes by name. MEASURED (P1) on the archived D1 reproduction (atomic\_elem\_newton, species mode, omega 1): a returned composition missing rho by  $1.768571764708571E-01$ , a nonfinite chemical refresh and exit status 1 become the entry composition restored at  $1.046888214610824E-15$ , an admissible refresh at  $1.10297235E-15$  and exit status 0. The first inadmissible operation of that failure, which the experiment report left unlocated, is now named: the composition solve drives  $X$  out of  $[0,1]$  and the range clip used to absorb it (`element_step_out_of_bounds`), and the mass discrepancy is downstream of the clip. Tests: `element_operator` 14/0 at entry, **24/0** after, **22/2** on a RED build with the four acceptance actions disabled (`an_unsolved_step_is_refused`, `an_unsolved_step_leaves_the_entry_composition` measured  $1.31725E-02$ ); `diffusion_tests` 35/0, `steady_species_rows` 194/0, `certification` 57/0 unchanged. Impact (MEASURED, P1, control built from the entry text of the one file): `mol_diffusion` and `lower_profile`, 300 steps, one thread, `Ion_species.txt` byte-identical and `Hydro_ioniz.txt` identical but the provenance `run=` timestamp, on both cases; the marching caller passes no status, so none of the new tests runs there. One fixture of the same suite was corrected while there (`the_relaxation_fixed_point_under_a_callers_ghost` built a column whose species missed rho by  $1.321E-01$ ): on the suite's own mass-closed column the worst interior element row of the relaxation fixed point moves from  $1.62384E-02$  to  $1.35062E-13$ , so the fixed point on a mass-closed column really is a zero of the stationary element row.
- P2 (defect D2): the carrier movement bound now constrains the returned state** (`diffusive_photochemistry.f90` +225/-81, `src/tests/carrier_retry/carrier_retry.f90` +261/-1). The displacement is defined once (the largest change of any solved carrier column over the physical cells, divided by the largest H2 mixing ratio of the outer entry state, a quantity that cannot move while the pass runs), it is enforced before acceptance on every trial, a refused trial is undone and halved down to  $2^{*(-10)}$  of the cell transport time, and the pass ends on one of five named outcomes. MEASURED (P2) on `carrier_elem_newton` through the archived driver: the entry text returns `drift = 2.8126757844770485e-02` at `trust = 0.01`, reproducing the 0.02812676 of docs/solver\_partition\_experiment\_20260911.md section 7.2 exactly; the

repaired routine returns  $9.9956436739254070e-03$  at the same setting and  $2.9845242387380568e-03$  at  $\text{trust} = 0.003$ , so the bound holds and it scales (a factor 3.33 smaller setting gives a factor 3.35 smaller movement). The halving ladder was measured on an instrumented copy: 0.4412 at the full cell crossing time, then 0.2523, 0.1445, 0.0779, 0.0404, 0.0205, 0.0103,  $5.155e-3$ ,  $2.581e-3$ ,  $1.291e-3$  and  $6.458e-4$  at the tenth halving, so the movement is proportional to the trial length and a bound of  $1e-4$  keeps nothing on this column. Tests: **carrier\_retry 111 PASS / 10 FAIL** on a RED control carrying the entry logic behind the new interface, **122/0** after; **carrier\_returned\_state\_acceptance 36/0**, **attempted\_step 70/0**, **carrier\_constraint\_attribution 10/0**, **carrier\_reference\_scales 14/0**, **species\_masses 9/0**, **certification 57/0**, **adv\_static\_limit 53/0**. Impact: **mol\_carrier**, 300 steps, one thread, **Ion\_species.txt** byte-identical and **Hydro\_ioniz.txt** identical but the provenance timestamp, which is the expected result because the marching loop calls **photochemical\_transport\_step** directly and never this routine. The cost is stated: a bounded pass advances less, so the frozen H2 residual after the composition update is larger ( $4.4708e-01$  at  $\text{trust} 0.01$  against the entry text's  $3.1897e-01$ , from  $4.9275e-01$  before), and the state handed on is correspondingly closer to the entry state.

- P3 (defect D3): the stop test of the stationary solve is now the acceptance test of the state** (**steady\_newton.f90** +203/-29, **src/tests/steady\_completion\_flag/run.sh** +61/-15, **src/tests/krylov\_and\_dogleg/krylov\_and\_dogleg\_tests.f90** +127/-1). The loop top of **solve\_steady\_jfnk** and of **solve\_steady\_ptc** no longer stops for success on **steady\_gates\_met** alone: it stops only when the gate AND every row the certification judges the carried system by are within their own tolerances, measured on the iterate by the same evaluation and the same formula the acceptance at return uses. A gate met while a certified row refuses prints one line naming the row, its measure, its own tolerance and its cell, and the iteration continues within its budget. MEASURED (P3) on the D3 reproduction (**carrier\_elem\_newton**, first hydrodynamic solve): the entry text stopped at the loop top of iteration 14 with **info = 0** and was refused at return on a mass row of  $5.954e-12$  against  $3e-12$ ; after the repair the same solve continues, 27 of its 40 iterations meet the gate with a row refusing, and it ends **info = 1** on its pass budget with mass  $5.945e-12$ , momentum  $5.991e-13$ , energy  $2.481e-09$ . So the second of the two admitted outcomes: the mass row of this solve does not come below  $3e-12$  in 27 further iterations (it stands at  $5.9e-12$  to  $7.9e-12$  throughout, consistent with the N33 rounding floor), and the flag is now honest about it. Cost of the added test, MEASURED with a temporary clock pair in a private build: 27 calls, mean 3.19 ms against a mean outer iteration of 0.73 s, 0.44 percent of one outer iteration, and only behind the gate, so an iteration below the gate pays nothing. Tests: **krylov\_and\_dogleg 279/0** on the control objects and **290/0** after (11 new rows asserting the predicate the stop test reads); **steady\_completion\_flag RED** on both real logs at entry (an **info = 0** log with no statement that the stop test read the certified rows; a refusal at return with no stop statement) and **GREEN** after; **certification 57/0**, **carrier\_returned\_state\_acceptance 37/0**, **attempted\_step 71/0** unchanged. Impact: the **wasp\_full\_newton** reload ends **info = 0**, **||R||**  $1.667e-09$ , flux spread  $6.233e-14$ , 8 outer iterations, **log10 Mdot 13.30** on both binaries, the certification block line for line identical, every output byte-identical but the provenance line of the two **Hydro\_ioniz\*** files and one extra loop-top stop: line in the log; the **carrier\_elem\_newton** reload at cap 40 gives every (JFNK) line identical because the gate is never met on that reload. No golden can move: READ from the ten **DEFAULT\_CASES**, none of their golden logs carries a

(JFNK) start or (PTC) start line at all.

- **P4: the momentum row's reference scale under Well balanced: True** (RK\_rhs.f90 +90/-2, steady\_residual.f90 +59/-2, src/tests/grid\_and\_gates/hydrostatic\_residual.f90 +318/-11, that suite's README.md +44/-1). Under the well-balanced option store\_row\_terms scaled the momentum row by  $\max(|dF(2)|, |S(2)|, |Smom|)$  with  $S(2) = 0$  by construction, so every scaled momentum residual was exactly 1 (N37, ISSUES 3.7). The reference is now  $\max(|dF\_2|, |S\_2|, |S\_visc|, \text{equilibrium\_pressure\_force}(j))$ , the last term the magnitude of the pressure force of the cell's own hydrostatic equilibrium, which is the right-hand side of whichever cancellation identity the assembled row belongs to (PLM  $|\rho_j| [A^+ (\phi_i(j) - \phi_c(j)) + A^- (\phi_c(j) - \phi_i(j-1))]/dV$ , WENO3  $|\rho_j| (\phi_i(j) - \phi_i(j-1))/dr$ ; Kaeppli and Mishra 2016, A&A 587, A94, eq. 16 in spherical geometry). It has one definition, equilibrium\_pressure\_force\_of\_state, called by RK\_rhs, by positivity\_limited\_fluxes and by assemble\_residual (the last so that the kind-generic EXHALE\_RESID\_QUAD assembly, which returns the well-balanced row but not the term it is read against, does not scale the row by a stale weight). The numerator is untouched; with no gravity the term is zero and the scale is the row's own terms exactly (MEASURED 0 for all four scheme/solver pairs). MEASURED (P4): wasp\_full\_newton reloaded with the key ON is **NOT CERTIFIED** on the entry text on exactly one entry, the momentum row at **1.000E+00** at cell 1 (which reproduces the N37 record to the digit, spread 1.539E-15, energy 9.911E-12, mass 1.228E-13, Mdot 13.30), and **CERTIFIED** after, info = 0, ||R|| 9.719E-10, momentum row 9.719E-10 against 1.0E-08 at cell 500, log10 Mdot 13.30. Tests: grid\_and\_gates hydrostatic\_residual **28 PASS / 12 FAIL** on the entry-text objects (the twelve failures at exactly 1.000 for all four pairs) and **40/0** after, of which 16 are new rows measuring the row divided by residual\_row\_scale(2,...), the one expression the certification reads; the whole suite 162/0 at entry (READ, N37) and **178 PASS / 0 FAIL** after; krylov\_and\_dogleg 279/0 and certification 57/0 unchanged. Default path (key absent): wasp\_full\_newton reload and mol\_carrier at 300 steps byte-identical on both binaries apart from the provenance line, every (JFNK) line identical, key-off info = 0, ||R|| 1.667E-09, Mdot 13.30. The option's effect on the two solver fixtures is large and in one case negative, and both are recorded: the carrier reload's handed-back ||R|| falls from 1.721 to **1.026E-01**, the key-off level (READ: N37 9.34E-02), with a measured momentum row of 1.404E-02 at cell 499 instead of a pinned 1; the atomic reload stops earlier (info = 2 at iteration 24) and hands back 1.702 against the control's 1.415 at cap 40, because the merit no longer carries 500 momentum rows pinned at one (||Fs|| 2 17.1 against 28.2). A defect found while there and fixed: the allocation guard in RK\_rhs tested face\_flux for the well-balanced arrays face\_q\_up/face\_q\_dn as well, and face\_flux/face\_p are also allocated by store\_the\_interface\_fluxes, so a stationary evaluation reached before any marching stage could write an unallocated array; the guards are now separate and each tests its own array, with no arithmetic on any path.
- **P5: the analysis document corrected** (docs/solver\_approach\_analysis\_20260910.md, +463/-115, 114 to 592 lines). All nine corrections the review established are in place, each with an inline [corrected 2026-09-11: ...] mark citing the review section or the experiment measurement behind it: the spatial truncation error is not a lower bound on the algebraic residual (the frozen-background H2 wind row measures 2.449021e-13 on the same

500-cell grid where the withdrawn argument put a  $5e-4$  floor); Ritz-magnitude ratios of a masked compression are neither condition numbers nor the species Schur complement; partitioning moves the coupled iteration outside the Krylov solve with outer matrix  $(1 - \omega) I + \omega D^{-1} C A^{-1} B$  and the outer loop is not new code (steady\_wind\_with\_element\_diffusion); the CETIMB route is renamed as informed by CETIMB and gains the two conditions the review adds; the well-balanced option removes the equilibrium's algebraic cancellation and not the floor; Koskinen et al. 2013 lower boundary is section 2.1.1; the three-unknown path uses its existing globalization and not the trust region (`use_tr = (nspec_row .gt. 0)`); section 7 recast with the 2026-09-11 numbers; a new section 9 indexing the twelve changes and a new section 5.5 on partitioned components inside a coupled method (Knoll and Keyes 2004, J. Comput. Phys. 193, 357). Three numbers of the original disagreed with their sources and are corrected, one of them wrong: the N4b row for the carrier reload read "1.4 -> 0.09" where the measurement is 0.151 -> 0.090. Citations were verified independently in the published text (Koskinen 2013 section headers by pdftotext -layout; the Knoll and Keyes bibcode by the ADS API).

- **P6: the outer iteration as a stated contract** (src/EXHALE\_main.f90, +448/-131). steady\_wind\_with\_element\_diffusion now says in its own header which equations are alternated with the wind and why (a species the run moves by an operator-split transport step, the diffusing elements always and the molecular carriers where Coupled carrier solve is off, has a stationary balance that no hydrodynamic solve touches), and it accepts a state only by the certification of the FULL set of active equations evaluated on the refreshed state it would hand back. A hydrodynamic `info = 0` while an alternated species row refuses is not an accepted state, and a nonzero hydrodynamic flag no longer ends the loop by itself, because both `info = 1` and `info = 2` hand back a state the composition update still makes progress from. The endings are named (`outer_certified`, `outer_state_not_finite`, `outer_element_update_refused`, `outer_no_progress`, `outer_pass_budget`), the P1 status is consumed, and the quantity the passes are controlled by is the worst gated species row of the certification: where it fails to fall over a pass both updates are shortened (the element under-relaxation  $\omega$  halved, floor 0.125; the carrier movement bound halved, floor `carrier_trust_floor = 1.0d-3`) and `outer_no_fall_max = 3` consecutive passes without a fall end the loop. The last pass takes no composition update, because an update exists to be consumed by the next hydrodynamic solve and taking it would replace a stationary wind by a state the update itself spoiled. The composition drift and the volume-weighted carrier residual are reported and are not tests. With no alternated species the body is one steady solve and the state refresh, and the flag is the solve's own, which is what an atomic run and a coupled carrier solve both do. Default path MEASURED on the P6 run pair (`wasp_full_newton.ctrl` against `.new`): `info = 0`, `||R|| 1.667E-09`, flux spread 6.233E-14, CERTIFIED, `log10 Mdot 13.30`, and `output/Ion_species.txt` and `output/Hydro_ioniz.txt` byte-identical between the two.
- **P7: the measurement.** Below.
- **P8: the corrections propagated to the documents and tests that carried them** (`docs/code_status_20260910.md`, `docs/ISSUES_20260909.md`, `docs/certification_tolerance_anchoring_20260910.md`, `src/tests/steady_selfconsistent_residual/run.sh`, `src/tests/steady_species_rows/steady_species_rows_tests.f90`). The withdrawn inference (a spatial truncation error of  $5e-4$  at  $N = 500$  floors the algebraic residual, so  $1e-5$  cannot certify) is corrected in place in

the three documents that still carried it, marked with the review section and the measured 2.449021e-13 beside it, and decision 22 (a) is kept as the standing decision in every one of them with only its justification changed; the grid-refinement anchor and its remap control are relabelled as consistency diagnostics; "permanently uncertified until the operator's order or the grid changes" is withdrawn and quoted where it stood; ISSUES 3.7's store\_row\_terms item is struck through and marked RESOLVED by P4, and the resolved table gains D1, D2 and D3 with their measured before and after. Two test contract errors, both RED before and GREEN after: steady\_selfconsistent\_residual assertion 2 read "a hydrodynamic row above its tolerance implies info = 2", which conflates not certified with one flag and already failed on the entry text for any budget-limited log (FAIL flag\_of\_carrier\_elem\_newton measured=info=1 reference=info=2 -> PASS, with wasp\_full\_newton still PASS and a synthetic info = 0 copy still FAILing); and relaxation\_drift\_covers\_every\_element built its column by swinging each metal mixing ratio about the reservoir on a fixed hydrogen fraction, so its species missed the density the column hands the operator by 1.66042E-02, which is now 6.58484E-16 on a column of mass fractions that sum to one, the suite going 194/0 to **195/0** with a new row measuring the closure by the code's own mass policy.

- **P9: this section**, docs/session\_handoff\_20260911.md, the status section of the plan, the dated sections of the two fixture READMEs, the reopening paragraphs of docs/code\_status\_20260910.md section 3 and docs/ISSUES\_20260909.md section 5, and the new entries of TO\_BE\_DONE.md.
- **P10: the best-iterate ledger reads the hydrodynamic rows against their own certification tolerances** (steady\_newton.f90 +131/-55, src/tests/krylov\_and\_dogleg/krylov\_and\_dogleg\_tests.f90 +135/-1, READ from the P10 report; the whole uncommitted diff of the two files now stands at +335/-84 and +268/-1, MEASURED by `git diff --numstat`, P3's share included). Items P10, P11 and P12 were added after the plan had run, on the user's instruction of 2026-09-11 to fix the open items the session had left to the user ("fix all of them", and "the pressure term too"); P10 is the third entry of the plan's "what remains". distance\_from\_certification divided the hydrodynamic slot by the run's Resid tol, so the ledger that chooses the state handed back could not see a mass row above 3e-12 while ||R|| was below 1e-8. The slot now arrives as a distance formed by ONE new public routine, hydrodynamic\_distance\_from\_certification, max(mass/3e-12, momentum/1e-8, energy/1e-6) on the windows resid\_relnorm uses, so  $d < 1$  is exactly the row condition the loop-top stop of P3 and the acceptance at return impose, and keep\_this\_iterate ranks on that distance on every route including the three-unknown one (the nspec\_row <= 0 branch that ranked on ||R|| is gone). certified\_row\_measures hands the plain ||R|| of the same evaluation back through a new optional argument rather than leaving a caller to re-derive it, and the iteration line prints the two side by side. resid\_tol\_of\_solve is KEPT, with one reader, the EXHALE\_RESID\_BRANCH\_REPORT block, where rmax(1) is deliberately the plain mixed-row measure and dividing it by one row's tolerance would be a ratio of two different rows. Tests: krylov\_and\_dogleg 290/0 at entry, **300 PASS / 9 FAIL** on a RED build in which both distances read maxval(rc)/1.0d-8 (the superseded reading, at the Resid tol of these fixtures), **309/0** after, the 19 new rows being the block judged\_distance\_reads\_each\_row\_against\_its\_own\_tolerance; certification 57/0, steady\_completion\_flag 3/0, steady\_selfconsistent\_residual 3/0 unchanged. Impact MEASURED (P10) against a private build of the entry text of the one file: wasp\_full\_newton reloaded

ends info = 0, ||R|| 1.667E-09, log10 Mdot 13.30 with **all eleven output files byte-identical**, the ledger handing back the same iterate and the only number that moves being the reported best-iterate distance, 1.749E-01 (which is ||R||\_best/1e-8) to 8.469E-02 (the same iterate's worst row over its own tolerance, so 8.5 percent of it). The carrier reload shows the re-ranking doing what it is for: pass 2 hands back a mass row of **4.85E-12 at ||R|| 2.459E-09** where the control took 6.98E-12 at 1.593E-09, and pass 3 **4.96E-12 at 2.005E-09** against 8.17E-12 at 1.723E-09, with the H2 carrier row and the worst gated species row identical pass by pass because the carrier relaxation sits outside the hydrodynamic solve. The HD 209458 b element reload runs its twelve passes in 1064 s against the control's 1043 s (2 percent on a shared machine, not a claim) with every one of the twelve worst gated species rows and its cell identical to the control, and ends on the same single refusing entry, the hydrodynamic mass row at **8.893E-10 at cell 3** against the control's 1.054E-09 at cell 16; the solves reporting a stalled best iterate fall 5 to 3 and those ending on "no descent direction exists" 11 to 10. What stays open, and what this measurement settles: the mass row does NOT reach 3e-12 and that state still does not certify, because at 1e-9 to 1e-12 the row IS the N33 rounding of the base-layer flux difference. MEASURED (P10, finding 3): at pass 1 the ledger's best distance is 4.407E+02, a mass row of 1.32E-09, while the same state re-evaluated at its own composition after the best-iterate restore carries **4.11E-09**, so ranking iterates by that row is ranking by the luckiest rounding of a quantity that moves when the state is evaluated again, which is why the pass-by-pass mass rows scatter rather than descend in the control and in the new run alike. The entry is an acceptance question, not a solver item, and no further solver item follows from it.

- **P11: the momentum row's reference scale is the largest PHYSICAL term of the momentum equation** (RK\_rhs.f90, steady\_residual.f90, src/tests/grid\_and\_gates/hydrostatic\_residual.f90 and that suite's README.md; the whole uncommitted diff of the four files stands at +285/-2, +151/-33, +736/-12 and +94/-1, MEASURED by `git diff --numstat`, with P4's +90/-2, +59/-2, +318/-11 and +44/-1 inside it). The fourth entry of the plan's "what remains", and the "pressure term" of the user's instruction. The scale was built from the pieces of the discretization,  $\max(|dF(2)|, |S(2)|, |S_{\text{mom}}|)$  plus P4's equilibrium pressure force under the well-balanced key; it is now built from the three left-hand terms of  $d(\rho \ v)/dt + \text{div}(\rho \ v \ v) + dp/dr + \rho \ d\phi/dr = S_{\text{visc}}$ , kept beside `face_flux/face_p` as module arrays of `RK_integration`, filled on both reconstructions, under both key values and on the positivity-repair path, and gathered by one routine, `momentum_row_terms_of_cell`, with `momentum_row_terms_of_state` filling a whole state and `reconstruction_continuation_rhs` blending the three terms with the same  $\lambda$  as `dF` and `S`. The residual  $R(2) = dF(2) - S(2) - S_{\text{visc}}$  is untouched on every path, and the split is a rearrangement of the assembled row and not a model of it:  $F(2) - p_{\text{out}}$  is the momentum flux without the pressure exactly for every flux function, and the geometric term is recovered from the source as  $p_{\text{geom}} = S(2) + \text{weight}$  so that no second definition of the cell pressure exists. MEASURED (P11), on a state with structure in density, pressure and velocity,  $\max_j |r_{\text{am}} + dp/dr + \rho \ g - (dF_2 - S_2)|$  over the scale is **0 under WENO3 and 1.7e-16 (default) / 5.3e-15 (key on) under PLM**, for both Riemann solvers, so the three terms add up to the row they scale. The defect is closed on the states where the terms are known in closed form: with no gravity and a uniform pressure at rest, where every term of the equation vanishes, the PLM scale read **1.000 of the geometric pressure term**  $(A^+ - A^-) p_j/dV$ , that is  $2 p/r$ , with both Riemann solvers at every N, and now reads the tiny floor 2.22507E-

308; on supersonic uniform flow at  $M = 3$ , where the ram divergence is the only term, it stood **6.66667E-02** = **1/(gamma M M)** above it and now agrees with it to  $1.9\text{e-}14$  (PLM) and  $4.2\text{e-}13$  (WENO3); on the discrete hydrostatic equilibrium the PLM scale was **2.79144E-01**, **28 percent BELOW the weight it balances**, because  $S(2)$  is the weight minus the geometric term and no term of the max was the weight itself, and the departure  $(w - s)/w$  is now 0. P4's own zero-gravity diagnostic closes with it: the departure of the well-balanced scale from the base scheme's on the same state falls from **9.43654E-01** to **3.74E-13** under PLM with ROE ( $2.76\text{E-}13$  with HLLC), the two having agreed under WENO3 all along ( $1.7\text{e-}13$  to  $2.7\text{e-}13$ ). Tests: hydrostatic\_residual **50 PASS / 10 FAIL** on the entry-text objects and **60/0** after, the ten being the four scheme/solver pairs of P4's corrected zero\_gravity\_momentum\_scale\_is\_the\_rows\_own\_terms (which had compared the option's scale with  $\max(|dF_2|, |S_2|)$ , the SUM of the two terms the row holds,  $7.50\text{E-}01$  FAIL to 0) and the PLM pairs of the three new identity statements; the whole grid\_and\_gates suite **188/10 to 198/0** ( $198 = P4\text{'s } 178 + 20$  new rows); krylov\_and\_dogleg 290/0, certification 57/0 and adv\_static\_limit 53/0 before and after. Two quantities are REPORTED and not asserted, with the reason: the two-sided departure of the scale from the weight on the well-balanced scheme's own equilibrium is not round-off for the base scheme (MEASURED  $8.5\text{e-}4$  at  $N = 250$  falling to  $2.3\text{e-}6$  at  $N = 2000$ , PLM with ROE), because that state is the discrete equilibrium of the WELL-BALANCED reconstruction and HLLC's face pressure carries a dissipative part of size  $dr/H$  times the weight; and the halving statement subtracts the unperturbed row of the same cell, since the undifferenced ratio is not 2 (MEASURED 1.0012 at  $N = 250$ , 1.586 at  $N = 2000$ ) while the imbalance the perturbation ADDS halves to 2.000000 within  $5\text{e-}6$ . Impact: nothing that was certified stops being certified. wasp\_full\_newton reloaded with the key OFF is **CERTIFIED** on both binaries, info = 0,  $\log_{10} Mdot$  13.30, with the momentum row moving **8.469E-10 to 1.165E-09** of its  $1.0\text{E-}08$  tolerance, the mass row  $1.453\text{E-}13$  to  $2.017\text{E-}12$  of its  $3.0\text{E-}12$  (so this reload now certifies with less margin on the mass row) and  $|R|$   $1.667\text{E-}09$  to  $1.495\text{E-}09$ ; the outputs are not byte-identical, because the merit and the loop-top gate both divide by the momentum scale and the two solves accept different iterates, and the states differ by at most **2.6E-09 relative** over every column ( $9.3\text{e-}10$  in Hydro\_ioniz.txt,  $2.6\text{e-}09$  in Ion\_species.txt). With Well balanced: True the same reload is **CERTIFIED** on both, the momentum row moving **9.719E-10 to 9.621E-11** with all eleven output files byte-identical and the  $|Fs|/2$  ladder identical iteration by iteration: at cell 500,  $r = 4.62$ , the ram divergence exceeds the cell's weight by about a factor ten, which is why the measure of the certified state falls by that factor on a supersonic outer cell. The marching path of mol\_carrier at 300 steps and one thread is **byte-identical in all seven output files**,  $\log_{10} Mdot$  7.95, the run.log differing in exactly eight lines, every one of them a report of the momentum measure (the worst-row entry  $1.8164\text{E+}00$  at cell 438 to  $1.9788\text{E+}00$  at cell 449) and none of them a state. On the carrier reload every pass line is identical but the momentum entry, which falls **1.32E-12 to 3.22E-13** after pass 1 ( $4.47\text{E-}13$  and  $3.54\text{E-}13$  at passes 2 and 3), the state as loaded reading  $3.622\text{E-}01$  to **3.311E-01** in its scaled momentum row, and all four output files byte-identical. What stays open: the kind-generic EXHALE\_RESID\_QUAD assembly still hands back neither the momentum terms nor the face departures, so under that option (default off, a rounding control experiment nothing is adopted from) both are read from the last RK\_rhs evaluation rather than from the state just assembled; the fix belongs in hydrodynamic\_rows.f90, which this item does not own.

- **P12 (with P12b): two test-hygiene defects, and the same shape in the sibling**

**scripts** (src/tests/test\_columns.f90 new, 125 lines; attempted\_step/run.sh +18/-5, carrier\_returned\_state\_acceptance/run.sh +18/-5, carrier\_retry/run.sh +18/-5, carrier\_reference\_scales/run.sh +22/-6, carrier\_constraint\_attribution/run.sh +22/-6, constrained\_network\_layout/run.sh +22/-6, species\_masses/run.sh +22/-8, element\_operator/run.sh +5/-0, steady\_species\_rows/run.sh +5/-0, and the two drivers, MEASURED by `git diff --numstat`; of the drivers' totals the item's own share is READ from the P12 report, the use line, the renamed calls and 66 lines out of `element_operator_tests.f90` and 43 out of `steady_species_rows_tests.f90`). The fifth entry of the plan's "what remains", both halves of it. The **build\_stamp fallback**: the trigger is narrower than "a tree that has never been built", it needs such a tree AND a work directory outside `$root/build`, which is the concurrency-safe usage the worker rules mandate, because `source_closure.py` walks `$root/build` recursively and a default work directory inside it swept the fallback into the closure by accident. Seven `run.sh` now select `${EXHALE_OBJDIR:-$root/build}/build_stamp.f90`, write the fallback into their own work directory only when that is absent, and compile whichever of the two is in force as the FIRST file of the closure with the closure filtered of any second copy, which is the shape `src/tests/physics_probe/run.sh` already had; `grep -ln 'if [ ! -f "$root/build/build_stamp.f90" ]' src/tests/*/run.sh` now returns nothing. Four of them gained a private object directory, `EXHALE_TEST_OBJDIR`, without which the no-build verification cannot be run on them at all (`species_masses`, which had no override of any kind, `carrier_reference_scales`, `carrier_constraint_attribution`, `constrained_network_layout`). RED and GREEN, MEASURED in four cases: at HEAD with an external work directory the compile dies on Cannot open module file 'build\_stamp.mod' and each script prints its one failure line, **0 PASS / 1 FAIL, exit 1**, in `attempted_step`, `carrier_returned_state_acceptance`, `carrier_reference_scales`, `carrier_constraint_attribution`, `carrier_retry` and `species_masses`; after the fix every suite gives the count it gives at HEAD with a default work directory, **70/0** (`attempted_step`), **36/0** (`carrier_returned_state_acceptance`), **14/0**, **10/0**, **122/0**, **9/0** and **8/0** (`constrained_network_layout`, which has no `utilities.f90` in its closure, so it has no RED to show and its change is the shape alone), in a tree with no `build/` and in the repository against a private build, and the diff of the PASS/FAIL lines before and after is empty for all of them. Which stamp each case used is MEASURED too: in the no-build case every work directory holds a `build_stamp.o` compiled from the fallback, and in the case with a build tree no fallback file was written at all, so the generated stamp was taken, which the superseded scripts could not do because they looked only in `$root/build`. The **column written twice**: the new module `test_columns` holds `column_carrying_its_own_density(f_c, q_h2, he_taper, metal_swing)`, named for what it builds, with its header stating the invariant `sum_i m_i n_i = rho` and the round-off it holds to, and `column_mass_closure(rho_c, f_c)`, the measurement moved out of `element_operator_tests.f90` unchanged; the two drivers call them at 15 and 2 sites and use `utils`, only: `calc_rho` is gone from both, it having been there only for the moved code. The shared construction reduces to each former copy exactly and not approximately (`metal_swing = 0` gives `swing = 1.0d0`, `q_h2 = 0` leaves `isp_H2` at zero, `he_taper = .false.` selects the constant the steady driver used, and `zsum` accumulates in the same element order), and the statement is the measurement: `element_operator`

**24/0** and `steady_species_rows` **195/0** before and after, with the diff of the two full logs, diagnostic lines included, empty in both, so every printed digit is unchanged (`the_column_is_mass_closed_as_built` 4.10591E-16, `drift_column_carries_its_own_density` 6.58484E-16, the relaxation's 27 steps and returned drift 1.7790E+00 among them). The file is in no source list of the Makefile (`grep -c test_columns Makefile` = 0) and no dependency generator sees it; the two `run.sh` that need it compile it immediately before their driver. Nothing stays open from this item: its own report found nothing further, and the four sibling scripts and the hardcoded work directory it reported are what P12b then closed.

Two changes outside the item list, made by the advisor while reading these three:

- `src/tests/steady_species_rows/steady_species_rows_tests.f90`, the two rows P10 reported as stating the superseded semantics. In `judged_distance_is_the_worst_row_over_its_tolerance` the hydrodynamic slot now arrives as a distance (`rows(1)` 2.0d0, 0.5d0 in place of 2.0d-8, 5.0d-9), with a comment saying so, and `judged_distance_follows_the_residual_target`, which asserted the property the item removes, is replaced by `hydrodynamic_distance_reads_each_row_against_its_own_tolerance`: a mass row of 1e-9 beside a momentum row of 1e-13 and an energy row of 1e-8 gives 1e-9/3e-12, the mass row being what the distance reports. The two `resid_tol_of_solve` assignments and that import are gone.
- the `Coupled carrier solve` key comment in `src/modules/files_I0/input_read.f90` (+17/-13, MEASURED). It stated that the alternation "was measured not to converge and not to be able to" (section 139), which is narrower than it reads after P7. It now names the default (False), says what the alternation is judged by (the certification of the refreshed state in `steady_wind_with_element_diffusion`), cites the two reloads of this section for what it reaches, and keeps True as the coupled measurement option.

Four more items ran in the afternoon of the same day: **P14** and **P15**, the two findings of the Codex adversarial review of the working tree (the review and the verification of each finding are the subsection below), and **P16** and **P17**, the user's first decision of the afternoon (the mass tolerance re-anchored on the fixture's measured rounding floor) and its consequence for the best-iterate ledger. **P18** followed and closed the same day.

- **P14: every ending of the stationary outer iteration hands back one consistent state** (`src/EXHALE_main.f90`, `steady_wind_with_element_diffusion` only, +84/-40 against the entry text; `src/tests/grid_and_gates/output_state_consistency.sh` +181/-4; MEASURED by P14 with `diff -u` and `git diff --numstat`). Codex finding 1, and it is real. On a stagnation ending the composition had been updated while the particle densities and the temperature written beside it had not, so the state the run wrote was one composition update ahead of its own particle count and temperature. MEASURED (P14) on the forced stagnation of the hot-Uranus carrier reload: re-evaluating the written state moves its temperature by **1.317089E-08** relative at cell 295, where the same fixture ending on its pass budget, which takes no update, moves by **5.921E-16**; the written temperature of the worst cell is 1.768302856793E+03 where the same conserved state and composition give 1.768302833502E+03, and the worst species column is 2.543E-03 (He 2<sup>3</sup>S, the stage answering to the temperature with the steepest exponential). Four changes. The progress control (the `sp_worst` test, `n_no_fall`, the omega and trust halving, the setting of the stagnation ending) now stands between the

certification of the pass and the update the next solve would consume, guarded so that a certified or non-finite pass reaches neither, and the update's own gate therefore excludes the stagnation ending by itself: the pass that ends the iteration takes no update. The ending is announced with the other endings, below the summary line and the diagnostics, so the pass that ends on it prints its summary, its carrier diagnostics and its H2 front like every other pass. A pass that takes no update says which of the two reasons it was. And the element path's update now refreshes the particle densities and the temperature between `relax_element_composition` and the equilibrium sweep (`get_species_densities`, `comp_T_from_p`), which is the shape the carrier path already had; before it, the sweep equilibrated the new composition at the old temperature. The header states the invariant. One behavior change to record because it is not a reordering: the shortened step of a pass whose measure did not fall is now consumed by THAT pass's update, where before it reached the next pass's. RED and GREEN on the new second case of `output_state_consistency.sh`, which runs the stagnation (`EXHALE_CARRIER_TRUST=1e-4` `EXHALE_OUTER_PASSES=8` `EXHALE_JFNK_MAXIT=5`, the setting that stagnates; the brief's cap of 40 does not, the worst gated H2 row falling through the hydrodynamic solve alone), asserts by name that the ending reached is the stagnation one, and then measures the Restart intent: stationary evaluate round trip over the 500 physical cells against **1e-12**: `measured=1.317089e-08 FAIL` on the entry text, `measured=6.499247e-14 PASS` after. The tolerance is justified by measurement and not by a snapshot: the states handed back by an ending that takes no update measure `6.5E-14` and `5.9E-16`, and the defect measures `1.3E-08`, four decades above the allowance. The suite goes **199 PASS / 1 FAIL** on the entry text to **200 / 0**. Impact: `wasp_full_newton` reloaded is unmoved (`info = 0`, CERTIFIED, `||R|| 1.495E-09`, flux spread `1.9113E-12`, `log10 Mdot 13.30` on both binaries, all eleven outputs identical apart from the provenance `run= timestamp`); the carrier reload at three passes is identical pass line by pass line to the control and to the P7 record; the element reload's pass 1 is identical in every digit and the ladder parts from pass 2, because the sweep of pass 1's update now runs at the relaxed composition's own temperature (momentum `1.60E-12` against `4.24E-14` at pass 2), both builds ending on the pass budget with the same 9 refusing equations and the same base-layer mass row as the worst refusing entry.

- **P15:** the kind-generic rows hand back the face departures their momentum row was built from (`src/modules/time_step/hydrodynamic_rows.f90` +75/-15, `src/modules/time_step/hydrodynamic_rows_body.inc` +51/-17, `src/tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90` +255/-1 measured against the file as it stood when the item started; MEASURED by P15). Codex finding 2, and the third entry of the plan's "what remains". Under Well balanced: True the momentum row's pressure-gradient term, the momentum row scale and every measure read against that scale are formed from the module arrays `face_q_up/face_q_dn` of `RK_integration`, which only `RK_rhs` and the positivity repair ever filled; the kind-generic rows kept their own departures as locals and `store_the_interface_fluxes` wrote back `face_flux` and `face_p` alone, so in the kind-generic rows those three quantities belonged to the last `RK_rhs` evaluation or to the zeros of the first allocation. The departures now leave the kind-generic text as two OPTIONAL intent(out) arguments of `hydrodynamic_flux_difference_and_source`, converted to double at the one rounding on the way out exactly as the flux and the face pressure are, both public wrappers ask for them, and

store\_the\_interface\_fluxes stores them under the key. The reconstruction continuation CAN reach the kind-generic rows (assemble\_residual chooses the text from generic\_precision\_rows\_arm() alone and the generic text carries its own blending branch), and on it the terms would be wrong for two reasons rather than one, the single-scheme weight `w_face_p` and the fact that the two schemes reconstruct different faces so no single pair of departures satisfies the blended row identity; the `generic_text_is_usable` therefore now refuses `recon_lambda_on` .and.  $0 < recon\_lambda < 1$ , ahead of every evaluation of that text. That branch ends in error stop and has no executed row, which is stated rather than worked around. Twelve new rows, each on a state the kind-generic rows assemble AFTER a residual was assembled on a different state, both reconstructions and both precision kinds: MEASURED RED **2.098877E+01** (PLM) and **2.032204E+01** (WENO3) on the pressure gradient against the production term, **3.262392E+00** and **3.428893E+00** on the row scale, **1.190363E+01** and **1.181676E+01** on the row rebuilt from the stored face data, and GREEN at **0 bitwise** for the generic-double instantiation, which IS the production operator bit for bit, and 2.6E-16 to 2.7E-15 for the quadruple one at its 1e-13 bound. krylov\_and\_dogleg **309/0 to 321/0**; certification 57/0 and steady\_species\_rows 195/0 unchanged. Impact: wasp\_full\_newton reloaded is unmoved with the key off, and with Well balanced: True and EXHALE\_RESID\_QUAD=2 it stays CERTIFIED at the certified momentum row **9.621E-11**, P11's key-on production value, on both binaries, because the JFNK and PTC loops reset the sweep kind to marching before the certification of the handed-back state and the kind-generic text is consulted only for a non-marching sweep kind (READ, steady\_newton.f90); the brief's expectation that the control's certification reads another state's departures is corrected by that measurement. What the defect did move is the measure INSIDE the loop, where the kind-generic rows are armed: the convergence measure of iteration 1 is **1.756E-03** on the control against **1.753E-03** after, the momentum row divided by the pressure-gradient term of another state and overstated by 0.17 percent, and that measure is what the loop-top stop of P3 and the ledger of P10 read on every iterate. `||Fs||2` is identical iteration by iteration, iterations 2 to 7 are identical line for line, the accepted state is the same state, and `mol_carrier` marching is byte-identical.

- **P16: the hydrodynamic mass-row tolerance is anchored on the row's measured rounding floor** (src/modules/time\_step/certification.f90 +212/-2, src/modules/time\_step/steady\_residual.f90 +177/-1, src/tests/certification/certification\_contexts.f90 +105/-0, docs/certification\_tolerance\_anchoring\_20260910.md +302/-0 appended; MEASURED by P16 with `git diff --numstat`, the share of the other uncommitted item in steady\_residual.f90 separated by attributing each hunk). The user's first decision of the afternoon, and it CLOSES the first entry of the plan's "what remains": the tolerance is now  $\max(\text{cert\_tol\_mass}, \min(\text{cert\_tol\_mass\_ceiling}, \text{cert\_mass\_round\_margin} * \text{floor}(j)))$  with `cert_mass_round_margin = 10`, the ceiling 1 (a row at 1 is a mass flux changing by the whole of itself across the cell, not an operative branch on any state measured), `cert_tol_mass = 3e-12` unchanged and now the FLOOR of the tolerance, and the verdict on that row taken cell by cell through the pure `mass_row_cell_verdict`. **The floor expression the item's brief specified could not serve, and the report of that is part of the result:**  $\text{eps} (|A+ F_{\text{up}}| + |A- F_{\text{dn}}|)/dV / \text{residual\_row\_scale}(1, j)$  is algebraically between one and two ulps at every cell, because the row's scale IS the larger of the two addends and the `dV` cancels, and MEASURED it is **4.4409E-16 on all 500 cells of all four**

**states** looked at, so ten times it never rises above  $3e-12$  and the item as written would have been inert. The cause is the one N33 established: the interface flux carries the last bit not of its own value  $\rho \ v$  but of the  $O(\rho \ c_s)$  quantities the Riemann solve differences, so the floor is about  $\epsilon_{\text{ps}}/\text{Mach}$  and not  $\epsilon_{\text{ps}}$ . The estimate adopted is  $\epsilon_{\text{ps}} \rho (|v| + c_s) A / dV$  over the row's own scale, and it bounds the row's measured non-smoothness within a factor **1.5931** (wasp\_full\_newton as loaded, base Mach  $3.937E-03$ ), **1.8857** (atomic\_elem\_newton as loaded,  $5.062E-06$ ), **2.6806** (carrier\_elem\_newton as loaded,  $2.400E-04$ ) and **1.7733** (the HD 209458 b state this item certifies,  $2.767E-06$ ), where the specified expression is short by two to six decades. The margin is measured, not chosen:  $c_{\text{round}} = 10$  is a round number 3.7 times above the largest of those four ratios and inside N33's independent reading of the same floor against its own bound, 5.5 to 12 (READ). The measurement itself is in the code: EXHALE\_MASS\_FLOOR\_SCAN=1 (new, default off) adds one ulp to the density of every physical cell, runs the whole flux assembly again and reports the STEP each mass row takes beside both estimates, and the state given is then assembled once more and reproduces its own mass rows to  $0.00E+00$  on every fixture. Tests: certification **57/0** on the entry driver against the entry objects, **0 PASS / 1 FAIL** with exit 1 on the new driver against the entry objects (the compile naming the three new symbols), **70/0** after, the eleven new rows asserting that a floor below the fixed value leaves exactly  $3e-12$  and names the fixed tolerance, that a floor above it gives exactly  $c_{\text{round}} * \text{floor}$  and names the rounding anchor of that cell, that a row at half the anchor is WITHIN where the superseded rule transcribed inside the test refuses it, that a row above the anchor still refuses, that the ceiling binds, and that the margin stands above every ratio measured. Seven further suites are identical row for row in both builds, and a -fcheck=bounds,do,mem build runs both reloads with no runtime trap (the floor estimate reads the ghost cell at  $j = 1$ , which is why it was checked). **Impact.** wasp\_full\_newton reloaded is unmoved: CERTIFIED in both builds, mass row  $1.212E-12$  at cell 500 with the verdict taken there against the FIXED  $3.0E-12$  at distance 0.404,  $\log_{10} \text{Mdot}$  13.30, outputs byte-identical apart from the provenance timestamp; its floor is below  $3E-13$  at every cell, so the anchor cannot reach that case, and the cold-start wasp\_full\_newton of make check cannot be affected either, its stop having no row binding (READ from backup/regression/wasp\_full\_newton/run.log) and a tolerance that only ever rises cannot move it. **The HD 209458 b element reload CERTIFIES, at pass 12**, outer pass 12: ACCEPTED -- every active equation of this state is within its own tolerance, where the control (a build of the entry text of the two files in the same tree) ends the same twelve passes NOT CERTIFIED on the one entry it had before, the hydrodynamic mass row at  $7.159E-10$  against  $3.0E-12$  at cell 13. On the certified state the largest mass row is  $7.703E-10$  at cell 2 and the verdict is taken at cell 34 ( $r = 1.00665$ ),  $6.542E-10$  against **4.6E-09, the rounding anchor of that cell**, distance 0.141; the eight elemental wind rows and the worst gated species row of all twelve passes are identical to the control to three digits, cell included, and the two builds part at pass 6, the first pass whose hydrodynamic solve stops on the loop-top test because its mass row is inside its own tolerance ( $\text{info} = 0$  where the control's is  $\text{info} = 2$ ). The anchored tolerance still refuses states: the mass row is admitted at 11 of the 12 passes and refused at pass 3 at distance 69.9. On the hot-Uranus carrier reload the base cells are admitted by their own anchor at every pass and the refusal moves OUT to cell 180 ( $r = 1.0707$ ), where the estimated floor is below  $3e-12$  and the row carries signal, refusing  $5.067E-12$  to  $9.549E-12$  against the fixed  $3.0E-12$ : the refusal has moved from rounding to signal. The H2 carrier row is unchanged

pass by pass in both builds (7.38E-02 at cell 290, 7.22E-02 at 289, 7.06E-02 at 288), and the three passes cost 15.47 s against the control's 69.39 s. The derivation, both candidate estimates, the measured ratios and all three impact measurements are the appended section "**Anchor (6), 2026-09-11: the mass row's rounding floor**" of docs/certification\_tolerance\_anchoring\_20260910.md; its nine earlier sections are byte-identical.

- P17: the best-iterate ledger reads the mass row against the cell's own tolerance** (src/modules/time\_step/steady\_newton.f90 +146/-32, src/tests/krylov\_and\_dogleg/krylov\_and\_dogleg\_tests.f90 +98/-0; MEASURED by P17). P16's own outside-scope finding, and the consequence of the anchor for the ledger: the hydrodynamic slot was formed as  $rc(1)/cert\_tol\_mass$  from a triple of row maxima, so the ledger ranked the mass row against  $3e-12$  while the stop test and the acceptance read the cell's own tolerance. The slot is now  $\max_j$  over cells  $j$  and rows  $k$  of  $|R_{kj}| / residual\_row\_scale(k,j) / tol\_k(j)$  over the whole physical column, with  $tol\_1(j)$  taken from the certification's own rule at that cell through P16's public `mass_row_cell_verdict` (so the anchor formula is not restated in the solver module) and the fixed momentum and energy tolerances on the other two rows; that is the same functional the acceptance applies cell by cell, so  $d < 1$  in the ledger is now exactly the hydrodynamic part of the verdict. `rnorm` is unchanged and is now formed only when the caller asks for it, which the two loop-top calls do and the three trial calls do not. The superseded `hydrodynamic_distance_from_certification` keeps its name, signature and body and is documented as what it is, a BOUND from above on the distance ( $tol\_mass\_at(j) \geq cert\_tol\_mass$  at every cell), because its removal would break the compile of a test file P17 does not own; P18 removes it. MEASURED RED twice: the new driver does not build against the entry-text objects (exit 1, Symbol 'hydrodynamic\_distance\_from\_certification\_by\_cell' ... not found in module 'steady\_newton'), and a private build in which the new function's body is replaced by the superseded reading and nothing else is altered gives **327 PASS / 7 FAIL** with the seven rows naming the column (the\_wind\_cell\_holds\_the\_mass\_rows\_distance 3.333333E+02 against 3.333333E+00, the\_base\_cell\_is\_a\_decade\_inside\_its\_own\_tolerance 3.333333E+02 against 1.000000E-01, is\_a\_certified\_hydrodynamic\_state F against T among them). GREEN krylov\_and\_dogleg **321/0 to 334/0**, and seven further suites identical in both builds. **Impact.** The single number that says the item worked, on the HD 209458 b element reload: the ledger's best judged iterate: distance over the twelve passes runs **2.2E+02 to 6.3E+05 in the control and 1.321E-01 to 1.868E-01 after**, while the acceptance CERTIFIES the pass-12 state in both builds at a mass-row distance of 0.141 (control) and 0.132 (new, verdict at cell 50,  $r = 1.00977$ ,  $4.245E-10$  against the  $3.2E-09$  anchor). The control's ledger therefore ranked every iterate of every solve as hundreds to hundreds of thousands of times outside certification, on a row its own acceptance admitted; the ledger and the acceptance now speak about one quantity. The trajectories part at pass 2 and the elemental relaxation is not damaged (the worst gated species row descends  $2.61E-03$  to  $7.73E-06$  against the control's  $2.61E-03$  to  $6.84E-06$ , both inside  $1e-5$  at pass 12, and the certified state's elemental rows are SMALLER in the new build on every element, Fe  $5.086E-05$  against  $8.966E-05$ , C  $3.70E-06$  against  $1.00E-05$ , O  $3.34E-06$  against  $1.24E-05$ ). The cost is not measurable: 909 outer iterations at 1.206 s against 861 at 1.216 s, so the 500 rounding-floor evaluations added per judged residual cost nothing this measurement can separate from noise. More restarts and fewer stagnations, which is the reading of a ledger that now stops improving where

||R||/3e-12 used to keep moving on rounding noise: "best iterate has not improved" fires in 7 solves against 4, "no descent direction exists" ends 11 against 8, and the control's one STAGNATED solve is gone, while no solve fails and every pass still returns a state. On the hot-Uranus carrier reload the consequence is a verdict: the hydrodynamic solve returns **info = 0 at all three passes** against the control's 0, 2, 2; the mass row's verdict is WITHIN at passes 2 and 3 (cell 1, 1.826E-11 against the 2.5E-11 anchor, and cell 2, 6.437E-12 against the same) where the control refuses at cell 180; the state of the last pass has **1 refusing entry, the H2 carrier row alone**, against the control's 2; and the best judged distance runs 0.5282, 0.8819, 0.2578 against 1.630, 3.501, 1.840. The H2 carrier row is identical pass by pass in both builds and in the P7 record (7.38E-02, 7.22E-02, 7.06E-02 at cells 290, 289, 288), which is what must hold with the carrier relaxation outside the hydrodynamic solve. wasp\_full\_newton reloaded is unmoved, and for a stated reason: its estimated floor is below 3e-13 at every cell, so the cell's tolerance IS the fixed value everywhere and the maximum of the ratios is the ratio of the maxima; the two run logs are identical line for line except the Execution Time, every written file is byte-identical except the provenance timestamp, and the best judged distance is 6.722E-01 in both builds. A separate fix in the same file, reported on its own: write\_gate\_met\_while\_a\_row\_refuses printed the largest measure of a row and its cell beside the tolerance of the BINDING cell, a pair of numbers belonging to two different cells, and now names the binding cell throughout where the row records one, which is what the certification's own refusing entry list already did; no suite parses that line and the carrier rerun on the final binary reproduces every reported digit.

- **P18: one entry point for the hydrodynamic distance** (steady\_newton.f90 +23/-50, krylov\_and\_dogleg\_tests.f90 +49/-25, steady\_species\_rows\_tests.f90 +22/-16; P17's own outside-scope finding). With the ledger reading the mass row cell by cell (P17), the fixed-tolerance form hydrodynamic\_distance\_from\_certification(rc) had no production reader left and survived only through test rows. It is DELETED, with its public export and the then-unused cert\_tol\_mass import of the module; every test row that read it restates its fact through hydrodynamic\_distance\_from\_certification\_by\_cell on a one-cell column whose rounding floor (1e-30) puts it under the fixed tolerance, so the restatement is the same double division and every literal reference is unchanged. MEASURED (P18): `grep -rn hydrodynamic_distance_from_certification src/ | grep -v _by_cell` returns nothing; krylov\_and\_dogleg 334/0 before and after with no row lost (two renamed: the\_fixed\_tolerance\_alone\_reports\_the\_base\_cell, and\_the\_fixed\_reading\_is\_never\_below\_the\_distance), steady\_species\_rows 195/0, certification 70/0, steady\_completion\_flag 3/0; a compile RED (the entry-text drivers against the delivered objects fail on the missing symbol), a semantic RED of the cell-by-cell fact (327/7, P17's seven rows) and a semantic RED of the restatement itself (329/5, the fixture's floor raised into the anchored regime) all discriminate. Impact: the wasp\_full\_newton reload gives info = 0, CERTIFIED, ||R|| 1.495e-9, log10 Mdot 13.30 in both builds, all eleven output files byte-identical, the two run logs differing in the execution-time line alone. Documents that named the deleted function as current fact (certification\_tolerance\_anchoring\_20260910.md, session\_handoff\_20260911.md) were corrected by the advisor.

• **P20:** the code-size appendices moved to this log and were re-measured (docs/Update\_EXHALE\_appendix.tex new, docs/Update\_EXHALE\_stage1.tex appendix removed and its \date fixed at 2026-09-04, src/utils/update\_log\_to\_tex.py postamble, pointer lines in README.md and README\_HOWTO.md). The two appendices, the code-size summary against the original ATES and the list of source inherited unchanged from it, were stranded in the stage-1 log and stale at 2026-08-30. They are now a hand-maintained fragment the generator \inputs after the converted body, so appending to this log keeps them. Re-measured 2026-09-11 with python3 src/utils/codesize.py (MEASURED, all of it): the Fortran tree is **205 files / 76,259 SLOC**, 21.7x the original ATES (47 files / 3,514 SLOC), against 111 files / 21,018 SLOC and 6.0x at the 2026-08-30 edition; the core excluding the Wind-AE port is 169 files / 71,503 SLOC, of which the production build is 95 files / 50,762 SLOC (14.4x) and the src/tests/ drivers are 74 files / 20,741 SLOC, the rule the script applies being every .f90 under src/ outside src/modules/wind\_ae/. 123 new core modules (49 production, 74 test drivers), 38 of the 47 original files modified, and the files **byte-identical with upstream are down from 13 to 8**: hybrd, hybrd1 and fdjac1 took the unit\_step\_floor difference-step argument, Source.f90 the well-balanced option (N37), J\_inc.f90 the single-spectrum-type decision, and speed\_estimate\_HLLC.f90 was deleted as dead code, so one upstream file is now removed rather than none. Update\_EXHALE\_stage1.pdf 464 -> 460 pages, Update\_EXHALE\_stage2.pdf 139 -> 158 pages (MEASURED, pdfinfo).

• **P21:** the carrier relaxation solves the chemistry of the composition it advances (src/modules/lower\_atmosphere/diffusive\_photochemistry.f90 +115/-27, src/EXHALE\_main.f90 +17/-7, src/tests/carrier\_retry/carrier\_retry.f90 +287/-16; these three counts are MEASURED by diff -u against the entry texts saved in the scratch directory before the first edit and NOT by git diff, for the reason in "An external commit during the session" below). Verdict: the design does what it was meant to do at the level it was meant to act. With the chemistry equilibrated after every transport step the pass keeps, the unbounded pass at a fixed wind converges to a state that really is stationary for the coupled operator: on the hot-Uranus carrier reload the carrier steady residual falls from **7.99E-02 to 1.42E-09** in the worst cell and from **2.32E-03 to 1.18E-11** volume-weighted (MEASURED, P21, M1 below). The frozen fixed point was not a fixed point of transport plus chemistry, and this is the measurement of that. Two negative results come with it and are in the same subsection: at the production bound nothing moves, and the joint fixed point at a fixed wind is a fully molecular column.

The design as implemented. A new private routine equilibrate\_chemistry\_at\_fixed\_pressure(rho, p, f\_sp, T, heat, cool, eta) takes get\_species\_densities (the particle count and the electron density of the composition), then comp\_T\_from\_p (its temperature at the conserved state's own pressure), then ioniz\_eq. No new module dependency was needed in either direction and ionization\_equilibrium.f90 is not edited. It is called inside the trial loop of relax\_photochemical\_composition immediately after a trial is ACCEPTED and before the fixed-point test, so a trial the operator refuses, or one that leaves the movement bound, is undone and halved before any sweep is taken and leaves the chemistry exactly where it found it. **The sweep IS the background reinstall**: ioniz\_eq writes bg\_cell(j) = ieq\_cell for every cell of a molecular sweep

(READ), while `install_background_of_adopted_state` belongs to the steady solver's best-iterate bookkeeping and is not involved here. The sweep is handed the transported partition and fixes  $x(\text{H}_2)$ , the oxygen carriers and, where the protons are carried, the ionized fraction from the composition it reads, so it returns the chemistry OF that partition and does not re-solve it. **The movement bound is read on the transport output, before the sweep:** the bound governs what the transport operator moved in one trial, and the chemistry then closes on the step that was kept. MEASURED consequence: the reported drift of every bounded production pass stays at  $9.98\text{E-}03$  to  $1.00\text{E-}02$  at trust  $1\text{e-}2$ , so the sweep does not carry the returned state out of its bound. The fixed-point test is now taken between two states the chemistry has closed on (`fnow` is recomputed after the sweep and becomes `fprev`), so `carrier_relax_fixed_point` means the carriers AND their chemistry stopped moving together. Interface: `relax_photochemical_composition(rho, v, p, T, f_sp, heat, cool, eta, trust, drift, nstep, outcome)`, with `p intent(in)`, the conserved state's own pressure untouched, and `T, heat, cool, eta intent(inout)`, written only where a refresh happened. The carrier branch of `steady_wind_with_element_diffusion` passes the new arguments and its own `ioniz_eq` call is DELETED as redundant; the `get_species_densities` and `comp_T_from_p` beside it are KEPT, because they are not a refresh but the filling of the density columns the routine keeps beside the composition, and they repeat the arithmetic of the pass's last internal call exactly.

Tests: `carrier_retry` **122/0 at entry, 124 PASS / 3 FAIL on a RED build** carrying the delivered module with the one call `equilibrate_chemistry_at_fixed_pressure` line removed, **127/0 after**. The new section (11) has five rows and states why the joint fixed point cannot be asserted on the synthetic 12-cell column at all: the shortest admissible trial still moves  $6.46\text{e-}4$  of the largest entry  $\text{H}_2$  mixing ratio, six decades above `relax_tol = 1\text{e-}10`. What it asserts instead is the consistency the joint fixed point rests on: the background third-body density read from the returned composition (GREEN  $4.7244\text{E-}07$  against the entry composition's  $5.5008\text{E-}03$ ; RED  $5.4938\text{E-}03$  against  $1.2191\text{E-}16$ ), the background temperature (GREEN  $2.5675\text{E-}07$  against  $5.4837\text{E-}03$ ; RED  $5.5241\text{E-}03$  against 0), and one further sweep moving the composition by  $1.3637\text{E-}10$  against `hybrdl`'s own `xtol = sqrt(eps) = 1.5\text{E-}08` (RED  $2.0110\text{E-}04$ ). Every other suite whose driver links `diffusive_photochemistry` is unchanged before and after: `carrier_returned_state_acceptance` 36/0, `attempted_step` 70/0, `steady_species_rows` 195/0, `certification` 70/0, `carrier_constraint_attribution` 10/0, `carrier_reference_scales` 14/0, `element_operator` 24/0, `species_masses` 9/0.

- **P22: the record of P21, of the terminology sweep and of the external commit** (this section with its two new subsections, the status rows of `docs/PLAN_20260911_partitioned_solver.md`, `docs/session_handoff_20260911.md`, the rewritten  $\text{H}_2$  front entry and the new outside-scope entry of `TO_BE_DONE.md`, and one sentence in `docs/code_status_20260910.md` section 3).
- **D1 to D3: docs/EXHALE\_physics\_and\_algorithms.tex, the physical equations and the algorithms as the code applies them** (user instruction 2026-09-11, evening: a document at the depth of Caldiroli et al. 2021 sections 2 and 3, written from the source, not from the memos). Main file with `natbib`; eleven section files in `docs/physics_overview/` (introduction and notation, hydrodynamics, radiation, ionization, thermal source,

molecules and metals, numerical scheme, boundary and initial conditions, stationary state, output and post-processing, the table of physics keys with the reference list, 69 entries). D1 wrote the hydrodynamic, numerical, boundary and stationary sections, D2 the radiation, ionization, thermal, molecular and post-processing ones, D3 made them one document (55 inline citations converted, one symbol for one quantity with a notation table, about 20 cross references, repetition and run narrative removed). MEASURED: latexmk rc 0, 57 pages, no LaTeX or hyperref warning, every `\texttt{}` file name present in `src/` (74 checked). Code facts established while writing, each stated in the document where it belongs: no escape-probability suppression acts on the Ly-alpha cooling channel (`lambda_coex_HI` in `Cool_coeff.f90` is the optically thin fit; the escape probability enters the metal fine-structure statistical equilibrium and the  $H(n=2)$  populations only); `equation_T` (`T_equation.f90`) is called by `post_process_adv.f90` and test drivers, the marching loop's temperature being `solve_energy_semi_implicit`; `Numerical_flux`: is a mandatory key with no default and every shipped input (70 of 70) says HLLC, ROE being refused with molecular chemistry unless `Caloric_EOS`: `monatomic`; 2D `approximate_method`: `alpha` applies neither the dayside dilution nor the mass-loss factor, which match the `Rate/...` strings only. Pointers added in `README.md` and the project `CLAUDE.md`.

- **The measurement of record after P14 to P17** (advisor, private snapshot build of the tree with P14 to P17, 8 threads, MEASURED): the HD 209458 b element reload, partitioned, 12 passes, ACCEPTED -- every active equation of this state is within its own tolerance at pass 12 (worst gated species row 7.73e-6, He/H partition at cell 289; mass 9.26e-10, momentum 2.44e-14, energy 2.17e-8); the hot-Uranus carrier reload, 3 passes, hydro info = 0 at every pass (mass 7.33e-12 / 1.29e-11 / 6.44e-12 admitted by the cell's anchor), the one refusing entry the H2 carrier row (7.06e-2 at cell 288).

Three changes outside the item list, made by the advisor while reading P14 to P17:

- `assert_written_state_is_the_accepted_one` (`src/EXHALE_main.f90`) carries an ABSOLUTE floor of 1e-12 beside its 1e-10 relative test. P14 reported the false alarm: the spread is a difference of  $O(1)$  numbers over their mean, so two evaluations of one state agree only to the rounding of that difference, and on the partitioned carrier reload a state flat to fourteen digits read **3.0982E-14 as written against 2.7487E-14 accepted** while the relative test alone called that a change of u. 1e-12 stands a decade above that rounding and four decades below the smallest spread a gate has ever accepted (READ, the comment at the test).
- the three rows of the `grid_and_gates` suite that wrote to a fixed directory under `$ROOT/build/tests/grid_and_gates` and ignored `EXHALE_TEST_OUT` now honor it (`momentum_row_from_fluxes_only.sh`, `base_level_single_statement.sh`, `sed_coverage_stop.sh`; READ, each now takes `${EXHALE_TEST_OUT:-$ROOT/build/tests/grid_and_gates}`). Two concurrent runs of the suite, which is what a before-and-after measurement is, collided there. P14 fixed `output_state_consistency.sh`, the fourth, because its item owned it.
- the paragraph at `assemble_residual` (`src/modules/time_step/steady_residual.f90`) that stated what the kind-generic rows do not hand back is rewritten to what P15 made true: the kind-generic rows return the momentum row and store the interface fluxes, the face pressures and, under the well-balanced option, the face departures they were built from, but neither the three terms of the momentum equation nor the equilibrium pressure force

the row is read against, so the one expression of each is evaluated there for the state just assembled; and a reconstruction continuation never reaches them, the generic text refusing it.

Two changes outside the item list, made by the advisor while reading P21:

- the H2 front: diagnostic of `src/EXHALE_main.f90` printed `r(max(j50,1))` and `r(max(j01,1))`, so a front that had left the domain was reported as the radius of the first cell. On P21's M1 run that read `x2=1e-2` at `r=1.0002` for a column in which no cell is below `1e-2` at all, beside a correct `x2=0.5` at `r=2.2367`. Each level is now printed as `> r(N)` when no cell of the column is below it.
- the noun of the 2026-09-11 terminology rule removed from the comment at `src/EXHALE_main.f90` line 2022 and from `diffusive_photochemistry.f90` line 248, and from the two fixture READMEs (about 26 lines), which item A4 left to whoever was editing those files.

### The pseudo-time start of the stationary restart route

Found while running P6 and settled by the advisor. `Restart intent: stationary` entered the stationary solve at the CFL interval of the reloaded state. A state written by a run is already relaxed, and a continuation started at the CFL step from it does not reach the root: MEASURED, 8 threads, the hot-Uranus carrier reload's hydrodynamic solve is handed back at `||R|| 2.685E-01` after 40 iterations and the next four passes enter at 1.133, 1.018, 1.026 and 1.025 and hand back 2.976E-01 to 3.579E-01 (`<scratchpad>/P6/runs/carrier_p6/run.log`), while the HD 209458 b element reload stops on its stagnation detector at `||R|| 1.490` and the pass is then refused because the element relaxation finds no admissible advance (`<scratchpad>/P6/runs/atomic_partition.new/run.log`). With the start at 1.0, the value the marching hand-off uses, the same two solves reach `||R|| 2.507E-09` at iteration 13 and hand back **1.498E-09** (carrier) and hand back **1.167E-08** (atomic). The advisor therefore made 1.0 the default of the `Restart intent: stationary` route (`src/EXHALE_main.f90` line 5234), with `EXHALE_PTC_DTAU0` as the override for measurement; the direct steady route keeps its CFL start, which is the pseudo-transient continuation it was designed as. Verified with no environment variable set: the run prints `stationary restart: dtau0 = 1.00E+00` and reproduces the first pass to every printed digit (`<scratchpad>/P7/carrier_default_dtau0/run.log`). No regression case sets `Restart intent: at all` (MEASURED: the `grep` over `backup/regression/*/input.inp` returns nothing), so this default moves no golden; it moves any user run that sets the key.

### P7: the measurement of record

One binary (`EXHALE_P6.x`, built from the merged tree), both fixtures, `Restart intent: stationary`, `EXHALE_PTC_DTAU0=1.0`, `OMP_NUM_THREADS=8`, 500 cells. The partitioned route is `Coupled carrier solve: False` with `EXHALE_OUTER_PASSES` setting the pass budget and `EXHALE_JFNK_MAXIT` the hydrodynamic cap; the coupled route is the same fixture with `Coupled carrier solve: True`. Wall times of the partitioned runs are the sum of the seconds the outer pass lines report, which agrees with the run directory's file timestamps to under a second; the coupled carrier control prints no total, so its time is the timestamp span of its run directory. Every number MEASURED from the log named.

| route           | fixture             | passes (hydro cap)                                                   | wall time | hydro mass / momentum / energy   | worst gated species row   | certified?            |
|-----------------|---------------------|----------------------------------------------------------------------|-----------|----------------------------------|---------------------------|-----------------------|
| state as loaded | carrier, hot Uranus | --                                                                   | --        | 2.284e-1 / 1.954 / 1.871         | H2 1.166e-1 at cell 492   | no, 4 entries         |
| partitioned     | carrier             | 8 (40)                                                               | 173 s     | 8.75e-12 / 1.66e-12 / 2.41e-9    | H2 6.217e-2 at cell 282   | no, 2 entries         |
| partitioned     | carrier             | 40 (40)                                                              | 831 s     | 8.121e-12 / 3.377e-12 / 2.730e-9 | H2 2.234e-2 at cell 443   | no, 2 entries         |
| coupled         | carrier             | one solve (250), stopped by the stagnation detector at iteration 105 | 321 s     | 1.231e-2 / 1.009e-1 / 5.801e-2   | H2 7.815e-2 at cell 283   | no, 4 entries         |
| state as loaded | atomic, HD 209458 b | --                                                                   | --        | 1.509 / 3.348e-1 / 1.892         | He/H 5.840e-1 at cell 291 | no, 11 entries        |
| partitioned     | atomic              | 5 (80)                                                               | 528 s     | 1.05e-9 / 4.40e-13 / 1.13e-8     | O 4.64e-4 at cell 268     | no, 9 entries         |
| partitioned     | atomic              | 12 (80)                                                              | 1043 s    | 1.054e-9 / 1.665e-13 / 1.501e-8  | He/H 5.468e-6 at cell 288 | <b>no, 1 entry</b>    |
| coupled         | atomic              | 3 (250), ended in the fourth                                         | 2392 s    | 1.16e-6 / 2.89e-6 / 2.83e-6      | He/H 8.29e-5 at cell 295  | no, run not completed |

Sources: <scratchpad>/P7/carrier\_elem\_newton\_part1/run.log, carrier\_part40/run.log, carrier\_coupled250/run.log, atomic\_elem\_newton\_part1/run.log, atomic\_part12/run.log, atomic\_coupled250/run.log. The atomic coupled control alternates the element relaxation too, because He\_diffusion is on, and its three completed passes cost 1380.59 s, 482.04 s and 528.98 s; it was ended in the fourth. The (EXHALE\_main) outer pass N: line of every pass carries the hydrodynamic flag, the worst gated species wind row with its cell, the three row maxima, omega, the carrier trust and the seconds, and the closing (certification) block names what refuses. Those <scratchpad>/ paths are the session scratch root and are TEMPORARY: they will be gone, and everything quoted from them is recorded here, in the two fixture READMEs and in docs/session\_handoff\_20260911.md. The runs reproduce from the pinned fixtures by the recipes in those READMEs.

What the ladders show. On the atomic fixture the worst gated species row falls  $2.61\text{e-}3$ ,  $3.81\text{e-}3$ ,  $1.77\text{e-}3$ ,  $9.45\text{e-}4$ ,  $4.64\text{e-}4$ ,  $2.36\text{e-}4$ ,  $1.20\text{e-}4$ ,  $5.97\text{e-}5$ ,  $3.07\text{e-}5$ ,  $1.50\text{e-}5$ ,  $7.94\text{e-}6$ ,  $5.47\text{e-}6$  over the twelve passes (the one rise, pass 1 to pass 2, is the alternation's own feedback across a hydrodynamic solve and is why a single rise does not end the loop), the refusing entries fall 9 to 1, and in the last pass every elemental wind row is within  $1\text{e-}5$ : He/H  $5.468\text{e-}6$  at cell 288, and the metals  $1.582\text{e-}6$  (Mg) to  $3.720\text{e-}6$  (O). On the carrier fixture the H2 wind row falls  $7.38\text{e-}2$  to  $2.23\text{e-}2$  over the 40 passes with the binding cell moving 290 to 443, which is the H2 front advancing about one cell in a pass and not a residual settling.

**The decision.** The plan's adoption rule is that the route is adopted only if it certifies where the coupled solve does not, or reaches a strictly lower joint residual in no more wall time. Neither fixture certifies, so the first branch is not met on either. The second branch is met on both: on the atomic fixture the partitioned route stands strictly below the coupled control on every judged row (mass  $1.054\text{e-}9$  against  $1.16\text{e-}6$ , momentum  $1.665\text{e-}13$  against  $2.89\text{e-}6$ , energy  $1.501\text{e-}8$  against  $2.83\text{e-}6$ , worst gated species row  $5.468\text{e-}6$  against  $8.29\text{e-}5$ ) in 1043 s against 2392 s for three passes; on the carrier fixture the partitioned route at 8 passes stands strictly below the coupled control on every judged row ( $8.75\text{e-}12$  against  $1.231\text{e-}2$ ,  $1.66\text{e-}12$  against  $1.009\text{e-}1$ ,  $2.41\text{e-}9$  against  $5.801\text{e-}2$ , H2  $6.217\text{e-}2$  against  $7.815\text{e-}2$ ) in 173 s against 321 s, and the 40-pass continuation goes on to  $2.23\text{e-}2$  for 831 s. The partitioned route is therefore what the production outer loop runs. Coupled carrier solve already defaults to False (`carrier_in_newton = .false.,` `src/modules/init/parameters.f90` line 419), so no default changes; True remains the coupled measurement option, and what this session added on the partitioned side is the joint acceptance contract of P6 and the three defect repairs P1 to P3 that made a pass mean something.

**What did not certify, and why.** Entry 1 was CLOSED the same afternoon by the user's first decision (P16, anchor (6)); entries 2 and 3 are stated as they stand:

1. **CLOSED 2026-09-11 by P16.** The atomic element reload refused on the hydrodynamic mass row alone,  $1.054\text{E-}09$  against  $3.0\text{E-}12$  at cell 16 ( $r = 1.0031$ ), on the three-unknown solve of this fixture. Every one of its twelve hydrodynamic solves ends with  $||R||$  between  $8.980\text{E-}09$  and  $1.501\text{E-}08$  against the run's `Resid tol` of  $1\text{e-}8$ , and the last one ends on no descent direction exists for the banded model at this state at  $||R||$   $1.501\text{E-}08$ . So the alternation has done its work (the element rows are inside their tolerance) and what is left is the base-layer mass row of the hydrodynamic solve, which is the N33 rounding floor territory and not a species-coupling question. **P10 settled which of the two it is.** With the best-iterate ledger reading that row against its own  $3\text{e-}12$  (P10), the same twelve passes end at  $8.893\text{E-}10$  at cell 3, the same order, and the row is MEASURED to carry no signal at that level: the ledger's best distance of pass 1 is  $4.407\text{E+}02$ , a mass row of  $1.32\text{E-}09$ , while the same state re-evaluated at its own composition after the best-iterate restore carries  $4.11\text{E-}09$ . At  $1\text{e-}9$  to  $1\text{e-}12$  the row is the rounding of the base-layer flux difference, so ranking iterates by it is ranking by rounding noise and no ledger can bring it down. The entry was therefore an ACCEPTANCE question, the  $3\text{e-}12$  tolerance against the base-layer rounding floor of this fixture, and not a solver item; the user decided it the same afternoon, and with the mass row judged at  $\max(3\text{e-}12, 10 * \text{floor}(j))$  the same twelve-pass recipe **CERTIFIES at pass 12** (MEASURED by P16 and again by P17, which certifies at pass 12 in both of its builds). The verdict is taken at a wind cell, not at the base: cell 34,  $r = 1.00665$ ,  $6.542\text{E-}10$  against the  $4.6\text{E-}09$  anchor of that cell, distance 0.141 (P16), and cell 50,  $r = 1.00977$ ,  $4.245\text{E-}10$  against  $3.2\text{E-}09$ , distance 0.132 (P17).

The base-layer floor of this fixture is what the anchor is measured on: see the decisions subsection below and docs/certification\_tolerance\_anchoring\_20260910.md, "Anchor (6)".

2. The carrier reload refuses on the H2 wind row, and the user's second decision of the afternoon ran the continuation that says what that row IS. MEASURED over 110 passes (the table and the recipe are in the decisions subsection below): the H2 front does NOT stop,  $x_2 = 0.5$  moving 1.0726 to 1.1230 R\_p over 108 passes at about  $5E-04$  R\_p in a pass with no sign of slowing, so the continuation is unfinished after 110 passes and 0.05 R\_p. And the worst gated row stops being the front after pass 40: it stands at **2.2E-02, nearly uniform in r**, and its cell walks outward through the wind, 443 to 500, reaching the outer boundary at pass 108 while the value does not fall. That floor is a property of the operator and not of the front: the bounded carrier pass (trust  $1e-2$  on the H2 mixing-ratio maximum, met by one transport step) advances the front but leaves the wind cells, whose H2 mixing ratio is orders of magnitude below the maximum, at one implicit step in a pass, so their own H2 balance never relaxes, and the drift measure, absolute in the grid maximum, does not see them. The consequence for the method, recorded and not adopted: a bounded front advance and a relaxation of the wind cells' H2 balance are two different operations, and the second needs its own step (a cell-relative measure, or a wind-window relaxation after the bounded pass) before the H2 wind row can be judged at  $1e-5$ . Whether the front stops at all inside 2 R\_p is a physical question and stays open (the P23 comparison against Koskinen et al. 2022 Model A).
3. The hot-Uranus mass row sits on the same floor throughout the carrier run,  $4.48E-12$  to  $1.57E-11$  against  $3.0E-12$  over the 40 passes, with the momentum row at  $4.14E-13$  to  $5.13E-12$  and the energy row at  $1.75E-09$  to  $4.56E-09$ . The mass row of that fixture is the rounding of the base-layer flux difference recorded in N33, which is the reading P16 then made operative: under anchor (6) the base cells of this fixture are ADMITTED by their own anchor at every pass and the refusal moves out to cell 180 ( $r = 1.0707$ ), where the floor is below  $3e-12$  and the row carries signal.

### The Codex adversarial review (2026-09-11)

Run on the working tree after P1 to P13, that is on the merged text of the morning's items and not on the committed tree. Two findings, both verified in the source by the advisor before any item was written, and both closed the same afternoon. The findings are quoted as they were written.

1. **[high]** "정체 종료 시 조성과 출력용 상태가 서로 달라집니다 (src/EXHALE\_main.f90:6503-6514)": on a stagnation exit the composition had been updated while the particle densities and the temperature were not refreshed on the element path, so the certified state and the written state differed; the recommendation was to move the progress control before the composition update. Verified in the source, reproduced as a measured inconsistency of  $1.317089E-08$  in the temperature of the written state, and **closed by P14** with exactly that ordering: the progress control now stands between the certification of the pass and the update, so a pass that ends the iteration takes no update, and the element path's update refreshes the particle densities and the temperature before its equilibrium sweep.
2. **[medium]** "고정밀 경로가 다른 상태의 압력 정보를 인증에 사용합니다 (src/modules/time\_step/steady\_residual.f90:256-258)": with EXHALE\_RESID\_QUAD the generic rows stored

only `face_flux` and `face_p`, so `momentum_row_terms_of_state` read the previous `RK_rhs` call's `face_q_up/face_q_dn` (zeros on the first call); the recommendation was to return the terms from the generic computation and to refuse the continuation combination. Verified in the source, and **closed by P15**, both halves: the kind-generic rows now hand their own face departures back along the route the fluxes take, and the continuation combination is refused, because the kind-generic rows CAN be reached on a continuation (`assemble_residual` chooses the text from `generic_precision_rows_arm()` alone and the generic text carries its own blending branch) and no single pair of departures satisfies the blended row identity.

Neither finding moves a certified state of the default path: P14's `wasp_full_newton` reload and P15's, with the key off and with `Well balanced: True` and `EXHALE_RESID_QUAD=2`, are CERTIFIED with every output byte-identical apart from the provenance timestamp. What each moves is stated in its own bullet above.

#### User decisions of 2026-09-11 (afternoon)

Two decisions were taken after the open items of the plan were put to the user.

**Decision 1: the mass tolerance is re-anchored above the fixture's measured rounding floor** (item P16, and the ledger consequence P17). The tolerance of the continuity row is now

$$\text{tol}(j) = \max(\text{cert\_tol\_mass}, \min(\text{cert\_tol\_mass\_ceiling}, \text{c\_round} * \text{floor}(j)) )$$

with `cert_tol_mass` =  $3\text{e-}12$  unchanged as the FLOOR, the ceiling 1, and `c_round` = `cert_mass_round_margin` = 10. The floor estimate is  $\text{eps} \cdot \rho(|v| + c_s) \cdot A / dV$  over the row's own scale, which is one ulp in the wind and about  $\text{eps}/\text{Mach}$  in a layer whose flux is a tiny residue of the momentum the cell carries at its own fastest signal speed. The brief's first expression for the floor,  $\text{eps} (|A+ F_{\text{up}}| + |A- F_{\text{dn}}|)/dV / \text{residual\_row\_scale}(1, j)$ , is inert, and that is a measured result and not a judgement: the row's scale IS the larger of the two addends and the  $dV$  cancels, so the expression is algebraically between one and two ulps at every cell, MEASURED at **4.4409E-16 on all 500 cells of all four states** examined, and ten times it never rises above  $3\text{e-}12$ . The adopted estimate bounds the step each mass row actually takes when one ulp is added to the density of every physical cell (`EXHALE_MASS_FLOOR_SCAN=1`, new and default off) within a factor

| state                                         | base Mach | step over the adopted<br>estimate | step over the brief's<br>expression |
|-----------------------------------------------|-----------|-----------------------------------|-------------------------------------|
| <code>wasp_full_newton</code> as<br>loaded    | 3.937E-03 | <b>1.5931</b>                     | 1.109E+02                           |
| <code>atomic_elem_newton</code> as<br>loaded  | 5.062E-06 | <b>1.8857</b>                     | 2.4365E+05                          |
| <code>carrier_elem_newton</code> as<br>loaded | 2.400E-04 | <b>2.6806</b>                     | 5.0915E+03                          |
| the HD 209458 b state P16<br>certifies        | 2.767E-06 | <b>1.7733</b>                     | 1.4201E+06                          |

All MEASURED by P16, largest ratio over the 500 cells of each state.  $c_{\text{round}} = 10$  is a round number 3.7 times above the largest of those ratios and inside N33's independent reading of the same floor against its own bound, 5.5 to 12 (READ). The outcome: the **HD 209458 b element reload CERTIFIED at pass 12**, where the control built from the entry text of the two files ends the same twelve passes NOT CERTIFIED on the hydrodynamic mass row ( $7.159\text{E-}10$  against  $3.0\text{E-}12$  at cell 13), and **wasp\_full\_newton unmoved** in both builds (CERTIFIED, mass row  $1.212\text{E-}12$  at cell 500 judged against the FIXED  $3.0\text{E-}12$  at distance 0.404,  $\log_{10} \text{Mdot}$  13.30, outputs byte-identical apart from the provenance timestamp), its floor being below  $3\text{E-}13$  at every cell. The anchor still refuses states: the HD 209458 b mass row is refused at pass 3 at distance 69.9, and on the hot-Uranus carrier reload the refusal moves from the base cells, admitted by their own anchor, out to cell 180 where the row carries signal. The derivation and all three impact measurements are the appended section "Anchor (6), 2026-09-11: the mass row's rounding floor" of docs/certification\_tolerance\_anchoring\_20260910.md.

**Decision 2: the long continuation of the hot-Uranus carrier reload.** The pass budget was set to 400 and the run went to pass **110**, where the progress control ended it ("the worst gated species row has not fallen in 3 consecutive passes"), the carrier trust having been halved  $1.0\text{E-}02$  to  $5.0\text{E-}03$  to  $2.5\text{E-}03$  at passes 109 and 110. Recipe: backup/regression/carrier\_elem\_newton, Load IC? True, Coupled carrier solve: False, Restart intent: stationary, EXHALE\_OUTER\_PASSES=400 EXHALE\_JFNK\_MAXIT=40 OMP\_NUM\_THREADS=8. Binary: the advisor's snapshot build of the merged tree after P1 to P12 and BEFORE P14 to P17, so every hydrodynamic solve of this run returns  $\text{info} = 1$  on the base-layer mass row ( $5\text{E-}12$  to  $1\text{E-}11$  against  $3\text{E-}12$ ) and the run predates anchor (6). All numbers MEASURED from its run.log.

| pass | worst gated H2 row | cell | H2 front $x_2 = 0.5$<br>[R_p] | $x_2 = 1\text{e-}2$ [R_p] |
|------|--------------------|------|-------------------------------|---------------------------|
| 1    | 7.38E-02           | 290  | 1.0726                        | 1.0966                    |
| 10   | 5.86E-02           | 281  | 1.0755                        | 1.1017                    |
| 20   | 4.07E-02           | 274  | 1.0796                        | 1.1070                    |
| 30   | 2.62E-02           | 267  | 1.0827                        | 1.1140                    |
| 40   | 2.23E-02           | 443  | 1.0871                        | 1.1199                    |
| 50   | 2.20E-02           | 473  | 1.0917                        | 1.1262                    |
| 70   | 2.19E-02           | 482  | 1.1017                        | 1.1413                    |
| 90   | 2.18E-02           | 490  | 1.1126                        | 1.1563                    |
| 100  | 2.17E-02           | 494  | 1.1184                        | 1.1643                    |
| 108  | 2.17E-02           | 500  | 1.1230                        | 1.1727                    |
| 110  | 2.17E-02           | 500  | 1.1230                        | 1.1727                    |

**Two regimes.** Passes 1 to 30: the worst gated row is at the front's outer edge, its cell moving 290 to 267 ( $r$  about 1.28 to 1.25), and it falls  $7.4\text{E-}02$  to  $2.6\text{E-}02$  as the front relaxes. From pass 40 the worst row is no longer at the front: it sits at  $2.2\text{E-}02$  and its cell walks outward through the wind, 443 to 500, reaching the outer boundary at pass 108 while the value does not fall. The front itself keeps moving outward at about  $5\text{E-}04 R_p$  in a pass,  $x_2 = 0.5$  from 1.0726 to 1.1230 over 108 passes, with no sign of stopping.

So three statements, the first two measured and the third a reading of them. (a) The front has NOT reached a stationary position in 110 passes: 0.05 R<sub>p</sub> traveled and the continuation unfinished. (b) The remaining wind residual of 2.2E-02 is nearly uniform in r and is NOT the front. (c) The reading: the bounded carrier pass, trust 1E-02 on the H2 mixing-ratio maximum and met by one transport step, advances the front but leaves the wind cells, whose H2 mixing ratio is orders of magnitude below the maximum, at one implicit step in a pass, so their own H2 balance never relaxes, and the drift measure, absolute in the grid maximum, does not see them. The progress control ended the run because after pass 40 the worst gated row measures that wind floor and not the front.

Consequence for the method, recorded and NOT adopted: a bounded front advance and a relaxation of the wind cells' H2 balance are two different operations, and the second needs its own step, a cell-relative measure or a wind-window relaxation after the bounded pass, before the H2 wind row can be judged at 1e-5. The physical question whether the front stops at all inside 2 R<sub>p</sub> is separate and stays open (the P23 comparison against Koskinen et al. 2022 Model A).

### P21: the fixed point at a fixed wind, and the production loop

All of this is MEASURED by P21 on backup/regression/carrier\_elem\_newton (Coupled carrier solve: False, Restart intent: stationary, EXHALE\_JFNK\_MAXIT=40, 8 threads, scratch copies), against a control binary built from the entry text of the three files.

**M1, the fixed point at fixed hydro.** EXHALE\_CARRIER\_TRUST=1e6, that is the movement bound disabled, and EXHALE\_OUTER\_PASSES=2 rather than 1, because the outer loop takes no composition update on its last pass (P6) and at 1 the relaxation never runs. Both binaries end the pass on carrier\_relax\_fixed\_point.

| quantity                                                | control (frozen chemistry)      | new (refreshed)               |
|---------------------------------------------------------|---------------------------------|-------------------------------|
| transport steps kept                                    | 38                              | 57                            |
| composition drift                                       | 9.79E-01                        | 9.95E-01                      |
| carrier steady residual after the pass, volume-weighted | 2.32E-03                        | 1.18E-11                      |
| worst cell                                              | 7.99E-02 at cell 266, r = 1.211 | 1.42E-09 at cell 2, r = 1.000 |
| H2 front, x2 = 0.5                                      | 1.1327 R <sub>p</sub>           | 2.2367 R <sub>p</sub>         |
| x2 = 1e-2                                               | 1.2056 R <sub>p</sub>           | nowhere in the domain         |
| largest x2 on the grid                                  | 0.9774                          | 0.9848                        |
| cells clamped onto the element budget                   | none                            | none                          |
| pass 1 wall time                                        | 18.64 s                         | 16.04 s                       |
| pass 2 wall time (hydrodynamics only)                   | 16.66 s                         | 37.06 s                       |

The wind H2 row before the pass is 7.38E-02 on both, at cell 290, reproducing the experiment's 7.380744e-2, and the control's 7.99E-02 reproduces its 7.989781e-2 to the digit. The experiment's three-number sequence has only two members in the production loop, the diagnostic being taken after the caller's refresh in the control, so the frozen intermediate (2.449021e-13) is not a state the production loop ever holds; in the new build there is no intermediate at all, which is the point. Wall time of the new pass is LOWER despite 50 percent more steps and one sweep a step; the plausible reading, offered as a reading, is that a transport step taken on the background of the composition it advances converges in fewer inner iterations than one taken far from a frozen background.

**M2, the production loop: at the production bound nothing moves.** Same fixture and recipe with the movement bound at its default  $1e-2$ , EXHALE\_OUTER\_PASSES=12 and a 60-pass continuation, against the control's first three passes and against the 110-pass ladder of the same day's decision 2 (both READ from the P7 run logs).

| pass | worst gated H2 row,<br>cell | recorded ladder     | $x_2=0.5 / x_2=1e-2$ | recorded        |
|------|-----------------------------|---------------------|----------------------|-----------------|
| 1    | 7.38E-02, 290               | 7.38E-02, 290       | 1.0726 / 1.0966      | 1.0726 / 1.0966 |
| 10   | 5.86E-02, 281               | 5.86E-02, 281       | 1.0755 / 1.1017      | 1.0755 / 1.1017 |
| 20   | 4.07E-02, 274               | 4.07E-02, 274       | 1.0796 / 1.1070      | 1.0796 / 1.1070 |
| 30   | 2.62E-02, 267               | 2.62E-02, 267       | 1.0827 / 1.1140      | 1.0827 / 1.1140 |
| 40   | 2.23E-02, 443               | 2.23E-02, 443       | 1.0871 / 1.1199      | 1.0871 / 1.1199 |
| 50   | 2.20E-02, 473               | 2.20E-02, 473       | 1.0917 / 1.1262      | 1.0917 / 1.1262 |
| 60   | 2.20E-02, 477               | (not in the ladder) | 1.0953 / 1.1327      |                 |

Twelve and sixty passes reproduce the recorded ladder to every printed digit: the worst gated H2 row, its cell and both front positions are identical at passes 1, 10, 20, 30, 40 and 50. Every pass ends on the movement bound with 3 to 6 kept steps and a drift of  $9.98E-03$  to  $1.00E-02$ , and every hydrodynamic solve returns  $info = 0$ . So **the 2.2E-02 wind floor does not go away and the frozen chemistry was not its cause**; the bound's measure, absolute in the grid maximum, remains the standing explanation. Cost is not a reason to hesitate: a production pass costs 3.07 to 3.72 s against the control's 3.10 to 3.54 s, and one equilibrium sweep on this 500-cell grid costs below 0.029 s at 8 threads (MEASURED from 300 marching steps of `mol_carrier` in 8.66 s including initialization), so 3 to 6 of them cannot be resolved against that scatter.

**M3 and M4, the default paths.** `mol_carrier` marching, EXHALE\_MAXSTEPS=300, one thread: `Hydro_ioniz.txt`, `Ion_species.txt` and both `_adv` files identical except the two `git=` provenance lines, and `Cooling_breakdown.txt`, `Heating_breakdown.txt` and `IC_dump.txt` byte-identical, which is the expected result because the marching loop calls `photochemical_transport_step` directly and never this routine. `wasp_full_newton` reloaded, one thread: the two run logs are byte-identical, both  $info = 0$  and CERTIFIED at  $||R|| = 1.920E-09$ , ten of the twelve written files byte-identical and the other two differing in the same provenance line alone. No molecules, so the branch is never entered.

**Does the front stop at fixed hydro, and where?** No, and the way it fails to stop is new information. At a wind held fixed and with the chemistry solved on the advancing composition, the pass runs to a genuine fixed point of the coupled operator ( $1.42E-09$ ) whose H2 fills the whole domain:  $x_2 = 2n(H_2)/n_H$  is 0.98 at 1.1  $R_p$ , 0.96 at 1.2, 0.84 at 1.5, 0.60 at 2.0 and 0.33 at 3.0  $R_p$ , with no cell below  $1e-2$  anywhere. The largest  $x_2$  is 0.9848, so this is NOT the element ceiling and no cell is clamped in either run, and P2's frozen artifact ( $x_2$  driven to the element ceiling in 123 cells) does not appear. A plausible reading, offered as a reading: the H2 column and its self-shielding grow together, so re-solving the photodissociating field on the advancing composition removes the sink that held the front, and nothing but the wind can stop it, which it cannot while it is held fixed. What this says is that **the front's position is set by the hydrodynamic response and not by the carrier operator alone**, consistent with the bounded advance being the only usable form of the update; the physical question, whether the front stops inside 2  $R_p$  in the coupled system, stays exactly where P23

has it.

**Does the wind floor go away?** No, as M2 records. The 2.2E-02 row that appears at pass 40 and walks outward through the wind is reproduced pass for pass and cell for cell with the chemistry refreshed. The frozen background was not its cause; the bound's measure, absolute in the grid maximum, still is.

### The terminology sweep (A1 to A4, 2026-09-11)

The user asked whether "arm" is a word physics uses for what this tree was using it for, and instructed that terms belonging to software engineering, clinical trials or management not be used as the names of physical or numerical concepts at all. The rule is now in `~/claude/CLAUDE.md` under "금지 표현", in two bullets. The first: avoid terms that physics, chemistry, mathematics and astronomy do not use, in documents, comments, log messages and identifiers alike, and write instead what the thing is, the standard term of those fields or plain English, naming what is compared ("control build", "measured build"), which physics option is meant ("the well-balanced option"), or which quantity ("the mass row", "the H2 front"); where such a term already stands, fix it where it is visible. The second: the NOUN "arm" is forbidden, for one side of a comparison, for a selectable code path and for one branch of a fixture alike, and code identifiers (ARM\_\*, \*\_arm) are held to the same rule; the VERB "arm / armed / arming / disarm", meaning to enable a device, is plain English and stays.

Four workers swept the tree. Every count is MEASURED by the worker that owns the files, by `grep -w -i -E "arms?"` against the git HEAD text, and no number, file name or measured statement was changed anywhere in the four diffs.

- **A1, the 51 top-level docs/\*.md outside the two update logs: 396 to 17.** All 17 that remain are allowed: 14 are the verb, 3 are the literal directory glob `arm*`. The replacements are by sense: a side of a comparison became "the two runs" or "both builds"; an optional code path became its key or its physics ("the well-balanced option", "the kind-generic residual assembly"); a fixture's solve became "the element reload" or "the carrier reload", which is the word those same documents already use; a rung of a ladder of runs became "rung"; one chemical network of a comparison became "network". One code span changed, the placeholder `<arm>` of a run-directory path in `vulcan_photochem_comparison.md`, which named nothing on disk and is now `<network>`, the names that do exist there being `vulcan/`, `ncho/`, `zahnle/`, `zahnle_s/` and `eqfit/` (READ from the tree).
- **A2, docs/Update\_EXHALE\_stage2.md: 176 to 1** (the one is the verb, "when both stellar keys arm  $H(n=2)$ "), and `Update_EXHALE_appendix.tex` 1 to 0. The largest class was "in both arms" of a two-binary comparison, now "in both builds"; then the optional paths ("the well-balanced option", "quad accumulation", "the kind-generic rows"), the fixtures ("the atomic element reload", "the carrier rows"), and the unknown space of B5j/B5k ("the density unknown", "the log unknown"). `Update_EXHALE_stage2.tex` was regenerated and the PDF rebuilt at **158 pages**, the same count as before the sweep (MEASURED, `pdfinfo`). Thirteen British spellings were corrected in the same pass, twelve in the Markdown and one in the appendix (modelling, centre/centred, labelled, cancelled, behaviour, travelled, NEIGHBOUR), as the American-spelling rule for documents asks.
- **A3, the hand-maintained LaTeX and the stage-1 log:** `Update_EXHALE_stage1.md` 211 to 4 and `Update_EXHALE_stage1.tex` 196 to 7 (verb uses and four `arm_heh1_x2matched` case-name literals), and

nine other docs/\*.tex files, of which lit\_wasp121.tex was deliberately not edited: its "Fe I (blue arm), VLT/UVES" is the spectrograph's own blue arm and its "spiral arms" are the outflow's, both standard astronomy and not the forbidden sense. Nine PDFs were rebuilt with every page count unchanged (MEASURED, pdfinfo: Update\_EXHALE\_stage1 460, EXHALE\_user\_manual 90, lhs1140b\_exhale\_vs\_pwinds 76, and six smaller memos).

- **A4, the source, the tests, the scripts and the top-level documents: 35 files, 328 to 12**, the 12 all the verb plus two references to the arm\* case directories by name. Identifiers renamed: generic\_precision\_rows\_arm to generic\_precision\_rows\_selected; ARM\_OFF, ARM\_QUADRUPLE, ARM\_GENERIC\_DOUBLE to ROWS\_PRODUCTION, ROWS\_QUADRUPLE, ROWS\_GENERIC\_DOUBLE; arm\_carrier\_log\_unknown to form\_carrier\_unknown\_space (named for what it forms, since it runs in both coordinates and always forms carrier\_log\_floor); arm\_departure\_of\_the\_rows to key\_departure\_of\_the\_rows; the local arm\_name to kind\_name; the local e\_arm to e\_wb; and eleven printed test-row names. Environment variables and input keys were NOT touched (EXHALE\_RESID\_QUAD, Well balanced:, EXHALE\_ELEMENT\_CONSTRAINT\_ROWS). Every old row name and changed message was grepped over the whole working copy and matched only the file that defines it, so nothing parses them. MEASURED: the eight suites that link the changed modules keep their counts exactly (krylov\_and\_dogleg 334/0, certification 70/0, grid\_and\_gates 200/0, steady\_species\_rows 195/0, residual\_determinism 5/1 by design, steady\_completion\_flag 3/0, steady\_selfconsistent\_residual 3/0, element\_operator 24/0); and the wasp\_full\_newton reload is byte-identical between a build of the delivered text and a control build of the entry text, both plain (info = 0, ||R|| = 1.495E-09, flux spread 1.911E-12) and under EXHALE\_RESID\_QUAD=2 with Well balanced: True (info = 0, ||R|| = 7.364E-10, flux spread 1.591E-13), so the renamed selector still selects. That the selection is not inert was measured beside it: EXHALE\_RESID\_QUAD=1 gives ||R|| = 7.363E-10 at a flux spread of 1.588E-13, different in the last digits from mode 2, while mode 2 against no selection at all is byte-identical, which is the design assertion for the generic-double instantiation.

Three further occurrences were removed by the advisor at the close, outside the four items: the noun in the comment at src/EXHALE\_main.f90 line 2022 and in diffusive\_photochemistry.f90 line 248, and about 26 lines of the two fixture READMEs (backup/regression/atomic\_elem\_newton/README.md, carrier\_elem\_newton/README.md), which A4 left to the item then editing them.

What stays by design, everywhere: the verb; the arm\* case directories under backup/regression/ and the two glosses that now point at them as file names; and the astronomy senses in lit\_wasp121.tex.

### An external commit during the session

Commit **a7c18a6**, "**remove build\_\***" was made by the user at 21:39:24 KST on 2026-09-11 from another session, while item P21 was mid-edit (READ, git log and git show --stat a7c18a6: 149 files changed, 39,652 insertions, 201 deletions). It removed the build\_dift/ and build\_fcheck/ dependency stamps; it added the 141 files of docs/audit\_20260905/partition\_experiment\_20260911/runs/, 38 MB of copied inputs, .state and .faces dumps and logs (MEASURED, du -sh); and it carried P21's in-progress edits of src/EXHALE\_main.f90,

diffusive\_photochemistry.f90 and carrier\_retry.f90 as they stood at that minute. Two consequences are recorded because they are visible in the measurements above. P21's line counts are diff -u against the entry texts saved in its scratch directory before the first edit, git diff no longer being able to produce that item's diff. And the provenance git= line of every output written by a build made after the commit reads a7c18a620f1b where the control binary's reads 0608382f639c, which is the only difference in P21's M3 and M4 comparisons. The "What the repository carries" paragraph of docs/audit\_20260905/partition\_experiment\_20260911/README.md, which said runs/ was kept in the working copy only, now records that it is committed since a7c18a6.

## Gates

Advisor verification of the merged tree (READ from the closing brief): snapshot build with the default toolchain; element\_operator 24/0, carrier\_retry 122/0, krylov\_and\_dogleg 290/0, certification 57/0, steady\_species\_rows 194/0 and 195/0 after P8; grid\_and\_gates 158 PASS with 3 FAIL, all three missing files of the snapshot copy (inputdata, examples) rather than assertions (P4 measured the same suite 178/0 in the live tree); the wasp\_full\_newton reload info = 0, ||R|| 1.667e-9, CERTIFIED, log10 Mdot 13.30, unchanged. make check (advisor, 2026-09-11 09:40 KST, the shared build rebuilt from this tree): REGRESSION PASS, every DEFAULT\_CASES output and wasp\_full\_newton byte-identical to the goldens; no golden moved, because no default case enters the stationary solve or the fixed-wind relaxations.

After P10 to P12 (advisor, merged tree, private snapshot build, 8 threads, MEASURED 2026-09-11): the thirteen suites that link the changed modules all PASS (krylov\_and\_dogleg 309/0, grid\_and\_gates 187/1 with the one FAIL a WASP-52b SED file absent from the snapshot copy, steady\_species\_rows 195/0, element\_operator 24/0, carrier\_retry 122/0, certification 57/0, attempted\_step 70/0, carrier\_returned\_state\_acceptance 36/0, species\_masses 9/0, carrier\_reference\_scales 14/0, carrier\_constraint\_attribution 10/0, constrained\_network\_layout 8/0, steady\_completion\_flag 3/0). The two fixtures re-run on that binary reproduce P10's and P11's records: the HD 209458 b element reload, 12 passes, worst gated species row 2.61e-3 -> 5.47e-6 with only the hydrodynamic mass row refusing (8.893e-10 against 3e-12 at cell 3, r = 1.0006); the carrier reload, 3 passes, H2 7.38e-2 -> 7.06e-2 at cells 290 -> 288, mass 8.12e-12 / 4.85e-12 / 4.96e-12, momentum 3.22e-13 / 4.47e-13 / 3.54e-13.

Second make check after P10 to P12 (advisor, 2026-09-11 15:15 KST, the shared build rebuilt from the merged tree): REGRESSION PASS, every DEFAULT\_CASES output and the cold-start wasp\_full\_newton byte-identical to the goldens; no golden moved.

After P14 to P17, each count MEASURED by the item that owns the file and READ from its report: grid\_and\_gates **200/0** (P14, the second case of output\_state\_consistency.sh; 199/1 on the entry text), krylov\_and\_dogleg **321/0** after P15 and **334/0** after P17, certification **70/0** after P16 (57/0 on the entry driver), with steady\_species\_rows 195/0, adv\_static\_limit 53/0 and the other suites each item ran unchanged in both of its builds. The two rows that FAIL by design or by fixture are unchanged: residual\_determinism's closure\_spread\_within\_the\_row\_tolerance\_atomic\_elem\_newton at 3.354E+01 (N33, by design) and steady\_selfconsistent\_residual's handback\_matches\_the\_accepted\_iterate on a reload log that prints no iterate norm.

Third make check after P14 to P18 (advisor, 2026-09-11 21:10 KST, the shared build rebuilt from the final tree): REGRESSION PASS, every DEFAULT\_CASES output and the cold-start wasp\_full\_newton byte-identical to the goldens; no golden moved.

Fourth make check after P21 and the terminology sweep (advisor, 2026-09-12 01:40 KST, the shared build rebuilt from the tree with P21, A1 to A4 and the closing edits): REGRESSION PASS, every DEFAULT\_CASES output and the cold-start wasp\_full\_newton byte-identical to the goldens; no golden moved.

### Reported by the workers and not fixed

- P1: relaxation\_drift\_covers\_every\_element in steady\_species\_rows builds a column whose species miss its own density by 1.660E-02 (closed the same day by P8); relax\_element\_composition damps helium alone, which belongs to the outer composition control; element\_diffusion\_step's trace-metal branch re-seeds an exhausted cell and project\_elements clamps grp\_new at zero, the only place in the step that can create or destroy mixture mass, now covered by the closure test and not investigated further.
- P2: the carrier relaxation reaches into the steady residual for its advecting flux (carrier\_rows\_advect = .true.) while the element operator is given its flux by the caller, so the routine cannot be called at all unless someone else assembled a mass row; residual\_row\_scale in steady\_residual.f90 carries a comment block that is the history of a change and not the physics; the fixed-point exit of the carrier relaxation is unreachable in production while the H2 front propagates, and loosening relax\_tol to make it reachable is not a fix.
- P3: ~~distance\_from\_certification divides the hydrodynamic slot by the run's Resid tol and not by each row's own certification tolerance, so the best-iterate ledger cannot see a mass row above 3e-12 while ||R|| is below 1e-8, which is exactly the state D3 is about (it did no harm on the D3 run, where the returned-iterate's mass row is 5.945e-12 against the last iterate's 7.914e-12, but the ranking does not guarantee it)~~ (fixed 2026-09-11, P10); steady\_selfconsistent\_residual assertion 2 (closed the same day by P8); attempted\_step/run.sh and carrier\_returned\_state\_acceptance/run.sh write a fallback build\_stamp.f90 into their own work directory, where the compile cannot find it, so a tree that has never been built fails with Cannot open module file 'build\_stamp.mod' (fixed 2026-09-11, P12, in those two and in five sibling scripts).
- P4: the kind-generic rows do not form the equilibrium pressure term themselves (closed from assemble\_residual, but the two places must stay one expression if the kind-generic rows are ever made to write their own face data back); ~~the default path's PLM momentum scale carries a coordinate artifact, the geometric pressure term (A+ - A-) p\_j/dV entering the max separately although it is one physical term with the flux divergence, so at zero gravity with a uniform pressure the scale is 2 p/r where the physical force is zero (MEASURED: the well-balanced scale, which is the row's true content, sits 0.944 below the base scheme's for both Riemann solvers at every N; under WENO3 the two agree to 2E-13). Fixing it moves the default momentum measure at every cell and therefore the acceptance of every recorded state, so it is a user decision.~~ (fixed 2026-09-11, P11, on the user's instruction: the scale is now the largest of the three physical terms, the zero-gravity artifact reads 2.22507E-308 in place of 1.000, and the acceptance movement is measured, wasp\_full\_newton still CERTIFIED.)

- P5: the same truncation-as-floor reasoning stood in three other documents (closed the same day by P8); the analysis document's own title is still stronger than its corrected conclusion; docs/session\_handoff\_20260910.md and docs/steady\_solver\_design.md carry statements of the same family, not read line by line; the experiment's private build is retained at /tmp/exhale\_partition\_20260911.ZxXJEB per its own section 9.
- P8: docs/certification\_tolerance\_anchoring\_20260910.md still opens with cert\_tol\_element and cert\_tol\_carrier at 1e-8, which decision 22 (a) superseded, and whether that memo is a dated record or a live reference is a decision; docs/code\_status\_20260910.md section 3 described the stage-2 solver program as closed (this session's item P9 marks it superseded); ~~mass\_closed\_column exists only in src/tests/element\_operator/element\_operator\_tests.f90~~ and two suites now build the same column by hand, so the recipe belongs in one place a test can call (fixed 2026-09-11, P12: src/tests/test\_columns.f90).
- P9: the Coupled carrier solve key comment in src/modules/files\_IO/input\_read.f90 states that the alternation "was measured not to converge and not to be able to" (section 139). The first half is what P7 measured against: the alternation with the joint acceptance contract of P6 brings every elemental wind row of the atomic fixture inside 1e-5 and moves the carrier front monotonically, so the comment's reading is at least narrower than it states. Not my file this increment; it needs the sentence rewritten to what section 139 actually measured (the old alternation, without the acceptance contract, and its sign error on the front's motion): (fixed 2026-09-11, advisor.)
- P10: two rows of src/tests/steady\_species\_rows/steady\_species\_rows\_tests.f90 state the superseded semantics, feeding a raw ||R|| into the slot that now takes a distance and asserting that the hydrodynamic slot follows the run's Resid tol (MEASURED 193 PASS / 2 FAIL on the new objects) (fixed 2026-09-11, advisor); dj that the ledger ranks with is the accepted trial's own rows while the judged rows line prints the loop-top evaluation, so the reported best distance can sit below the minimum of the printed lines (wasp\_full\_newton: 8.469E-02 against a printed minimum of 1.178E+00), which is entry-text behavior and not a defect; in the EXHALE\_RESID\_BRANCH\_REPORT block one divisor for the whole class stays, because the block aggregates spreads across seeds whose maximum can sit in different rows.
- P11: the kind-generic rows still hand back neither the momentum terms nor the face departures (hydrodynamic\_rows.f90's store\_the\_interface\_fluxes writes face\_flux/face\_p and nothing else), so under EXHALE\_RESID\_QUAD the well-balanced face departure and, on a reconstruction continuation with  $0 < \lambda < 1$ , which of the two pressure conventions the blended flux carries are read from the last RK\_rhs evaluation instead of from the state just assembled; the shape of the fix is two weights in place of the use\_plm branch, and the option is default off and a rounding-control experiment nothing is adopted from. (fixed 2026-09-11, P15, which was also the Codex review's second finding: the kind-generic rows store their own face departures, and the continuation combination is refused rather than weighted, because the two schemes reconstruct different faces and no single pair of departures satisfies the blended row identity. The momentum terms are still not handed back by the kind-generic rows, and they are not read from a previous evaluation either: the one expression of each is evaluated at the call site for the state just assembled, which is the P4 entry above.) The marching stop of mol\_carrier reports a momentum measure of 1.979E+00 against a scale that is now the largest physical term in the layer, which is commensurable with the mass and energy entries in a way it was not; whether the marching

stop should quote it at all is an acceptance question.

- P12: nothing further. The four sibling scripts carrying the same dead fallback and the work directories hardcoded with no override, which the P12 report listed as outside its own two files, are what P12b closed the same day.
- P14: ~~assert\_written\_state\_is\_the\_accepted\_one raises a false alarm whenever the flux spread reaches the rounding level (MEASURED on the partitioned carrier reload, 3.0982E-14 as written against 2.7487E-14 accepted on a state whose mass flux is flat to fourteen digits, while the same run at a spread of 1.7736E-03 passes the test); it needs an absolute floor beside the relative one, or the warning will keep accusing the best states the solver produces and three rows of the grid\_and\_gates suite write to a fixed directory under \$ROOT/build/tests/grid\_and\_gates and ignore EXHALE\_TEST\_OUT, so two concurrent runs of the suite collide there (both fixed 2026-09-11, advisor, the floor at 1e-12 and the three rows named in the paragraph above).~~ What is left is one line of documentation: src/tests/grid\_and\_gates/README.md gives one line for each test name, and the output\_state row now asserts two things where the README says one.
- P15: quad\_rows\_calls and quad\_rows\_seconds are incremented in hydrodynamic\_rows\_in\_quadruple\_precision and public, and no file reads them (grep over src/), so the quadruple-precision rows' measured cost, which the module header calls a measured number and not an estimate, reaches no run summary; relatedly, nothing in run.log says whether EXHALE\_RESID\_QUAD took effect, so P15 had to establish that its quad run was a quad run indirectly, from the iteration-1 measure moving between the two binaries and from the sweep-kind reset READ in steady\_newton.f90. A line in the setup report, or those two counters reaching it, would make a quad run self-documenting. And the continuation refusal the item added has no executed row: it ends in error stop, which cannot be asserted from inside the driver process, and no production path reaches it, the key being default off and the continuation controller disarming before the JFNK finish (READ).
- P16: the control build of the HD 209458 b reload ends with a state its own certification refuses and **exits 0**, because the outer loop's ending is the pass budget and no stationary claim is made. That is the run-mode contract working as written, but a reader of the shell status cannot tell that case from a certified one.
- P17: ~~hydrodynamic\_distance\_from\_certification has no production reader left and cannot be deleted without breaking the compile of a test file the item does not own (item P18, above).~~ Two fixture observations, no action taken: running grid\_and\_gates twice at once with distinct EXHALE\_TEST\_OUT makes the flux\_spread sub-test report mass\_flux\_spread\_recomputed\_from\_output measured=no\_dump\_line, so the two invocations share something outside EXHALE\_TEST\_OUT (each invocation on its own gives 200/0), which is worth a line in that suite's README for whoever next runs two builds side by side; and the log-driven block of steady\_species\_rows requires a (species\_jac\_test) line in the log, so the best judged iterate <= min judged rows seen invariant it states is checked only on logs made with the species Jacobian probe armed, and P17 had to apply that comparison by hand (0 of the 12 solves of its new build report a best above their own printed minimum, against 1 of 12 in the control). A fixture log carrying that probe would be worth pinning.
- P21: the archived experiment drivers under docs/audit\_20260905/partition\_experiment\_20260911/ (transport\_wind\_experiment.f90 and the driver\_snapshot.f90 copies beside the runs) call relax\_photochemical\_composition with its old six-argument signature. They are not in the Makefile SRC list and no test run.sh builds them, so nothing compiles them; they are the record of the run they belong to

and are left as they stand. The H2 front: diagnostic P21 reported beside it was fixed by the advisor (above).

## 9. PLAN\_20260912: the seven findings of the stage-2 review (2026-09-12)

Plan of record: docs/PLAN\_20260912\_review\_fixes.md. User instruction of 2026-09-12: analyze docs/Update\_EXHALE\_stage2\_review\_20260912.md against the code, and where its analysis is right, write the plan and start the corrections. All seven findings R1 to R7 were confirmed in the source before the plan was written (the confirming lines are listed at the head of the plan); the corrections are items Q1 to Q4, Q1 and Q4 by the advisor, Q2 and Q3 by workers under docs/worker\_rules.md. Every number below is MEASURED on the tree of this section unless marked READ.

### The state contract at a fixed hydrodynamic state (R2, R3)

The decision behind Q1: at a fixed hydrodynamic state the quantities the solve holds are the conserved variables  $u = (\rho, \rho v, E)$ . A composition update at fixed hydro keeps  $u$  and recomputes the pressure and the temperature from the unchanged thermal energy  $\rho e = E - (\rho v)^2 / (2 \rho)$  through the caloric EOS of the new composition (pressure\_from\_energy\_density after get\_species\_densities, then comp\_T\_from\_p). A held pressure is not a state the solve knows: the review's 500-cell probe showed that keeping  $p$  while the main loop keeps  $E$  describes two molecular states  $3.3e-3$  apart.

### The items

- **Q1 (R1, R2, R3, R6, R7 and the outer progress measure), advisor.** diffusive\_photochemistry.f90 (+206/-204), the carrier branch and the summary of steady\_wind\_with\_element\_diffusion in EXHALE\_main.f90, src/tests/carrier\_retry/carrier\_retry.f90 (+195/-77) and its run.sh.
  - R1: photochemical\_transport\_step takes an optional trial argument; a trial of the stationary relaxation returns before the history mark, the message and the stop, so a rejected trial (an uncovered interval, a chemistry that did not close, a displacement beyond the movement bound) leaves carrier\_history\_certifiable\_flag as it found it. The marching path passes no trial and is unchanged.
  - R2, R3: relax\_photochemical\_composition( $u, v, f_{sp}, p, T, \text{heat}, \text{cool}, \text{eta}, \text{trust}, \text{drift}, \text{nstep}, \text{outcome}$ ) now takes the conserved state;  $p$  and  $T$  are outputs. pressure\_and\_temperature\_at\_fixed\_conserved\_state is the one expression of the contract above, and equilibrate\_chemistry\_at\_fixed\_conserved\_state cycles ( $p, T$  at fixed  $u$ ; sweep;  $p, T$  again) until the largest relative change of  $T$  falls below  $1e-6$  (at most five cycles), reads the sweep ledger (n\_nonfinite) and a finite composition, and reports ok; a refresh that does not close refuses the trial (carrier\_relax\_chemistry\_refused, a new named ending "the chemistry of the shortest admissible trial did not close") and the entry composition, background,  $p, T$ , heating, cooling and  $\text{eta}$  are restored. The caller in EXHALE\_main.f90 rebuilds  $W$  from the unchanged  $u$  and takes the returned composition; nothing is reconstructed from a held pressure any more.

- R6: the carrier retry suite's column is built by `column_carrying_its_own_density` of `src/tests/test_columns.f90` (the mass-closed constructor of the element suites; `run.sh` compiles it before the driver) with the ions moved out of the neutrals; a row asserts the column reconstructs its own density to  $1e-13$  before any step. The consistency rows are absolute: the returned pressure is that of the unchanged thermal energy (0.0), the returned temperature is that of the returned composition (0.0), the background third-body density and temperature the sweep stored are those of the returned composition ( $6.53e-7$  and  $2.28e-7$  relative, allowance  $10 \text{ } ieq\_res\_tol$ ), and one further sweep leaves the returned composition within the sweep tolerance ( $8.83e-7$  against  $ieq\_res\_tol$ ). Nine rows for R1: a rejected trial keeps the history certifiable, keeps no step, hands back the entry composition bit for bit and names the refused interval; a pass after the refusals keeps a step and stays eligible; a refused trial in phys mode reports exhausted, leaves the history certifiable and writes no composition. RED on the entry text (the trial marking the history, the refresh not closed): five rows fail (pressure  $1.2e-7$ , temperature  $3.5e-7$ , further sweep  $1.2e-6$ , the two R1 rows 0 of 1); GREEN on the tree: 138 PASS, 0 FAIL.
  - R7 and the progress measure: the outer summary of `steady_wind_with_element_diffusion` ranks every entry by its binding distance (`dist_bind`, `row_at_bind`, `jbind` where the entry carries them), so the mass row's refusal is attributed to the cell that binds it and not to the largest row over another cell's tolerance; and the progress control halves omega and the carrier movement bound on the JOINT distance of the state (the largest entry over its own tolerance, `prog_worst`), not on the worst species row alone. An entry that is not finite or not available counts as an infinite distance.
  - The review's 500-cell probe (`docs/audit_20260905/stage2_review_20260912/molecular_wind_state_probe.f90`, adapted to the new signature, hot-Uranus carrier reload, one hydro solve and one bounded relaxation at trust 0.01): returned temperature against the EOS of the returned state 0.0 (was  $5.5997819e-9$ ), pressure of the unchanged thermal energy against the returned pressure 0.0 (was  $3.2993e-3$ ), thermal-energy change a held pressure would need  $6.9e-16$  (was  $3.3250e-3$ ); mass closure  $8.33e-15$  and the five retained steps at drift  $9.9955e-3$  unchanged.
  - Impact on the routes: the unbounded pass of P21 M1 repeated on the new contract (`EXHALE_CARRIER_TRUST=1e6`, cap 40) reaches the fixed wind's fixed point in 61 transport steps at drift 0.995 with the H2 front ( $x_2 = 0.5$ ) at 3.079 R\_p (P21: 57 steps, 2.237 R\_p; both fully molecular fixed points, the front leaving the domain). The 12-pass production loop ends at H2 row  $5.49e-2$  at cell 279, front 1.0755 R\_p (the 110-pass ladder of P16 at pass 12:  $5.50e-2$  at cell 279, 1.0765 R\_p); no pass triggered the halving. `atomic_elem_newton` at three passes: pass lines identical to the control build. `wasp_full_newton` reload and the `mol_carrier` case (12000 steps, one thread): data rows identical to the control build and to the golden.
- Q2 (R4 in the operator), worker (Sonnet). `binary_element_diffusion.f90` (+108/-43), `src/tests/element_operator/element_operator_tests.f90` (+88/-1). `element_diffusion_step` judges its candidate on every call: the entry composition is kept, the four tests (solved, finite, X inside [0,1] to `element_fraction_bound_tol`, mass closure no worse than the entry's by `element_mass_closure_tol`) run whether or not status is present, and the entry composition is restored on a refusal. The outcome is always

left in the new public `element_step_last_status`, and `element_step_outcome_text` names it. Two new rows repeat the ordinary and the NaN-residual steps with no status at the call site: RED on the entry text (the refused candidate with departure 0.86 adopted, the status left at accepted), GREEN after (`element_operator` 28/0). A second, independent defect found by the change: the Newton solve of the helium mass fraction (`solve_mass_fraction`) read its arithmetic floor only on a DESCENDING pass, so a step whose residual had already reached the floor and whose next pass found no improving trial reported unsolved (residuals  $1e-12$  to  $1.1e-5$  at the refusal, MEASURED on the closed-column relaxations of `diffusion_tests` T1a, T7c, T7d and T10, which went 35/0 to 27/8 under the unconditional judgment alone); the no-descent exit now reads the same floor, and `newton_drop` moved from  $1e-6$  to  $2e-5$  so that the  $\sim 1e-5$  floor of the  $K_{zz} = 2e12$  homopause column (T7), which the module header already documented, lies inside it (`diffusion_tests` 35/0 again). Cost on `mol_diffusion` at 300 steps: 54.4 s against 54.2 s, inside the noise. `mol_diffusion` and `lower_profile` at 300 steps: data rows identical to the control build. `atomic_elem_newton`: pass 1 identical; at pass 2 the judged rows are identical and `omega` differs (0.500 control, 0.250 Q2), attributable to the floor change turning a floor-level residual from unsolved to solved (READ from the Q2 report; the advisor's own three-pass run of the merged tree, above, has `omega` 0.250 at pass 2 on both builds because the control binary of that run already carried Q2).

- Q3 (R5), worker (Sonnet).** `certification.f90` (+105/-17), `src/tests/certification/certification_contexts.f90` (+106), `docs/certification_tolerance_anchoring_20260910.md` (addendum). `mass_row_cell_verdict` returns a third, optional outcome: RESOLVED (`cert_mass_round_margin * floor < cert_tol_mass_ceiling`, judged as P16 anchored it) or UNRESOLVED (the margin times the floor reaches the ceiling, or the floor is not finite): an unresolved cell never reads within, its distance is  $\max(1, q/\text{ceiling})$ , so the review's counterexample ( $q = 0.5$  at floor 1) reads `within = F`, distance 1.0 (entry text: `within = T, 0.5`). The column loop is the pure `mass_row_column_verdict`, which counts the unresolved cells and names the first (`n_mass_unresolved`, `j_first_mass_unresolved`), and the report names that cell even when a different cell binds the verdict. `certification` 84/0 (70/0 before the new rows), `krylov_and_dogleg` 334/0, `steady_species_rows` 195/0 (READ from the Q3 report; the advisor's run of `certification` on the merged tree 84/0). The addendum records that the margin of 10 was measured on three converged or converging fixtures only, none a near-zero-flow or molecular-EOS column: the UNRESOLVED outcome exists for the regime the fixtures do not reach.
- Q4 (R4 in the marching caller), advisor.** The marching loop of `EXHALE_main.f90` reads `element_step_last_status` after the element step: in phys mode a refused step refuses the whole attempted step (`as_reject_element` at the diffusion operation, the verdict text naming the test that refused it), which the controller restores and retakes at half `dt` like a carrier interval left uncovered; in init mode the march goes on from the composition the step restored and the refusals are counted (the final report prints "element steps refused inside the initialization march: n"). A test knob, `element_refuse_leading_steps_for_test` (`EXHALE_ELEMENT_REFUSE_STEPS`, zero in every production run), refuses the first `n` element steps so the path can be exercised: on `mol_diffusion` in phys mode with two refusals injected, attempt 1 at  $1.47451E-04$  and attempt 2 at  $7.37253E-05$  are refused "at the element diffusion step: an element invariant broke" and attempt 3 runs at  $3.68626E-05$ ; in init mode the two refusals are counted and the run exits 0. (The phys-mode

run of that fixture then fails at step 0 on the coupled source step's fixed point with or without the knob, exit 2: a property of that fixed-step snapshot in phys mode, not of Q4.) Found beside it and fixed: the refusal census of attempted\_step\_note\_step stopped one reason short (as\_reject\_source\_fixed\_point), so a source-energy refusal was never counted by reason; the bound is now as\_reject\_source\_energy.

- **The stagnation row of grid\_and\_gates.** On the joint progress measure the configuration output\_state\_consistency.sh used to force the stagnation ending (EXHALE\_JFNK\_MAXIT=5, EXHALE\_CARRIER\_TRUST=1e-4) runs out its budget instead: the starved hydrodynamic rows keep falling (energy 4.9E-01 to 3.9E-02 over 8 passes, 1.1E-01 at 20) while the H2 row stands at 7.3E-02. The row now uses EXHALE\_JFNK\_MAXIT=40 with EXHALE\_CARRIER\_TRUST=1e-6: the hydro solves reach their roots, the H2 row stands at 7.38E-02 for four passes and the ending is announced at pass 4; the second assertion of that script (the ending hands back one state) measures 7.1e-16 against 1e-12. The script's header and the suite README say why the configuration moved.

## Gates

Advisor verification of the merged tree (private build build\_Q1 / EXHALE\_Q1.x of the default toolchain, deleted at the close): carrier\_retry 138/0, certification 84/0, element\_operator 28/0, attempted\_step 70/0, steady\_species\_rows 195/0, krylov\_and\_dogleg 334/0, diffusion\_tests 35/0, carrier\_returned\_state\_acceptance 36/0, carrier\_reference\_scales 14/0, carrier\_constraint\_attribution 10/0, species\_masses 9/0, steady\_completion\_flag 3/0; grid\_and\_gates 198/1, the one FAIL the stagnation row on its old configuration (the suite was started before the script was re-anchored), and the re-anchored output\_state\_consistency.sh 4/4 on its own afterwards. make check (advisor, 2026-09-12 10:35 KST, the shared build/ rebuilt from this tree with the gfortran on PATH, conda-forge 16.2): **REGRESSION PASS (all cases byte-identical)**, 64 file comparisons PASS with data identical. A first launch of the gate with the system gfortran 13.1 on PATH was stopped before its verdict and is not counted: the goldens are snapshots of the 16.2 build and a verdict against another compiler would say nothing about this change.

## Reported by the workers and not fixed

- Q2: the out-of-bounds outcome of the element step (element\_step\_out\_of\_bounds) is not reachable on a synthetic column of the suite's size (the drift flux vanishes at both ends of the composition axis in the discrete operator), so the new rows cover the solve-failed and nonfinite outcomes only; the one measured out-of-bounds excursion of record is P1's, on the atomic reload.
- Q3: grid\_and\_gates row mass\_flux\_spread\_recomputed\_from\_output failed once on the worker's control run ("no\_dump\_line") and passed on its measured run and on the advisor's; the P17 note on two concurrent invocations of that suite sharing something outside EXHALE\_TEST\_OUT stands.

**The oxygen rows in the physics document (2026-09-12, user report)**

The user read page 18 of docs/EXHALE\_physics\_and\_algorithms.pdf and found that "The molecular rows" wrote the four  $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$  balances and nothing of the oxygen-bearing molecules. Confirmed against the code: with `thereis_oxychem` the cell system of `System_HeH_mol_metals` carries two more unknowns,  $n_{\text{OH}}/n_0$  and  $n_{\text{H}_2\text{O}}/n_0$ , whose balances `oxygen_carrier_rows` writes on `O1/O1r`, `O2/O2r`, the three  $\text{H}_2\text{O}$  photolysis channels and the OH photolysis, with the  $\text{H}_2$  row gaining the exchange with the water cycle, free atomic oxygen closing the family ( $n_0 - n_{\text{OH}} - n_{\text{H}_2\text{O}} - n_{\text{CO}}$ ) and the H closure gaining  $n_{\text{OH}} + 2 n_{\text{H}_2\text{O}}$ ; the reverses come from detailed balance on the NIST-JANAF Shomate Gibbs energies; the  $\text{O}_4 + \text{O}_6$  pair is one channel with  $\text{O}(1\text{D})$  eliminated exactly; CO is not a row of the cell system but a transported carrier destroyed one-sidedly by  $\text{He}^+ + \text{CO}$  (Langevin 1.6e-9, UMIST RATE22 4068) and by Lyman-Werner photodissociation with the Visser et al. (2009) shielding, the  $\text{He}^+$  row taking the  $\text{He}^+$  it consumes at the frozen CO of the sweep. A paragraph "The oxygen rows" with the two balance equations now follows "The molecular rows" in docs/physics\_overview/ionization.tex, the oxygen-option subsection cross-references it, the carrier-transport subsection got a label, and three references were added (Baulch et al. 2005, Chase 1998, Millar et al. 2024). Rebuilt with `latexmk`: 58 pages, no undefined citation or reference, no overfull box.

**ESWENO assessment memo (2026-09-12, user question)**

The user asked what the energy-stable WENO schemes gain and pointed at the three papers under references/. New memo docs/esweno\_assessment\_20260912.{tex,pdf} (5 pages), written from the published versions of Yamaleev & Carpenter (2009a, JCP 228, 3025; 2009b, JCP 228, 4248) and Fisher et al. (2011, JCP 230, 3727) and from `src/modules/states/Reconstruction.f90`: the energy estimate comes from an added nonlinear dissipation term (2009a Eqs. 9, 35 to 37) whose proof needs only  $0 \leq w \leq 1$  and  $\sum w = 1$ ; the new weight functions (Eqs. 18, 21, 22) with  $\epsilon = O(\Delta x^2)$  hold the design order at smooth extrema and are separable from the estimate; the proof covers linear hyperbolic systems in characteristic form on periodic grids, with no proof for the Euler equations or a component-wise reconstruction (their Sections 5 and 6.2); the 2011 closures keep it on a finite domain with nonuniform flux points near the boundaries; the cost is 15 to 20 percent (2009b Table 1). EXHALE's WENO3 already uses the 2009a weights and floor and not the dissipation term, and three properties of the code (finite-volume state reconstruction on a stretched grid, component-wise, nonlinear Euler) keep the estimate from carrying over. No code change; nothing measured on this code attributes an instability to the reconstruction's symmetric part.

**The continuous transfer equations in the physics document (2026-09-12, user request)**

The user asked for the flux and rate formulas of Caldiroli et al. (2021) Eqs. (3) and (4) in docs/EXHALE\_physics\_and\_algorithms.tex, to be added only after checking them against what the code computes. Checked in `photoionization_field_at_cell_HHe` (`util_ion_eq.f90`): the outside column times cross sections gives `tau_out`, the cell's own densities give `dtau`, the working flux is  $F_{\text{XUV}} * \text{cell\_mean\_attenuation}(\text{tau\_out}, \text{dtau}) * \text{opa\_pf}(j)$ , the rate is  $\sum(\text{int\_f} * \sigma / E * dE) * 1e-18 * \text{erg2eV}$  and the one-particle heating  $\sum(\text{int\_f} * (1 - E_{\text{th}}/E) * f_{\text{heat}} * \sigma * dE) * 1e-18$ , contracted with the densities in

photoheating\_of\_cell; the discrete equations already in the document are these lines term by term. Added at the head of "Transfer along the radial ray": the continuous form  $F_E(r) = F_E(a) \exp(-\tau_E)$ ,  $\tau_E = \sum_a \sigma_a \int_{r_{\text{top}}}^r n_a dr'$ ,  $\Gamma_a$  and  $H(r)$  as Caldiroli's Eq. (3), with the one difference stated (the column starts at zero at the outer edge of the grid,  $r(N+N_g)$ , not at  $a$ ). Rebuilt: 59 pages, no undefined reference, no overfull box.

### FUV bands B3 and B4 integrated from the spectrum file (2026-09-12, user instruction)

The user asked why Stellar LW flux and Stellar FUV B3/B4 flux are separate inputs when a spectrum is given. READ from the code: with Spectrum type: Load the Lyman-Werner flux was already integrated from the file when its key was absent, but B3 (1231-1450 Å) and B4 (1451-2304 Å) came only from their keys; the analytic spectrum types cannot state those bands (a power law is the XUV fit extrapolated, a Planck the photosphere) and the ionizing photon grid does not carry them, which is why the keys exist. Instruction: integrate B3 and B4 from the file as well; Ly-alpha (band B2) stays a key, a reconstructed line flux being the better number.

Done: sed\_read.f90's lyman\_werner\_band\_flux\_from\_sed became sed\_band\_integrated\_flux(w\_lo, w\_hi) (one trapezoid over the file's rows clipped to any band); input\_read.f90 calls it with the band edges of oxygen\_rates (fuv\_band\_lo\_A/hi\_A) for LW on any molecular run and for B3 and B4 with the oxygen chemistry on, when the key is absent and a file is loaded; a stated key, zero included, always wins (fuv\_b3\_flux\_stated, fuv\_b4\_flux\_stated, fuv\_b3\_from\_spectrum, fuv\_b4\_from\_spectrum in parameters.f90); the setup report names the source and a stated zero. Fixed beside it: the derivation ran AFTER the oxygen-chemistry warnings, so a loaded-spectrum run without the LW key was told "Stellar LW flux is zero" while the file supplied it; the block now runs before those warnings. Manual rows of the two keys updated (EXHALE\_user\_manual.tex, rebuilt).

MEASURED (private build, hot-Uranus oxygen case with inputdata/sed/hot\_uranus\_solar\_koskinen2022.txt loaded, 3 steps): with the keys removed the run reads B3  $4.696\text{E}+02$  and B4  $4.445\text{E}+05 \text{ erg cm}^{-2} \text{ s}^{-1}$  against a hand trapezoid of the same file  $4.696068\text{E}+02$  and  $4.444851\text{E}+05$  (LW without its key  $3.687\text{E}+02$  against  $3.687009\text{E}+02$ ); with the keys stated (694.8, 658782.2) the stated values stand; with B3 stated as zero the band is off and the report says so. No regression case combines a loaded spectrum with the oxygen chemistry, so the new path fires in none of them; make check on the merged tree is recorded below. Documentation of the rule (user instruction, same day): the physics document's "Bands stated by their integrated flux" paragraph (docs/physics\_overview/radiation.tex) now states the two sources of each band flux, the band integral (its Eq. for  $F_{\text{band}}$ ), why the analytic spectrum types cannot supply the bands, and why Ly-alpha stays a key; the manual's Stellar LW flux row, the Oxygen chemistry row and the Tier 2b section (docs/EXHALE\_user\_manual.tex) say the same at the keys. Both rebuilt.

### The H2 extent against Koskinen et al. (2022) Model A (2026-09-12, user instruction)

The user asked how far out Model A keeps its H2, then for a digitization of their Figure 8 and a full record of the present state in docs/koskinen2022\_model\_a\_comparison.tex. Their text: "H becomes the dominant species at around  $r = 2 R_p$ , but the density of H2 remains significant at all altitudes" (their Figure 16: outflow supplies the H2 the chemistry destroys). Figure 8 digitized a second time, independently of the 2026-09-05 readings (400 dpi, ticks from the print, RGB isolation, every  $0.1 R_p$ ): benchmarks/koskinen2022\_model\_a/model\_a\_fig8\_digitized.txt; the two readings agree to 0.06 dex (H2) and 0.02 dex (H, He);  $q_0(\text{H}_2)$  0.837 and  $q_0(\text{H})$  0.028 recovered against the stated

0.84 and 0.026.  $f(\text{H}_2)$  of Model A: 0.93 / 0.69 / 0.53 / 0.43 / 0.39 at 1.2 / 1.5 / 2.0 / 3.0 / 4.0  $r_{\text{base}}$ ,  $f = 0.5$  at 2.15  $r_{\text{base}}$ , never below  $1e-2$ .

New section 5 of the memo ("The  $\text{H}_2$  extent"), figure docs/figures/k22\_model\_a/h2\_extent.pdf (docs/k22\_h2\_figure.py), the two stationary-route states pinned under benchmarks/koskinen2022\_model\_a/carrier\_reload\_state. MEASURED: the matched run matched\_hnu\_minus\_I (spherical domain to 7.24  $r_{\text{base}}$ , Rate/4 +  $\dot{M}$ ) has Model A's shape, no front,  $f(\text{H}_2)$  0.88 / 0.59 / 0.41 / 0.31 / 0.27 at the same radii, i.e. 0.05 to 0.12 below, and  $n(\text{H}_2)$  a uniform 0.23 to 0.27 dex below theirs from 1.2 to 7  $r_{\text{base}}$ . The carrier reload fixture of the solver series (backup/regression/carrier\_elem\_newton) is NOT that configuration: Roche domain to 4.71  $r_{\text{base}}$ , power-law spectrum at 0.048 au around a 1.2  $M_{\text{sun}}$  star, Rate/2 +  $\dot{M}/2$ , He  $2^3\text{S}$  on, no ionization transport, no conduction, default atomic rates (a first reading listed five differences from a diff against the superseded root input.inp; corrected against matched\_hnu\_minus\_I/input.inp), pinned from a B5g state; in it  $f(\text{H}_2)$  is zero above 1.1  $r_{\text{base}}$  ( $f = 0.5$  at 1.072,  $1e-2$  at 1.097), 12 bounded passes move that to 1.076 / 1.103, and the fixed-wind fixed point fills the domain (0.97 / 0.91 / 0.76 / 0.51 / 0.39). So the "walking  $\text{H}_2$  front" of sections 8 and 9 is a statement about the fixture, not about EXHALE's Model A solution; the next measurement is the fixture re-pinned on the matched keys from the matched run's state.

The user caught, from the figure, that the first version of the comparison was drawn on benchmarks/koskinen2022\_model\_a/output, the superseded Roche-domain state of the morning of 2026-09-05 (top 4.72  $r_{\text{base}}$ ), and that the fixture's domain differs from Model A's: the section and the figure were redrawn on matched\_hnu\_minus\_I, both figure scripts now default to that run, and benchmarks/koskinen2022\_model\_a/README.md (new) says which directory is which.

Gate of the tree carrying the band-flux change, the  $\text{H}_2$ -extent memo and the ESWENO memo: make check (advisor, 2026-09-12 15:38 KST, the shared build/ rebuilt with the gfortran on PATH): **REGRESSION PASS (all cases byte-identical)**, 64 file comparisons PASS with data identical.

### The Model A fixture re-pinned on the matched configuration (2026-09-12, user instruction "proceed")

New fixture backup/regression/carrier\_model\_a\_newton (README there): the matched run's input.inp with the stationary keys appended and its final state as the IC pair. Two changes were needed for it to run:

- input\_read.f90: the refusal of Ionization transport: True with Solver: Newton and no Coupled carrier solve (2026-09-10) is lifted. READ in ionization\_equilibrium.f90: the x\_hp\_fixed block imposes the transported proton fraction on every sweep whenever the carriers are transported and the state is a restart or carries a background, so in the partitioned alternation (P6) the hydrodynamic solve's sweeps are handed the transported proton and the carrier relaxation transports it; the refusal's premise (the last sweep re-solving the proton locally) held for the plain Newton finish it was written for and not for the alternation. MEASURED on the fixture: over 12 passes  $x(\text{H}^+)$  moves by at most 0.011, downward, not to the local root. Manual key row and README\_HOWTO.md updated.
- src/utis/map\_state\_to\_grid.py (new; closes the interp\_ic.py housekeeping item): the matched state was written on the 2026-09-05 grid and the mixed grid moved on 2026-09-10 (3414478) by up to  $4.4e-4$  in the

centers, which load\_IC refuses at  $1e-10$ . The tool maps a state onto another grid's centers (rho, p, heat, cool in log against  $\ln r$ ; v, T linear; every species as its ratio to the hydrogen nuclei, so He/H is exact in every mapped cell; the loader's He/H check at  $1e-6$  failed on a first version that interpolated the densities themselves,  $1.04e-5$ ). Round trip onto the old centers: rho, p, T to  $2e-4$ , v to  $5e-3$ .

MEASURED (12 bounded passes, EXHALE\_JFNK\_MAXIT=40, 8 threads, 3 to 7 s a pass, hydro info=0 every pass): the H+ carrier row falls  $2.1e-1 \rightarrow 7.4e-4$  (gate  $1e-5$ ), the H2 row ends at  $7.0e-4$ , and f(H2) moves TOWARD Model A: 0.884  $\rightarrow$  0.930 at 1.2  $r_{\text{base}}$  (Model A 0.933), 0.588  $\rightarrow$  0.650 at 1.5 (0.691), 0.408  $\rightarrow$  0.431 at 2.0 (0.530), unchanged above 3; f = 0.5 from 1.676 to 1.785  $r_{\text{base}}$  (Model A 2.15); no cell below  $1e-2$ . The confined H2 of the solver fixture was that fixture's configuration. State pinned as benchmarks/koskinen2022\_model\_a/matched\_fixture\_states/pass12, figure h2\_extent.pdf redrawn with it, memo section 5.4.

FOUND, reproduced, NOT diagnosed: a 60-pass run stops at pass 13 on the stagnation ending because the hydrodynamic energy row of cell 1 reads  $2.073e-4$  (tolerance  $1e-6$ ) on every pass while the solve's own residual there is  $-1.0e-11$ ; with Conduction: False it is  $4e-9$ . The conduction source at cell 1 is  $7.2e-5$ , so the certification's assembly (after the outer loop's own sweep) and the solve's disagree by 0.5 percent of it. Candidates: the ghost cell's particle count behind the base-face conductive flux, formed from different ghost compositions on the two paths, or the sweep's local re-resolution of the ghosts. Recorded in docs/TO\_BE\_DONE.md and docs/ISSUES\_20260909.md section 3.7 with the instrumentation it needs. Because the joint progress measure (Q1) is dominated by that constant entry (207 tolerances against the species rows' 74), the matched fixture cannot be run past the stagnation guard until it is removed.

Gate of the tree carrying the lifted refusal and the state mapper: make check (advisor, 2026-09-12 18:55 KST): **REGRESSION PASS (all cases byte-identical)**, 64 file comparisons PASS with data identical.

#### The base cell's energy row: diagnosed and fixed (2026-09-12, user instruction)

Instrumented on a private copy of the tree: the JFNK's final residual evaluation and the outer loop's certification assembly were captured on ONE state and subtracted cell by cell. At cell 1 every input (particle count, temperature, heating, cooling) was identical and only the energy residual differed ( $3.5e-7$ ); the conduction source of cell 1 was  $5.406e-5$  in the JFNK against  $5.371e-5$  in the outer loop; and the one input that differed was the particle count of ghost cell -1 (JFNK 0.545985, outer 0.546468,  $8.8e-4$  relative), traced to the ghost DENSITY the JFNK's count was formed from: 1.22394867 (the ghost of the state that entered the sweep pass) against 1.22503037 (the ghost after the pass's own Apply\_BC). newton\_residual computed rho and the particle count at the top of each sweep pass and refreshed the ghosts at its end (base\_boundary\_states reads n\_part\_cell1 of the composition the sweep just left), then handed assemble\_residual the refreshed u with the stale count; the base-face conductive flux, formed from the ghost temperature  $W(3)/n_{\text{part}}$ , was one composition stale. The certification forms the count from the state it is handed and was right. The ghost-fill hypothesis (Apply\_BC missing in the outer loop) was tested first and refuted (ghost state identical to  $1e-11$ ).

Fix (steady\_newton.f90, newton\_residual): after the sweep loop the density and the particle count are re-formed from the refreshed ghosts before the assembly. MEASURED on the matched fixture: the cell-1 energy row in the certification  $7.8e-9$  at pass 1 and  $3.5e-9$  at pass 3 (was  $2.073e-4$ ); the 60-pass run no longer stops at pass 13 and spends

its budget with the H2 row at  $4.1\text{e-}4$  (cell 500, the top) and the H+ row at  $3.7\text{e-}4$  (cell 135), hydro info=0 every pass, f(H2) at 1.2 / 1.5 / 2.0 / 3.0 / 4.0 r\_base 0.931 / 0.660 / 0.460 / 0.346 / 0.296 against Model A's 0.933 / 0.691 / 0.530 / 0.430 / 0.389 (the loaded marching state 0.884 / 0.588 / 0.408 / 0.310 / 0.267),  $f = 0.5$  at 1.851 r\_base (Model A 2.15), x(H+) 0.061 / 0.133 at 2.0 / 3.0 (Model A 0.020 / 0.080). State pinned as matched\_fixture\_states/pass60, figure and memo section 5.4 updated. Without conduction or viscosity the count enters no row: wasp\_full\_newton reload data rows identical to the control; krylov\_and\_dogleg 334/0, steady\_species\_rows 195/0, certification 84/0, carrier\_returned\_state\_acceptance 36/0, steady\_completion\_flag 3/0. make check recorded below.

Gate of the tree carrying the ghost particle-count fix: make check (advisor, 2026-09-12 22:57 KST): **REGRESSION PASS (all cases byte-identical)**, 64 file comparisons PASS with data identical.

## 10. PLAN\_20260913: the findings of the two reviews of 2026-09-12 (2026-09-13)

Plan of record: docs/PLAN\_20260913\_review\_fixes.md. User instruction: read docs/code\_bug\_optimization\_review\_20260912.md and docs/parallel\_code\_review\_20260912.md, confirm each finding in the code, plan the corrections and carry them out, running the binary in parallel where that shortens the verification. Every finding (B1 to B6, P1 to P3) was confirmed in the source before the plan was written (the confirming lines are in the plan's table); items S1 to S9. Every number below is MEASURED on this tree unless marked READ.

- **S1 (B1, B2, O3): the band integral of a loaded spectrum.** sed\_read.f90: sed\_band\_fluxes(nb, w\_lo, w\_hi, F, covered) replaces sed\_band\_integrated\_flux; the file is read once for the three bands, validated as it is read (positive, finite, strictly increasing wavelengths; finite nonnegative flux; a malformed row stops the run with the row named), and each band gets the exact integral of the piecewise-linear table (the flux interpolated at the clipped edges before the trapezoid); covered(b) says whether the table reaches both edges, and input\_read (band\_from\_spectrum) STOPS when a band the run needs is not covered, naming the key (a stated zero is the run's decision; a silent zero is not). MEASURED on the hot-Uranus oxygen case with the Koskinen file: B3  $4.686\text{E}+02$  and B4  $4.443\text{E}+05 \text{ erg cm}^{-2} \text{ s}^{-1}$  against the exact piecewise-linear integrals  $4.685913\text{E}+02$  and  $4.443075\text{E}+05$  (the entry text gave  $4.696\text{E}+02$  and  $4.445\text{E}+05$ , 0.2 and 0.05 percent high). New test fuv\_band\_quadrature in spectrum\_type (linear spectrum on unequal nodes with edges inside segments, on nodes and at the ends, exact to  $1\text{e-}12$ ; one covering segment; constant spectrum; an uncovered band): 14/0. wasp121b\_sed\_sensitivity moved to the new routine with its own transcription corrected the same way (spectrum\_type 170/0 over the whole suite). docs/input\_schema.md, the manual's key rows and the physics document's band paragraph say the new rule.

- **S2 (B3, B4, O2 counters): the thermochemical closure contract.** equilibrate\_chemistry\_at\_fixed\_conserved\_state starts with `ok = .false.` and sets it only when a cycle's temperature increment is below `chem_cycle_tol` AND the sweep ledger reports no nonfinite and no off-simplex cell AND `p`, `T`, `heat`, `cool`, `eta` are finite with 'p

0,  $T > 0$ ; the reason (converged, cycle budget spent, nonfinite composition, a cell left the element simplex, not admissible) and the last increment are returned and the run-wide

counts kept (`n_chem_closure_cycles`, `n_chem_closures_reached`, `n_chem_closures_refused`). The relaxation's refusal branch was already restoring the trial. Test knob `chem_cycles_cap_for_test`; four new rows in `carrier_retry` (an exhausted closure refuses the trial, names the spent budget, keeps no step, hands back the entry composition bit for bit). One existing row was restated: on the synthetic column the pass after the refused trials is refused by the CHEMISTRY (one molecular clamp, budget violation  $6.5\text{e-}4$ , on the background seeded at that state), which the old contract had hidden; the row now asserts what the history can be judged by (the pass is not refused at the interval and the history stays certifiable), and the section-11 pass, prepared on the background of the composition it relaxes from, keeps its two steps as before. `carrier_retry` 142/0. On the Model A fixture (3 passes) every closure is reached: pass lines identical to the day before.

- **S3 (B5, B6): the state mapper.** `src/utils/map_state_to_grid.py` validates finite, positive, strictly increasing radii in both source files and the target, their agreement to  $1\text{e-}12$ , the column labels, a positive hydrogen-nucleus density, and that every physical target center lies inside the source's rows (ghosts included: the lower ghosts carry the inflow reservoir composition and are interpolated like every other row, never rebuilt from the physical cells; only a target ghost may lie outside and is then extrapolated linearly in  $\ln r$ ); a violation is refused with exit 2 and nothing written. The output is an initialization seed: `# coupling: mode=init t_phys=0 certified=F ...`, the source line kept as `# mapped-from-coupling: .` New suite `src/tests/state_mapper` (identity mapping exact; a  $2\text{e-}4$ -shifted grid round trip  $1.8\text{e-}5$  on a 400-row column; He/H invariant to  $1\text{e-}12$ ; mismatched species radii, out-of-support physical targets and non-monotone radii refused; ghosts extrapolated; metadata rewritten): 17/0. The Model A fixture's IC/ regenerated with it (physical rows identical to the earlier mapping, ghosts within  $1\text{e-}4$ ; the 3-pass run reproduces the day-before pass lines).
- **S4 (P1): the continuation dump.** `cce_dump_paths_read` reads the two environment paths once, serially, from `input_read`; `write_continuation_` dump takes the path by argument, keeps every read and write of the once-only flag inside the critical region, sets the flag only after the file is open, and records the cell (`ieq_cell%jcell`, a new field set by the sweep; dump format 2, the probe reads 1 and 2).
- **S5 (P2): explicit initialization.** `keq_H_H_to_H2` no longer builds the H2 thermochemistry table lazily (it stops with a message if the table is not ready); the table is built serially by the main program, by the sweep's serial prologue (`ioniz_eq`) and by `carrier_set_init`, so every entry point that precedes a parallel read builds it. The Newton solver's EXHALE\_FORCE\_HYBRD1 switch is read once by `newton_solver_read_environment` from `input_read`; a driver that never calls it runs with the switch off. `init.f90` switches dynamic team adjustment off, reports the team obtained (an `omp single` block, the master construct being deprecated since OpenMP 5.1), says whether the count came from `OMP_NUM_THREADS` or the default `min(cores, 16)`, and its coverage comment now lists the present parallel regions (the old thread line was cut by its fixed format and said "default" whatever the source).
- **S6 (P3): the BLAS thread policy.** New `blas_thread_policy.f90`: the OpenBLAS entry points are resolved at run time (`dlopen/dlsym`, `-ldl` on the GNU link and in every test script), the count is set to 1 unless

OPENBLAS\_NUM\_THREADS is stated, and the setup report says what was found and set; a build without OpenBLAS is left alone. MEASURED on the wasp\_full\_newton reload at 8 OpenMP threads: 16.33 s with 1 BLAS thread against 16.32 s with 8 (the library's default here follows OMP\_NUM\_THREADS; unset, it is the machine's 72): the band factorizations gain nothing from a BLAS team, and the policy costs nothing.

- **S7: verification, in parallel.** Suites on the private build, run concurrently: spectrum\_type 170/0, carrier\_retry 142/0, carrier\_returned\_state\_acceptance 36/0, attempted\_step 70/0, certification 84/0, steady\_species\_rows 195/0, krylov\_and\_dogleg 334/0, element\_operator 28/0, species\_masses 9/0, steady\_completion\_flag 3/0, grid\_and\_gates 198/0, state\_mapper 17/0; the Model A fixture 3 passes; the two BLAS timing runs; make check (the full matrix, OMP\_NUM\_THREADS=1, run on the merged tree while the suites ran): REGRESSION PASS, every case byte-identical against the goldens, as the plan required (no option defaults changed; the band integral is not exercised by the gate's power-law spectra, the closure contract reports the same result on every gate case).
- **S8: measured-first items, not changed** (O1, O2 beyond the counters, A, B, C, E of the parallel review): recorded in docs/TO\_BE\_DONE.md with the measurement each needs; the counters of S2 and the team/BLAS report of S5/S6 are the instrumentation for them.

## Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-09-11)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines (SLOC)* exclude blank lines and comments (the comment stripper is string-aware, so a ! inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is src/utils/codesize.py, which reprints every number of this appendix and the byte-identical list of Table 11. Measured 2026-09-11, i.e. after the whole of stage 1 and the stage-2 series through item P19 of section 8: the acceptance and certification contracts, the attempted-step controller, the partitioned stationary solver, the oxygen and carbon chemistry, the molecular infrared bands and the test suites those series brought with them.

**What the script counts as *core*.** Every .f90 file under src/ except those in src/modules/wind\_ae/, which is the ported Wind-AE IC generator and is listed separately. That rule puts the standalone drivers under src/tests/ in the core figure, and they are now a large part of it, so the table splits the core row into the production build and those drivers. Only the production files are compiled into EXHALE.x; each test suite builds its own driver from src/tests/<suite>/run.sh.

|                                      | Fortran files | code lines (SLOC)                |
|--------------------------------------|---------------|----------------------------------|
| Original ATES (ATES-Code-main)       | 47            | 3 514                            |
| EXHALE core (excl. Wind-AE port)     | 169           | 71 503                           |
| of which the production build        | 95            | 50 762 ( $\approx 14.4\times$ )  |
| of which the src/tests/ drivers      | 74            | 20 741                           |
| net change (core)                    | +122          | +67 989 ( $\approx 20.3\times$ ) |
| Wind-AE IC-generator port (wind_ae/) | +36           | +4 756                           |
| EXHALE total                         | 205           | 76 259 ( $\approx 21.7\times$ )  |

**Line-by-line diff, core** (comments + blanks excluded; whitespace/case normalized): **69 179** code lines added in EXHALE (new modules + the new side of modified lines) and **1 190** removed from the original (deleted + the old side of modified lines), for **70 369** total changed lines. At the file level, of the 47 original files **38 were modified**, **8 are unchanged** (all eight byte-for-byte; the list is Table 11) and **one was removed**: flux/speed\_estimate\_HLLC.f90, Toro's adaptive  $p_*$  estimate, was used by Num\_Fluxes.f90 and never called, because the HLLC branch builds its own Davis/Einfeldt speeds inline. **123 new core modules** were added. Of those, 49 are part of the simulation build:

| New module                                                   | Purpose                                                                                                                                                                  | SLOC |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| time_step/steady_newton.f90                                  | stationary solve of the hydrodynamic rows: residual $F(Y) = 0$ on the physical cells, Newton/Krylov with pseudo-transient continuation, line search, best-iterate ledger | 9435 |
| lower_atmosphere/diffusive_photochemistry.f90                | vertical transport of the molecular carriers ( $H_2$ , OH, $H_2O$ , CO) solved together with the chemistry that makes and destroys them                                  | 2485 |
| lower_atmosphere/molecular_infrared_data.f90                 | band and line data of the molecular infrared coolant (generated by cooling_data/molecular_infrared_bands.py)                                                             | 2182 |
| functions/binary_element_diffusion.f90                       | conservative binary H/He element transport (stage-resolved pair coefficients, donor-cell drift, metals slaved to H)                                                      | 1898 |
| time_step/attempted_step.f90                                 | the attempted-step controller: one checkpoint, one trial, one adoption boundary, and the rollback contract                                                               | 1608 |
| time_step/certification.f90                                  | one evaluator of the equations a run actually solves, and the contexts that read its verdict                                                                             | 1273 |
| nonlinear_system_solver/constrained_chemical_equilibrium.f90 | chemical equilibrium of the molecular H/He network in logarithmic number densities, element conservation as explicit residual rows                                       | 1173 |
| lower_atmosphere/h2_self_shielding_table.f90                 | $H_2$ Lyman–Werner dissociation cross section with overlapping lines, and its two branching ratios                                                                       | 952  |
| time_step/energy_semi_implicit.f90                           | implicit source step for the thermal energy of a cell, at fixed composition and heating                                                                                  | 545  |
| time_step/steady_residual.f90                                | finite-volume stationary residual $R = du/dt$ and its norms, shared by the diagnostic, the convergence monitor and the stationary solver                                 | 509  |
| time_step/viscous_conduction.f90                             | Navier–Stokes viscous momentum force, its dissipation, and thermal conduction (Crank–Nicolson)                                                                           | 504  |
| radiation/excited_hydrogen.f90                               | non-LTE $H(n=2)$ 2s/2p equilibrium (Christie, Arras & Li 2013) and the Balmer feedback                                                                                   | 474  |
| lower_atmosphere/element_inventory.f90                       | one stoichiometric map of the H, He, O and C nuclei of a cell, and the three constraints a composition state is judged by                                                | 448  |

| New module                                        | Purpose                                                                                                                                             | SLOC |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| radiation/charge_exchange.f90                     | charge-exchange network (Huang et al. 2023, Table 4)                                                                                                | 446  |
| files_IO/lower_atmosphere_profile.f90             | lower-atmosphere profile reader: log- $p$ interpolation, elemental reservoirs, $K_{zz}(p)$                                                          | 367  |
| states/caloric_eos.f90                            | caloric equation of state: internal energy density, pressure and temperature, with the H <sub>2</sub> level ladder                                  | 340  |
| lower_atmosphere/molecular_reaction_heat.f90      | net chemical heat of the collisional reactions of the H <sub>2</sub> /He network                                                                    | 339  |
| lower_atmosphere/oxygen_rates.f90                 | rate coefficients and thermodynamic data of the oxygen-hydrogen chemistry, with their validity ranges                                               | 316  |
| functions/element_census.f90                      | one definition of the elemental and charge content of a composition state, and the invariant checks built on it                                     | 307  |
| nonlinear_system_solver/System_HeH_mol.f90        | coupled H/He + molecular (H <sub>2</sub> /H <sub>2</sub> <sup>+</sup> /H <sub>3</sub> <sup>+</sup> /HeH <sup>+</sup> ) system, turnover-scaled rows | 303  |
| radiation/electron_energy_degradation.f90         | degradation of a fast photoelectron in a partly ionized H/H <sub>2</sub> /He gas: the heat fraction and the ionizations of each target              | 287  |
| radiation/lya_rt.f90                              | Ly $\alpha$ transfer by the escape-probability closure, evaluated in line each step                                                                 | 257  |
| lower_atmosphere/molecular_infrared_cooling.f90   | infrared exchange of the molecular layer: H <sub>2</sub> O and CO vibration-rotation bands, H <sub>2</sub> quadrupole and magnetic-dipole lines     | 231  |
| states/base_boundary.f90                          | the lower boundary as a characteristic condition at the face, the number of conditions set by the Mach number there                                 | 199  |
| functions/h2_photo_channels.f90                   | H <sub>2</sub> photoabsorption resolved into mutually exclusive final-state channels, with one stoichiometric table                                 | 199  |
| lower_atmosphere/lyman_werner.f90                 | H <sub>2</sub> photodissociation in the Lyman and Werner bands (the Solomon process)                                                                | 195  |
| radiation/opacity_models.f90                      | opacity-model dispatcher for the photoionization cross sections (A/C/P/T)                                                                           | 180  |
| files_IO/metals_input_read.f90                    | parse metals.inp: trace-metal abundances at runtime                                                                                                 | 177  |
| functions/composition.f90                         | single point turning ( $\rho, f_{sp}$ ) into every species density, $n_e$ and $n_{tot}$ , and converting pressure and temperature                   | 162  |
| flux/species_face_flux.f90                        | advection of the species mass fractions on the same faces, with the same mass flux, as the density                                                  | 162  |
| time_step/hydrodynamic_rows.f90                   | the hydrodynamic rows of the stationary residual in double precision, instantiated from the kind-generic text                                       | 161  |
| nonlinear_system_solver/System_HeH_metals.f90     | coupled H/He+metals ionization (+ analytic Jacobian)                                                                                                | 150  |
| lower_atmosphere/mol_rates.f90                    | molecular reaction rates (Koskinen et al. 2022, Table 1)                                                                                            | 150  |
| lower_atmosphere/h3p_cooling.f90                  | optically thin H <sub>3</sub> <sup>+</sup> infrared cooling (Miller et al. 2013)                                                                    | 148  |
| nonlinear_system_solver/System_HeH_mol_metals.f90 | molecular + metals merged ionization system                                                                                                         | 141  |
| nonlinear_system_solver/newton_solver.f90         | damped Newton with an analytic Jacobian for the ionization systems, with the MINPACK hybrd1 retry                                                   | 128  |
| init/species_table.f90                            | metadata table of the metal stages and of the base, molecular and oxygen species                                                                    | 128  |
| lower_atmosphere/co_photodissociation.f90         | CO photodissociation on the 912–1201 Å band                                                                                                         | 127  |
| nonlinear_system_solver/ion_residual_core.f90     | shared ionization-residual rows and analytic-Jacobian pieces of the System_* family                                                                 | 127  |

| New module                                                | Purpose                                                                                                     | SLOC   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|
| lower_atmosphere/water_photolysis.f90                     | H <sub>2</sub> O and OH photodissociation in the four far-ultraviolet bands of the oxygen option            | 111    |
| lower_atmosphere/co_self_shielding_table.f90              | CO self-shielding function $\Theta[N(\text{CO}), N(\text{H}_2)]$ (Visser, van Dishoeck & Black 2009)        | 110    |
| nonlinear_system_solver/ion_cell_state.f90                | named-field cell state of the ionization residuals, in place of the position-packed params                  | 93     |
| lower_atmosphere/lower_column.f90                         | analytic lower-atmosphere column from the 1-bar level to the base pressure (Koskinen et al. 2022)           | 90     |
| flux/low_mach_dissipation.f90                             | gated fourth-difference stress damping the two-cell velocity mode a contact-resolving flux leaves undamped  | 87     |
| files_IO/opacity_input_read.f90                           | parse opacity.inp                                                                                           | 85     |
| radiation/hydrogen_n2_rates.f90                           | atomic data and rate coefficients of the H $n=2$ manifold and of the Ly $\alpha$ line, in one place         | 76     |
| nonlinear_system_solver/System_HeH_TR_metals.f90          | merged He 2 <sup>3</sup> S+metals system                                                                    | 68     |
| lower_atmosphere/h2_vibrational_relaxation.f90            | the fate and the size of the vibrational energy left in an H <sub>2</sub> molecule, for both of its drivers | 55     |
| nonlinear_system_solver/constrained_equilibrium_probe.f90 | standalone re-evaluation of one saved constrained-equilibrium state, without the hydrodynamics              | 15     |
| production new-module total                               |                                                                                                             | 30 003 |

The other 74 are the standalone test drivers under `src/tests/`, which did not exist in ATES at all. They state what a routine is allowed to return and are run by suite, not by the production binary:

| Test driver                                             | What it asserts                                                                                 | SLOC |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------|------|
| tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90     | what the linear solve and the step geometry of the stationary solver are allowed to return      | 2675 |
| tests/steady_species_rows/steady_species_rows_tests.f90 | the unknown space of the stationary system, one row per transported balance                     | 1806 |
| tests/diffusion_tests.f90                               | the binary H/He element-diffusion operator: analytic limits and conservation                    | 1051 |
| tests/grid_and_gates/hydrostatic_residual.f90           | the momentum row returns zero on a hydrostatic state                                            | 967  |
| tests/carrier_retry/carrier_retry.f90                   | the carrier-local checkpoint and the rejection-and-retry controller, on a real molecular column | 842  |
| tests/certification/certification_contexts.f90          | the certification evaluator and its contexts, against the predicates they replace               | 790  |
| tests/species_face_flux/species_face_flux_tests.f90     | species advection on the face mass fluxes, and the element rows of the diffusion operator       | 557  |
| tests/physics_probe/weno3_reconstruction_order.f90      | convergence order of the face reconstruction against the exact face value                       | 482  |
| tests/element_operator/element_operator_tests.f90       | the boundary the element transport operator is measured under                                   | 471  |
| tests/spectrum_type/loaded_sed_table_semantics.f90      | what a row of a loaded SED table means, and what the production consumers make of it            | 417  |

| Test driver                                                                       | What it asserts                                                                                        | SLOC |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------|
| tests/attempted_step/attempted_step_<br>tests.f90                                 | the checkpoint and the step-size policy of the attempted-step controller                               | 415  |
| tests/a2_m1/a2_m1_rate_check.f90                                                  | the oxygen rate coefficients against their published sources                                           | 406  |
| tests/spectrum_type/balmer_band_<br>quadrature.f90                                | quadrature of the Balmer-continuum band integrals of<br>excited_hydrogen                               | 367  |
| tests/energy_update/conduction_floor_<br>tests.f90                                | what the Crank–Nicolson transport stage returns, refuses and conserves                                 | 360  |
| tests/physics_probe/metal_photoionization_<br>fits.f90                            | every metal photoionization cross section against the published fit, and<br>only where it is valid     | 343  |
| tests/element_census_tests.f90                                                    | the elemental and charge invariants of a composition state                                             | 340  |
| tests/energy_update/energy_update_<br>tests.f90                                   | the residual-controlled temperature solve of the implicit energy source<br>step                        | 334  |
| tests/spectrum_type/loaded_sed_threshold_<br>edges.f90                            | the photon grid built from a loaded SED, and the rates formed on it                                    | 304  |
| tests/physics_probe/species_formation_<br>energy_table.f90                        | the formation-energy table of the energy ledger, and the reaction energies<br>that are its differences | 275  |
| tests/physics_probe/lya_band_<br>transmission.f90                                 | the stellar Ly $\alpha$ band reaching a cell through the atomic hydrogen column<br>above it            | 272  |
| tests/spectrum_type/wasp121b_sed_<br>sensitivity.f90                              | the production band quantities of the three WASP-121 b spectrum<br>candidates                          | 262  |
| tests/coupled_source_step/coupled_source_<br>tests.f90                            | the coupled temperature-composition source step                                                        | 260  |
| tests/physics_probe/lyman_werner_cell_<br>mean.f90                                | the Lyman–Werner rate a cell carries is the mean of the local rate over<br>that cell                   | 234  |
| tests/physics_probe/lya_beam_cell_mean.f90                                        | the stellar Ly $\alpha$ field of a cell is the beam mean over the cell’s own optical<br>depth          | 229  |
| tests/spectrum_type/balmer_continuum_<br>field.f90                                | the two scalar Balmer-continuum rates of H( $n=2$ )                                                    | 227  |
| tests/physics_probe/positivity_limiter_<br>scaling.f90                            | the positivity scaling of the reconstructed face states (Zhang & Shu 2010)                             | 223  |
| tests/carrier_returned_state_acceptance/<br>carrier_returned_state_acceptance.f90 | what makes a returned carrier state a solution, and what does not                                      | 222  |
| tests/grid_and_gates/photon_grid_<br>threshold_edges.f90                          | every active absorber’s threshold against the bin partition of the photon<br>grid                      | 213  |
| tests/physics_probe/photoionization_field_<br>substitution.f90                    | the order in which the field and the composition that attenuates it are<br>brought into agreement      | 209  |
| tests/adv_static_limit/stationarity_by_<br>the_mass_row.f90                       | which test decides that the state the post-process was handed is stationary<br>in a cell               | 199  |
| tests/physics_probe/co_destruction.f90                                            | the CO destruction rates, the shielding table, the deposited energies and<br>the cell mean             | 195  |
| tests/physics_probe/recombination_<br>coefficient_fits.f90                        | every recombination coefficient against the published fit                                              | 189  |
| tests/spectrum_type/photon_grid_n2_<br>floor.f90                                  | the photon grid built when the excited-hydrogen coupling is armed                                      | 184  |
| tests/carrier_reference_scales/carrier_<br>reference_scales.f90                   | the element a transported carrier is made of, and whether its chemistry<br>rows can be differentiated  | 181  |

| Test driver                                                                 | What it asserts                                                                                                       | SLOC |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------|
| tests/carrier_constraint_attribution/<br>carrier_constraint_attribution.f90 | which residuals belong to the solver and which to a constraint                                                        | 179  |
| tests/adv_static_limit/enthalpy_flux_of_<br>mass_divergence.f90             | the third term of the advected energy equation                                                                        | 178  |
| tests/grid_and_gates/free_outflow_<br>boundary.f90                          | the outer free-outflow boundary on a supersonic wind, and its ghost state                                             | 174  |
| tests/spectrum_type/planck_photon_<br>field.f90                             | the incident flux put on the photon grid for the photospheric blackbody type                                          | 173  |
| tests/a2_m2/a2_m2_kinetics_check.f90                                        | the kinetics quantities the oxygen option adds on top of the rate audit                                               | 163  |
| tests/physics_probe/lya_escape_<br>probability.f90                          | the escape probability of the two-level closure against the published slab solution                                   | 163  |
| tests/e1_h2/e1_h2_channel_check.f90                                         | the mutually exclusive H <sub>2</sub> photoabsorption channels                                                        | 161  |
| tests/adv_static_limit/upstream_caloric_<br>energy.f90                      | the upstream energy of the advected energy equation                                                                   | 154  |
| tests/physics_probe/photoionization_field_<br>self_consistency.f90          | the field one cell is solved in, against the composition that cell returns                                            | 148  |
| tests/grid_and_gates/photon_grid_<br>quadrature.f90                         | the photoionization rates of one unattenuated cell against the band quadrature                                        | 146  |
| tests/physics_probe/co_helium_ion_sink.f90                                  | He <sup>+</sup> + CO as a species sink, not only as a heat source                                                     | 145  |
| tests/physics_probe/riemann_wave_<br>speeds.f90                             | signal speeds at a face, and the state one conservative update with that flux leaves behind                           | 143  |
| tests/physics_probe/helium_level_cooling_<br>ledger.f90                     | which helium level each cooling channel of eval_cool is charged to                                                    | 139  |
| tests/physics_probe/heating_channel_<br>closure.f90                         | one assembly of the volumetric heating rate, and the closure of its channels                                          | 136  |
| tests/acceptance_classes/acceptance_class_<br>contract.f90                  | one meaning of the ionization acceptance classes for every consumer                                                   | 132  |
| tests/physics_probe/opacity_model_p_<br>ledger.f90                          | the radiative ledger of a cell whose cross sections are broadened by opacity model P                                  | 124  |
| tests/physics_probe/photoevent_energy_<br>ledger.f90                        | the energy an isolated photoabsorption event has to account for                                                       | 117  |
| tests/physics_probe/xuv_cell_mean_<br>attenuation.f90                       | the beam attenuation the photoionization rate of a cell is evaluated with                                             | 115  |
| tests/grid_and_gates/grid_window_<br>indices.f90                            | the two index statements that decide which cells a quantity is read from                                              | 101  |
| tests/physics_probe/h2_rovibrational_<br>identity.f90                       | the identities tying the H <sub>2</sub> partition function of the chemistry to the internal energy of the caloric EOS | 96   |
| tests/adv_static_limit/stationary_flux_<br>refusal.f90                      | the sensitivity of the advected energy correction to a non-stationary mass flux                                       | 91   |
| tests/physics_probe/ionization_threshold_<br>turn_on.f90                    | one definition of each ionization threshold, and every cross section turning on exactly there                         | 90   |
| tests/grid_and_gates/grid_width_<br>identity.f90                            | the cell-width and total-width identities of the radial grid                                                          | 83   |
| tests/physics_probe/h2_channel_detector_<br>ratio.f90                       | the measured proton-to-H <sub>2</sub> <sup>+</sup> ratio of Chung et al. (1993) against the four channels             | 83   |

| Test driver                                                               | What it asserts                                                                                               | SLOC   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|
| tests/physics_probe/fine_structure_escape_probability.f90                 | the escape probability of the ground-term fine-structure lines, and the depth it is fed                       | 80     |
| tests/physics_probe/atomic_mass_and_radius_constants.f90                  | the helium mass and the astronomical constants of the input readers, one definition each                      | 79     |
| tests/physics_probe/h3p_cooling_limits.f90                                | the limits, the continuities and the published values of the $H_3^+$ cooling rate                             | 79     |
| tests/physics_probe/assertion_report.f90                                  | one printed verdict line per assertion, and the failure count the driver exits on                             | 74     |
| tests/adv_static_limit/thermal_damkohler_limit.f90                        | the validity condition of the advection-corrected energy solve, as a function of the flow speed               | 72     |
| tests/a2_m2/a2_m2_g3_run_check.f90                                        | with every photolysis rate at zero, the solved O/OH/H <sub>2</sub> O partition is the chemical equilibrium    | 71     |
| tests/physics_probe/absorbed_fraction_switch.f90                          | $(1 - e^{-d})/d$ , the fraction of the photons entering a cell that the cell absorbs                          | 71     |
| tests/constrained_network_layout/transported_proton_constraint_layout.f90 | which species are unknowns of the continuation, and whether the system built is square                        | 71     |
| tests/test_columns.f90                                                    | synthetic columns whose species carry the mass density they are measured against                              | 67     |
| tests/physics_probe/h2_infrared_line_populations.f90                      | the LTE populations weighting the H <sub>2</sub> quadrupole lines are those of the caloric EOS ladder         | 65     |
| tests/physics_probe/wind_ae_bridge_composition.f90                        | the composition an EXHALE input.inp hands the Wind-AE solver                                                  | 54     |
| tests/physics_probe/voronov_carbon_ionization.f90                         | the electron-impact ionization of C <sup>0</sup> and C <sup>+</sup> against the published Voronov (1997) rows | 51     |
| tests/physics_probe/water_photolysis_lyman_alpha_yields.f90               | the H <sub>2</sub> O branching on the band that contains Ly $\alpha$ , against the published yields           | 49     |
| tests/physics_probe/h3p_recombination_branching.f90                       | the two product channels of $H_3^+$ dissociative recombination                                                | 44     |
| tests/energy_update/cooling_models.f90                                    | analytic cooling laws for the tests of the implicit energy update                                             | 43     |
| tests/species_masses/carrier_background_masses.f90                        | the mass the carrier transport gives each of its background collision partners                                | 36     |
| test-driver total                                                         |                                                                                                               | 20 741 |
| new-module total                                                          |                                                                                                               | 50 744 |

So the 123 new core modules contribute 50 744 of the 69 179 added code lines; the remaining 18 435 added (and 1 121 of the 1 190 removed, the other 69 being the deleted HLLC estimator) are edits spread across the 38 modified files. The ten heaviest, by added code lines, are the main driver (EXHALE\_main.f90, 3 498 added, 246  $\rightarrow$  3 625 SLOC: the marching loop, the attempted-step adoption boundary, the run-mode contracts and the written-state assertions), the ionization equilibrium and its diagnostics (ionization\_equilibrium.f90 2 274 and util\_ion\_eq.f90 1 803: the metal, molecular and oxygen networks, the acceptance classes, the photoionization field and the heating channels), the cooling module (Cool\_coeff.f90, 2 250 added, 442  $\rightarrow$  3 966 raw lines: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molecular rate coefficients), input parsing (input\_read.f90, 1 753), the state reader and writer (load\_IC.f90 1 063, write\_output.f90 928, write\_setup\_report.f90 751), the advection-corrected post-processor (post\_process\_adv.f90, 647) and the cross sections (cross\_sec.f90, 438). Those ten carry

15 405 of the 18 435. The rest of the edits are in the reconstruction, the fluxes, the boundary conditions, the grid, the time step, the energy equation and the `System_*` ionization systems, which wire the metals, the molecules, the opacity dispatcher, the Roche potential, the transonic/auto/Wind-AE IC, the excited-H and  $\text{Ly}\alpha$  physics, the element diffusion and the Brent/Newton solver upgrades into the original solver.

In total the EXHALE core is now  $\approx 20.3\times$  the original code base (3 514  $\rightarrow$  71 503 SLOC), or  $\approx 14.4\times$  counting only the production build (50 762 SLOC), touching almost all of the original files (38 of 47 modified, one deleted, plus 123 new); with the ported Wind-AE IC generator the full Fortran tree is 205 files / 76 259 SLOC ( $\approx 21.7\times$ ). Beyond Fortran, the tree also carries the Python tooling (GUI, the `EXHALE_transit.py` transmission post-processor, plotting/analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and 181 816 lines of documentation (187 md/tex files: the memos, the manual and this log) under `docs/`.

## Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (ATES/ATES-Code-main), **8 are byte-for-byte identical** in EXHALE (re-measured 2026-09-11): the numerics EXHALE reuses without touching, which are now the linear-algebra part of the included MINPACK nonlinear solver plus the gravity pre-evaluation. Table 11 lists them with their role. No file is identical after whitespace and case normalization but not byte-identical, so the normalized-unchanged set and the byte-identical set are the same eight.

Five files left the byte-identical set since the 2026-08-30 edition of this table, each for a documented reason rather than drift.

- `nonlinear_system_solver/hybrd.f90`, `hybrd1.f90` and `fdjac1.f90` carry a new `unit_step_floor` argument threaded from `hybrd` to `fdjac1`, which selects the forward-difference step of the Jacobian: `.false.` is MINPACK's own  $h = \varepsilon |x_j|$  with the exact-zero fallback, `.true.` floors it at unit magnitude,  $h = \varepsilon \max(1, |x_j|)$ , which is the rule for unknowns already scaled so that unity is their natural size (the logarithmic unknowns of the constrained chemical equilibrium). The `params` dummy also became assumed-size, so one transport array serves systems of different length. `hybrd1` passes `.false.` and is therefore unchanged in behavior.
- `states/Source.f90` gained the well-balanced option of 2026-09-10 (item N37 of the stage-2 solver program, `docs/ISSUES_20260909.md`): under `Well balanced: True` the gravitational source and the geometric pressure term are not evaluated here at all, because they cancel the equilibrium part of the momentum flux difference identically and `RK_rhs` assembles the row from what is left (Käppeli & Mishra 2014, J. Comput. Phys. 259, 199, eq. 2.26). The file also reads its pressure through `caloric_eos` instead of a constant  $\gamma$ .
- `radiation/J_inc.f90` carries the single-spectrum-type decision: `Spectrum type`: now states the spectrum of the whole photon grid, the XUV and the part below 13.6 eV alike, and no band is filled from a type the input did not select. The file also became the one definition of the eV-to-Hz conversion and of the Balmer-continuum head  $e_{\text{th,H I}}/4$ .

The two wave-speed estimators are no longer in the “effectively untouched” category either. `flux/speed_estimate_HLLC.f90` was deleted as dead code, as recorded above. `flux/speed_estimate_ROE.f90` was rewritten (99 code lines

added, 43 removed) into a star-state estimate that returns a status: the primitive-variable, two-rarefaction and two-shock branches of Toro (2009) chapter 9 are selected by the pressure ratio, the vacuum condition of his eq. 4.76 is tested first, and a face with no admissible star state is reported rather than given a square root of a negative ratio.

Table 11: The 8 Fortran files byte-identical between the original ATES (ATES-Code-main) and EXHALE (measured 2026-09-11), grouped by role.

| File                                                                                | Purpose                                                                |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <i>MINPACK linear algebra and constants</i> (Argonne; dir nonlinear_system_solver/) |                                                                        |
| qrfac.f90                                                                           | QR factorization via Householder transformations                       |
| qform.f90                                                                           | Accumulate the orthogonal factor $Q$ from the QR factorization         |
| r1updt.f90                                                                          | Rank-one update of the QR factorization                                |
| r1mpyq.f90                                                                          | Apply the Givens-rotation sequence of the rank-one update              |
| dogleg.f90                                                                          | Dogleg (trust-region) step combining Gauss–Newton and steepest descent |
| enorm.f90                                                                           | Euclidean norm of a vector (overflow/underflow safe)                   |
| dpmpar.f90                                                                          | Machine floating-point precision / range constants                     |
| <i>Grid setup</i>                                                                   |                                                                        |
| init/set_gravity_grid.f90                                                           | Pre-evaluate gravity at cell centers and interfaces                    |